# A PROOF OF THE ROY–WALDSCHMIDT THEOREM WITH PERIOD TERM, WITH EXPLICIT CONSEQUENCES AT CM AND BETA PERIODS

DRAFT — NOT FOR CIRCULATION

ABSTRACT. Let $G$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$, $K$ a number field, $\Lambda_K(G)$ and $\Omega_K(G)$ the $K$-spans of the logarithms of algebraic points and of the periods of $G$, $\Psi$ a $\overline{\mathbb{Q}}$-rational linear map on the Lie algebra of $G$, and $W \subset V$ subspaces of its target, $W$ defined over $\overline{\mathbb{Q}}$. Roy and Waldschmidt (1993) stated that, if $V$ contains no nonzero $\Psi$-image of an algebraic Lie subalgebra of the appropriate kind, then $\dim_K(V \cap \Psi(\Lambda_K(G))) \le (\alpha_K(G) - \kappa)\dim(V/W)$, where $\kappa = \dim_K(V \cap \Psi(\Omega_K(G)))$ is the period term. Their proof reduces to $K = \mathbb{Q}$ and then, to our knowledge (we could not consult Roy's paper), refers to a refinement of an argument of Roy that is not written out; the period-refined case $K = \mathbb{Q}$ is in print for hyperplanes (Waldschmidt 1988), for linear groups (Waldschmidt's book of 2000; Roy 1992) and with $\kappa$ replaced by 0 (Roy). We prove the theorem, with the sharper inner weight $\alpha_{K,\Psi}(G) \le \alpha_K(G)$ that appears inside their reduction (Theorem R), at two levels. (R-W): for every group and in every codimension, Theorem R follows from Waldschmidt's algebraic subgroup theorem with period term (1988, Thm. 4.1) as printed, by short arguments: subadditivity of the weight, a corner lemma for hyperplanes settled by Baker's theorem, an induction over subgroups, and a derivation of Wüstholz's analytic subgroup theorem from Theorem 4.1. For groups with a torus in codimension $\ge 2$ this uses Theorem 4.1 where one step of its published proof is only a pointer; there the proof is complete only relative to Theorem 4.1 as printed. (R-P): we write out that step of Waldschmidt's auxiliary function (multiplicities along $W$ with the period saving; our Lemma U) under a numerical condition. Lemma U is complete, and formally verified, for groups isogenous to $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times E_1 \times \cdots \times E_m$ with $E_j$ elliptic curves, and a proof sketch for general groups. Using Theorem 4.1 only through Lemma U, Theorem R is proved, modulo Lemma U, for every group when the relevant subspace is a hyperplane, for every group without a torus in every codimension, and for the remaining groups under a condition at their reduced subgroups; groups with a torus in codimension $\ge 2$ where this condition fails remain open at this level.

As applications we give explicit statements, with proofs, at periods of CM elliptic curves and of CM abelian varieties, i.e. at Beta values. For a nonzero eigen-period $P$ of a CM abelian variety the four exponentials conjecture holds at $t \in \overline{\mathbb{Q}}^\times P^{\pm 1}$ and $t \in \overline{\mathbb{Q}}^\times (P/\pi)^{\pm 1}$: for instance $2^{\varpi}$ and $3^{\varpi}$ are not both algebraic, nor are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, where $\varpi$ is the real period of $y^2 = 4x^3 - 4x$, nor $2^t$ and $3^t$ for $t = \Gamma(1/5)^2/\Gamma(2/5)$. The strong four exponentials conjecture holds at these $t$; it has elliptic analogues on CM curves, for instance $\wp(\pi/\varpi^2)$ and $\wp(\pi u/\varpi^2)$ are not both algebraic, where $\wp$ is the Weierstrass function of $y^2 = 4x^3 - 4x$ and $u$ the real elliptic logarithm of its point $(2, \sqrt{24})$. For two non-isogenous CM curves we obtain a bilinear statement and the linear independence over $\overline{\mathbb{Q}}$ of $\pi, \varpi, \Omega_\rho, \varpi\Omega_\rho$, where $\Omega_\rho$ is a period of $y^2 = 4x^3 - 4$; a quadratic relation among first-kind periods on the Jacobian of $y^7 = x(1-x)$ gives that 1, $B(\frac{2}{7}, \frac{2}{7})$, $B(\frac{3}{7}, \frac{3}{7})$, $B(\frac{3}{7}, \frac{3}{7})/B(\frac{1}{7}, \frac{1}{7})$ are linearly independent over $\overline{\mathbb{Q}}$. Further applications include Waldschmidt's Conjecture 42 at six classes of CM ratios ($\wp(\log 2/\varpi)$ and $\wp(\log 3/\varpi)$ are not both algebraic) and, from Theorem 4.1 in codimension $\ge 2$, that one of $e^{\pi^2}$ and $\wp(1/\pi)$ is transcendental. In the cases left open by the six exponentials theorems and complex conjugation (pairs with a conjugation-stable span at real $t$) none of these statements was, as far as we found, in print.

The main applications on $y^2 = 4x^3 - 4x$, including both four exponentials examples at $\varpi$, the strong four exponentials theorem and the elliptic example $\wp(\pi/\varpi^2)$, $\wp(\pi u/\varpi^2)$ (given the linear independence of 1, $\varpi$, $u$), are checked in Lean 4 with Mathlib (standard axioms only), relative to named classical inputs where indicated; the results on $y^2 = 4x^3 - 4$ in addition assume that $g_3$ of its period lattice is algebraic. Theorem R itself is not formalised. The other results are proved modulo published theorems and Lemma U.

*Date*: 31 October 2026 (draft; restructured around Theorem R).

## 1. Introduction

**The Roy–Waldschmidt theorem with period term.** Let $G$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$, $T_G$ its tangent space at the origin with its $\overline{\mathbb{Q}}$-structure, $\Omega(G) = \ker \exp_G$ its periods and $\Lambda(G) = \exp_G^{-1}(G(\overline{\mathbb{Q}}))$ the logarithms of its algebraic points. For a number field $K$ of degree $n$, $\Omega_K(G)$ and $\Lambda_K(G)$ are the $K$-spans of $\Omega(G)$ and $\Lambda(G)$ in $T_G(\mathbb{C})$. Fix a basis $\theta = (\theta_1, \dots, \theta_n)$ of $K/\mathbb{Q}$ and let $p_\theta \colon T_{G^n} = T_G^n \to T_G$, $(z_j) \mapsto \sum_j \theta_j z_j$. Let $d_0(G)$ (resp. $d_1(G)$) be the largest $d$ with a surjection $G \to \mathbb{G}_{\mathrm{a}}^d$ (resp. $G \to \mathbb{G}_{\mathrm{m}}^d$) over $\overline{\mathbb{Q}}$, and $\alpha(G) = 2\dim G - 2d_0(G) - d_1(G)$. Let $\mathcal{E}$ be a finite-dimensional complex vector space with a $\overline{\mathbb{Q}}$-structure, $\Psi \colon T_G(\mathbb{C}) \to \mathcal{E}$ a $\overline{\mathbb{Q}}$-rational linear map, and $\Psi_\theta = \Psi \circ p_\theta$. Let $H_0$ (resp. $H_\Psi$) be the largest connected algebraic subgroup of $G^n$ with $p_\theta(T_{H_0}) = 0$ (resp. $T_{H_\Psi} \subset \ker \Psi_\theta$), and put

$$\alpha_K(G) = \alpha(G^n/H_0)/n \ \ [29,\ \text{p. } 360], \qquad \alpha_{K,\Psi}(G) = \alpha(G^n/H_\Psi)/n \ \ (\text{the } \textit{inner weight});$$

neither depends on $\theta$, and $\alpha_{K,\Psi}(G) \le \alpha_K(G)$ (§2.1). Finally $S_K(G)$ is the set of subspaces $p_\theta(T_{G'})$ of $T_G$, for $G'$ an algebraic subgroup of $G^n$.

**Theorem 1.1** (Theorem R)**.** *Assume:*

- (G) *$G$ is a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of positive dimension;*
- (K) *$K$ is a number field;*
- ($\Psi$) *$\mathcal{E}$ is a finite-dimensional $\mathbb{C}$-vector space with a $\overline{\mathbb{Q}}$-structure, and $\Psi \colon T_G(\mathbb{C}) \to \mathcal{E}$ is a $\overline{\mathbb{Q}}$-rational linear map;*
- (V) *$V \subset \mathcal{E}$ is a complex subspace, and $W \subset V$ is a subspace defined over $\overline{\mathbb{Q}}$;*
- (i) *$V$ contains no nonzero subspace $\Psi(S)$ with $S \in S_K(G)$.*

*Put $\kappa = \dim_K\big(V \cap \Psi(\Omega_K(G))\big)$. Then $V \cap \Psi(\Lambda_K(G))$ is a finite-dimensional $K$-vector space, and*

$$\dim_K\big(V \cap \Psi(\Lambda_K(G))\big) \ \le\ \big(\alpha_{K,\Psi}(G) - \kappa\big)\dim_{\mathbb{C}}(V/W) \ \le\ \big(\alpha_K(G) - \kappa\big)\dim_{\mathbb{C}}(V/W).$$

With $\alpha_K$ in place of $\alpha_{K,\Psi}$ this is the implication (i)⇒(iii) of the Théorème of Roy and Waldschmidt [29, p. 363] (Corollary 2.2). By [29, Remarque 1], (i) holds whenever $V$ has no nonzero $\overline{\mathbb{Q}}$-rational point (and then $W = 0$); this is the only case used in our applications. Write $c_V = \dim \Psi(T_G)/(V \cap \Psi(T_G))$; it is the codimension, in $T_{G^n/H_\Psi}$, of the subspace to which the proof reduces the theorem (Proposition 3.4).

*Status:* We prove Theorem 1.1 at two levels (§3.8).

- (R-W) [Proved, given [37, Thm. 4.1] as printed]. Theorem 1.1 holds as stated, for every $G$ and every codimension. The proof (§3) uses Waldschmidt's Theorem 4.1 (Theorem 2.3) and nothing else from transcendence theory: the reduction to $K = \mathbb{Q}$ on $G^n/H_\Psi$ (Proposition 3.4), the weight lemmas of §3.1, the induction over subgroups in every codimension (Proposition 3.11) and a derivation of Wüstholz's analytic subgroup theorem from Theorem 4.1 (Proposition 3.5; we include it, but it is plausibly folklore). For groups with a torus in codimension $c_V \ge 2$ this level uses Theorem 4.1 with $W \ne 0$ and $\kappa > 0$ outside condition (2) of §2.2, which is where its published proof gives the multiplicity step only by a pointer (§2.4, Remark 3.12); there the proof is complete only relative to Theorem 4.1 as printed.
- (R-P) [Proved modulo published theorems and Lemma A.1], using Theorem 4.1 only through Lemma A.1 (Appendix A), i.e. with [37, Prop. 6.1] supplied by Lemma A.1, and, at linear nodes, through Waldschmidt's complete proof for linear groups (Lemma 3.8). Theorem 1.1 holds, modulo Lemma A.1, which is complete (formally verified; the written proof is a sketch) for $G$ isogenous to $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times E_1 \times \cdots \times E_m$ and a proof sketch otherwise:

($\alpha$) for every $G$ when $c_V = 1$ (Proposition 3.9, with the corner lemma and Baker's theorem);
($\beta$) in every codimension, for every $G$ with trivial maximal torus, i.e. every extension of an abelian variety by a vector group (such as $\mathbb{G}_{\mathrm{a}}^{d_0} \times A$ or $A^{\natural}$);
($\gamma$) in every codimension, for every other $G$, under condition (NR$^*$) of §3.7 at the reduced non-linear subgroups of $G^n/H_\Psi$.

The written proof of Lemma A.1 is a sketch in every case. It is formally verified (in the companion project `lean_lsg`, §12.3) when $G$ is isogenous to $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times E_1 \times \cdots \times E_m$, $E_j$ elliptic curves; this class contains $G^n$ and $G'' = G^n/H_\Psi$ and is closed under connected subgroups and quotients (property (P) of §4.1), so for such $G$ it covers every node of $G''$. For all other $G$, (R-P) is at the level of the sketch (Remark A.3): groups with a simple abelian factor of dimension $\geq 2$, vector extensions (formally verified only in the restricted cases listed at the end of Appendix A), non-split semi-abelian groups (such as a nontrivial extension of an elliptic curve by $\mathbb{G}_{\mathrm{m}}$), and mixed extensions of an abelian variety by $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b$.

The remaining case at level (R-P) is that of groups with a torus, in codimension $c_V \geq 2$, at reduced subgroups where (NR$^*$) fails. This happens at every reduced non-linear subgroup $H$ with $d_1(H) \geq 1$ and $\dim(V'' \cap T_H) = d_0(H)$, where $V'' \subset T_{G''}$ is the subspace of codimension $c_V$ above; for example at every reduced line in $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E$, at a reduced line in $\mathbb{G}_{\mathrm{m}}^2 \times E$ containing a period, and at a reduced line in $\mathbb{G}_{\mathrm{m}}^3 \times E$. We state this case as open (Remark 3.12). Theorem 1.1 is not formalised.

*Credit.* Roy and Waldschmidt prove their theorem by reduction to $K = \mathbb{Q}$, $\Psi = \mathrm{id}$ (their step (b), written in full; the inner weight is the inequality inside it, before its last line), and derive that case "de manière purement formelle" from Waldschmidt's Theorem 4.1 through Roy's categorical method of [28], with a refinement of [28, §6] that is not written out [29, p. 363, Remarque 2]. Parts of the case $K = \mathbb{Q}$ are in print: for hyperplanes it is [37, Thm. 1.1] under its condition (5.4), proved in [37, §5d]; for linear groups it is proved in every codimension in Waldschmidt's book [40, §11.7.3] (see also Roy [27]); and with $\kappa$ replaced by 0 it is [28, §6, remarque (i)], as [29, Remarque 2] records. In 1997 Waldschmidt called the period refinement of his effective subgroup theorem "accessible" [39, §7(f)]. To our knowledge, no complete written proof of the $\kappa$-refined statement in codimension $\geq 2$ exists for groups with an abelian part; we could not consult [28]. Waldschmidt's proof of Theorem 4.1 treats the multiplicity step for $W \neq 0$ (and exponent $b > 1$) by a pointer [37, p. 396]; Lemma A.1 writes it out. §2.4 and §11.1 give the details.

**Applications.** All other results of this paper are explicit consequences of Theorem 1.1 in its hyperplane case ($c_V = 1$, $\dim V/W = 1$), with proofs written in full modulo published theorems and Lemma A.1, except Theorems 10.4 and 10.5 and Theorem 10.8 except its part (U2) for $r = 2$, of §10, which use Theorem 4.1 on subspaces of codimension $\geq 2$. For each we also give a direct proof: an application of Theorem 4.1 to the explicit group $G'' = G^n/H_\Psi$ in which every connected algebraic subgroup is excluded by hand. That is the hyperplane induction of §3.6 carried out for one explicit group, and it shows which subgroups are tight (§4.4).

Let $\overline{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $\mathcal{L} = \{z \in \mathbb{C} : e^z \in \overline{\mathbb{Q}}^\times\}$ and $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. For an elliptic curve $E\colon y^2 = 4x^3 - g_2x - g_3$ with $g_2, g_3 \in \overline{\mathbb{Q}}$ we write $\wp, \zeta$ for its Weierstrass functions, $\Lambda_E$ for its period lattice, $\eta(\lambda) = \zeta(z+\lambda) - \zeta(z)$ for the quasi-periods, and $\mathcal{L}_E = \{u \in \mathbb{C} : u \in$

$\Lambda_E$ or $\wp(u) \in \overline{\mathbb{Q}}\}$. We use two curves throughout:

$$E_i\colon y^2 = 4x^3 - 4x, \qquad \Lambda = \varpi\,\mathbb{Z}[i], \qquad \varpi = \tfrac{\Gamma(1/4)^2}{2\sqrt{2\pi}}, \qquad \eta(\varpi) = \tfrac{\pi}{\varpi};$$
$$E_\rho\colon y^2 = 4x^3 - 4, \qquad \Lambda = \Omega_\rho\mathbb{Z}[\rho], \qquad \Omega_\rho = \tfrac{\Gamma(1/3)^3}{2^{4/3}\pi}, \qquad \eta(\Omega_\rho) = \tfrac{2\pi}{\sqrt{3}\,\Omega_\rho}.$$

(The quasi-periods follow from the Legendre relation and $\zeta(iz) = -i\zeta(z)$, $\zeta(\rho z) = \bar\rho\zeta(z)$. The real periods are Beta integrals, $\varpi = \int_1^\infty (x^3-x)^{-1/2}dx = \frac{1}{2}B(\frac{1}{4},\frac{1}{2})$ and $\Omega_\rho = \int_1^\infty (x^3-1)^{-1/2}dx = \frac{1}{3}B(\frac{1}{6},\frac{1}{2})$, and the $\Gamma$-forms follow from the reflection and duplication formulas; these are instances of the Chowla–Selberg formula [18].)

For a CM curve $E$ let $E^\natural$ be its universal vector extension and $\Lambda(E^\natural)$ the logarithms of its algebraic points, written in coordinates $(z, w)$ (base, and conjugate-eigen fibre; §4.2). For $E_i$, $E_\rho$ the fibre coordinate can be normalised so that $X = (u, \zeta(u)) \in \Lambda(E^\natural)$ for every $u \in \mathcal{L}_E \setminus \Lambda_E$, and in general $w(X) - \zeta^*(z(X)) \in \overline{\mathbb{Q}}$ for $X \in \Lambda(E^\natural)$ with $z(X) \notin \Lambda_E$, where $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$ and $\eta^* = \eta - \kappa_1\,\mathrm{id}$ are the conjugate-eigen versions of $\zeta$ and $\eta$ ($\kappa_1 \in \overline{\mathbb{Q}}$, $\kappa_1 = 0$ for $E_i, E_\rho$; §4.2).

**Four exponentials at CM periods.** The *four exponentials conjecture* says: if $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $t \notin \mathbb{Q}$, then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. The *strong four exponentials conjecture* [44, Conj. 1.5] says: if $y_1, y_2$ are linearly independent over $\overline{\mathbb{Q}}$ and $t \notin \overline{\mathbb{Q}}$, then $y_1, y_2, ty_1, ty_2$ are not all in $\widetilde{\mathcal{L}}$. Roy's strong six exponentials theorem [27] is the analogous statement with three numbers $y_j$. The four exponentials conjecture is known for algebraic $t$ (Gel'fond–Schneider) and in conjugation-generic situations (Diaz [13]; Waldschmidt [46]); see the paragraph after Corollary 1.5 and §11.2.

**Theorem 1.2.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication, let $\Omega \neq 0$ be a period of $E$, $c \in \overline{\mathbb{Q}}^\times$ and $\alpha, \beta \in \overline{\mathbb{Q}}^\times$. Fix logarithms $\log\alpha$, $\log\beta$ and put $\alpha^z = e^{z\log\alpha}$, $\beta^z = e^{z\log\beta}$.*

(a) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega}$ and $\beta^{c\Omega}$ are not both algebraic.*
(b) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^{c\Omega/\pi}$ and $\beta^{c\Omega/\pi}$ are not both algebraic.*

*Status:* [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$ (every $c \in \overline{\mathbb{Q}}^\times$); [Proved modulo the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times\Omega_\rho$; [Proved modulo published theorems and Lemma A.1] for every other CM period (§12.3).

For example (with the real logarithms), $2^\varpi$ and $3^\varpi$ are not both algebraic; nor are $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$, nor $2^{\Omega_\rho}$ and $3^{\Omega_\rho}$. Since the four exponentials statement is symmetric under $t \leftrightarrow 1/t$ (if $\alpha^{1/t}$, $\beta^{1/t}$ are algebraic, apply the statement at $t$ to them), the same holds at $t \in \overline{\mathbb{Q}}^\times/\Omega$ and $t \in \overline{\mathbb{Q}}^\times\pi/\Omega$. Theorem 1.2 consists of Corollaries 5.5 and 5.2 of Theorems 5.4 and 5.1 (§5.1). These are dimension bounds that also allow elliptic logarithms or fibre values in the second row. Theorem 5.1 also gives an elliptic statement, Corollary 1.4 below (stated after Theorem 1.3, next to its abelian analogue).

**Four exponentials at CM abelian periods.** Let $A/\overline{\mathbb{Q}}$ be a simple abelian variety with complex multiplication by a CM field $F$ of degree $2g$, with CM type $\Phi$. For an embedding $\sigma$ of $F$, an *eigen-differential* is a class $\omega_\sigma \in H^1_{\mathrm{dR}}(A/\overline{\mathbb{Q}})$ with $[x]^*\omega_\sigma = \sigma(x)\omega_\sigma$; it is of the first kind (holomorphic) iff $\sigma \in \Phi$. An *eigen-period* is a nonzero number $P_\sigma = \int_\gamma \omega_\sigma$, $\gamma \in H_1(A, \mathbb{Z})$. By Riemann's bilinear relations $P_\sigma P_{\bar\sigma} \in \pi\overline{\mathbb{Q}}^\times$. For a CM abelian variety that is not simple, every nonzero eigen-period is an algebraic multiple of an eigen-period of a simple factor, of the same kind (§4.1). For $N \geq 3$ and $0 < a, b < N$ with $a + b \neq N$, the Beta value $B(a/N, b/N) = \Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$ is, up to $\overline{\mathbb{Q}}^\times$, an eigen-period of a CM abelian variety (a factor of the Jacobian of the Fermat curve of degree $N$), of the first kind iff $a + b < N$ (§4.1).

**Theorem 1.3** (Theorem A)**.** *Let $P$ be a nonzero eigen-period of a CM abelian variety over $\overline{\mathbb{Q}}$, let $c \in \overline{\mathbb{Q}}^\times$ and $t \in \{cP,\ c/P,\ cP/\pi,\ c\pi/P\}$. Let $\alpha, \beta \in \overline{\mathbb{Q}}^\times$, fix logarithms $\log\alpha$, $\log\beta$ and put $\alpha^t = e^{t\log\alpha}$, $\beta^t = e^{t\log\beta}$. If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$, then $\alpha^t$ and $\beta^t$ are not both algebraic.*

*Status:* [Proved modulo published theorems and Lemma A.1] for $g \geq 2$ (§12.3 lists the instances checked in Lean conditionally on Theorem 2.5 for CM abelian varieties of dimension 2 and 3), except at $t \in \overline{\mathbb{Q}}^\times B(\frac{1}{5},\frac{1}{5})^{\pm1} \cup \overline{\mathbb{Q}}^\times (B(\frac{1}{5},\frac{1}{5})/\pi)^{\pm1}$, where it is [Proved; formally verified, standard axioms]. For $g = 1$ it consists of Theorem 1.2, which has the status stated there, and its $t \leftrightarrow 1/t$ extension to $t \in \overline{\mathbb{Q}}^\times/\Omega \cup \overline{\mathbb{Q}}^\times\pi/\Omega$, which follows in one line and is not formalised ([Proved modulo published theorems and Lemma A.1]).

In particular Theorem 1.3 holds at $t = B(a/N, b/N)^{\pm1}$ and $t = (B(a/N, b/N)/\pi)^{\pm1}$ for $0 < a, b < N$, $a + b \neq N$. For example, $2^t$ and $3^t$ are not both algebraic (for any choice of $\log 2$, $\log 3$; any two such choices are $\mathbb{Q}$-independent) for

$$t = \frac{\Gamma(1/5)^2}{\Gamma(2/5)}, \qquad t = \frac{\Gamma(1/5)^2}{\pi\,\Gamma(2/5)}, \qquad t = \frac{\Gamma(4/5)^2}{\Gamma(3/5)}, \qquad t = \frac{\Gamma(1/7)^2}{\Gamma(2/7)}.$$

The strong form holds at these $t$ as well (Theorem 6.8 in §6.2, label (c)): for instance $(2e)^t$ and $3^t$ are not both algebraic for $t = \Gamma(1/5)^2/\Gamma(2/5)$, and at least one of $e^{\pi^2}$, $e^{\pi P}$, $e^{P/\pi}$ is transcendental for every first-kind eigen-period $P$ (Corollary 6.11). Theorem 1.3 consists of Theorems 5.7 and 5.9 (§5.3), which are dimension bounds of the shape of Theorems 5.1 and 5.4, with the CM elliptic curve replaced by one eigen-coordinate of a CM abelian variety. Theorem 5.1 (for a CM elliptic curve) gives Corollary 1.4, and Theorem 5.7 its abelian analogue, Corollary 1.5:

**Corollary 1.4.** *For every CM curve with period $\Omega$, at least one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ is transcendental. On $y^2 = 4x^3 - 4x$: one of $e^{\pi^2} \approx 19333.69$, $e^{\pi\varpi} \approx 3779.84$, $e^{\pi^2/\varpi} \approx 43.12$ is transcendental.*

*Status:* on $y^2 = 4x^3 - 4x$: [Proved modulo Wüstholz's theorem in the form $\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}}$; formally verified otherwise] (Corollary 9.1); [Proved modulo published theorems and Lemma A.1] for every other CM period.

**Corollary 1.5.** *For every first-kind CM eigen-period $P$, at least one of $e^{\pi^2}$, $e^{\pi P}$, $e^{\pi^2/P}$ is transcendental. For instance, one of $e^{\pi^2}$, $e^{\pi\Gamma(1/5)^2/\Gamma(2/5)}$, $e^{\pi^2\Gamma(2/5)/\Gamma(1/5)^2}$ is transcendental.*

*Status:* [Proved modulo published theorems and Lemma A.1]; for $P = (1 - \zeta_5)B(\frac{1}{5},\frac{1}{5})$ (the example): [Proved modulo Wüstholz's theorem in the form $P \notin \widetilde{\mathcal{L}}$; formally verified otherwise]. The proof (Corollary 9.3) does not use the algebraic independence of $\pi$ and $P$ (which is not known for abelian periods); for $g = 1$ it is the proof of the elliptic case, Corollary 9.1, which therefore does not need Chudnovsky's theorem either (Remark 9.4).

Part of Theorems 1.2 and 1.3 also follows from the six exponentials theorem, Roy's strong six exponentials theorem and complex conjugation (Diaz [13]; Waldschmidt [46, Cor. 2.7, Thm. 4.1]). Every class $\overline{\mathbb{Q}}^\times t$ occurring in them contains a real number, and these arguments together cover every pair at every $t \notin \mathbb{R} \cup i\mathbb{R}$, and, at every $t$, every pair whose logarithms span a space that is not stable under complex conjugation (§11.2). They do not apply to real or purely imaginary $t$ — such as $t = \varpi$, $\varpi/\pi$, $\Omega_\rho$, or the Beta values themselves — with pairs whose logarithms span a conjugation-stable space, such as $(2, 3)$, which is where our examples lie, nor to Corollaries 1.4 and 1.5. Independently of conjugation, all of these are consequences of Theorem 1.1 (§4.4); for $g = 1$ the printed weight of [29] already suffices (with Wüstholz's theorem), and for $g \geq 2$ the inner weight is used (Remark 5.8, Remark 11.1).

**Strong four exponentials.** For a CM elliptic curve $E/\overline{\mathbb{Q}}$ with period $\Omega \neq 0$ let Fib be the set of fibre values (§4.2): $\zeta^*(u) + \overline{\mathbb{Q}}$ for $u \in \mathcal{L}_E \setminus \Lambda_E$ and $\eta^*(\lambda) + \overline{\mathbb{Q}}$ for $\lambda \in \Lambda_E$; they are the conjugate-eigen fibre coordinates of logarithms of algebraic points of the universal vector extension $E^\natural$.

**Theorem 1.6** (Theorem S)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication, $\Omega \neq 0$ a period of $E$, and $t \in \overline{\mathbb{Q}}^\times\Omega \cup \overline{\mathbb{Q}}^\times\Omega^{-1} \cup \overline{\mathbb{Q}}^\times\Omega/\pi \cup \overline{\mathbb{Q}}^\times\pi/\Omega$. If $y_1, y_2 \in \mathbb{C}$ are linearly independent over $\overline{\mathbb{Q}}$, then $y_1, y_2, ty_1, ty_2$ are not all in $\widetilde{\mathcal{L}}$.*

*Status:* [Proved modulo Wüstholz's theorem in the form $\Omega, \eta \notin \widetilde{\mathcal{L}}$ (Lemma 6.1); formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$; [Proved modulo that form of Wüstholz's theorem and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times\Omega_\rho$; [Proved modulo published theorems and Lemma A.1] for every other CM period. For the paper proof (trust point T), two internal referees graded the first write-up a proof sketch modulo this trust point, and a third, after the subgroup list was written out in full (§6), found no case missing.

It follows from the affine dimension bounds SMG and SMG$^\natural$ (Theorem 6.2), whose second rows may also contain elliptic logarithms, resp. fibre values. For instance (Corollary 6.5, which has label (c)):

- $\varpi\log 2$ and $\varpi\log 3$ are not both in $\widetilde{\mathcal{L}}$; $(2e)^\varpi$ and $3^\varpi$ are not both algebraic; $e^{\varpi/\pi}$ and $2^{\varpi/\pi}$ are not both algebraic (this recovers Corollary 9.7(ii), which §9.4 derives from a proof sketch); $e^{1/\varpi}$ and $2^{1/\varpi}$ are not both algebraic. Here $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62205755429211981046\ldots$ is the real period of $y^2 = 4x^3 - 4x$.
- For every CM period $\Omega$, one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\Omega/\pi}$ is transcendental (this recovers Corollary 9.7(i), which §9.4 derives from a proof sketch), and one of $e^{\pi^2}$, $e^{\pi^2/\Omega}$, $e^{1/\Omega}$ is transcendental.

Theorem 1.6 is the case $g = 1$ of Theorem 6.8 (§6.2): strong four exponentials at every $t \in \overline{\mathbb{Q}}^\times\{P^{\pm1}, (P/\pi)^{\pm1}\}$ for every eigen-period $P$ of a CM abelian variety (label (c)). In §6.3 the same count, read on a CM curve $E$ instead of $\mathbb{G}_\mathrm{m}$, gives strong *elliptic* four exponentials at $t \in \overline{\mathbb{Q}}^\times(\pi/\Omega^2)^{\pm1}$ and, for a CM curve with another CM field and period $\Omega'$, at $t \in \overline{\mathbb{Q}}^\times(\Omega'/\Omega)^{\pm1}$ and $\overline{\mathbb{Q}}^\times(\pi/\Omega\Omega')^{\pm1}$ (Corollary 6.13, label (c)): for instance, for every $u \in \mathcal{L}_{E_i} \setminus \{0\}$, $\wp_i(\pi/\varpi^2)$ and $\wp_i(\pi u/\varpi^2)$ are not both algebraic ($\wp_i$ the Weierstrass function of $E_i$). The classes $\overline{\mathbb{Q}}^\times(\Omega'/\Omega)^{\pm1}$ give, e.g., that $\wp_i(\Omega_\rho/\varpi)$ and $\wp_i(\Omega_\rho u/\varpi)$ are not both algebraic. Theorem 6.16 is the analogue on eigen-slots of CM abelian varieties, at slopes such as $B(\frac{2}{7},\frac{2}{7})/B(\frac{1}{7},\frac{1}{7})$.

**Two CM curves.** The same mechanism, applied to two non-isogenous CM curves, gives a statement with no logarithm of $\mathbb{G}_\mathrm{m}$ at all (and a twisted variant, Theorem 7.6 in §7.6, of which we give a proof sketch). Applied to periods, it also gives two statements of degree two about periods of the two curves (Corollary 9.5): $\pi, \varpi, \Omega_\rho, \varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$, and so are $1, \varpi, \Omega_\rho, \varpi\Omega_\rho/\pi$. These are not, as far as we can see, consequences of the analytic subgroup theorem ([Heuristic]: period conjecture), and we flag them, with Corollary 1.12(b), as the most delicate claims of the paper. Like Theorem 1.7, all three are consequences of Theorem 1.1 combined with Wüstholz's theorem (for them the printed weight of [29] already suffices, Remark 11.1); we found none of them stated in print.

**Theorem 1.7** (Theorem MX)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, t \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z(X) + z(Y) + a\big)\cdot\beta\,\eta(\Omega_\rho) \;=\; \big(w(X) + w(Y) + t\big)\cdot\varpi,$$

*then $a = 0$, $z(X) \in \mathbb{Q}(i)\varpi$ and $z(Y) \in \mathbb{Q}(\rho)\Omega_\rho$; in particular $\exp_{E_i}(z(X))$ and $\exp_{E_\rho}(z(Y))$ are torsion points. In particular, for $u \in \mathcal{L}_{E_i} \setminus \Lambda$ and $u' \in \mathcal{L}_{E_\rho} \setminus \Lambda_\rho$ not both torsion, $(u + u' +$*

*a)* $\beta\,\eta(\Omega_\rho) \neq (\zeta_i(u) + \zeta_\rho(u') + t)\,\varpi$. *An analogous statement holds for any two non-isogenous CM elliptic curves (Remark 7.3).*

*Status:* [Proved modulo Wüstholz's theorem (in the forms $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$ and the linear independence of 1 and elliptic logarithms of $E_i$, $E_\rho$), Masser's theorem, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. The analogue for other pairs of curves (Remark 7.3) is [Proved modulo published theorems and Lemma A.1]. (The dimension bound behind it, Proposition 7.1, is formally verified only in its single-point form; see its status line.)

**Corollary 1.8.** *For $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$ and $a, t \in \overline{\mathbb{Q}}$,*

$$\frac{\varpi\,\Omega_\rho\,(\zeta_i(u)+t)}{\pi\,(u+a)} \notin \overline{\mathbb{Q}}, \qquad \textit{equivalently} \qquad \frac{\zeta_i(u)+t}{u+a} \notin \overline{\mathbb{Q}} \cdot \frac{\pi^{5/2}}{\Gamma(1/4)^2\,\Gamma(1/3)^3}.$$

*Symmetrically for non-torsion $u' \in \mathcal{L}_{E_\rho}$ with $\zeta_\rho$.*

*Status:* for $u \in \mathcal{L}_{E_i}$: [Proved modulo the inputs of Theorem 1.7 and Laurent's theorem that $1, u, \zeta(u)$ are $\overline{\mathbb{Q}}$-linearly independent [21]; also a consequence of [52] on $E^\natural \times \mathbb{G}_{\mathrm{a}}$; formally verified otherwise]; the symmetric statement on $E_\rho$ is [Proved modulo published theorems and Lemma A.1].

For example, for the point with $\wp(u) = 2$ one has $u = 0.72694593546890819853\ldots$ and $\varpi\Omega_\rho\zeta(u)/(\pi u) = 3.76379062156897182773\ldots$ (computed independently twice, by theta functions and by the Laurent series of $\zeta$; an illustration only). The case $X = Y = 0$ of Theorem 1.7 gives:

**Corollary 1.9.** $\Gamma(1/4)^2\,\Gamma(1/3)^3/\pi^{5/2}$ *is transcendental.*

*Status:* [Proved modulo published theorems and Lemma A.1] (only Wüstholz's theorem is used, Remark 7.4; in the Lean certificate it is a named hypothesis).

This is immediate from Wüstholz's theorem (Remark 7.4), and probably also from the Wolfart–Wüstholz theorem on Beta values [51] (not checked); we found it stated nowhere, but do not claim it as new. Why Corollary 1.8 does not follow from known algebraic independence results is said in §11.4.

The single-curve version of the same count gives the following theorem, which turns out to be classical in substance.

**Theorem 1.10** (Theorem M)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication by $k$, and let $\Omega \neq 0$ be a period. If $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ and $\ell \in \widetilde{\mathcal{L}}$ satisfy $i\pi\,u = \Omega\,\ell$, then $u = s\Omega$ and $\ell = s\,i\pi$ for some $s \in \overline{\mathbb{Q}}$; if $u \in \mathcal{L}_E$ then $s \in k$.*

*Status:* for $\Omega \in \overline{\mathbb{Q}}^\times\varpi$: [Proved; formally verified, standard axioms] for the first conclusion, and [Proved modulo Masser's theorem; formally verified otherwise] for $s \in k$ when $u \in \mathcal{L}_E$; for $\Omega \in \overline{\mathbb{Q}}^\times\Omega_\rho$: [Proved modulo the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for the first conclusion, and [Proved modulo published theorems and Lemma A.1] for $s \in k$; [Proved modulo published theorems and Lemma A.1] for every other CM period.

**Corollary 1.11.** *On $E_i$ (and analogously on every CM curve): (M1) $\wp(c\varpi/\pi) \notin \overline{\mathbb{Q}}$ for $c \in \overline{\mathbb{Q}}^\times$; (M2) $\wp(\varpi \log\alpha/\pi) \notin \overline{\mathbb{Q}}$ for $\alpha \in \overline{\mathbb{Q}}^\times$ not a root of unity; (M3) at least one of $e^{\pi^2}$, $\wp(\pi\varpi)$ is transcendental; more precisely, if $e^{\pi^2} \in \overline{\mathbb{Q}}$ then $\wp(c\pi\Omega) \notin \overline{\mathbb{Q}}$ for every $c \in \overline{\mathbb{Q}}^\times$ and every period $\Omega$ of every CM curve; (M3′) for $\alpha \in \overline{\mathbb{Q}} \setminus \{0,1\}$ and any branch, one of $e^{i\pi\log\alpha}$, $\wp(\varpi\log\alpha)$ is transcendental; (Mη) $\eta(\varpi)\,u \notin \widetilde{\mathcal{L}}$ for $u \in \mathcal{L}_{E_i} \setminus \mathbb{Q}(i)\varpi$.*

*Status:* (M1) and (M3) on $E_i$ (with $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ in (M3)): [Proved; formally verified, standard axioms]; (M3′): [Proved modulo the Hermite–Lindemann theorem; formally verified otherwise]; (M$\eta$): [Proved modulo Masser's theorem; formally verified otherwise]; (M2), (M3) for other CM periods, and the analogues on other CM curves: [Proved modulo published theorems and Lemma A.1].

*Why MX is different.* After multiplying by $\pi$, every relation in Theorem 1.10 with $u \in \mathcal{L}_E$ becomes a $\overline{\mathbb{Q}}$-linear relation among periods of 1-motives, because of Laurent's identity (6); this is why the analytic subgroup theorem proves it too. The hypothesis of Theorem 1.7 instead contains the products $\Omega_\rho\,\zeta_i(u)$ and $\eta(\varpi)\,u'$, each pairing a period of one curve with a logarithm of the other. No CM identity linearises these; by the period conjecture they are genuine degree-two periods (§7.8). The extra strength comes from a second true degree-two period relation (the Legendre relation of the second curve), which the count sees as a second period on the line.

**First-kind period relations.** In Theorem 1.7, the period term $\kappa = 2$ on a line came from two Legendre relations, i.e. from two eigen-pairs $\omega\,\eta^*(\omega) \in \overline{\mathbb{Q}}^{\times}\pi$; this forces universal vector extensions into the group. Products of CM abelian varieties involving a simple factor of dimension $\geq 3$ can have homogeneous quadratic relations among first-kind periods of four different eigen-slots (Remark 8.9). On the Jacobian $A$ of $C\colon y^7 = x(1-x)$ (genus 3, CM by $\mathbb{Q}(\zeta_7)$), with $E$ an elliptic curve with CM by $\mathbb{Q}(\sqrt{-7})$, the Beta cocycle and one reflection give

$$B(\tfrac{1}{7},\tfrac{1}{7})\,B(\tfrac{2}{7},\tfrac{2}{7}) = \frac{\sin(3\pi/7)}{\sin(\pi/7)}\,B(\tfrac{3}{7},\tfrac{3}{7})\,B(\tfrac{1}{7},\tfrac{2}{7}),$$

where $B(j/7, j/7)$ ($j = 1, 2, 3$) are the periods of the holomorphic eigen-differentials $dx/y^{7-j}$ on $C$ and $B(1/7, 2/7) \in \overline{\mathbb{Q}}^{\times}\Omega_E$ (Chowla–Selberg [18]). Theorem 8.2 (§8) uses such a relation as the period term. On $y^7 = x(1-x)$ it gives:

**Corollary 1.12** (Corollary X7). (a) *For $O, Q \in C(\overline{\mathbb{Q}})$ with $Q - O$ of infinite order in $A$,*

$$\int_O^Q \frac{dx}{y^5} \Big/ \int_O^Q \frac{dx}{y^6} \;\notin\; \overline{\mathbb{Q}} \cdot \frac{B(3/7,3/7)}{B(1/7,1/7)}.$$

(b) *The numbers 1, $B(\frac{2}{7},\frac{2}{7})$, $B(\frac{3}{7},\frac{3}{7})$, $B(\frac{3}{7},\frac{3}{7})/B(\frac{1}{7},\frac{1}{7})$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are $\pi$, $\Gamma(\frac{2}{7})^2\Gamma(\frac{3}{7})$, $\Gamma(\frac{1}{7})\Gamma(\frac{3}{7})^2$, $\Gamma(\frac{2}{7})\Gamma(\frac{3}{7})^2/\Gamma(\frac{1}{7})$.*

*Status:* [Proved modulo published theorems and Lemma A.1], for the dimension statement (Proposition 8.1), for Theorem 8.2 (whose deduction from the proposition is the standard analytic-subgroup step, written out in §8) and for (a) and (b). That (a) and (b) are not linearisations, i.e. not consequences of the analytic subgroup theorem, is [Heuristic] (period conjecture).

The relations at levels 9 and 24 (Lemma 8.12, [Proved]) give two more instances (Theorem 8.13), and families of CM types of $\mathbb{Q}(\zeta_N)$ give $\kappa = 2^{g-2}$ parallel pairs at once (Theorem 8.15): for instance 1, $B(\frac{4}{15},\frac{4}{15})/B(\frac{1}{15},\frac{2}{15})$, $B(\frac{4}{15},\frac{4}{15})$, $B(\frac{1}{15},\frac{7}{15})$, $B(\frac{2}{15},\frac{6}{15})$, $B(\frac{1}{3},\frac{1}{3})$ are linearly independent over $\overline{\mathbb{Q}}$ (label (c)).

**More applications.** Waldschmidt's Conjecture 42 [45] (four exponentials on $E \times \mathbb{G}_{\mathrm{m}}$) holds at six classes of CM column ratios (Theorem 10.1): for instance $\wp(\log 2/\varpi)$ and $\wp(\log 3/\varpi)$ are not both algebraic on $y^2 = 4x^3 - 4x$. On subspaces of codimension $\geq 2$, Theorem 4.1 applied subgroup by subgroup is stronger than the uniform bound of Theorem 1.1 (at $G' = 0$ it divides by the codimension, [37, (4.2)]); this gives that one of $e^{\pi^2}$, $\wp(1/\pi)$ is transcendental on $y^2 = 4x^3 - 4x$ (Theorem 10.4), a collinearity statement for period points of three CM curves (Theorem 10.5), and statements on abelian varieties with imaginary-quadratic multiplication of

signature $(r, s)$, such as Picard Jacobians, where the period term comes from a single eigenspace (Theorem 10.8).

**How the parts fit together.** Theorem 1.1 is proved in §§2–3. Every application in §§5–9 has $V$ a line in $\mathbb{C}^2$ with transcendental slope, $\Psi$ onto $\mathbb{C}^2$ and $W = 0$, so it is an instance of the hyperplane case, and the bound $\alpha_{K,\Psi}(G) - \kappa$ is computed from the explicit quotient $G''$ (§4.4, with a table). Theorem 1.3 contains Theorem 1.2 ($g = 1$), and its proof is that of Theorems 5.1 and 5.4 with new numbers: the point is that the quotient group in the count has abelian part of dimension $n/2$ for every $g$ (Lemma 4.1), so the inner weight is that of the elliptic case. Theorem 8.2 is a second source of the period term: Theorems 1.7 and 7.6 (and the abelian-surface sketch of §7.7) obtain $\kappa = 2$ on a line only from eigen-pair (Legendre) relations, which force vector extensions into the group; in Theorem 8.2 the group is an abelian variety times $\mathbb{G}_{\mathrm{a}}^2$ and the margin of the count is $n$ instead of $n - 2$. Theorem 1.6 at $t \in \overline{\mathbb{Q}}^{\times}(\Omega/\pi)^{\pm 1}$ contains the strong statement of Corollary 9.7(iii), which §9.4 derives from an affine version of Theorem 5.1 stated as a proof sketch with the subgroup list not written out (Theorem 9.6); §6 writes that list out (Theorem SMG) and adds the fibre version SMG$^{\natural}$, which gives the classes $\overline{\mathbb{Q}}^{\times}\Omega^{\pm 1}$.

**Organisation.** §2 sets up Theorem 1.1: notation and weights, Waldschmidt's Theorem 4.1 (completed by Lemma A.1 of Appendix A), Statement R (the case $K = \mathbb{Q}$, $\Psi = \mathrm{id}$) with its hyperplane form, and the position of Theorem 1.1 in the literature. §3 proves Theorem 1.1. §4 gives the linear algebra of CM abelian varieties of any dimension and of their universal vector extensions, and explains how the applications follow from Theorem 1.1. §5 proves Theorems 1.2 and 1.3 (and, in §5.4, four exponentials for quasi-logarithms, Theorem 5.16), §6 Theorems 1.6 and 6.8 and the elliptic and eigen-slot analogues, §7 Theorems 1.10 and 1.7 and Proposition 7.5 (with a proof sketch of Theorem 7.6), and §8 Theorem 8.2 with its instances at levels 7, 9, 24 and Theorem 8.15. §9 collects the corollaries MG1, MG3, A2, the degree-two statements and the affine version. §10 gives Theorems 10.1, 10.4, 10.5 and 10.8. §11 discusses credit, the literature and the overlap with known results, and §12 the consistency checks, the dependencies and the formal verification.

**Conventions on labels.** Each main application carries a status line with one of the following labels (or the label [Proof sketch], described after them); where the label depends on the period (typically when a result is formally verified for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ only), the status line says which label applies to which case. Theorem 1.1 carries the two levels stated after it.

(a) [Proved; formally verified, standard axioms]: a deduction of the statement (possibly by another route than the proof given here; see §12.3), including the instances of Waldschmidt's theorem (Theorem 2.3, or its hyperplane consequence Theorem 2.5, with the step of Lemma A.1) on which that deduction rests, is checked in the Lean certificate described in §12.3, from Lean's standard axioms only. These are not always the instances used in our paper proofs. The group models used there were refereed, not derived in Lean from algebraic geometry.

(b) [Proved modulo X; formally verified otherwise]: as (a), except that the named classical input X (for example Wüstholz's theorem in a stated form, Baker's theorem, the Hermite–Lindemann theorem, Masser's theorem, Laurent's theorem on $\zeta$, or the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$) enters the Lean proof as an explicit hypothesis.

(c) [Proved modulo published theorems and Lemma A.1]: the proof is complete given Waldschmidt's Theorem 4.1 [37] as printed (Theorem 2.3), the other published results cited, and Lemma A.1, the one step of [37, Prop. 6.1] that is only indicated in print. The written proof of Lemma A.1 is a proof sketch (checked by two internal

referees, not peer-reviewed); it is formally verified (`lean_lsg`) for groups isogenous to $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times E_1 \times \cdots \times E_m$. A result that uses fewer inputs (for instance only Wüstholz's theorem) also carries this label when it is not formally verified. In particular, each such result, except those that use only Wüstholz's theorem, also follows from the hyperplane case of Theorem 1.1 at level (R-P), or, for the results of §10 in codimension $\geq 2$, from Theorem 4.1 completed by Lemma A.1.

A main result one step of whose proof is only sketched carries the status label [Proof sketch] instead, and its status line names that step; this applies to Theorem 7.6 and Corollary 7.7 only. (Other proof sketches, such as Theorem 9.6, are not main results and say so in their headings.) We call the results with label (b) "formally verified modulo named inputs": the named classical theorems X are hypotheses of their Lean proofs, so these results are not unconditional.

For steps and remarks we use: [Proved]; [Lit] (a published result, cited); [Proof sketch] (the argument is written at the level of the case check, with some routine steps compressed, or the internal referees graded it so); [Heuristic] (depends on the period conjecture or the Rohrlich–Lang conjecture); [Numerical] (computed, illustrates only). "Internal referee" means an independent check within this project, not peer review.

## 2. Theorem R: setting, inputs and position

This section fixes the notation of Theorem 1.1 (§2.1), states the one transcendence input of its proof, Waldschmidt's algebraic subgroup theorem with period term (§2.2), states the case $K = \mathbb{Q}$, $\Psi = \mathrm{id}$ (Statement R) and its hyperplane form (§2.3), and says precisely what is in print (§2.4).

2.1. **Notation, weights, and the printed statement.** For a connected commutative algebraic group $G$ over $\overline{\mathbb{Q}}$ we use the notation of the introduction: $T_G$, $\exp_G$, $\Omega(G)$, $\Lambda(G)$, $\Omega_K(G)$, $\Lambda_K(G)$. $\Lambda(G)$ is a $\mathbb{Q}$-vector space (division points of algebraic points are algebraic) containing $\Omega(G)$, and $\Omega(G)$ is a discrete subgroup of $T_G(\mathbb{C})$, free of finite rank; so $\Lambda_K(G) = \sum_j \theta_j \Lambda(G)$ and $\Omega_K(G) = \sum_j \theta_j\, \mathbb{Q}\Omega(G)$. Algebraic subgroups are taken over $\overline{\mathbb{Q}}$ (in characteristic 0 every algebraic subgroup of $G$ is). For a connected algebraic subgroup $G' \subset G$ we have $\Omega(G) \cap T_{G'} = \Omega(G')$ and $\Lambda(G) \cap T_{G'} = \Lambda(G')$. The weights $\alpha$, $\alpha_K$, $\alpha_{K,\Psi}$ and the set $S_K(G)$ are defined in the introduction; the subspaces in $S_K(G)$ are $\overline{\mathbb{Q}}$-rational, since $p_\theta$ has coefficients in $K$ [29, p. 360].

**Remark 2.1** (The inner weight)**.** (1) $H_0 \subset H_\Psi$, so $G^n/H_\Psi$ is the quotient of $G^n/H_0$ by $H_\Psi/H_0$, and $\alpha_{K,\Psi}(G) \leq \alpha_K(G)$ by Lemma 3.2; this is the inequality "On peut aussi majorer $\alpha(G')$ par $n\alpha_K(G)$" of [29, p. 364], which ends their step (b). (2) $\alpha_{K,\Psi}(G)$ does not depend on $\theta$. If $\theta' = \theta M$ with $M \in \mathrm{GL}_n(\mathbb{Q})$, then $p_{\theta'} = p_\theta \circ M$ on $T_G \otimes \mathbb{Q}^n$. Choose $N \in \mathbb{Z}_{>0}$ with $NM$ integral; the isogeny $\varphi = [NM]$ of $G^n$ has $d\varphi = NM$, so $T \subset \ker \Psi_{\theta'}$ iff $d\varphi(T) \subset \ker \Psi_\theta$, hence $H_\Psi(\theta') = \varphi^{-1}(H_\Psi(\theta))^\circ$, and $\varphi$ induces an isogeny $G^n/H_\Psi(\theta') \to G^n/H_\Psi(\theta)$. $\alpha$ is invariant under isogeny (Lemma 3.1). The same argument with $\Psi = \mathrm{id}$ applies to $\alpha_K$, as in [29, p. 360]. (3) $\alpha_{K,\Psi}$ can be much smaller than $\alpha_K$: for $G = \mathbb{G}_{\mathrm{m}}^2 \times A$ with $A$ a simple CM abelian variety of dimension $g$ read through one eigen-coordinate, $\alpha_K(G) = 2 + g$ but $\alpha_{K,\Psi}(G) = 3$ (Remark 5.8).

**Corollary 2.2** ([29, Théorème, (i)⇒(iii), p. 363])**.** *Under the hypotheses of Theorem 1.1,*

$$\dim_K\big(V \cap \Psi(\Lambda_K(G))\big) \leq (\alpha_K(G) - \kappa)\dim_{\mathbb{C}}(V/W).$$

This is the second inequality of Theorem 1.1. Roy and Waldschmidt also prove (iii)⇒(ii) (finiteness)⇒(i); we do not need these. By [29, Remarque 1], (i) holds when $V$ has no nonzero $\overline{\mathbb{Q}}$-rational point (and then $W = 0$), since every $\Psi(S)$, $S \in S_K(G)$, is rational. We use only this case in the applications. Corollary 2.2 is also used for the first proof of Theorem 1.10, in the sketch of §7.7, and to explain the counts (Remarks 7.2 and 5.8).

2.2. **Waldschmidt's subgroup theorem with period term.**

**Theorem 2.3** ([37, Thm. 4.1, pp. 385–386])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d = d_0 + d_1 + d_2 \geq 1$, with projections $\pi_0, \pi_1$ onto the first two factors ($d_0, d_1$ need not be maximal). Let $V \subset T_G(\mathbb{C})$ be a subspace of dimension $N < d$, $W \subset V$ a subspace defined over $\overline{\mathbb{Q}}$, and $Y \subset V$ a finitely generated subgroup with $\Gamma = \exp_G Y \subset G(\overline{\mathbb{Q}})$; put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker \exp_G)$. Then there is a connected algebraic subgroup $G' \neq G$ defined over $\overline{\mathbb{Q}}$ such that, with $\delta = \dim G/G'$, $\delta_0 = \dim \mathbb{G}_{\mathrm{a}}^{d_0}/\pi_0(G')$, $\delta_1 = \dim \mathbb{G}_{\mathrm{m}}^{d_1}/\pi_1(G')$, $\delta_2 = \delta - \delta_0 - \delta_1$, $\lambda = \mathrm{rk}_{\mathbb{Z}}\, \Gamma/(\Gamma \cap G')$ and $\tau = \dim W/(W \cap T_{G'})$, one has $\delta > \tau$ and*

$$(\lambda + \delta_1 + 2\delta_2)(d - N) \leq (\delta - \tau)(d_1 + 2d_2 - \kappa). \tag{1}$$

($G' = 0$ is allowed.) The proof [37, §§6–7] combines an auxiliary function [37, Prop. 6.1] with Philippon's zero estimate [26]. Prop. 6.1 asks for a polynomial that vanishes with multiplicity $\geq T$ along $W$ at the points of $\exp_G(Y(S))$, with the saving $\kappa$ in the parameters. The $\kappa$-saving is proved in [36, Prop. 2.1 and 2.4] (an auxiliary function constructed with periods), for $W = 0$ and $b = 1$ [37, p. 396]; for the rest W88 refers to [36] and to Bertrand [34, App. I, Lemme 7 and its Corollaire, with Lemme 1] (cited in [37, p. 396] as "Lemma 7"). All our applications use both $W \neq 0$ and $\kappa > 0$, and so does the proof of Theorem 1.1: after the reduction of §3.2 the subspace $W''$ contains $\ker \Psi''$, which is nonzero as soon as $\Psi''$ is not injective, even when $W = 0$. We write out the multiplicity step in Appendix A (Lemma A.1). It holds under the condition

$$2N \geq d_1 + 2d_0 + \kappa, \tag{2}$$

which reads $d_1 + 2d_2 - \kappa \geq 2(d - N)$ (Lemma 3.3). It is satisfied in all our applications: for Theorem 5.1: $5n - 2 \geq 3n$; Theorem 5.4: $5n \geq 3n$; Theorem 1.7: $4n + 2 \geq 2n + 2$; Proposition 7.5: $4n + 2 \geq 2n$; Theorem 9.6: $5n + 2 \geq 3n + 4$; Theorems 5.7 and 5.9: $5n - 2 \geq 3n$ and $5n \geq 3n$; Proposition 8.1: $4n + 2 \geq 2n + 4$; Theorem 6.2: $5n + 2 \geq 3n + 4$ and $5n + 2 \geq 3n + 2$; the results of §§6.2, 6.3, 8.6 and 10 and Theorem 5.16 check it in their proofs. In the proof of Theorem 1.1, (2) holds automatically at the non-linear nodes of a hyperplane induction (Lemma 3.6) and at the reduced nodes of a group without torus (Lemma 3.10).

When $d_0 \geq 1$, the proof of Theorem 2.3 uses an exponent $b > 1$ in the case $\delta = \tau$; the last paragraph of the proof of Lemma A.1 covers this for every $d_0$. Proposition 8.1 and Theorem 6.2(a) use $d_0 = 2$ (as do their analogues Theorems 6.9 (SMG-A), 6.12(1), 6.16 (EA), 8.13, 8.15 and 10.4), whereas the other proved results use $d_0 \leq 1$. This is legitimate, for three reasons.

- Theorem 2.3 is stated for any $d_0$ ("we do not assume that $d_0$ and $d_1$ are maximal", p. 385), and $d_0$ enters (1) only through $\delta_2 = \delta - \delta_0 - \delta_1$.
- Waldschmidt applies his main theorem with $d_0 = 2$ himself: in the proof of [37, Cor. 2.1] (p. 377, $G = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2$, with the same two-row affine mixing as in Theorem 6.2(a), without $E$). He also applies it with $d_0 = n$ and $W \neq 0$ in the second proof of [37, Cor. 3.3] (p. 384, $G = \mathbb{G}_{\mathrm{a}}^n \times \mathbb{G}_{\mathrm{m}}$, $W = \mathbb{C}(1, 0, \ldots, 0, 1)$). Both of these are applications of Theorem 1.1 there, which is the case $\kappa = 0$.
- Lemma A.1 is stated for general $d_0$. Its only $d_0$-dependent hypothesis is (2), which we check in each application; its other hypothesis $d_2 \geq 1$ holds in every application below ($d_2 \geq n/2$). The compensation of the exponent $b$ by the factor $(\log S)^{bd_0}$ in its last paragraph is uniform in $d_0$. (This last point is a two-sentence step of Lemma A.1, so it is part of trust point T; Theorem 9.6, a proof sketch, uses the same bookkeeping with $d_0 = 2$, which is consistent with it but is not independent support.)

So $d_0 = 2$ adds no risk beyond trust point T itself. We call "trust point T" the use of Theorem 2.3 with $W \neq 0$ and $\kappa > 0$, completed by Lemma A.1; the direct proofs of Theorems 5.1, 5.4, 1.7, 5.7, 5.9, 6.2 and of Proposition 8.1 use exactly this, as do the proofs of Theorems 5.16, 6.9,

6.12, 6.16, 8.13, 8.15 and the results of §10, and so does level (R-P) of Theorem 1.1. In all our applications in §§5–9 $V''$ is a hyperplane, $N = d-1$, and $W'' = \ker \Psi''$, so that $\tau = \delta + r' - 2$ where $r' = \dim \Psi''(T_{G'})$; thus only $G'$ with $r' \le 1$ can satisfy $\delta > \tau$, and then $\delta - \tau = 2 - r'$.

2.3. **Statement R and the hyperplane statement.** For a subspace $V \subset T_G(\mathbb{C})$ write $(\mathrm{i})_V$ for the condition that $V$ contains no nonzero $T_{G'}$, $G' \subset G$ a connected algebraic subgroup. For $K = \mathbb{Q}$ and $\Psi = \mathrm{id}$, $S_{\mathbb{Q}}(G)$ is the set of Lie algebras of algebraic subgroups of $G$, so condition (i) of Theorem 1.1 is $(\mathrm{i})_V$, and $\alpha_{\mathbb{Q},\mathrm{id}} = \alpha_{\mathbb{Q}} = \alpha$.

**Statement 2.4** (Statement R)**.** Let $G$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d \ge 1$, let $V \subset T_G(\mathbb{C})$ satisfy $(\mathrm{i})_V$, and let $W \subset V$ be defined over $\overline{\mathbb{Q}}$. Then for every finitely generated subgroup $Y \subset V \cap \Lambda(G)$,

$$\mathrm{rk}_{\mathbb{Z}}\, Y \le \big(\alpha(G) - \mathrm{rk}_{\mathbb{Z}}(Y \cap \Omega(G))\big)\, \dim_{\mathbb{C}}(V/W).$$

Statement R is Theorem 1.1 for $K = \mathbb{Q}$, $\Psi = \mathrm{id}$: $V \cap \Omega(G)$ is finitely generated, so taking $Y \supseteq V \cap \Omega(G)$ gives $\dim_{\mathbb{Q}}(V \cap \Lambda(G)) \le (\alpha(G) - \dim_{\mathbb{Q}}(V \cap \mathbb{Q}\Omega(G)))\dim_{\mathbb{C}}(V/W)$. Conversely, Theorem 1.1 follows from Statement R by Proposition 3.4. Statement R is [29, proof, step (a), p. 363]. We prove it in §3, for hyperplanes in Proposition 3.9 and in every codimension in Proposition 3.11. For hyperplanes it is the following theorem of Waldschmidt, in his formulation.

**Theorem 2.5** ([37, Thm. 1.1, pp. 375–376, under (5.4) of §5d, p. 394])**.** *Let $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$ of dimension $d \ge 1$, with $d_0$, $d_1$ maximal and $d_2 = d - d_0 - d_1$. Let $V$ be a hyperplane of $T_G(\mathbb{C})$, $W \subset V$ a subspace of dimension $t \ge 0$ defined over $\overline{\mathbb{Q}}$, and $Y = \mathbb{Z}y_1 + \cdots + \mathbb{Z}y_m \subset V$ a subgroup of rank $m$ with $\exp_G Y \subset G(\overline{\mathbb{Q}})$. Put $\kappa = \mathrm{rk}_{\mathbb{Z}}(Y \cap \ker \exp_G)$. If*

$$m > (d_1 + 2d_2 - \kappa)(d - 1 - t), \tag{5.4}$$

*then $V$ contains a nonzero algebraic Lie subalgebra of $T_G(\mathbb{C})$ defined over $\overline{\mathbb{Q}}$, that is, $T_{G'} \subset V$ for some connected algebraic subgroup $G' \ne 0$ of $G$ defined over $\overline{\mathbb{Q}}$.*

Since $\alpha(G) = d_1 + 2d_2$ for a maximal decomposition, Theorem 2.5 is the contrapositive of Statement R for hyperplanes. The printed Theorem 1.1 has the hypothesis $m > (d_1 + 2d_2)(d - 1 - t)$, without $\kappa$; [37, §5d] proves it under the weaker hypothesis (5.4) (p. 394 prints $\kappa = \ell - m$, a misprint for $m - \ell$, cf. Lemma 5.2 there). The case $W = 0$ ($t = 0$) of Theorem 2.5 is also printed as [44, Thm. 5.1], where it is attributed to [37, Thm. 1.1] and to [38, Thm. 4.1]. The proof in [37, §5d] applies Theorem 2.3 (in the form [37, Cor. 5.3]) to $G$ with $N = d - 1$, and then again to an auxiliary subgroup $G'$ given by [37, Lemma 5.1]. If Theorem 4.1 is used only through Lemma A.1, condition (2) can fail at that second application, but only at a linear corner such as $G' = \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \subset \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E$ (Lemma 3.6). Proposition 3.9 gives a self-contained proof of Theorem 2.5, with Theorem 4.1 through Lemma A.1 at the non-linear nodes, Waldschmidt's proof for linear groups at the linear nodes (Lemma 3.8) and Baker's theorem at the corners (Lemma 3.7). So Theorem 2.5 carries the label [Proved modulo published theorems and Lemma A.1]; the applications that use it (Proposition 7.5, the second derivations of Remarks 5.3, 5.6, 7.2 and 8.5, and the formal verification of §12.3) no longer rest on [37, §5d] as published.

2.4. **What is in print, and what is new.** *The statement.* Roy and Waldschmidt state Corollary 2.2 for every connected commutative $G$ over $\overline{\mathbb{Q}}$, every number field $K$ and every $V$ [29, Théorème, p. 363]. Their proof has two steps. Step (b) (p. 364) reduces to $K = \mathbb{Q}$, $\Psi = \mathrm{id}$ on $G' = G^n/H$, where $H$ is the largest connected subgroup killed by $\Psi \circ p_\theta$; it is written in full, and it contains the inner weight: the inequality obtained before the final "On peut aussi majorer $\alpha(G')$ par $n\alpha_K(G)$" is the first inequality of Theorem 1.1. Step (b) also uses case (a)

to see that $\Psi'$ is injective on $\Lambda(G')$; this is not needed for the inequality (Proposition 3.4). Step (a) (p. 363), which is Statement R, is derived "de manière purement formelle" from [37, Thm. 4.1] by applying Roy's categorical Proposition 2.1 of [28] to a category $\mathcal{G}_w$ and functions $a', b, c, d, r$ of [28, §6]. Their Remarque 2 says that Statement R with the $\kappa$-term was known when $\dim V = \dim G - 1$ (citing [37, Thm. 1.1]) or with $\kappa$ replaced by 0 ([28, §6, remarque (i)]), and is proved in general "en raffinant les arguments du §6 de [R1]" (with the function $a'$ in place of $a''$).

*In print.* (1) Hyperplanes: Theorem 2.5, with the proof of [37, §5d], for every $G$ up to isogeny (Lemma 3.1). Together with step (b) this proves Theorem 1.1 whenever $c_V = 1$, given Theorem 4.1 as printed. (2) Linear groups $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$: the period-refined subgroup theorem is proved in every codimension in [40, §11.7.3] (Lemma 11.24 with Theorem 11.21 gives [37, Thm. 4.1] for linear groups, and §11.7.2 deduces the other formulations, following Roy). Roy [27, Thm. 1, p. 25] restates [37, Thm. 4.1] for linear groups with the period term explicit: with $\Omega = 0 \times 2\pi i\,\mathbb{Q}^{d_1}$, the term $\dim_{\mathbb{Q}}(Y \cap \Omega)$ is subtracted from $d_1$ on the right-hand side; his Theorem 2 (p. 25, deduced from Theorem 1 in his §§2–3) refines it, in every codimension, by a choice of the quotient, keeping that term. [29, Remarque 3] derives the torus case with $\Psi = \mathrm{id}$ from [27, Thm. 5]. (3) $\kappa = 0$ in every codimension: [28, §6, remarque (i)], cited through [29, Remarque 2]; Waldschmidt's review of [28] (zbMATH 0763.11031) states its main inequality without a period term. (4) Waldschmidt's effective subgroup theorem [39] does not contain the period term; he writes in [39, §7(f)] that a refinement with periods seems "accessible" and states it as a conjecture.

*New here.* To our knowledge, no complete written proof of the $\kappa$-refined statement in codimension $\geq 2$ exists for groups with an abelian part; we could not consult [28], whose §6 may contain the verification of Roy's Proposition 2.1 for the function $a'$ (§11.1). This paper does not supply one in full either: at level (R-P) the case of groups with a torus in codimension $\geq 2$ outside (NR$^*$) is open (Remark 3.12), and outside the groups isogenous to $\mathbb{G}_{\mathrm{a}}^{a} \times \mathbb{G}_{\mathrm{m}}^{b} \times \prod_j E_j$ Lemma A.1 is a proof sketch. What §3 adds is: a deduction of Theorem 1.1, with the inner weight, from Theorem 4.1 as printed, in every codimension (level (R-W)); a subadditivity lemma for $\alpha$ that replaces the citation of [28, Lemme 4.3] in [29, p. 359] (Lemma 3.2); and, using Theorem 4.1 only through Lemma A.1, the cases $(\alpha)$–$(\gamma)$ of level (R-P), with a corner lemma (Lemma 3.6) that confines the failure of (2) in hyperplane inductions to linear groups, where Baker's theorem or Waldschmidt's proof for linear groups applies. We also include a derivation of Wüstholz's analytic subgroup theorem from Theorem 4.1 (Proposition 3.5); it is short, and plausibly folklore, since the route from Philippon's zero estimate to the analytic subgroup theorem is standard. The substantive gap in the published proofs is the multiplicity step of Theorem 4.1 for $W \neq 0$ and $b > 1$, which [37, p. 396] treats by a one-sentence pointer to [36, Prop. 2.4] (proved there for $W = 0$, $b = 1$) and to Bertrand's Lemme 7; Lemma A.1 writes it out under (2). Everything else in §3 is a short argument.

## 3. Proof of Theorem R

A *node* is a quadruple $(G, V, W, Y)$ as in Statement 2.4: $G$ connected commutative over $\overline{\mathbb{Q}}$ of dimension $d \geq 1$, $V \subset T_G(\mathbb{C})$ satisfying $(\mathrm{i})_V$, $W \subset V$ defined over $\overline{\mathbb{Q}}$, and $Y \subset V \cap \Lambda(G)$ finitely generated. We write $N = \dim V$, $t = \dim W$, $m = \mathrm{rk}\, Y$, $\kappa = \mathrm{rk}(Y \cap \Omega(G))$ and $\Gamma = \exp_G(Y) \subset G(\overline{\mathbb{Q}})$, so that $\mathrm{rk}\,\Gamma = m - \kappa$. Note that $N < d$, by $(\mathrm{i})_V$ and $d \geq 1$. For a subgroup $Y$ of $T_G(\mathbb{C})$ let $H_Y$ be the smallest connected algebraic subgroup of $G$ with $Y \subset T_{H_Y}$; it exists because $T_{(H_1 \cap H_2)^\circ} = T_{H_1} \cap T_{H_2}$, and $H_{\mathbb{Z}y_1 + \cdots + \mathbb{Z}y_m} = H_{y_1} + \cdots + H_{y_m}$. A node is *reduced* if $H_Y = G$.

The proof has three parts: the reduction of Theorem 1.1 to Statement R on $G^n/H_\Psi$ (§3.2), and Statement R for hyperplanes (§3.6) and in every codimension (§3.7), each by induction over subgroups with Theorem 2.3 at every node. §3.1 collects the structure facts and §§3.3–3.5 the inputs at special nodes. The levels of Theorem 1.1 are assembled in §3.8. The arguments of this section were checked by two independent internal referees, not peer-reviewed. The first supplied the corrections incorporated here (the linear nodes of §3.5, the Baker corner, the trivial decomposition in Proposition 3.5, and the form of (NR$^*$)); the second checked every step against [29], [37] and [40] and found no fatal error (only the trivial case $Y = 0$ in the proof of Lemma 3.8, now added).

### 3.1. **Structure and weights.**

**Lemma 3.1** (Structure)**.** *Let $G$ be a connected commutative algebraic group over $\overline{\mathbb{Q}}$.*

(a) *$G$ is isogenous over $\overline{\mathbb{Q}}$ to $\mathbb{G}_{\mathrm{a}}^{d_0(G)} \times \mathbb{G}_{\mathrm{m}}^{d_1(G)} \times G_2$, where $G_2$ is connected and has no surjection onto $\mathbb{G}_{\mathrm{a}}$ or $\mathbb{G}_{\mathrm{m}}$.*

(b) *An isogeny $\varphi\colon G \to G'$ identifies $T_G$ with $T_{G'}$ (with their $\overline{\mathbb{Q}}$-structures), maps $\Lambda(G)$ onto $\Lambda(G')$ and $\Omega(G)$ onto a subgroup of finite index of $\Omega(G')$, and induces a bijection between connected algebraic subgroups compatible with tangent spaces. It preserves $d_0$, $d_1$, $\alpha$, the ranks $m$, $\kappa$ of a node, and the conditions* $(\mathrm{i})_V$.

(c) *If $0 \to U \times T \to G \to B \to 0$ is the Chevalley decomposition ($U$ unipotent, $T$ a torus, $B$ an abelian variety), then $\operatorname{rk}\Omega(G) = \dim T + 2\dim B$, and $\operatorname{rk}\Omega(G) \le d_1(G) + 2\dim G_2 = \alpha(G)$.*

*Proof.* (a) If $G \to \mathbb{G}_{\mathrm{a}}^{d_0}$ and $G \to \mathbb{G}_{\mathrm{m}}^{d_1}$ are onto, so is $G \to \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$: its image is a connected subgroup projecting onto both factors, and connected subgroups of $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$ are products since $\operatorname{Hom}(\mathbb{G}_{\mathrm{a}}, \mathbb{G}_{\mathrm{m}}) = \operatorname{Hom}(\mathbb{G}_{\mathrm{m}}, \mathbb{G}_{\mathrm{a}}) = 0$. Take $d_0 = d_0(G)$, $d_1 = d_1(G)$ and such a surjection $\phi$. Since $\operatorname{Hom}(B, \text{linear groups}) = 0$, $\phi$ is already onto from $U \times T$, so $\phi|_U\colon U \to \mathbb{G}_{\mathrm{a}}^{d_0}$ and $\phi|_T\colon T \to \mathbb{G}_{\mathrm{m}}^{d_1}$ are onto ($\operatorname{Hom}(T, \mathbb{G}_{\mathrm{a}}) = \operatorname{Hom}(U, \mathbb{G}_{\mathrm{m}}) = 0$). The first has a linear section and the second a section up to isogeny; together they give $s\colon \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \to G$ with $\phi \circ s$ an isogeny. With $G_2 = (\ker\phi)^\circ$ the map $(s, \mathrm{incl})\colon \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2 \to G$ has finite kernel and is onto by dimension. $G_2$ has no $\mathbb{G}_{\mathrm{a}}$ or $\mathbb{G}_{\mathrm{m}}$ quotient by maximality of $d_0, d_1$.

(b) $d\varphi$ is a $\overline{\mathbb{Q}}$-rational isomorphism and $\exp_{G'} \circ d\varphi = \varphi \circ \exp_G$. Since $\varphi$ has finite fibres, $\varphi(x) \in G'(\overline{\mathbb{Q}})$ iff $x \in G(\overline{\mathbb{Q}})$, so $d\varphi(\Lambda(G)) = \Lambda(G')$; and $d\varphi(y) \in \Omega(G')$ iff $\exp_G(y) \in \ker\varphi$, which is killed by some $e \ge 1$, so $e\, d\varphi^{-1}(\Omega(G')) \subset \Omega(G)$. Connected subgroups correspond by $H \mapsto \varphi(H)$, $H' \mapsto \varphi^{-1}(H')^\circ$. If $\ker\varphi$ is killed by $e$, there is $\psi\colon G' \to G$ with $\psi \circ \varphi = [e]$, so surjections onto $\mathbb{G}_{\mathrm{a}}^d$, $\mathbb{G}_{\mathrm{m}}^d$ transfer in both directions.

(c) $0 \to \Omega(U \times T) \to \Omega(G) \to \Omega(B)$ is exact with image of finite index (a period of $B$ lifts to a vector whose exponential lies in $(U \times T)(\mathbb{C}) = \exp(T_{U\times T})$), $\operatorname{rk}\Omega(U \times T) = \dim T$ and $\operatorname{rk}\Omega(B) = 2\dim B$. For the inequality use (a) and (b): $\operatorname{rk}\Omega(G) = d_1 + \operatorname{rk}\Omega(G_2)$, $\Omega(G_2)$ is discrete in $T_{G_2}(\mathbb{C})$, and $\alpha(G) = 2\dim G - 2d_0 - d_1 = d_1 + 2\dim G_2$. □

By (b), Theorem 2.3, which is stated for literal products, applies to every $G$ after transport along an isogeny; [37, p. 389] handles subgroups "up to isogeny" in the same way. From now on we identify $G$ with $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ ($d_0, d_1$ maximal), write $\pi_0, \pi_1$ for the projections, and call this the *maximal decomposition*.

**Lemma 3.2** (Weights; subadditivity)**.** *Let $G$ have maximal decomposition $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$, and let $H \subset G$ be a connected algebraic subgroup. Put $d_0' = \dim\pi_0(H)$, $d_1' = \dim\pi_1(H)$, $d_2' = \dim H - d_0' - d_1'$, $\delta = \dim G/H$, $\delta_0 = d_0 - d_0'$, $\delta_1 = d_1 - d_1'$, $\delta_2 = \delta - \delta_0 - \delta_1$ (the quantities*

*of Theorem 2.3 for $G' = H$). Then $d'_2, \delta_2 \geq 0$ and*

$$\alpha(H) \leq d'_1 + 2d'_2, \qquad \alpha(G/H) \leq \delta_1 + 2\delta_2,$$
$$\operatorname{rk}\Omega(G/H) \leq \delta_1 + 2\delta_2, \qquad (d'_1 + 2d'_2) + (\delta_1 + 2\delta_2) = \alpha(G).$$

*In particular $\alpha(H) + \alpha(G/H) \leq \alpha(G)$, and $\alpha(H), \alpha(G/H) \leq \alpha(G)$.*

*Proof.* The image $(\pi_0, \pi_1)(H)$ is a connected subgroup of $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$, hence equal to $\pi_0(H) \times \pi_1(H)$ (as in the proof of Lemma 3.1(a)); so $d'_0 + d'_1 \leq \dim H$. $H$ surjects onto the vector group $\pi_0(H)$ and onto the torus $\pi_1(H)$, so $d_0(H) \geq d'_0$, $d_1(H) \geq d'_1$, and $\alpha(H) \leq 2\dim H - 2d'_0 - d'_1 = d'_1 + 2d'_2$. Next, $G \to \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \to (\mathbb{G}_{\mathrm{a}}^{d_0}/\pi_0(H)) \times (\mathbb{G}_{\mathrm{m}}^{d_1}/\pi_1(H)) \cong \mathbb{G}_{\mathrm{a}}^{\delta_0} \times \mathbb{G}_{\mathrm{m}}^{\delta_1}$ is onto and kills $H$, so $G/H$ surjects onto $\mathbb{G}_{\mathrm{a}}^{\delta_0} \times \mathbb{G}_{\mathrm{m}}^{\delta_1}$. Hence $\delta \geq \delta_0 + \delta_1$, $\alpha(G/H) \leq 2\delta - 2\delta_0 - \delta_1 = \delta_1 + 2\delta_2$, and, the kernel $K'$ of this surjection having dimension $\delta_2$, $\operatorname{rk}\Omega(G/H) \leq \operatorname{rk}\Omega(\mathbb{G}_{\mathrm{m}}^{\delta_1}) + \operatorname{rk}\Omega(K'^{\circ}) \leq \delta_1 + 2\delta_2$ (a period of $G/H$ mapping to 0 lies in $T_{K'^{\circ}}$ and is a period of $K'^{\circ}$). Finally $d'_1 + \delta_1 = d_1$ and $d'_2 + \delta_2 = d - d_0 - d_1 = d_2$, and $\alpha(G) = d_1 + 2d_2$. □

This is the subadditivity $\alpha(G') + \alpha(G/G') \leq \alpha(G)$ that [29, p. 359] cites from [28, Lemme 4.3]. It also gives $\alpha(G''_1) \leq \alpha(G''_2)$ for a quotient $G''_1$ of $G''_2$, which is Remark 2.1(1).

**Lemma 3.3.** *For any decomposition $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ (not necessarily maximal) put $\alpha'' = d_1 + 2d_2$. Then (2) holds iff $\alpha'' - \kappa \geq 2(d - N)$. For the maximal decomposition $\alpha'' = \alpha(G)$.*

*Proof.* $d_1 + 2d_0 = 2d - \alpha''$. □

### 3.2. The reduction to $K = \mathbb{Q}$ (step (b) of Roy–Waldschmidt).

**Proposition 3.4.** *Let $G, K, \Psi, V, W$ satisfy the hypotheses of Theorem 1.1, including (i). Let $G'' = G^n/H_\Psi$ with quotient map $q$, let $\Psi'' \colon T_{G''} \to \mathcal{E}$ be the map with $\Psi'' \circ dq = \Psi_\theta$, and put $V'' = \Psi''^{-1}(V)$, $W'' = \Psi''^{-1}(W)$. Then:*

(a) *$\Psi''(\Lambda(G'')) = \Psi(\Lambda_K(G))$ and $\Psi''(\mathbb{Q}\,\Omega(G'')) = \Psi(\Omega_K(G))$;*
(b) *if $G'' \neq 0$, then $V''$ satisfies $(\mathrm{i})_{V''}$ in $G''$;*
(c) *$W''$ is defined over $\overline{\mathbb{Q}}$, and $\dim V''/W'' \leq \dim V/W$, with equality if $V \subset W + \Psi(T_G)$;*
(d) *$\alpha(G'') = n\,\alpha_{K,\Psi}(G) \leq n\,\alpha_K(G)$;*
(e) *the codimension of $V''$ in $T_{G''}$ is $c_V = \dim \Psi(T_G)/(V \cap \Psi(T_G))$.*

*If Statement 2.4 holds for $(G'', V'', W'')$ (or $G'' = 0$), then Theorem 1.1 holds for $(G, K, \Psi, V, W)$.*

*Proof.* $G''$ is a connected commutative algebraic group over $\overline{\mathbb{Q}}$, and $\Psi''$ is $\overline{\mathbb{Q}}$-rational because $p_\theta$ has coefficients in $K$ and $dq$ is a $\overline{\mathbb{Q}}$-rational surjection.

(a) $\Lambda(G'') = dq(\Lambda(G^n))$: let $\exp(x) = P \in G''(\overline{\mathbb{Q}})$; lift $P$ to $\tilde{P} \in G^n(\overline{\mathbb{Q}})$ and $x$ to $y$; then $\exp(y) - \tilde{P} \in H_\Psi(\mathbb{C}) = \exp(T_{H_\Psi})$ since $H_\Psi$ is connected, say $= \exp(h)$, and $y - h \in \Lambda(G^n)$ with $dq(y - h) = x$. The same argument with $P = 0$ gives $\Omega(G'') = dq(\Omega(G^n))$. Hence $\Psi''(\Lambda(G'')) = \Psi_\theta(\Lambda(G)^n) = \Psi(\sum_j \theta_j \Lambda(G)) = \Psi(\Lambda_K(G))$, and likewise for the $\mathbb{Q}$-span of the periods.

(b) A connected algebraic subgroup of $G''$ is $\tilde{G}/H_\Psi$ with $\tilde{G} \supseteq H_\Psi$ connected, and $\Psi''(T_{\tilde{G}/H_\Psi}) = \Psi_\theta(T_{\tilde{G}}) = \Psi(S)$ with $S = p_\theta(T_{\tilde{G}}) \in S_K(G)$. If $T_{\tilde{G}/H_\Psi} \subset V''$, then $\Psi(S) \subset V$, so $\Psi(S) = 0$ by (i), i.e. $T_{\tilde{G}} \subset \ker \Psi_\theta$, and $\tilde{G} = H_\Psi$ by maximality.

(c) $W''$ is the preimage of a $\overline{\mathbb{Q}}$-rational subspace under a $\overline{\mathbb{Q}}$-rational map. $\Psi''$ induces an injection $V''/W'' \to V/W$. If $V \subset W + \Psi(T_G)$ then $V = W + (V \cap \Psi(T_G))$, and $\Psi''$ maps $V''$ onto $V \cap \Psi(T_G)$ (since $\Psi''(T_{G''}) = \Psi_\theta(T_{G^n}) = \Psi(T_G)$), so the injection is onto ([29, (3)]).

(d) The equality is the definition; the inequality is Remark 2.1(1).

(e) $\Psi''(T_{G''}) = \Psi(T_G)$ and $V'' = \Psi''^{-1}(V \cap \Psi(T_G))$.

*Deduction.* If $G''=0$ then $\Psi_\theta=0$, $\Psi(\Lambda_K(G))=0$ and there is nothing to prove. Otherwise $V''\neq T_{G''}$ by (b). $V\cap\Psi(\Omega_K(G))$ is a $K$-space of finite dimension $\kappa$ (it lies in the $K$-span of the finitely many $\Psi(\omega)$, $\omega$ in a basis of $\Omega(G)$). Let $X$ be a $K$-subspace of $V\cap\Psi(\Lambda_K(G))$ of finite dimension $x$ containing $V\cap\Psi(\Omega_K(G))$. By (a) we may choose a $\mathbb{Q}$-basis of $V\cap\Psi(\Omega_K(G))$ of the form $\Psi''(\omega_i'')$ with $\omega_i''\in\Omega(G'')$ ($1\le i\le n\kappa$; clear denominators), and extend it to a $\mathbb{Q}$-basis of $X$ by elements $\Psi''(y_j'')$ with $y_j''\in\Lambda(G'')$. All these $\omega_i'',y_j''$ lie in $V''$. Let $Y''$ be the group they generate: $\operatorname{rk}Y''=nx$ (their images are $\mathbb{Q}$-independent), and $\kappa'':=\operatorname{rk}(Y''\cap\Omega(G''))\ge n\kappa$. Statement R for $(G'',V'',W'',Y'')$, (c), (d) and $\alpha(G'')-\kappa''\ge0$ (Lemma 3.1(c)) give

$$nx\le(\alpha(G'')-\kappa'')\dim V''/W''\le(n\alpha_{K,\Psi}(G)-n\kappa)\dim V/W.$$

So every such $X$ has $x\le(\alpha_{K,\Psi}(G)-\kappa)\dim V/W$; hence $V\cap\Psi(\Lambda_K(G))$ is finite-dimensional and satisfies the first inequality of Theorem 1.1. The second is (d). □

Only the surjections in (a) are used; the injectivity of $\Psi''$ on $\Lambda(G'')$, for which [29, p. 364] use case (a) again, is not needed. The subspace $W''\supseteq\ker\Psi''$ is nonzero whenever $\Psi''$ is not injective, even for $W=0$; and $\ker\Psi''$ contains no algebraic Lie subalgebra, by maximality of $H_\Psi$, so it cannot be divided out. Any connected $H\subset H_\Psi$ could replace $H_\Psi$, at the cost of a larger weight (Lemma 3.2); so a proof that uses a subgroup $U\subset H_\Psi$ found by hand (as in the applications) needs only an upper bound for $\alpha(G^n/U)$.

### 3.3. **The analytic subgroup theorem from Theorem 4.1.**

**Proposition 3.5** (Wüstholz's analytic subgroup theorem)**.** *Let $R\subset T_G(\mathbb{C})$ be a subspace defined over $\overline{\mathbb{Q}}$ and $Y\subset R\cap\Lambda(G)$ a finitely generated subgroup. Then $T_{H_Y}\subset R$.*

This is Wüstholz's theorem [52] (for $Y=\mathbb{Z}y$: the smallest algebraic subgroup whose Lie algebra contains a logarithm $y$ of an algebraic point has its Lie algebra in every $\overline{\mathbb{Q}}$-rational subspace containing $y$). We derive it from Theorem 2.3, so that level (R-W) of Theorem 1.1 rests on Theorem 4.1 alone.

*Proof.* Induction on $d=\dim G$. Replacing $(G,R)$ by $(H_Y,R\cap T_{H_Y})$ we may assume $H_Y=G$, and must show $R=T_G$. Suppose not. Apply Theorem 2.3 with $V=W=R$ (so $N=t<d$) and the *trivial decomposition* $d_0=d_1=0$, $G_2=G$, which Theorem 2.3 allows (cf. [36, pp. 636–637]). Then $\delta_2=\delta$, and (1) reads $(\lambda+2\delta)(d-N)\le(\delta-\tau)(2d-\kappa)$ for some connected $G'\neq G$ with $\delta>\tau$. If $G'=0$, then $\delta=d$, $\tau=N$ and $\lambda=m-\kappa$, so $(m-\kappa+2d)(d-N)\le(d-N)(2d-\kappa)$ gives $m\le0$: $Y=0$, and then $H_Y=0\neq G$, a contradiction. If $G'\neq0$, let $q\colon G\to\bar G=G/G'$. Then $\bar R=dq(R)$ is $\overline{\mathbb{Q}}$-rational with $\dim\bar R=\tau<\delta=\dim\bar G$, and $\bar Y=dq(Y)\subset\bar R\cap\Lambda(\bar G)$. By induction $T_{H_{\bar Y}}\subset\bar R\neq T_{\bar G}$, so $H_{\bar Y}\neq\bar G$, and $q^{-1}(H_{\bar Y})^\circ$ is a proper connected subgroup of $G$ whose tangent space contains $Y$, contradicting $H_Y=G$. □

For the trivial decomposition, condition (2) reads $2N\ge\kappa$, which always holds because $Y\cap\Omega(G)$ is discrete in $R$; and $d_2=d\ge1$, $d_0=0$ (so the device $b>1$ is not needed). So at level (R-P) Proposition 3.5 needs Lemma A.1 for $G_2=G$, i.e. for a $G_2$ that may contain factors $\mathbb{G}_{\mathrm a}$, $\mathbb{G}_{\mathrm m}$ (Remark A.3); alternatively one may cite [52] itself. Hermite–Lindemann is a check: on $\mathbb{G}_{\mathrm a}\times\mathbb{G}_{\mathrm m}$ with $R=\mathbb{C}(1,1)$ the only admissible $G'$ is 0. The trivial decomposition does not help elsewhere: absorbing $\mathbb{G}_{\mathrm a}$ or $\mathbb{G}_{\mathrm m}$ factors into $G_2$ restores (2) but raises the weight, and the resulting bounds are too weak for Statement R (on $\mathbb{G}_{\mathrm a}^{d_0}\times\mathbb{G}_{\mathrm m}$ one gets $m\le2(d_0-t)$ instead of $d_0-t$).

### 3.4. **Corners.**

**Lemma 3.6** (Corner lemma)**.** *Let $G$ have maximal decomposition $\mathbb{G}_{\mathrm a}^{d_0}\times\mathbb{G}_{\mathrm m}^{d_1}\times G_2$, let $V\subset T_G(\mathbb{C})$ be a hyperplane satisfying* (i)$_V$*, and put $\rho=\operatorname{rk}(V\cap\Omega(G))$. If $\rho\ge\alpha(G)-1$, then $G_2=0$, $d_1\le1$*

*and $\rho = 0$. In particular, if (2) fails at a hyperplane node $(G, V, W, Y)$ (that is, $\kappa \geq \alpha(G) - 1$, by Lemma 3.3), then $G \cong \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$ with $d_1 \leq 1$ and $\kappa = 0$.*

*Proof.* Let $0 \to U \times T \to G_2 \to B \to 0$ be the Chevalley decomposition of $G_2$. By Lemma 3.1(c), $\rho \leq \operatorname{rk} \Omega(G) = d_1 + \dim T + 2 \dim B$, while $\alpha(G) = d_1 + 2(\dim U + \dim T + \dim B)$. So $\rho \geq \alpha(G) - 1$ forces $2 \dim U + \dim T \leq 1$: $U = 0$ and $\dim T \leq 1$.

*Case* $\dim T = 0$, $B \neq 0$. Then $G_2 = B$ and $\operatorname{span}_{\mathbb{R}} \Omega(G) = i\mathbb{R}^{d_1} \oplus T_B$. The real span $L \subset V$ of $V \cap \Omega(G)$ has real codimension $\leq 1$ in it, so $L \cap T_B$ has real codimension $\leq 1$ in $T_B \cong \mathbb{C}^g$, $g \geq 1$. A real subspace of real codimension $\leq 1$ of $\mathbb{C}^g$ spans $\mathbb{C}^g$ over $\mathbb{C}$ (a proper complex subspace has real dimension $\leq 2g - 2$). So $T_B \subset V$, contradicting $(\mathrm{i})_V$.

*Case* $\dim T = 1$. Then $B \neq 0$ (otherwise $G_2 = \mathbb{G}_{\mathrm{m}}$, against the maximality of $d_1$), and $\rho \geq \alpha(G) - 1 = \operatorname{rk} \Omega(G)$. So $V \cap \Omega(G)$ has finite index in $\Omega(G)$, and $V$ contains $\operatorname{span}_{\mathbb{C}} \Omega(G_2)$, which is $T_{G_2}$ (it contains the periods of $T$ and maps onto $T_B$, since $\Omega(G_2) \to \Omega(B)$ has image of finite index). This contradicts $(\mathrm{i})_V$.

*Case* $G_2 = 0$. If $\rho \geq 1$, a nonzero sublattice of $\Omega(G) = 0 \times 2\pi i \mathbb{Z}^{d_1}$ lies in $V$; its complex span is the Lie algebra of a subtorus, contradicting $(\mathrm{i})_V$. So $\rho = 0$, and $\alpha(G) = d_1 \leq \rho + 1 = 1$. □

**Lemma 3.7** (Corners by Baker's theorem)**.** *Statement R holds for $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$ with $d_1 \leq 1$ and $V$ a hyperplane.*

*Proof.* Here $\alpha(G) = d_1$, and $\kappa = 0$ by Lemma 3.6; we must show $m \leq d_1(d - 1 - t)$. If $d_1 = 0$, then $V \cap \Lambda(G) = V \cap \overline{\mathbb{Q}}^{d_0}$, and a nonzero element would span a $\overline{\mathbb{Q}}$-rational line, the Lie algebra of a subgroup; so $Y = 0$. Let $d_1 = 1$, with coordinates $(z, x) \in \mathbb{C}^{d_0} \times \mathbb{C}$. By $(\mathrm{i})_V$, $0 \times \mathbb{C} \not\subset V$, so the hyperplane $V$ is the graph $\{(z, L(z))\}$ of a linear form $L$ on $\mathbb{C}^{d_0}$. An element of $Y$ is $(b, L(b))$ with $b \in \overline{\mathbb{Q}}^{d_0}$ and $L(b) \in \mathcal{L}$. The map $y \mapsto L(b(y))$ is injective on $Y$ (a kernel element would be a $\overline{\mathbb{Q}}$-point $(b, 0)$ of $V$), so for a basis $y_1, \dots, y_m$ of $Y$ the logarithms $\ell_j = L(b_j)$ are $\mathbb{Q}$-linearly independent, and by Baker's theorem [3] the numbers $1, \ell_1, \dots, \ell_m$ are $\overline{\mathbb{Q}}$-linearly independent. The rational subspace $W \subset V$ is the graph of $L$ over a $\overline{\mathbb{Q}}$-rational $W_0 \subset \mathbb{C}^{d_0}$ of dimension $t$, with $L(W_0(\overline{\mathbb{Q}})) \subset \overline{\mathbb{Q}}$. If $\sum_j c_j b_j \in W_0(\overline{\mathbb{Q}})$ with $c \in \overline{\mathbb{Q}}^m$, then $\sum_j c_j \ell_j \in \overline{\mathbb{Q}}$, so $c = 0$. Hence $b_1, \dots, b_m$ are $\overline{\mathbb{Q}}$-independent modulo $W_0(\overline{\mathbb{Q}})$, and $m + t \leq d_0 = d - 1$. □

So at a corner, Statement R costs a citation of Baker's theorem, not of Theorem 4.1 for linear groups.

### 3.5. **Linear nodes.**

**Lemma 3.8** (Theorem 4.1 for linear groups)**.** *If $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1}$ and $\kappa = 0$, the conclusion of Theorem 2.3 holds, without condition (2). If $(G, V, W, Y)$ is a node with $G$ linear, then $\kappa = 0$.*

*Proof.* [Lit] For $d \geq 2$, $1 \leq N < d$ and $Y \neq 0$ the first sentence is Waldschmidt's [40, Thm. 11.21, p. 404] (§11.7.1), which is written in full, for $W \subset V$ rational over $\overline{\mathbb{Q}}$, $Y \neq 0$ of finite rank in $V \cap \Lambda(G)$, and $\lambda = \operatorname{rk} \Gamma / (\Gamma \cap G'(\mathbb{C}))$; its conclusion $\delta > \tau$, $(\lambda + \delta_1)/(\delta - \tau) \leq d_1/(d - N)$ is (1) with $d_2 = \delta_2 = 0$ and $\kappa = 0$. ([40, §11.7.3] adds the periods through Lemma 11.24 and notes that the result is [37, Thm. 4.1] for linear groups.) If $Y = 0$ (in particular if $N = 0$), $G' = 0$ works: then $\lambda = 0$, $\delta_1 = d_1$, $\delta = d$, $\tau = t$, so $\delta > \tau$ as $t \leq N < d$, and (1) reads $d_1(d - N) \leq (d - t)d_1$, which holds as $t \leq N$. The case $d = 1$ is covered by these cases. For the second sentence: if $\kappa \geq 1$, a nonzero sublattice of $\Omega(G) = 0 \times 2\pi i \mathbb{Z}^{d_1}$ lies in $V$, and its complex span is the Lie algebra of a subtorus, contradicting $(\mathrm{i})_V$. □

This is the input that a Lemma U with $d_2 = 0$, $\kappa = 0$ would provide. We use the linear subgroup theorem of [40] for it. (Alternatively, [36, p. 641] treats $d_2 = 0$ inside the proof of Prop. 2.4: only the last argument there can fail, and for $\kappa = 0$ it is replaced by the Remarque of p. 636;

we do not use this route.) Linear nodes occur in hyperplane inductions away from the corners: for a hyperplane in $\mathbb{G}_{\mathrm{m}}^2 \times E$, Theorem 2.3 can return $G' = \mathbb{G}_{\mathrm{m}}^2$, and Statement R at $(\mathbb{G}_{\mathrm{m}}^2, \text{a line})$ is the six exponentials theorem ($t = 0$) or the Gel'fond–Schneider theorem ($t = 1$).

### 3.6. **Hyperplanes.**

**Proposition 3.9** (Statement R for hyperplanes)**.** *Statement 2.4 holds whenever $V$ is a hyperplane of $T_G(\mathbb{C})$: for every node $(G, V, W, Y)$ with $\dim V = d - 1$,*

$$m \le (\alpha(G) - \kappa)(d - 1 - t).$$

*The proof uses Theorem 2.3 only at non-linear nodes satisfying (2) (through Lemma A.1), Lemma 3.8 at linear nodes, and Lemma 3.7 at corners.*

*Proof.* Induction on $d$. If $d = 1$ then $V = 0$, $Y = 0$ and the claim is $0 \le 0$. Let $d \ge 2$, with the maximal decomposition of $G$, and $\alpha = \alpha(G) = d_1 + 2d_2$.

*Case 1: (2) fails.* By Lemma 3.6, $G$ is a corner, and Lemma 3.7 applies.

*Case 2: (2) holds.* Apply Theorem 2.3 (by Lemma A.1 if $d_2 \ge 1$, by Lemma 3.8 if $d_2 = 0$; then $\kappa = 0$). With $d - N = 1$ it gives a connected $G' \ne G$ with $\delta > \tau$ and

$$\lambda + \bar\alpha \le (\delta - \tau)(\alpha - \kappa), \qquad \bar\alpha := \delta_1 + 2\delta_2.$$

If $G' = 0$, then $\lambda = m - \kappa$, $\bar\alpha = \alpha$, $\delta - \tau = d - t$, and $m - \kappa + \alpha \le (d - t)(\alpha - \kappa)$ is the claim. Let $G' \ne 0$, of dimension $d'$, and $q\colon G \to \bar G = G/G'$. By $(\mathrm{i})_V$, $T_{G'} \not\subset V$, so $V' = V \cap T_{G'}$ is a hyperplane of $T_{G'}$, and it satisfies $(\mathrm{i})_{V'}$. Put $W' = W \cap T_{G'}$, of dimension $t'$, and $Y' = Y \cap T_{G'} \subset V' \cap \Lambda(G')$, with $m' = \operatorname{rk} Y'$ and $\kappa' = \operatorname{rk}(Y' \cap \Omega(G'))$. By induction and Lemma 3.2,

$$m' \le (\alpha(G') - \kappa')(d' - 1 - t') \le (d_1' + 2d_2' - \kappa')(d' - 1 - t').$$

The group $\bar Y = dq(Y) \cong Y/Y'$ maps onto $q(\Gamma) \cong \Gamma/(\Gamma \cap G')$ with kernel $\bar Y \cap \Omega(\bar G)$, of rank $\bar\kappa$; so $m - m' = \lambda + \bar\kappa$. The map $dq$ sends $Y \cap \Omega(G)$ to $\bar Y \cap \Omega(\bar G)$ with kernel $Y' \cap \Omega(G')$, so $\kappa \le \kappa' + \bar\kappa$; and $\bar\kappa \le \operatorname{rk}\Omega(\bar G) \le \bar\alpha$ (Lemma 3.2). Since $(d_1' + 2d_2') + \bar\alpha = \alpha$, we get $d_1' + 2d_2' - \kappa' \le \alpha - \bar\alpha - \kappa + \bar\kappa \le \alpha - \kappa$. With $t = t' + \tau$, $d = d' + \delta$ and $d' - 1 - t' \ge 0$:

$$m = m' + \lambda + \bar\kappa \le (\alpha - \kappa)(d' - 1 - t') + (\delta - \tau)(\alpha - \kappa) = (\alpha - \kappa)(d - 1 - t). \qquad \square$$

*Proof of Theorem 2.5.* It is the contrapositive of Proposition 3.9, since $\alpha(G) = d_1 + 2d_2$ for the maximal decomposition. $\square$

Every node of this induction is a hyperplane node, so either (2) holds there or the node is a corner; this replaces [37, Lemmas 5.1, 5.2] and the second application of Theorem 4.1 in [37, §5d]. Small cases: on $\mathbb{G}_{\mathrm{m}}^2 \times E$ the nodes are $\mathbb{G}_{\mathrm{m}}$, $E$ (trivial), $\mathbb{G}_{\mathrm{m}}^2$ (linear, Lemma 3.8) and $T' \times E$ (Lemma A.1), and $\kappa \ge 3$ forces $T_E \subset V$; on $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E$ the nodes are $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}$ (a corner), $\mathbb{G}_{\mathrm{a}} \times E$ and $\mathbb{G}_{\mathrm{m}} \times E$, where $\kappa \ge 1$, resp. $\kappa \ge 2$, forces $T_E \subset V$.

### 3.7. **Arbitrary codimension.**

**Lemma 3.10** ((2) at reduced nodes)**.** *Let $(G, V, W, Y)$ be a reduced node, with maximal decomposition $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times G_2$. Then $d\pi_0\colon V \to \mathbb{C}^a$ is onto, and $\operatorname{rk}(V \cap \Omega(G)) \le 2(N - a)$. Hence (2) holds at the node if $b + \operatorname{rk}(V \cap \Omega(G)) \le 2(N - a)$, and in particular whenever $b = d_1(G) = 0$. If $G$ is unipotent, then $G = 0$.*

*Proof.* $d\pi_0(Y) \subset \overline{\mathbb{Q}}^a$. If its $\overline{\mathbb{Q}}$-span were a proper subspace $U_0$, then $\pi_0^{-1}(U_0)^\circ = U_0 \times \mathbb{G}_{\mathrm{m}}^b \times G_2$ would be a proper connected subgroup whose tangent space contains $Y$, against $H_Y = G$. So $d\pi_0(V) \supseteq d\pi_0(Y)$ spans $\mathbb{C}^a$. Periods lie in $\ker d\pi_0$, and $V \cap \Omega(G)$ is discrete in $V \cap \ker d\pi_0$, of dimension $N - a$. Then (2), $2N \ge b + 2a + \kappa$, follows from $\kappa \le \operatorname{rk}(V \cap \Omega(G))$. If $G = \mathbb{G}_{\mathrm{a}}^a$, then $Y \subset V \cap \overline{\mathbb{Q}}^a = 0$ by $(\mathrm{i})_V$, so $G = H_Y = 0$. $\square$

**Condition (NR$^*$).** For a group $G''$ and a subspace $V'' \subset T_{G''}$ we say that (NR$^*$) holds if for every connected algebraic subgroup $H \neq 0$ of $G''$ of the form $H = H_{Y'}$, with $Y' \subset V'' \cap \Lambda(G'')$ finitely generated, that is not linear,

$$d_1(H) + \mathrm{rk}\big(V'' \cap \Omega(H)\big) \leq 2\big(\dim(V'' \cap T_H) - d_0(H)\big). \tag{3}$$

By Lemma 3.10 this is exactly (2) at the reduced non-linear nodes, with $\kappa$ replaced by the larger $\mathrm{rk}(V'' \cap \Omega(H))$. It holds automatically if $G''$ has trivial maximal torus ($d_1(H) = 0$ for every $H$), and if $V''$ is a hyperplane (then $V'' \cap T_H$ is a hyperplane of $T_H$ for every $H \neq 0$, and Lemma 3.6, stated with $\rho = \mathrm{rk}(V'' \cap \Omega(H))$, shows that (3) can fail only for linear $H$). An earlier form of (NR$^*$), with all subgroups $H$ and a clause for linear $H$, was wrong: it fails for $H = \mathbb{G}_{\mathrm{a}}$.[1]

**Proposition 3.11** (Statement R in every codimension)**.** *Let $(G, V, W)$ be as in Statement 2.4. Assume that the conclusion of Theorem 2.3 holds at every reduced node $(H, V \cap T_H, W \cap T_H, Y')$ with $H \subset G$, and that Proposition 3.5 holds for the subgroups of $G$. Then Statement R holds for $(G, V, W)$: $m \leq (\alpha(G) - \kappa)(N - t)$ for every finitely generated $Y \subset V \cap \Lambda(G)$.*

*Proof.* Induction on $d$; put $A = \alpha(G) - \kappa$, which is $\geq 0$ by Lemma 3.1(c).

*Reduction.* If $H = H_Y \neq G$, pass to the node $(H, V \cap T_H, W \cap T_H, Y)$: (i) is inherited, $\kappa$ is unchanged ($\Omega(H) = \Omega(G) \cap T_H$), $\alpha(H) \leq \alpha(G)$ (Lemma 3.2), and $(V \cap T_H)/(W \cap T_H)$ injects into $V/W$. By induction, $m \leq (\alpha(H) - \kappa)\dim(V \cap T_H)/(W \cap T_H) \leq A(N - t)$. So assume $H_Y = G$.

*Theorem 4.1.* With the maximal decomposition, Theorem 2.3 gives a connected $G' \neq G$ with $\delta > \tau$ and $(\lambda + \bar\alpha)(d - N) \leq (\delta - \tau)A$, $\bar\alpha = \delta_1 + 2\delta_2$. If $G' = 0$, then $\lambda = m - \kappa$, $\bar\alpha = \alpha(G)$ and $\delta - \tau = d - t$, so $m \leq A(N - t)/(d - N) \leq A(N - t)$.

Let $G' \neq 0$, $q\colon G \to \bar G = G/G'$, and write $\bar V = dq(V)$, $\bar W = dq(W)$, $V' = V \cap T_{G'}$, $W' = W \cap T_{G'}$, $Y' = Y \cap T_{G'}$, with $m', \kappa'$ as before. Put

$$e = \dim \bar V - \dim \bar W, \quad f = \dim V'/W', \quad p = \dim T_{G'} - \dim V', \quad q_0 = \delta - \dim \bar V.$$

Then $N - t = e + f$, $d - N = p + q_0$, $\tau = \dim \bar W$, $\delta - \tau = e + q_0$, and $p \geq 1$ by (i)$_V$. As in the proof of Proposition 3.9, $m = m' + \lambda + \bar\kappa$, $\kappa \leq \kappa' + \bar\kappa$, $\bar\kappa \leq \bar\alpha$, and $\alpha(G') - \kappa' \leq A - (\bar\alpha - \bar\kappa)$. The node $(G', V', W', Y')$ satisfies (i)$_{V'}$, so by induction $m' \leq (\alpha(G') - \kappa')f$.

*Case $e \geq 1$.* From (1), $\lambda \leq A(e + q_0)/(p + q_0) - \bar\alpha$. Hence

$$m \leq \big(A - (\bar\alpha - \bar\kappa)\big)f + A\frac{e + q_0}{p + q_0} - \bar\alpha + \bar\kappa \leq Af + A\frac{e + q_0}{p + q_0} \leq A(f + e) = A(N - t),$$

using $\bar\alpha \geq \bar\kappa$, $f \geq 0$, $A \geq 0$ and $(e + q_0)/(p + q_0) \leq e$ for $p, e \geq 1$.

*Case $e = 0$.* Then $V \subset W + T_{G'}$, a $\overline{\mathbb{Q}}$-rational subspace containing $Y$. By Proposition 3.5, $T_G = T_{H_Y} \subset W + T_{G'}$, so $\tau = \dim(W + T_{G'})/T_{G'} = \delta$, contradicting $\delta > \tau$. □

Theorem 2.3 is applied only at reduced nodes $H = H_{Y'}$ ($H \subset G$, $Y' \subset V \cap \Lambda(G)$), because the reduction step comes first at every node. For hyperplanes the case $e = 0$ cannot occur. For $V$ of codimension $\geq 2$ the case $G' = 0$ gives the bound $A(N - t)/(d - N)$, which is stronger than Statement R by the factor $d - N$; the subgroups $G' \neq 0$ force the codimension-free form. This is why Theorem 4.1 applied subgroup by subgroup can give more than Theorem 1.1 in codimension $\geq 2$ (§10).

[1] Pointed out by an internal referee.

3.8. **Proof of Theorem R.** By Proposition 3.4, Theorem 1.1 for $(G, K, \Psi, V, W)$ follows from Statement 2.4 for $(G'', V'', W'')$, where $G'' = G^n/H_\Psi$ and $V''$ has codimension $c_V$ in $T_{G''}$.

*Level (R-W).* Proposition 3.11 applies to $(G'', V'', W'')$: Theorem 2.3 as printed has no condition (2), and Proposition 3.5 follows from it. This proves Theorem 1.1 as stated, given [37, Thm. 4.1] as printed.

*Level (R-P).* ($\alpha$) If $c_V = 1$, Proposition 3.9 applies to $(G'', V'', W'')$; its inputs are Lemma A.1 at the non-linear nodes (where (2) holds by Lemma 3.6), Lemma 3.8 at the linear nodes, and Baker's theorem at the corners. ($\beta$) If $G$ has trivial maximal torus, so have $G^n$, $G''$ and all their connected subgroups: the torus of a quotient $\bar{G} = G/H$ is the image of the torus of $G$ (the linear part of $\bar{G}$ is the image of that of $G$, since a connected group that is both a quotient of a linear group and an abelian variety is trivial). By Lemma 3.10, (2) holds at every reduced node, and the reduced linear nodes are 0. So Proposition 3.11 applies with Theorem 2.3 through Lemma A.1, and Proposition 3.5 through Lemma A.1 for the trivial decomposition (or [52]). ($\gamma$) Under (NR$^*$) for $(G'', V'')$, (2) holds at the reduced non-linear nodes, and the reduced linear nodes are handled by Lemma 3.8; the conclusion follows as in ($\beta$). □

**Remark 3.12** (The remaining case)**.** At level (R-P) the case left open is a reduced node $H$ with a torus quotient and $d_1(H) + \kappa_H > 2(N_H - d_0(H))$, in codimension $\geq 2$. This includes every reduced non-linear node with $d_1(H) \geq 1$ and $N_H = d_0(H)$. The simplest example is a line in $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times E$ with $H_Y = G$ ($2 \geq 1 + 2 + \kappa$ fails); others are a line in $\mathbb{G}_\mathrm{m}^2 \times E$ with $H_Y = G$ containing a period ($2 \geq 2 + \kappa$ fails for $\kappa \geq 1$) and a line in $\mathbb{G}_\mathrm{m}^3 \times E$ with $H_Y = G$ ($2 \geq 3 + \kappa$ fails). There Theorem 2.3 is used outside the range of Lemma A.1. For $W = 0$, [36, Prop. 2.4] handles exactly the complement of (2) by its reductions (a)–(c) (pp. 638–639), which end in "either $d_2 = 0$ or $2n \geq d_1 + 2d_0 + \kappa$" and pass to the linear quotient $G/G_2$. A version with multiplicities along $W$ would prove Theorem 1.1 in full at level (R-P). Two points must be checked: the reduced construction has to be run at the scale $\Delta$ of Lemma A.1, not at the smaller scale $\Delta'$ of W83's reduction (b), because Steps 2–3 of Lemma A.1 need smallness $e^{-\nu^6\Delta}$ against derivatives of order $T$ with $T \log T \asymp \Delta$, and with degrees fitted to $\Delta' \ll \Delta$ Liouville's inequality fails; and reduction (a) (dividing out $\kappa$ when $d_2 = 0$) needs $W$ compatible with the splitting $\mathbb{C}^n/\ker\phi \cong \mathbb{C}^{n-\kappa} \times (\mathbb{C}^\times)^\kappa$. We have not done this. In the other direction, at nodes where the periods in $Y$ are linearly independent over $\mathbb{C}$ (for instance $\kappa = 0$), W83's Remarque on p. 636 should give Lemma A.1 under the weaker condition $d_1 + 2d_2 - \kappa \geq d - N$, without Masser's interpolation theorem [Proof sketch; not used].

## 4. CM abelian varieties, universal vector extensions, and the applications of Theorem R

This section gives the linear algebra of CM abelian varieties of any dimension $g$ and of their universal vector extensions (§§4.1–4.2), the weights (§4.3), and the way the applications of §§5–9 follow from Theorem 1.1 (§4.4).

4.1. **CM abelian varieties.** Let $A/\overline{\mathbb{Q}}$ be a simple abelian variety of dimension $g$ with CM by the CM field $F$, $[F : \mathbb{Q}] = 2g$, with (primitive) CM type $\Phi$. For $\sigma \in \Phi$ let $z_\sigma \colon \mathrm{Lie}(A) \to \mathbb{C}$ be the $\overline{\mathbb{Q}}$-rational eigen-coordinate dual to a holomorphic eigen-differential $\omega_\sigma$, so that $z_\sigma([x]v) = \sigma(x) z_\sigma(v)$ for $x \in \mathcal{O}_F$. Since $H_1(A, \mathbb{Q}) = F\gamma_0$ for any $0 \neq \gamma_0 \in \Omega(A)$, we have $z_\sigma(x\gamma_0) = \sigma(x) P_\sigma$ with $P_\sigma = z_\sigma(\gamma_0)$, so $z_\sigma(\Omega(A)) \subset \sigma(F) P_\sigma$.

*Reduction to the simple case.* A CM abelian variety over $\overline{\mathbb{Q}}$ is isogenous to $\prod_i B_i^{m_i}$ with $B_i$ simple CM abelian varieties of primitive type, with CM fields $F_i$. By its *CM algebra* we mean any commutative semisimple $\mathbb{Q}$-subalgebra $\mathcal{E} \subset \mathrm{End}^0(A)$ of degree $2 \dim A$ (so $\mathcal{E} = \prod_i \mathcal{E}_i$ with $\mathcal{E}_i \supseteq F_i$ maximal commutative in $M_{m_i}(F_i)$), and eigen-differentials of $A$ are taken with respect

to $\mathcal{E}$; the argument below does not depend on this choice. Being maximal commutative, $\mathcal{E}$ contains the centre $\prod_i F_i$ of $\mathrm{End}^0(A)$ and its central idempotents. An eigen-character $\mathcal{E} \to \overline{\mathbb{Q}}$ is nonzero on exactly one of these, so the eigen-differential lives on one isotypic part $B_i^{m_i}$, and the character restricts to an embedding $\sigma_0$ of $F_i$, which acts on $B_i^{m_i}$ diagonally. An eigen-differential is therefore, on that part, a $\sigma_0$-eigen class for $F_i$, i.e. a $\overline{\mathbb{Q}}$-combination of pull-backs $\mathrm{pr}_k^*\omega_{\sigma_0}$ of one $\sigma_0$-eigen differential of $B_i$, of the same kind. So every nonzero eigen-period lies in $\overline{\mathbb{Q}}\,\sigma_0(F_i)P_{\sigma_0}(B_i) \setminus \{0\} = \overline{\mathbb{Q}}^\times P_{\sigma_0}(B_i)$, and it suffices to prove Theorem 1.3 for simple $A$ of primitive type.

**Lemma 4.1** (Lemma K′)**.** *Let $F \subset K$, where $K$ is a number field containing the Galois closure of $F$, $n = [K : \mathbb{Q}]$, and let $\theta_1, \dots, \theta_n$ be a basis of $K/\mathbb{Q}$. For an embedding $\sigma\colon F \to K$ put $\psi_\sigma(m) = \sum_j \theta_j \sigma(m_j)$ on $F^n$. Then $(\psi_\sigma)_\sigma\colon F^n \to K^{2g}$ is a $\mathbb{Q}$-linear bijection, each $\psi_\sigma$ is $\sigma$-semilinear and onto $K$, and for every set $\Sigma$ of embeddings the common kernel of $(\psi_\sigma)_{\sigma\in\Sigma}$ is an $F$-subspace of $F$-dimension $n - |\Sigma|\, n/(2g)$.*

*Proof.* $F^n \cong F \otimes_{\mathbb{Q}} K$ via $m \mapsto \sum m_j \otimes \theta_j$, and $F \otimes_{\mathbb{Q}} K \cong K^{2g}$ via $x \otimes y \mapsto (\sigma(x)y)_\sigma$, since $K$ contains the Galois closure; the composite is $(\psi_\sigma)_\sigma$. Semilinearity is clear. The common kernel is the preimage of $\{0\}^\Sigma \times K^{2g-|\Sigma|}$, of $\mathbb{Q}$-dimension $(2g - |\Sigma|)n$, i.e. of $F$-dimension $(2g - |\Sigma|)n/(2g)$. □

For $g = 1$, $F = k$ is an imaginary quadratic field, $\psi_{\mathrm{id}}$ and $\psi_-$ are the maps $\psi$, $\phi$ below, and Lemma 4.1 becomes the following lemma, which we use throughout for CM elliptic curves (we keep its direct proof).

**Lemma 4.2.** *Let $k \subset K$ be an imaginary quadratic field inside a number field of degree $n$, $\theta_1, \dots, \theta_n$ a basis of $K/\mathbb{Q}$, and on $k^n$ put $\psi(m) = \sum \theta_j m_j$, $\phi(m) = \sum \theta_j \bar{m}_j$. Then $(\psi, \phi)\colon k^n \to K \times K$ is a $\mathbb{Q}$-linear bijection;* $\ker\psi$ *and* $\ker\phi$ *are $k$-subspaces of dimension $n/2$, and* $\ker\psi \cap \ker\phi = 0$.

*Proof.* $k^n \cong K \otimes_{\mathbb{Q}} k$ via $m \mapsto \sum \theta_j \otimes m_j$, and $K \otimes_{\mathbb{Q}} k \cong K \times K$ via $x \otimes y \mapsto (xy, x\bar{y})$; the composite is $(\psi, \phi)$. $\psi$ is $k$-linear and $\phi$ is $k$-semilinear, and both are onto $K$, which has $k$-dimension $n/2$. □

An *eigen-slot* is a pair $(A, \sigma)$ with $A/\overline{\mathbb{Q}}$ a simple CM abelian variety and $\sigma$ an embedding of its CM field; its periods are the eigen-periods $P_\sigma$ of $\omega_\sigma$ (on $A^\natural$ if $\sigma \notin \Phi$). Two eigen-slots are equal iff the abelian varieties are isogenous and the embeddings correspond under the isogeny. An eigen-slot is of the first kind iff $\sigma \in \Phi$, and of the second kind otherwise. We use the following standard facts.

- (S1) [Lit] (standard; see Shimura [32]). $\mathrm{End}^0(A^n) = M_n(F)$, and Hom between non-isogenous simple abelian varieties is 0. Abelian subvarieties of $A^n$ are the $B_S$, $S \subset F^n$ an $F$-subspace, with $\dim B_S = g \dim_F S$ and $T(B_S) = \{(\sigma(M)u_\sigma)_{\sigma\in\Phi} : u_\sigma \in \mathbb{C}^{\dim S}\}$ in the eigen-coordinates ($M$ a matrix whose columns span $S$). On $T(B_S)$ the form $\theta^{\mathsf{T}} z_\sigma$ vanishes iff $\psi_\sigma(S) = 0$, and different $\sigma \in \Phi$ carry independent $u_\sigma$.
- (P) [Proved] Every connected algebraic subgroup of $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times B$, $B$ an abelian variety, is a product $(H' \cap \mathbb{G}_{\mathrm{a}}^a) \times (H' \cap \mathbb{G}_{\mathrm{m}}^b) \times B'$; subgroups of $\mathbb{G}_{\mathrm{a}}^a$ are $\overline{\mathbb{Q}}$-subspaces and of $\mathbb{G}_{\mathrm{m}}^b$ rational subtori. (The image of $H'$ in $B$ is an abelian subvariety $B'$, and $H'/(H' \cap M)$, $M$ the linear part, maps to the linear group $M/(H' \cap M)$ trivially; so $H' = (H' \cap M)^0 \times B'$ by dimension, and $H' \cap M$ splits since $\mathrm{Hom}(\mathbb{G}_{\mathrm{a}}, \mathbb{G}_{\mathrm{m}}) = \mathrm{Hom}(\mathbb{G}_{\mathrm{m}}, \mathbb{G}_{\mathrm{a}}) = 0$.)
- (Ch) By Dedekind's independence of characters, the vectors $(\sigma(x))_{\sigma\in\Phi}$, $x \in F$, span $K^g$ over $K$. Hence $\Lambda_K(A)$, read in the eigen-coordinates, is $\prod_{\sigma\in\Phi} K\, z_\sigma(\Lambda(A))$, and likewise for $\Omega_K(A)$: the slot coordinates of $K$-combinations of logarithms (or periods) can be prescribed independently.

(W) [Lit] (Wüstholz [52]). The numbers 1, $\pi$ and nonzero periods of pairwise distinct eigenslots (first or second kind, on one or several simple CM abelian varieties) are linearly independent over $\overline{\mathbb{Q}}$. In particular $P_\sigma \notin \overline{\mathbb{Q}}$ and $P_\sigma/\pi \notin \overline{\mathbb{Q}}$. A nonzero eigen-period of $A$ (resp. a second-kind eigen-period of $A^\natural$) is not in $\overline{\mathbb{Q}}\mathcal{L}$.

All of these follow from Wüstholz's theorem on $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times \prod_i A_i^\natural$ (resp. with $\mathbb{G}_{\mathrm{m}}^r$ added), applied to the point with coordinates 1, $2\pi i$ (resp. the logarithms) and the cycles, as in the following example. For example, $P/\pi \notin \overline{\mathbb{Q}}$: otherwise the point $(\gamma_0, 2\pi i)$ of $\mathrm{Lie}(A \times \mathbb{G}_{\mathrm{m}})$ lies in a $\overline{\mathbb{Q}}$-hyperplane $z_\sigma = a\,x$; the smallest algebraic subgroup whose Lie algebra contains it is $A \times \mathbb{G}_{\mathrm{m}}$ ($A$ is simple and the subgroup is a product by (P)), whose Lie algebra is not in the hyperplane.

*Universal vector extension.* For $\sigma \notin \Phi$ the eigen-coordinate $w_\sigma$ lives on $\mathrm{Lie}(A^\natural)$. We use the following facts. The input from the literature is [Lit] (Mazur–Messing [25]): $\mathrm{Lie}(A^\natural) = H^1_{\mathrm{dR}}(A)^\vee$ functorially, and $A^\natural$ is universal among extensions of $A$ by vector groups (every such extension of $B$ by $U$ is the push-out of $B^\natural$ along a unique linear map $W_B \to U$, where $W_B$ is the fibre of $B^\natural$). The rest is [Proved] by the one-line arguments given.

(F1) $\mathrm{Lie}(A^\natural) = H^1_{\mathrm{dR}}(A)^\vee$ with functorial $\mathcal{O}_F$-action. In the dual eigenbasis $(e_\tau)$ the fibre Lie algebra is $\bigoplus_{\tau \notin \Phi} \mathbb{C} e_\tau$, and $w_\sigma$ ($\sigma \notin \Phi$) is nonzero exactly on $e_\sigma$, which lies in the fibre. *Proof:* the Lie algebra sequence of $0 \to V \to A^\natural \to A \to 0$ is the dual of $0 \to H^0(A, \Omega^1) \to H^1_{\mathrm{dR}}(A) \to H^1(A, \mathcal{O}) \to 0$, so the fibre is the annihilator of $H^0(A, \Omega^1) = \bigoplus_{\tau \in \Phi} \overline{\mathbb{Q}} \omega_\tau$.

(F2) For $B \subset A^n$, $B^\natural \to (A^n)^\natural$ is injective on fibres and $\pi^{-1}(B) = B^\natural + V^n$ ($V$ the fibre). *Proof:* on fibres the map is dual to the restriction $H^1(A^n, \mathcal{O}) \to H^1(B, \mathcal{O})$, which is onto because $B$ is a direct factor of $A^n$ up to isogeny (Poincaré reducibility); and $B^\natural$ maps onto $B$.

(F3) $\mathrm{Ext}(B, \mathbb{G}_{\mathrm{a}}) \cong H^1(B, \mathcal{O})$ and $\mathrm{Hom}(B, \mathbb{G}_{\mathrm{a}}) = 0$: a connected $H$ over $B$ with given intersection $W'$ with the fibre exists iff $W' \supseteq W_B$, and then $H = B^\natural + W'$ is unique. *Proof:* such an $H$ is a splitting over $B$ of $\pi^{-1}(B)/W'$, which by (F2) and universality is the push-out of $B^\natural$ along $W_B \to V^n/W'$; it splits iff this map is 0, i.e. $W_B \subseteq W'$, and the splitting is unique since $\mathrm{Hom}(B, V^n/W') = 0$.

(F4) $T(B_S^\natural) = \bigoplus_\tau \tau(S) \otimes \mathbb{C}$ and $W_{B_S} = \bigoplus_{\tau \notin \Phi} \tau(S) \otimes \mathbb{C}$. *Proof:* under the comparison isomorphism $H^1_{\mathrm{dR}}(B_S)^\vee \cong H_1(B_S, \mathbb{Q}) \otimes \mathbb{C}$, with $H_1(B_S, \mathbb{Q}) = S\gamma_0 \subset F^n\gamma_0$, the $\tau$-eigen component is $\tau(S) \otimes \mathbb{C}$; the fibre part is $\tau \notin \Phi$ by (F1).

The periods of $A^\natural$ satisfy $\Omega(A^\natural) \otimes \mathbb{Q} = F\tilde{\gamma}_0$ with $w_\sigma(x\tilde{\gamma}_0) = \sigma(x)P_\sigma$, and the values $w_\sigma(\Lambda(A^\natural))$ ("$\sigma$-fibre values") are second-kind abelian integrals $\int \omega_\sigma + \overline{\mathbb{Q}}$.

*Beta values* [Lit] (Weil [50]; Rohrlich's appendix to [18]; Koblitz–Ogus [19]). For $N \geq 3$ and $0 < a, b < N$ with $a+b \not\equiv 0 \bmod N$, $B(a/N, b/N)$ is, up to $\overline{\mathbb{Q}}^\times$, the period of an eigen-differential on the Fermat curve of degree $N$, with Koblitz–Ogus label $\Phi_{a,b} = \{u \in (\mathbb{Z}/N)^\times : \langle ua/N\rangle + \langle ub/N\rangle + \langle u(N-a-b)/N\rangle = 1\}$ (after reducing $a/N, b/N$ to lowest common denominator); the corresponding isotypic factor of the Jacobian is a power of a simple CM abelian variety, and the differential is of the first kind iff $a + b < N$. Deligne's theorem (stated by Koblitz–Ogus [19]; it rests on Deligne's theorem that Hodge cycles on abelian varieties are absolutely Hodge [12]) gives: if $\mathbf{1}_{\Phi_a} + \mathbf{1}_{\Phi_b} = \mathbf{1}_{\Phi_c} + \mathbf{1}_{\Phi_d}$ on $(\mathbb{Z}/N)^\times$, then $P_aP_b \in \overline{\mathbb{Q}}^\times P_cP_d$. Together with the rule that $\pi$ corresponds to the constant function 1, and $\mathbf{1}_\Phi + \mathbf{1}_{-\Phi} \equiv 1$, the same dictionary contains $P_\sigma P_{\bar{\sigma}} \in \pi\overline{\mathbb{Q}}^\times$. That these are all multiplicative relations is the Rohrlich–Lang conjecture, used below only in remarks labelled [Heuristic].

4.2. **CM elliptic curves and their universal vector extensions.** For a CM elliptic curve $E$ let $E^\natural$ be its universal vector extension, with base coordinate $z$ and conjugate-eigen fibre

coordinate $w$: $\mathcal{O} = \mathrm{End}(E)$ acts on $T(E^\natural) = H^1_{\mathrm{dR}}(E)^\vee$ by its identity character on $z$ and by the conjugate character on $w$ (the eigenline of $H^1_{\mathrm{dR}}$ for $\bar a$ is spanned by $x\,dx/y + c\,dx/y$ with $c \in \overline{\mathbb{Q}}$, so these coordinates are rational; this is (F1) for $g = 1$). In these coordinates the periods are $(\lambda, \eta^*(\lambda))$, $\lambda \in \Lambda_E$, where $\eta^*(\lambda) = \eta(\lambda) - \kappa_1\lambda$ satisfies $\eta^*(a\lambda) = \bar a\,\eta^*(\lambda)$; the logarithms of points of $E^\natural(\overline{\mathbb{Q}})$ are $(u, \zeta^*(u) + t)$ with $u \in \mathcal{L}_E$, $t \in \overline{\mathbb{Q}}$ and $\zeta^* = \zeta - \kappa_1\,\mathrm{id}$. On $E_i$ and $E_\rho$, $\kappa_1 = 0$. Since the characters $a \mapsto a, \bar a$ of $k$ are $K$-independent for $K \supseteq k$,

$$\Omega_K(E^\natural) = K\lambda_0 \times K\eta^*(\lambda_0) \qquad (\Lambda_E \otimes \mathbb{Q} = k\lambda_0). \tag{4}$$

We call *fibre values* the $w$-coordinates of $\Lambda(E^\natural)$, that is, $\zeta^*(u) + \overline{\mathbb{Q}}$ for $u \in \mathcal{L}_E \setminus \Lambda_E$ and $\eta^*(\lambda) + \overline{\mathbb{Q}}$ at lattice points, and write Fib for the set of fibre values. By Legendre and $\eta^*(a\lambda) = \bar a\eta^*(\lambda)$, $\Omega\eta^*(\Omega) \in \overline{\mathbb{Q}}^\times\pi$ for every CM period $\Omega$.

4.3. **Weights.** $\alpha_K(\mathbb{G}_{\mathrm{a}}) = 0$, $\alpha_K(\mathbb{G}_{\mathrm{m}}) = 1$, and for an elliptic curve $\alpha_K(E) = 1$ if $E$ has CM by $k$ and $K \supseteq k$, else 2 [29, Lemme, p. 361]; more generally $\alpha_K(A) = g$ for a simple CM abelian variety $A$ of dimension $g$ when $K$ contains the Galois closure of its CM field ([29, Lemme, p. 361]; it is also Lemma 4.1 with $\Sigma = \Phi$); $\alpha_K$ is additive on products [29, (2)]. For the universal vector extension $E^\natural$ of a CM curve and $K \supseteq k$ we need:

**Lemma 4.3.** $\alpha_K(E^\natural) = 2 + 2/n$.

*Proof.* In the coordinates $(z, w)$ of §4.2, let $H \subset (E^\natural)^n$ be connected with $p_\theta(T_H) = 0$, with abelian image $B \subset E^n$. Then $B = B_S$ for a $k$-subspace $S \subset k^n$ on which both $\psi$ (base) and $\phi$ (fibre) of Lemma 4.2 vanish; since $\ker\psi \cap \ker\phi = 0$, $B = 0$. Hence $H$ lies in the fibre $\mathbb{G}_{\mathrm{a}}^n$, and the largest such $H$ is the rational hyperplane $\sum\theta_j t_j = 0$, of dimension $n-1$. The quotient has dimension $n+1$ and admits no surjection to $\mathbb{G}_{\mathrm{a}}$ or $\mathbb{G}_{\mathrm{m}}$ (it is a non-split extension of $E^n$ by $\mathbb{G}_{\mathrm{a}}$), so $\alpha = 2(n+1)$ and $\alpha_K = 2 + 2/n$. □

The inner weights $\alpha_{K,\Psi}$ of the applications are computed from the explicit quotients $G''$ built in their proofs (Remark 5.8 and §4.4). For $K \supseteq k$, the endomorphisms $[a]$, $a \in \mathcal{O}_k$, act on $T(E^\natural)$ by $d[a](z, w) = (az, \bar a w)$ and preserve $\Lambda(E^\natural)$; for $v = (u, f) \in \Lambda(E^\natural)$ and $a \neq \bar a$, $a \cdot v - d[a]v = (0, (a - \bar a)f)$, so

$$\Lambda_K(E^\natural) = K\mathcal{L}_E \times K \cdot \mathrm{Fib} \qquad (K \supseteq k), \tag{5}$$

the logarithm analogue of (4) [Proved]. So a single factor $E^\natural$ can be read through its base in one coordinate of $\Psi$ and through its fibre in another.

4.4. **From Theorem R to the applications.** Every main result of §§5–9 has the following shape. There are a group $G$ (a product of copies of $\mathbb{G}_{\mathrm{a}}$, $\mathbb{G}_{\mathrm{m}}$, CM elliptic curves, CM abelian varieties and their universal vector extensions), a $\overline{\mathbb{Q}}$-rational map $\Psi\colon T_G \to \mathbb{C}^2$ (onto), a line $V = \mathbb{C}(1, s)$ (or $\mathbb{C}(p_1, p_2)$) whose slope is transcendental, and $W = 0$. So condition (i) of Theorem 1.1 holds by [29, Remarque 1], $c_V = 1$ and $\dim V/W = 1$. A period datum $w$ on $V$ gives $\kappa \geq 1$ (for Theorems 1.7, 8.2 and 8.15, two or more period relations give $\kappa \geq 2$, resp. $\kappa$), and a counterexample to the statement gives further vectors on $V$, $K$-independent of the period data. Theorem 1.1 at level (R-P)($\alpha$) bounds their total number by $\alpha_{K,\Psi}(G) - \kappa$, and the inner weight is $\alpha(G'')/n$ for the quotient $G'' = G^n/U$ constructed in each proof ($U = H_\Psi$ there, and an upper bound for $\alpha(G'')$ would suffice, Proposition 3.4). The table lists the numbers (for $n = [K : \mathbb{Q}]$ large enough; "data" counts the period data together with the vectors a counterexample would give).

| result | $G''$ | $\alpha_{K,\Psi}$ | $\kappa$ | data | bound |
|---|---|---|---|---|---|
| Thms. 5.1, 5.7 (MG, MG-A) | $\mathbb{G}_{\mathrm{m}}^{2n} \times A''$ | $3$ | $1$ | $3$ | $2$ |
| Thms. 5.4, 5.9 (MG$^\natural$, MG$^\natural$-A) | $\mathbb{G}_{\mathrm{m}}^{2n} \times Q''$ | $3+2/n$ | $1$ | $3$ | $2+2/n$ |
| Thm. 6.2(a), Thm. 6.9 (SMG, SMG-A) | $\mathbb{G}_{\mathrm{a}}^{2} \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ | $3$ | $1$ | $3$ | $2$ |
| Thm. 6.2(b), SMG$^\natural$-A | $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$ | $3+2/n$ | $1$ | $3$ | $2+2/n$ |
| Thm. 6.12(1), Thm. 6.16 (EA) | $\mathbb{G}_{\mathrm{a}}^{2} \times A_1'' \times A_2'' \times A'''$ | $3$ | $1$ | $3$ | $2$ |
| Thm. 6.12(2), (3), EA$^\natural$ | $\mathbb{G}_{\mathrm{a}} \times A_1'' \times A_2'' \times Q$ | $3+2/n$ | $1$ | $3$ | $2+2/n$ |
| Props. 7.1, 7.5 (MX, MX′, deg2) | $\mathbb{G}_{\mathrm{a}} \times G_\ell$, $\dim = 2n+2$ | $4+2/n$ | $2$ | $3$ | $2+2/n$ |
| Prop. 8.1 (X, X7, X9, X24) | $\prod_j A_j'' \times \mathbb{G}_{\mathrm{a}}^{2}$ | $4$ | $2$ | $3$ | $2$ |
| Thm. 8.15 ($\kappa$ pairs) | $\prod_j A_j'' \times \mathbb{G}_{\mathrm{a}}^{2}$ | $2\kappa$ | $\kappa$ | $\kappa+1$ | $\kappa$ |
| Thm. 5.16 ($X = \mathbb{G}_{\mathrm{m}}$) | $Q_1 \times X'' \times Q_3$ | $3+4/n$ | $1$ | $3$ | $2+4/n$ |
| Thm. 10.1, rows (iii), (iv) | $\mathbb{G}_{\mathrm{m}}^{n} \times A'' \times A'''$ | $3$ | $1$ | $3$ | $2$ |
| Thm. 10.1, fibre rows | $\mathbb{G}_{\mathrm{m}}^{n} \times A'' \times Q''$ | $3+2/n$ | $1$ | $3$ | $2+2/n$ |

In each row the bound is $<$ the number of data (for $n \geq 3$, resp. $n \geq 5$ for QL), which is the contradiction. Theorem 1.10 is Corollary 2.2 on $E \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$ ($\alpha_K = 2$, $\kappa = 1$). The statements of the applications then follow by the deductions written after each dimension bound (Baker's theorem, Wüstholz's theorem, Masser's theorem, Laurent's theorem, as stated there).

The proofs we write out apply Theorem 2.3 once, directly to $G''$, and exclude every connected algebraic subgroup by an explicit case check, with (2) checked at $G''$. They need neither the reduction of §3.2 nor the induction of §3.6, nor Lemma 3.8 or Baker's theorem at corners; they show which subgroups are tight and by how much, and they are the proofs whose instances of Theorem 4.1 are formally verified in §12.3 (there through Theorem 2.5). Several applications are already consequences of the printed weight $\alpha_K$ of Corollary 2.2 on a suitable product group; which ones is recorded in §11.3.

## 5. Four exponentials at CM eigen-periods

This section proves Theorems 1.2 and 1.3. §§5.1–5.2 treat a CM elliptic curve (Theorems MG and MG$^\natural$, the $K$-forms behind Theorem 1.2); §5.3 treats one eigen-coordinate of a CM abelian variety of any dimension (Theorems MG-A and MG$^\natural$-A, behind Theorem 1.3). The abelian case checks are those of the elliptic ones, because the quotient group has abelian part of dimension $n/2$ for every $g$.

5.1. **Theorem MG.** Throughout this section $E/\overline{\mathbb{Q}}$ has complex multiplication by $k$, $\Omega \neq 0$ is a period of $E$, and for a number field $K$ we write $K\mathcal{L}$, $K\mathcal{L}_E$ for the $K$-spans of $\mathcal{L}$ and $\mathcal{L}_E$. By Baker's theorem $K\mathcal{L} \cong K \otimes_{\mathbb{Q}} \mathcal{L}$, so elements of $\mathcal{L}$ that are $\mathbb{Q}$-linearly independent are $K$-linearly independent.

**Theorem 5.1** (Theorem MG)**.** *Let $c \in \overline{\mathbb{Q}}^{\times}$ and $s = c\,\Omega/\pi$. For every number field $K \supseteq k$,*

$$\mathcal{S}_K := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\mathcal{L}_E\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K \leq 2,$$

*and $2\pi i \in \mathcal{S}_K$ when $ic \in K$ (since then $s \cdot 2\pi i = 2ic\,\Omega \in K\mathcal{L}_E$).*

*Status:* [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ and $K \ni i, c$ (the passage to an arbitrary $K \supseteq k$ uses Baker's theorem and is not formalised); [Proved modulo published theorems and Lemma A.1] in general.

*Proof.* Enlarging $K$ does not change the conclusion (by Baker, $K$-independence in $K\mathcal{L}$ is preserved under base change), so assume $K \supseteq k(i, c)$; $n = [K : \mathbb{Q}]$ is even (the count below closes

for every such $n$). Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times E$ with coordinates $(x, y, z)$, $\Psi(x, y, z) = (x, y + z)$, $V = \mathbb{C}(1, s)$. Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$, put $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, and $\Psi_\theta = \Psi \circ p_\theta$.

*The group.* Connected algebraic subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times E^n$ are products $T_L \times B_S$ of a subtorus (given by a subspace $L \subset \mathbb{Q}^{2n}$) and an abelian subvariety (given by a $k$-subspace $S \subset k^n$). Such a subgroup has tangent space in $\ker \Psi_\theta$ iff $L = 0$ and $\psi(S) = 0$, where $\psi(m) = \sum \theta_j m_j$; by Lemma 4.2, $\ker \psi$ has $k$-dimension $n/2$. Put $U = B_{\ker \psi}$ and $G'' = G^n/U = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $A'' = E^n/U$, of dimension $n/2$. Then $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2$. The induced map $\Psi''$ is onto $\mathbb{C}^2$, $V'' = \Psi''^{-1}(V)$ is a hyperplane, and $W'' = \ker \Psi'' \subset V''$ is $\overline{\mathbb{Q}}$-rational of dimension $d - 2$. For a connected $G' \subset G''$ put $r' = \dim \Psi''(T_{G'})$; then $\tau = \delta + r' - 2$, so only $r' \le 1$ can satisfy $\delta > \tau$, and $\delta - \tau = 2 - r'$.

*The data.* $w = (2\pi i, 2ic\Omega) \in \Psi(\Omega_K(G))$. Suppose $x_1, x_2 \in \mathcal{S}_K$ with $2\pi i, x_1, x_2$ $K$-independent, and put $v_j = (x_j, s x_j) \in V \cap \Psi(\Lambda_K(G))$. Let $Y''$ be generated by the images in $T_{G''}$ of elements of $\Lambda(G^n)$ with $\Psi_\theta$-images $D\theta_h w$ (taken in $\Omega(G^n)$, $D$ a common denominator), $\theta_h v_1$, $\theta_h v_2$ ($h = 1, \dots, n$). Then $\operatorname{rk} Y'' = 3n$, $\Gamma = \exp Y'' \subset G''(\overline{\mathbb{Q}})$, and $\kappa = n$: a period in $Y''$ has first coordinate in $2\pi i K$, so its $v$-parts vanish. The inequality of Theorem 2.3 reads $\lambda + \delta_1 + 2\delta_2 \le (2 - r') \cdot 2n$.

*No obstruction.*

- $G' = 0$: $\lambda = 2n$, so the left side is $2n + 2n + n = 5n > 4n$.
- $r' = 1$ with image $\mathbb{C}(0, 1)$ (this covers every $G'$ with an abelian part, since $\psi$ does not vanish on any nonzero abelian subvariety of $A''$): an element of $Y''$ lying in $T_{G'} + \Omega(G'')$ has first coordinate in $2\pi i K$, so $\operatorname{rk}(Y'' \cap (T_{G'} + \Omega(G''))) \le n$ and $\lambda \ge 2n$; the left side is $\ge 3n > 2n$.
- $r' = 1$ with image a line $\mathbb{C}(p, q) \ne \mathbb{C}(0, 1)$: then $G' = T_L$ with $\dim L \le n$. Two $K$-independent data in $T_{G'} + \Omega(G'')$ would put $x_1, x_2$ in a $K$-space of dimension $\le 2$ containing $2\pi i$ (namely $K \cdot 2\pi i + K\pi^2/\Omega$ if $q = 0$, and $K \cdot 2\pi i + K \cdot p\,\Omega/(sp - 1)$ otherwise, using $s \notin K$), against the $K$-independence of $2\pi i, x_1, x_2$. So $\lambda \ge n$, $\delta_1 \ge n$, $\delta_2 = n/2$, and the left side is $\ge 3n > 2n$.
- $r' = 0$ forces $G' = 0$.

So no connected $G' \ne G''$ satisfies the conclusion of Theorem 2.3, a contradiction. □

*Controls.* If $E$ has no CM then $U = 0$, $d_2 = n$, and $G' = 0$ satisfies the inequality ($6n \le 6n$); so does $G' = 0$ with $\kappa$ deleted ($5n \le 6n$), with only one datum, or with $k \not\subset K$. If $s \in K$ (a rational line) the full subtorus over $V$ satisfies it with equality; replacing $E$ by $\mathbb{G}_{\mathrm{m}}$ makes $s$ algebraic, and then the statement is false ($s = 2$, $x = \log 2$).

*Provenance.* By §4.4, Theorem 5.1 is a consequence of Theorem 1.1 ($\alpha_{K,\Psi}(G) - \kappa = 3 - 1 = 2 < 3$); here the printed weight already suffices ($\alpha_K(\mathbb{G}_{\mathrm{m}}^2 \times E) = 3$; Remark 5.8, Remark 11.1). We found no explicit statement of it in print; the proof above applies Theorem 2.3 directly to $G''$.

**Corollary 5.2** (MG2: four exponentials at $t \in \overline{\mathbb{Q}}^\times \Omega/\pi$)**.** *Let $t \in \overline{\mathbb{Q}}^\times \cdot \Omega/\pi$ and $\alpha, \beta \in \overline{\mathbb{Q}}^\times$ with $\log\alpha, \log\beta$ linearly independent over $\mathbb{Q}$. Then $\alpha^t = e^{t\log\alpha}$ and $\beta^t = e^{t\log\beta}$ are not both algebraic. If moreover $\log\alpha, \log\beta, 2\pi i$ are linearly independent over $\mathbb{Q}$, then $t\log\alpha$ and $t\log\beta$ are not both in $\overline{\mathbb{Q}}\mathcal{L} + \overline{\mathbb{Q}}\mathcal{L}_E$. For example, $2^{\varpi/\pi}$ and $3^{\varpi/\pi}$ are not both algebraic.*

*Status:* the first sentence is Theorem 1.2(b) and has its status. The second sentence is [Proved modulo published theorems and Lemma A.1] (for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ it is the formally verified $K$-form of Theorem 5.1 combined with Baker's theorem).

*Proof.* If $\log\alpha, \log\beta, 2\pi i$ are $\mathbb{Q}$-independent, take $x_1 = \log\alpha$, $x_2 = \log\beta$ in Theorem 5.1: with $2\pi i$ they are three $\mathbb{Q}$-independent logarithms, hence $K$-independent. Otherwise $m\log\beta = a\log\alpha + 2\pi i\, b$ with integers $m \ne 0$, $b \ne 0$. If $\alpha^t$ and $\beta^t$ are algebraic, so is $e^{2\pi i b t} = (\beta^t)^m(\alpha^t)^{-a}$, i.e. $2\pi i b t = 2ib\,c\,\Omega \in \mathcal{L}$ (with $t = c\Omega/\pi$). Then $\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, which the analytic subgroup theorem

[52] excludes (on $\mathbb{G}_{\mathrm{m}} \times E$, the smallest algebraic subgroup whose tangent space contains $(\ell, \Omega)$ is $\mathbb{G}_{\mathrm{m}} \times E$). (Remark 5.3 gives a proof that needs no such step.) □

The determinations matter: $\log \alpha = \log 2$ and $\log \beta = \log 2 + 2\pi i$ are $\mathbb{Q}$-independent, and for the determinations $e^{t \log 2}$ of both powers the statement would say $2^t \notin \overline{\mathbb{Q}}$, a one-exponential statement that we do not prove.

**Remark 5.3** (Second derivation via §5d)**.** Theorem 2.5 gives Theorem 5.1 with a shorter subgroup analysis, and the first sentence of Corollary 5.2 without the case distinction. Keep $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $\Psi''$, $V'' = \Psi''^{-1}(V)$ and $W'' = \ker \Psi''$ from the proof of Theorem 5.1: $d = 5n/2$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, and $W''$ is $\overline{\mathbb{Q}}$-rational of dimension $t_W = d - 2$ (the $t$ of Theorem 2.5; we write $t_W$ to keep $t$ for the exponent of Corollary 5.2), so $d - 1 - t_W = 1$. Let $Y''$ be generated by the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ $(h = 1, \dots, n)$. The $n$ period lifts lie in $Y'' \cap \ker \exp$, so $\kappa \geq n$ (in fact $\kappa = n$), and condition (5.4) of [37] reads $3n > (2n + n - \kappa) \cdot 1$, i.e. $3n > 2n$.

*The rank $m = 3n$.* For the data of Theorem 5.1 ($2\pi i, x_1, x_2$ $K$-independent) the first coordinates of the $\Psi''$-images $D\theta_h \cdot 2\pi i$, $\theta_h x_j$ are $\mathbb{Q}$-independent. For the data of Corollary 5.2 ($x_j = \log \alpha_j$, $s x_j = t \log \alpha_j \in \mathcal{L}$) only the $\mathbb{Q}$-independence of $\log \alpha, \log \beta$ is needed. Take the period lifts with toric part $(2\pi i D e_h, 0)$ and $E^n$-part $m^{(h)}\Omega \in \Lambda_E^n$, where $\psi(m^{(h)}) = 2icD\theta_h$ (possible by Lemma 4.2, as $ic \in K$; $D$ a common denominator), and the data lifts toric, $(x_j e_h,\ s x_j e_h)$. The linear form induced by $\theta^{\mathsf{T}} z$ on $T_{A''}$ takes the $\mathbb{Q}$-independent values $2icD\theta_h\Omega$ on the period lifts and 0 on the data lifts, so a $\mathbb{Q}$-relation among the $3n$ lifts involves only the data lifts; their first-block coordinates $x_j e_h$ are $\mathbb{Q}$-independent.

*No subgroup in $V''$.* Theorem 2.5 gives a connected $G' \neq 0$ with $T_{G'} \subset V''$, and $G' = T_L \times B_S$. If $B_S \neq 0$, then $T_{G'}$ contains a vector $(0, b)$ with $\theta^{\mathsf{T}} b \neq 0$ (since $\psi$ does not vanish on any nonzero abelian subvariety of $A''$), whose image $(0, \theta^{\mathsf{T}} b)$ is not on $V = \mathbb{C}(1, s)$. If $B_S = 0$, then $L \neq 0$, and for $0 \neq l = (l_1, l_2) \in L$ the image $(\theta^{\mathsf{T}} l_1, \theta^{\mathsf{T}} l_2) \in K^2 \setminus 0$ lies on $\mathbb{C}(1, s)$; so $\theta^{\mathsf{T}} l_1 \neq 0$ and $s \in K$, which is false ($\Omega/\pi \notin \overline{\mathbb{Q}}$ [23]).

For $K = \mathbb{Q}(i)$, $n = 2$, $\theta = (1, i)$: $A'' \cong E$ via $(z_1, z_2) \mapsto z_1 + i z_2$, so $G'' = \mathbb{G}_{\mathrm{m}}^4 \times E$, $d = 5$, $t_W = 3$, $m = 6$, $\kappa = 2$, and (5.4) is $6 > 6 - \kappa$. This route uses neither the $K$-independence of $2\pi i, \log \alpha, \log \beta$ nor Wüstholz's theorem: it proves the first sentence of Corollary 5.2 directly, also when $2\pi i$ lies in the $\mathbb{Q}$-span of $\log \alpha, \log \beta$. Its subgroup exclusion uses only $s \notin K$. It rests on Theorem 2.5, proved in §3.6 (and in print as [37, §5d]).

5.2. **The fibre twin: Theorem MG$^\natural$.** Let $E^\natural$ be read on its conjugate-eigen fibre coordinate $w$ (§4.2), with $\eta^* = \eta^*(\Omega)$; recall that $\Omega\eta^* \in \overline{\mathbb{Q}}^\times \pi$ and that the fibre values are the $w$-coordinates of $\Lambda(E^\natural)$.

**Theorem 5.4** (Theorem MG$^\natural$)**.** *Let $c \in \overline{\mathbb{Q}}^\times$ and $s = c\,\eta^*/\pi \in \overline{\mathbb{Q}}^\times/\Omega$. For every number field $K \supseteq k$, $\dim_K \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K \cdot (\textit{fibre values})\} \leq 2$.*

*Status:* [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ and $K \ni c$ with $[K : \mathbb{Q}] \geq 4$ (the passage to an arbitrary $K \supseteq k$ uses Baker's theorem and is not formalised); [Proved modulo published theorems and Lemma A.1] in general.

*Proof.* As in the proof of Theorem 5.1 we may enlarge $K$; we take $K \supseteq k(i, c)$ with $n = [K : \mathbb{Q}] \geq 4$ (for $n = 2$ the count below does not close). Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times E^\natural$, $\Psi(x, y, X) = (x,\ y + w(X))$, $V = \mathbb{C}(1, s)$, and $\theta$, $p_\theta$, $\Psi_\theta = \Psi \circ p_\theta$ as before.

*The group.* There are no nonzero homomorphisms between tori and connected subgroups of $(E^\natural)^n$ in either direction, so connected subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times (E^\natural)^n$ are products. A connected $H \subset (E^\natural)^n$ over $B_S$ contains the canonical lift $B_S^\natural$ (since $\mathrm{Hom}(B_S^\natural, \mathbb{G}_{\mathrm{a}}) = 0$), whose tangent space is $\{(\sigma(M)z, \bar{\sigma}(M)w)\}$; so $\theta^{\mathsf{T}} w$ vanishes on $T_H$ only if $\phi(S) = 0$. Hence the largest connected

$U$ with $T_U \subset \ker \Psi_\theta$ is $U = B^\natural_{\ker\phi} + \{\theta^{\mathsf T} w = 0\}$, and $G'' = \mathbb{G}_{\mathrm m}^{2n} \times Q''$ with $Q'' = (E^\natural)^n/U$, an extension of $A'' = E^n/B_{\ker\phi}$ (dimension $n/2$) by $\mathbb{G}_{\mathrm a}$ (coordinate $\varsigma = \theta^{\mathsf T} w$). Over an abelian subvariety $B' \neq 0$ of $A''$, coming from $S' \supsetneq \ker\phi$, we have $\phi(S') \neq 0$, so the image of $B^\natural_{S'}$ contains the fibre. Thus $Q''$ has no splitting over any $B' \neq 0$, and its connected subgroups are 0, the fibre $F$, and the $\pi^{-1}(B')$ (dimension $\leq n/2+1$). $Q''$ is a quotient of an anti-affine group, so $\mathrm{Hom}(Q'', \mathbb{G}_{\mathrm a}) = 0$: $d_0 = 0$, $d_1 = 2n$, $G_2 = Q''$, $d_2 = n/2+1$. The induced map is $\Psi''(t_1, t_2, q) = (\theta^{\mathsf T} t_1, \theta^{\mathsf T} t_2 + \varsigma(q))$, $V''$ is a hyperplane, $W'' = \ker\Psi''$, and $\tau = \delta + r' - 2$ as before.

*The data.* The period datum is $(2\pi i, 2ic\eta^*) \in \Psi(\Omega_K(G))$ (using $ic \in K$ and that $\phi$ is onto $K$). With $x_1, x_2$ as in Theorem 5.1, $\mathrm{rk}\, Y'' = 3n$ and $\kappa = n$ (a period of $G''$ has first coordinate in $2\pi i K$). The inequality reads $\lambda + \delta_1 + 2\delta_2 \leq (2 - r')(2n+2)$.

*No obstruction.* Every $G'$ is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$, and $H \neq 0$ forces $\mathbb{C}(0,1) \subset \Psi''(T_{G'})$.

- $G' = 0$: $\lambda = 2n$, and the left side is $2n + 2n + (n+2) = 5n + 2 > 4n + 4$.
- Image $\mathbb{C}(0,1)$ ($H \neq 0$, or $L$ in the second block; $l = \dim L \leq n$, $h = \dim H \leq n/2+1$): elements of $Y''$ in $T_{G'} + \Omega(G'')$ have first coordinate in $2\pi i K$, so $\lambda \geq 2n$, and the left side is $\geq 2n + (2n - l) + 2(n/2 + 1 - h) \geq 3n > 2n + 2$.
- $H = 0$ and image a line $\mathbb{C}(1, q)$, $q \in K$: a datum $X$ in $T_{G'} + \Omega(G'')$ satisfies $X(s - q) \in K \cdot 2\pi i + K \cdot 2\pi i s$, so $X \in K \cdot 2\pi i + K \cdot 2\pi i/(s-q)$ (here $s \notin K$ because $\Omega$ is transcendental). So $\lambda \geq n$, and the left side is $\geq n + n + (n+2) = 3n + 2 > 2n + 2$.
- $r' = 0$ forces $G' = 0$; $r' = 2$ fails $\delta > \tau$.

The minimum margin is $n - 2$. □

**Corollary 5.5** (MG$^\natural$2: four exponentials at $t \in \overline{\mathbb{Q}}^\times\Omega$)**.** *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and $c \in \overline{\mathbb{Q}}^\times$, then $\alpha^{c\Omega} = e^{c\Omega\log\alpha}$ and $\beta^{c\Omega} = e^{c\Omega\log\beta}$ are not both algebraic. For example, $2^{\varpi}$ and $3^{\varpi}$ (real logarithms) are not both algebraic.*

*Status:* as Theorem 1.2(a).

*Proof.* Put $x_j = c\Omega\log\alpha_j$ (with $\alpha_1 = \alpha$, $\alpha_2 = \beta$). If $\alpha^{c\Omega}, \beta^{c\Omega}$ are algebraic then $x_j \in \mathcal{L}$, and $s x_j \in \overline{\mathbb{Q}}\log\alpha_j \subset \overline{\mathbb{Q}}\mathcal{L}$. A $K$-relation among $2\pi i, x_1, x_2$ without the $2\pi i$ term contradicts Baker; one with it gives $0 \neq \ell \in \overline{\mathbb{Q}}\mathcal{L} \cap \overline{\mathbb{Q}}^\times \eta^*$, impossible by the analytic subgroup theorem on $\mathbb{G}_{\mathrm m}^r \times E^\natural$ [52]. (Remark 5.6 gives a proof that needs no such step.) □

**Remark 5.6** (Second derivation via §5d)**.** As in Remark 5.3, keep $G'' = \mathbb{G}_{\mathrm m}^{2n} \times Q''$, $\Psi''$, $V''$ and $W'' = \ker\Psi''$ from the proof of Theorem 5.4: $d = 5n/2 + 1$, $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2 + 1$, $t_W = \dim W'' = d - 2$. With the $3n$ lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$ one has $m = 3n$ and $\kappa \geq n$ (in fact $\kappa = n$), and condition (5.4) of [37] reads $3n > (2n + n + 2 - \kappa)\cdot 1$, i.e. $3n > 2n + 2$. It holds as soon as $\kappa \geq 3$, so for every admissible $n \geq 4$ (at $n = 2$ it would read $6 > 6$). For $K = \mathbb{Q}(i, \sqrt{2})$, $n = 4$: $G'' = \mathbb{G}_{\mathrm m}^8 \times Q''$, $d = 11$, $t_W = 9$, $m = 12$, $\kappa = 4$, and (5.4) is $12 > 14 - \kappa$.

*The rank.* For the data of Theorem 5.4 it follows from the first coordinates, as in Remark 5.3. For Corollary 5.5 take $s = 1/(c\Omega) \in \overline{\mathbb{Q}}^\times \eta^*/\pi$, so that $s x_j = \log\alpha_j \in \mathcal{L}$, and take the data lifts toric, $(x_j e_h,\ \log\alpha_j\, e_h)$. Take the period lifts with toric part $(2\pi i D e_h, 0)$; the second coordinate $D\theta_h \cdot 2\pi i s$ of their image is then carried by the fibre coordinate $\varsigma$ of $Q''$. So $\varsigma$ separates the period lifts from the data lifts, and the first-block coordinates $x_j e_h$ of the data lifts are $\mathbb{Q}$-independent as soon as $\log\alpha, \log\beta$ are.

*No subgroup in $V''$.* A connected $G' \neq 0$ with $T_{G'} \subset V''$ is $T_L \times H$ with $H \in \{0, F, \pi^{-1}(B')\}$. If $H \neq 0$, then $F \subset H$, and $T_F$ contains a vector with image $(0,1) \notin V$. If $H = 0$, then $L \neq 0$ and, as in Remark 5.3, a nonzero $l \in L$ gives $s \in K$, which is false since $s \in \overline{\mathbb{Q}}^\times/\Omega$ and $\Omega$ is transcendental.

So this route, too, needs only the $\mathbb{Q}$-independence of $\log\alpha, \log\beta$ and no Wüstholz step. It rests on Theorem 2.5 (§3.6).

By the symmetry $t \leftrightarrow 1/t$, Corollaries 5.2 and 5.5 give the four exponentials conjecture at every $t \in \overline{\mathbb{Q}}^\times\Omega/\pi \cup \overline{\mathbb{Q}}^\times\pi/\Omega$ and at every $t \in \overline{\mathbb{Q}}^\times\Omega \cup \overline{\mathbb{Q}}^\times/\Omega$, for every CM period $\Omega$.

*Proof of Theorem 1.2.* Part (a) is Corollary 5.5, and part (b) is Corollary 5.2. □

5.3. **CM abelian varieties: Theorems MG-A and MG$^\natural$-A.** Throughout this subsection $A$, $F$, $\Phi$ are as in §4.1, $F^c$ is the Galois closure of $F$, and $K\mathcal{L}$ is the $K$-span of $\mathcal{L}$. By Baker's theorem $K\mathcal{L} \cong K \otimes_{\mathbb{Q}} \mathcal{L}$, so $\mathbb{Q}$-independent elements of $\mathcal{L}$ are $K$-independent, and enlarging $K$ does not change the conclusions below.
By the reduction to the simple case in §4.1, we may and do assume $A$ simple of primitive type.

**Theorem 5.7** (Theorem MG-A)**.** *Let $\sigma \in \Phi$, $P = P_\sigma \neq 0$, $c \in \overline{\mathbb{Q}}^\times$ and $s = cP/\pi$. For every number field $K \supseteq F^c(i, c)$,*

$$\mathcal{S}_K := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\,z_\sigma(\Lambda(A))\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K \le 2,$$

*and $2\pi i \in \mathcal{S}_K$.*

*Status:* [Proved modulo published theorems and Lemma A.1]; two internal referees (correct with fixes, applied; exact independent enumerations for $g = 1, \dots, 6$). For $P = (1-\zeta_5)B(\frac{1}{5}, \frac{1}{5})$ on the Jacobian of $y^5 = x(1-x)$, with $z_\sigma(\Lambda(A))$ replaced by $\mathbb{Z}P$: [Proved; formally verified, standard axioms] (`Bridge.mgA5_K_uncond`).

*Proof.* Let $n = [K : \mathbb{Q}]$ (a multiple of $2g$), $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times A$ with coordinates $(x, y, z)$, $\Psi(x, y, z) = (x,\ y + z_\sigma(z))$, $V = \mathbb{C}(1, s)$, and $\theta$, $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, $\Psi_\theta = \Psi \circ p_\theta$ as in the proof of Theorem 5.1 ($\Phi$ is the CM type).

*The group.* By (P) and (S1), connected algebraic subgroups of $\mathbb{G}_{\mathrm{m}}^{2n} \times A^n$ are $T_L \times B_S$, and $T_{T_L \times B_S} \subset \ker\Psi_\theta$ iff $L = 0$ and $\psi_\sigma(S) = 0$. By Lemma 4.1 with $\Sigma = \{\sigma\}$, $\ker\psi_\sigma$ has $F$-dimension $n - n/(2g)$. Put $U = B_{\ker\psi_\sigma}$, of dimension $gn - n/2$, and $G'' = G^n/U = \mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $A'' = A^n/U$. Then $\dim A'' = n/2$ *for every* $g$, and $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2$: the numbers of Theorem 5.1. An abelian subvariety $B \neq 0$ of $A''$ comes from some $S \supsetneq \ker\psi_\sigma$, so $\psi_\sigma(S) \neq 0$ and $\Psi''(T_B) = \mathbb{C}(0, 1)$; and $\dim B \le n/2$.

*The data.* Since $\psi_\sigma$ is onto $K$, the $\theta$-span of $z_\sigma(\Omega(A^n))$ is $KP$, so $w = (2\pi i, 2icP) \in \Psi(\Omega_K(G))$ (using $ic \in K$); in particular $2\pi i \in \mathcal{S}_K$. Suppose $x_1, x_2 \in \mathcal{S}_K$ with $2\pi i, x_1, x_2$ $K$-independent, and put $v_j = (x_j, sx_j) \in V \cap \Psi(\Lambda_K(G))$. Let $Y''$ be generated by the images in $T_{G''}$ of lifts of $D\theta_h w$ (in $\Omega(G^n)$, $D$ a common denominator), $\theta_h v_1$, $\theta_h v_2$, $h = 1, \dots, n$. Then $\mathrm{rk}\,Y'' = 3n$, $\Gamma = \exp Y'' \subset G''(\overline{\mathbb{Q}})$ and $\kappa = n$: since $U$ is connected, a period of $G''$ is the image of a period of $G^n$, so its $\Psi''$-image has first coordinate in $2\pi i K$, and its $v$-components vanish. Inequality (1) reads $\lambda + \delta_1 + 2\delta_2 \le (2 - r') \cdot 2n$.

*No obstruction.* We need $s \notin K$, i.e. $P/\pi \notin \overline{\mathbb{Q}}$, which is (W). The case check is that of the proof of Theorem 5.1 verbatim, because the only possible dependence on $g$ is through $\dim A''$ and the bound $\dim B \le \dim A''$, both independent of $g$: $G' = 0$ gives $5n > 4n$; image $\mathbb{C}(0, 1)$ gives $\lambda \ge 2n$ and left side $\ge 3n > 2n$ (here a subtorus $T_L \subset G'$ lies in the second $\mathbb{G}_{\mathrm{m}}^n$ block, because $\theta^{\mathsf{T}}$ vanishes on no nonzero rational vector, so $\dim L \le n$; and $\dim B \le n/2$); a subtorus with image a line $\mathbb{C}(p, q) \neq \mathbb{C}(0, 1)$ has $\dim L \le n$, and two $K$-independent data in $T_{G'} + \Omega(G'')$ would put $x_1, x_2$ in a 2-dimensional $K$-space containing $2\pi i$ (using $s \cdot 2\pi i \in KP$ and $s \notin K$), so $\lambda \ge n$ and the left side is $\ge 3n > 2n$; and $r' = 0$ forces $G' = 0$ (maximality of $U$). The minimum margin is $n$. Condition (2) reads $5n - 2 \ge 3n$. □

**Remark 5.8** (The Roy–Waldschmidt count)**.** Roy–Waldschmidt's theorem [29, Théorème (iii)] as stated uses $\alpha_K(G)$, and $\alpha_K(A) = g$ for CM $A$ [29, Lemme, p. 361]; for $G = \mathbb{G}_{\mathrm{m}}^2 \times A$ it gives the bound $(2+g-1)\cdot 1 \geq 3$, no contradiction with three data once $g \geq 2$. Theorem 1.1 uses instead the *inner weight* (the inequality inside step (b) of their proof) $\alpha_{K,\Psi}(G) := \alpha(G^n/H_\Psi)/n$, with $H_\Psi$ the largest connected subgroup killed by $\Psi \circ p_\theta$; it is never larger than the printed weight $\alpha_K(G)$ ([29, p. 364]: $\alpha(G^n/H_\Psi) \leq n\alpha_K(G)$). Here $\alpha_{K,\Psi}(G) = \alpha(G'')/n = (2n + 2\cdot n/2)/n = 3$, and $(3-1)\cdot 1 = 2 < 3$. The saving comes from $\Psi$ reading a single eigen-slot. For $g = 1$ the two weights agree, $\alpha_K(\mathbb{G}_{\mathrm{m}}^2 \times E) = 3$, and the theorem as stated gives $(3-1)\cdot 1 = 2 < 3$: Theorem 5.1 already follows from Corollary 2.2 (Remark 11.1).

**Theorem 5.9** (Theorem MG$^\natural$-A)**.** *Let $\sigma \notin \Phi$, $P = P_\sigma \neq 0$ the corresponding (second-kind) eigen-period, $c \in \overline{\mathbb{Q}}^\times$ and $s = cP/\pi$, so $s \in \overline{\mathbb{Q}}^\times/P_{\bar\sigma}$. For every number field $K \supseteq F^c(i,c)$,*

$$\mathcal{S}_K^\natural := \{x \in K\mathcal{L} :\ s\,x \in K\mathcal{L} + K\,w_\sigma(\Lambda(A^\natural))\} \quad \textit{satisfies} \quad \dim_K \mathcal{S}_K^\natural \leq 2,$$

*and $2\pi i \in \mathcal{S}_K^\natural$ (as $s \cdot 2\pi i = 2icP_\sigma \in K\,w_\sigma(\Omega(A^\natural))$).*

*Status:* [Proved modulo published theorems and Lemma A.1]; two internal referees, one of whom built $U$ and $Q''$ explicitly for $g = 2$ ($n = 8, 16$) and $g = 3$ ($n = 12$). For $P = (1-\zeta_5^4)B(\frac{4}{5}, \frac{4}{5})$ on the Jacobian of $y^5 = x(1-x)$, with $w_\sigma(\Lambda(A^\natural))$ replaced by $\mathbb{Z}P$: [Proved; formally verified, standard axioms] (`Bridge.mgAn5_K_uncond`).

*Proof.* Prove it over $K' \supseteq K$ with $[K' : \mathbb{Q}] \geq 4$ and descend by Baker, as in the proof of Theorem 5.4. Let $G = \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{m}} \times A^\natural$, $\Psi = (x,\, y + w_\sigma)$, $V = \mathbb{C}(1, s)$. By (F1)–(F4), the largest connected subgroup with tangent space in $\ker \Psi_\theta$ is $U = B^\natural_{\ker \psi_\sigma} + \{\theta^{\mathsf{T}} t_\sigma = 0\}$, where $t_\sigma$ is the $\sigma$-fibre coordinate, of dimension $(gn - n/2) + (gn - 1)$. The quotient $Q'' = (A^\natural)^n/U$ is an extension of $A''$ (dimension $n/2$) by $\mathbb{G}_{\mathrm{a}}$; it is a quotient of an anti-affine group, so $\mathrm{Hom}(Q'', \mathbb{G}_{\mathrm{a}}) = 0$, and it does not split over any abelian subvariety $B \neq 0$ of $A''$ (the image of $B_S^\natural$, $S \supsetneq \ker \psi_\sigma$, contains the fibre since $\psi_\sigma(S) \neq 0$). Its connected subgroups are 0, the fibre, and the $\pi^{-1}(B)$, and every nonzero one maps onto $\mathbb{C}(0,1)$. So $G'' = \mathbb{G}_{\mathrm{m}}^{2n} \times Q''$ with $d_0 = 0$, $d_1 = 2n$, $d_2 = n/2+1$, connected subgroups of $G''$ are $T_L \times H''$, the period datum is $w = (2\pi i, 2icP_\sigma)$, $\kappa = n$, $\mathrm{rk}\, Y'' = 3n$, and the subtorus case needs $s \notin K$, i.e. $P_{\bar\sigma} \notin \overline{\mathbb{Q}}$ (W). This is exactly the situation of Theorem 5.4 (same type of quotient, same dimensions, same subgroup list, same data), with $d_1 + 2d_2 - \kappa = 2n + 2$, so (1) reads $\lambda + \delta_1 + 2\delta_2 \leq (2 - r')(2n + 2)$. Its case table:

| $G'$ | $r'$ | left side | right side |
|---|---|---|---|
| $0$ ($\lambda = 2n$) | 0 | $5n + 2$ | $4n + 4$ |
| $T_L \times H''$ with image $\mathbb{C}(0,1)$ ($\lambda \geq 2n$, $\dim L \leq n$, $\dim H'' \leq n/2 + 1$) | 1 | $\geq 3n$ | $2n + 2$ |
| subtorus with image a line $\neq \mathbb{C}(0,1)$ ($\lambda \geq n$, $\dim L \leq n$, $H'' = 0$) | 1 | $\geq 3n + 2$ | $2n + 2$ |

and $r' = 0$ forces $G' = 0$. So there is no obstruction iff $n \geq 3$, hence for every admissible $n$ (which is even and $\geq 4$: $n$ is a multiple of $2g$, and $[K' : \mathbb{Q}] \geq 4$); the minimum margin is $n - 2$. Condition (2) reads $5n \geq 3n$. □

*Proof of Theorem 1.3.* By the reduction in §4.1, $P$ is an algebraic multiple of an eigen-period of the same kind of a simple CM abelian variety of primitive type, and the classes in the statement depend only on $\overline{\mathbb{Q}}^\times P$. Write $P^+$ for a first-kind and $P^-$ for a second-kind eigen-period; $P_\sigma P_{\bar\sigma} \in \pi\overline{\mathbb{Q}}^\times$ gives $\pi/P^+ \in \overline{\mathbb{Q}}^\times P^-$ and $\pi/P^- \in \overline{\mathbb{Q}}^\times P^+$ (for the conjugate slots).

*Direct families*: $t = cP^+/\pi$ (Theorem 5.7 with $s = t$) and $t = cP^-/\pi$ (Theorem 5.9 with $s = t$). Take $x_1 = \log\alpha$, $x_2 = \log\beta$; if $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$ then $sx_j \in \mathcal{L}$. If $2\pi i, \log\alpha, \log\beta$ are

$\mathbb{Q}$-independent they are $K$-independent (Baker), contradicting the theorem (which also puts $2\pi i$ in the set). Otherwise $m\log\beta = a\log\alpha + 2\pi i b$ with integers $m, b \neq 0$, so $e^{2\pi i b t} \in \overline{\mathbb{Q}}$ and $t\cdot 2\pi i \in \overline{\mathbb{Q}}\mathcal{L}$; but $t\cdot 2\pi i = 2ic\,P^+$ (resp. $2ic\,P^-$) is a nonzero algebraic multiple of an eigen-period, which is not in $\overline{\mathbb{Q}}\mathcal{L}$ by (W) (on $A\times\mathbb{G}_{\mathrm{m}}^r$, resp. $A^\natural\times\mathbb{G}_{\mathrm{m}}^r$).

*Inverted families*: the four exponentials statement at $t$ for all $\mathbb{Q}$-independent pairs is equivalent to the statement at $1/t$ (if $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$, apply the statement at $1/t$ to $\alpha^t, \beta^t$, whose logarithms $t\log\alpha, t\log\beta$ are $\mathbb{Q}$-independent). The inverses of the direct classes are $\overline{\mathbb{Q}}^\times\pi/P^+ = \overline{\mathbb{Q}}^\times P^-$ and $\overline{\mathbb{Q}}^\times\pi/P^- = \overline{\mathbb{Q}}^\times P^+$. So all of $t \in \overline{\mathbb{Q}}^\times P^{\pm1}$, $\overline{\mathbb{Q}}^\times(P/\pi)^{\pm1}$ are covered, for $P$ of either kind ($1/P^+ \in \overline{\mathbb{Q}}^\times P^-/\pi$, $1/P^- \in \overline{\mathbb{Q}}^\times P^+/\pi$).

*Beta values.* By §4.1, $B(a/N, b/N)$ is a first-kind eigen-period up to $\overline{\mathbb{Q}}^\times$ if $a+b<N$ and a second-kind one if $a+b>N$. Of the four examples, $\Gamma(1/5)^2/(\pi\Gamma(2/5)) = B(\frac15,\frac15)/\pi$ and $\Gamma(4/5)^2/\Gamma(3/5) \in \overline{\mathbb{Q}}^\times B(\frac45,\frac45)$ (in fact $B(\frac45,\frac45) = \Gamma(\frac45)^2/\Gamma(\frac85) = \frac53\Gamma(\frac45)^2/\Gamma(\frac35)$) are covered by Theorem 5.7 (the latter inverted; the two are inverse to each other up to $\overline{\mathbb{Q}}^\times$), and $B(\frac15,\frac15)$, $B(\frac17,\frac17)$ by Theorem 5.9 (inverted). □

**Remark 5.10** (Are the examples new instances?)**.** [Heuristic] Theorem 1.3 for a given Beta value $t$ would already follow from Theorem 1.2 if $t$ lay in $\overline{\mathbb{Q}}^\times\Omega^{\pm1}$ or $\overline{\mathbb{Q}}^\times(\Omega/\pi)^{\pm1}$ for a CM elliptic period $\Omega$. For $t \in \overline{\mathbb{Q}}^\times\{P^{\pm1}, (P/\pi)^{\pm1}\}$ this happens iff $P \in \overline{\mathbb{Q}}^\times\{\Omega,\ \pi/\Omega,\ 1/\Omega,\ \Omega/\pi,\ \pi\Omega,\ \pi^2/\Omega\}$. Wüstholz's theorem excludes the two linear coincidences $P \in \overline{\mathbb{Q}}^\times\Omega$ and $P \in \overline{\mathbb{Q}}^\times\pi/\Omega = \overline{\mathbb{Q}}^\times\eta^*$ (for $A$ not isogenous to a power of the curve); the other four are quadratic period relations, excluded only by the Rohrlich–Lang conjecture. So each named $t$ is a new instance only conditionally.

**Remark 5.11** (Not a linearisation)**.** [Heuristic] The hypothesis of Theorem 5.7 on the data $x_j = \log\alpha_j$ reads $cP\cdot\log\alpha = \pi\cdot\log\gamma$: a period times a logarithm against $\pi$ times a logarithm. The abelian analogue of Laurent's identity (6) (an internal note, not reproduced here) linearises products of a period with the logarithm of the conjugate slot of the same variety, not $P\log\alpha$. So we do not expect Theorem 1.3 to follow from the analytic subgroup theorem.

**Remark 5.12** (Strong form)**.** The $\mathbb{G}_{\mathrm{a}}$-bookkeeping of Theorem 6.2(a) below transfers to Theorems 5.7 and 5.9. This gives the strong four exponentials conjecture, for eigen-periods $P$ of either kind, at every $t$ in the classes $\overline{\mathbb{Q}}^\times P^{\pm1}$ and $\overline{\mathbb{Q}}^\times(P/\pi)^{\pm1}$ (Theorem 6.8 in §6.2, with the mixed forms SMG-A and SMG$^\natural$-A, Theorem 6.9).

5.4. **The general form, and elliptic four exponentials.** Theorems 5.1 and 5.4 are the cases $s_1 = s_3 = \mathbb{G}_{\mathrm{m}}$ of the following count. Let $s_1, s_2, s_3$ be *factors*: $\mathbb{G}_{\mathrm{m}}$ (period $2\pi i$), the base of a CM curve (period $\Omega$), or the fibre of the universal vector extension of a CM curve (period $\eta^*$). In the count of Theorem 2.3 the first two have inner weight $\alpha_{K,\Psi}$ (Remark 5.8) equal to 1 per factor and a fibre factor has inner weight $1+2/n$ (its printed weight is $\alpha_K(E^\natural) = 2+2/n$, Lemma 4.3). All cases are consequences of Theorem 1.1; the fibre-free cases (every factor $\mathbb{G}_{\mathrm{m}}$ or the base of a CM curve) already follow from Corollary 2.2 ($\alpha_K \le 3$, $\kappa = 1$). Write $\Lambda_K(s)$ for the $K$-span of the corresponding logarithms ($\mathcal{L}$, $\mathcal{L}_E$, or fibre values). Let $p_j$ be the period of $\sigma_j$, $r = p_2/p_1$, and take $\Psi$ reading $s_1$ into the first coordinate and $s_2+s_3$ into the second.

**Theorem 5.13** (General form; proof sketch)**.** *Assume:*

- *$s_1, s_2$ are not of the same kind on isogenous groups (so $r \notin \overline{\mathbb{Q}}$);*
- *if $s_2$ and $s_3$ are of the same kind on isogenous groups, they are merged into one factor;*
- *(allowing $r \in \overline{\mathbb{Q}}^\times p_2/p_1$) $K$ contains this constant, the CM fields, and the Legendre and isogeny constants, and $[K:\mathbb{Q}] \ge 2f+2$, where $f$ is the number of fibre factors;*
- *$s_1, s_3$ are not both fibre factors.*

*Then*

$$\dim_K\{\lambda \in \Lambda_K(s_1) :\ r\lambda \in \Lambda_K(s_2) + \Lambda_K(s_3)\} \le 2.$$

The count is that of Theorem 5.1, with the threshold $n > 2f$ coming from $G' = 0$ ($5n + 2f > 4n + 4f$). An internal referee checked all 27 factor-kind triples by an exact integer case check. The load-bearing step (a bound on the data in subtori with image a $K$-line) was done by hand, and the subgroup lists are not written out in this paper. The excluded case is "four exponentials for quasi-logarithms": $s_1, s_3$ fibre factors of isogenous curves (for non-isogenous curves we have not written it out). It has mixed subgroups whose fibre slope $\mu$ is algebraic but possibly outside $K$, and it needs the $K(\mu)$-linear independence of $K$-independent fibre values. This gap is now closed: Lemma 5.15 below shows that the fibre slope of a mixed subgroup is $K$-rational ($\mu \in K$), and Theorem 5.16 proves four exponentials for quasi-logarithms, with the second factor $\mathbb{G}_{\mathrm{m}}$, the base of a CM curve, the fibre of a CM curve with another CM field, or a first-kind eigen-coordinate of a CM abelian variety, for $[K : \mathbb{Q}] \ge 6 = 2f + 2$. When $s_1 = s_3$ is the base of a CM curve $E$, Theorem 5.13 is an elliptic analogue of four exponentials:

**Corollary 5.14** (Elliptic four exponentials at CM slopes; proof sketch)**.** *Let $E$ have CM, with period $\Omega$, and let $t$ be one of $\pi/\Omega^2$, $\Omega'/\Omega$, $\eta^*(\Omega')/\Omega$, where $\Omega'$ is a period of a CM curve $E'$ not isogenous to $E$, all up to $\overline{\mathbb{Q}}^\times$. If $u_1, u_2 \in \mathcal{L}_E$ and $\Omega$ are linearly independent over $k$, then $tu_1$ and $tu_2$ do not both lie in $\overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\Lambda(s_2)$. Here $\Lambda(s_2)$ is the set of fibre values of $E$ (for $t = \pi/\Omega^2$), $\mathcal{L}_{E'}$ (for $t = \Omega'/\Omega$), or the set of fibre values of $E'$ (for $t = \eta^*(\Omega')/\Omega$). In particular $tu_1, tu_2$ are not both elliptic logarithms of algebraic points of $E$.*

Corollary 5.14 is now contained in Theorem 6.12 of §6.3, which is proved (label (c)), in a stronger form ($u_j \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$, and the target of (SE) also contains $\overline{\mathbb{Q}}$; $k$-independent elements of $\mathcal{L}_E$ are $\overline{\mathbb{Q}}$-independent by Masser's theorem [23]). Its abelian analogue, on eigen-slots of CM abelian varieties, is Theorem 6.16.

*Four exponentials for quasi-logarithms.* Let $E$ have CM by $k$, $\Omega \ne 0$ a period, $\eta = \eta^*(\Omega)$, and $\mathrm{Fib}_E$ the set of fibre values of $E$ (§4.2: constants, quasi-logarithms $\zeta^*(u) + \overline{\mathbb{Q}}$, $u \in \mathcal{L}_E$, and $\eta^*(\Lambda_E)$). The following lemma is the missing step of the excluded case.

**Lemma 5.15** (Fibre images are rational)**.** *Let $K \supseteq k$ have degree $n$, $\theta$ a $\mathbb{Q}$-basis of $K$, $\phi(m) = \sum \theta_j \bar{m}_j$ on $k^n$, and $Q = (E^\natural)^n/(B^\natural_{\ker\phi} + \{\theta^{\mathsf{T}} w = 0\})$, with fibre coordinate $\varsigma = \theta^{\mathsf{T}} w$ (the quotient $Q''$ of the proof of Theorem 5.4). Put $Q_1 = Q_3 = Q$. If $H \subset Q_1 \times Q_3$ is a connected algebraic subgroup which is not a vector group and $(\varsigma_1, \varsigma_3)(T_H)$ is a line $\ell$, then $\ell$ is spanned by a vector of $K^2$, and the abelian image of $H$ has dimension at most $n/2$.*

*Proof.* Let $\tilde{H} \subset (E^\natural)^{2n}$ be the connected preimage, with abelian image $B_S$, $S \subset k^{2n}$, and let $M$ be a matrix over $\mathcal{O}_k$ whose columns span $S$. Since $\mathcal{O}_k$ acts on $\mathrm{Lie}(E^\natural)$ by $\mathrm{diag}(a, \bar{a})$, $\mathrm{Lie}(B^\natural_S) = \{(Mu, \bar{M}u')\}$, and $\tilde{H} \supseteq B^\natural_S$ by the universal property of vector extensions ((F3) for $g = 1$). The fibre vectors $(0, \bar{M}u')$ have fibre coordinates $(\phi(M_1)u', \phi(M_3)u')$, which span the $K$-rational space $R_S = \mathrm{span}\{(\phi(s_1), \phi(s_3)) : s \in S\}$; $R_S = 0$ exactly when $S \subset \ker\phi \oplus \ker\phi$, i.e. when $H$ is a vector group. So $0 \ne R_S \subseteq \ell$ and $\ell = R_S$. The map $S/(\ker\phi)^2 \to K^2$, $s \mapsto (\phi(s_1), \phi(s_3))$, is injective with image in the $K$-line $\ell \cap K^2$, whence the dimension bound. □

**Theorem 5.16** (Theorem QL)**.** *Let $p_2$ be one of: $2\pi i$; a period $\Omega'$ of a CM curve $E'$ ($E' = E$ allowed); the quasi-period $\eta' = \eta^*(\Omega')$ of a CM curve $E'$ with CM field $k' \ne k$; a first-kind eigen-period $P$ of a simple CM abelian variety $A$. Let $X_2$ be the corresponding set of values $\mathcal{L}$, $\mathcal{L}_{E'}$, $\mathrm{Fib}_{E'}$, $z_\sigma(\Lambda(A))$. Let $c \in \overline{\mathbb{Q}}^\times$ and $t = cp_2/\eta$. If $f_1, f_2 \in \overline{\mathbb{Q}}\mathrm{Fib}_E$ and $\eta, f_1, f_2$ are linearly independent over $\overline{\mathbb{Q}}$, then $tf_1$ and $tf_2$ are not both in $\overline{\mathbb{Q}}\mathrm{Fib}_E + \overline{\mathbb{Q}}X_2$.*

*Status:* [Proved modulo published theorems and Lemma A.1]; two internal referees (Lemma 5.15 correct, with an independent derivation and an exact falsity hunt over about 500 subspaces; Theorem 5.16 correct with fixes, applied).

**Corollary 5.17.** *Let $t \in \overline{\mathbb{Q}}^\times\{\Omega,\ \Omega\Omega'/\pi,\ \Omega/\Omega',\ \Omega P/\pi\}^{\pm 1}$, with $\Omega'$, $P$ as in Theorem 5.16 (in the class $\Omega/\Omega'$ the curve $E'$ has CM field $k' \neq k$). If $f_1, f_2 \in \overline{\mathbb{Q}}\mathrm{Fib}_E$ are linearly independent over $\overline{\mathbb{Q}}$, then $tf_1, tf_2$ are not both in $\overline{\mathbb{Q}}\mathrm{Fib}_E$. For example (class $\Omega/\Omega'$ with $E = E_i$, $E' = E_\rho$), if $u_1, u_2 \in \mathcal{L}_{E_i}$ and $\varpi$ are linearly independent over $\mathbb{Q}(i)$, then $(\varpi/\Omega_\rho)\zeta_i(u_1)$ and $(\varpi/\Omega_\rho)\zeta_i(u_2)$ are not both $\overline{\mathbb{Q}}$-linear combinations of $1$ and quasi-logarithms $\zeta_i(u)$, $u \in \mathcal{L}_{E_i} \setminus \varpi\mathbb{Z}[i]$.*

*Status:* [Proved modulo published theorems and Lemma A.1].
In the class $\overline{\mathbb{Q}}^\times\Omega^{\pm 1}$ ($p_2 = 2\pi i$) the statement is of degree one: by Laurent's identity (6), $\Omega\zeta^*(u) = \lambda(u,\Omega) + \eta u$ with $\eta u \in \overline{\mathbb{Q}}^\times(\pi/\Omega)u$ a $\overline{\mathbb{Q}}$-combination of third-kind periods, so the hypothesis $tf_1, tf_2 \in \overline{\mathbb{Q}}\mathrm{Fib}_E$ is a pair of $\overline{\mathbb{Q}}$-linear relations among periods of 1-motives, and that case is within reach of the analytic subgroup theorem (we keep it in Theorem 5.16 as a calibration case). The bilinear content is in the other three classes, where a period of another slot multiplies a fibre value, as in the example.

*Proof of Theorem 5.16.* Let $K$ be a number field of degree $n \geq 6$ containing $k$, $c$, the CM field of the second factor (the Galois closure of $F$ if $X_2$ comes from $A$) and all coefficients needed to write $f_j \in \overline{\mathbb{Q}}\mathrm{Fib}_E$ and $tf_j \in \overline{\mathbb{Q}}\mathrm{Fib}_E + \overline{\mathbb{Q}}X_2$ ($j = 1, 2$), so that $v_j \in \Psi(\Lambda_K(G))$. Take $G = E^\natural \times X \times E^\natural$ with $X = \mathbb{G}_\mathrm{m}, E', E'^\natural, A$ respectively, $\Psi = (w_1,\ x_2 + w_3)$ ($w_1, w_3$ the fibre coordinates, $x_2$ the coordinate of $X$ read by $X_2$) and $V = \mathbb{C}(1,t)$. If $X = E'^\natural$, the two fibre coordinates read in the second target are first merged (the vector subgroup $\{\theta^\mathsf{T}w' + \theta^\mathsf{T}w_3 = 0\}$ lies in the kernel). Then $G'' = Q_1 \times X'' \times Q_3$ with $Q_1, Q_3$ as in Lemma 5.15 (for $X = E'^\natural$: $Q_1 \times (X'' \times Q_3)/D$, $D = \{\varsigma' + \varsigma_3 = 0\}$, with one fibre $\mathbb{G}_\mathrm{a}$ over the abelian part of $X'' \times Q_3$). The data are $w = (\eta, cp_2)$ and $v_j = (f_j, tf_j)$; $m = 3n$, $\kappa = n$ (a period in $Y''$ has first coordinate in $K\eta$), and $\Psi(\Omega_K(G)) = K\eta \times (K\eta + Kp_2)$. Here $t \notin K$: $t \in \overline{\mathbb{Q}}$ would put $p_2$ in $\overline{\mathbb{Q}}\eta \subset \overline{\mathbb{Q}}\mathrm{Fib}_E$, which Wüstholz's theorem [52] excludes. For $r' = 1$, (1) reads $\operatorname{rk} Y_{G'} + l + 2\dim(G'/T') \geq 4n$, where $T'$ is the torus part of $G'$ and $l = \dim T'$. Subgroups with image $\mathbb{C}(0,1)$ have $\operatorname{rk} Y_{G'} \leq n$ and $l + 2\dim(G'/T') \leq 2n + 2$. Subgroups of $Q_1$ have image $\mathbb{C}(1,0)$, dimension $\leq n/2 + 1$ and $\operatorname{rk} Y_{G'} \leq 2n$. Mixed subgroups of $Q_1 \times Q_3$ that are not vector groups have, by Lemma 5.15, a $K$-rational image line, dimension $\leq n/2 + 1$, and $\operatorname{rk} Y_{G'} \leq 2n$: if $v_1, v_2 \in \ell + \Psi(\Omega_K(G))$ with $\ell = \mathbb{C}(p,1)$, $p \in K$, then $f_j(1 - pt) \in K\eta + Kp_2 = \eta(K + Kt)$ (period datum $p_2 = t\eta/c$), and if $\ell = \mathbb{C}(1,0)$ then $tf_j \in \eta(K + Kt)$; either way $\eta, f_1, f_2$ lie in a 2-dimensional $K$-space, against their $K$-independence. Vector lines have dimension 1 and $\operatorname{rk} Y_{G'} \leq 3n$. These cases are complete. If $X''$ has an abelian part isogenous to the base of $E$ ($E' = E$, or $A$ isogenous to a power of $E$), abelian subvarieties may mix $Q_1$, $X''$ and $Q_3$ (and, for $X = E'^\natural$, $Q_1$ and the merged factor); if the abelian image of $G'$ projects nontrivially to $X''$ (resp. to the $E'$-part of the merged factor), its image contains $(0,1)$, so $r' = 1$ forces image $\mathbb{C}(0,1)$, hence $\phi(S_1) = 0$ and $G' \subset X'' \times Q_3$, which is the case of image $\mathbb{C}(0,1)$ above; otherwise Lemma 5.15 applies verbatim (to $Q_1 \times Q_3$, resp. to $Q_1$ and the merged factor). So every subgroup with $r' = 1$ has total at most $3n + 2 < 4n$. For $G' = 0$ the inequality fails with margin $n - 4 > 0$, since $n \geq 6$ (it obstructs exactly for $n \leq 4$); this is the minimum margin. No subgroup satisfies (1). Condition (2) holds ($d_0 = 0$, $2N = 2d - 2 \geq 3n + 2 \geq d_1 + \kappa$). □

*Proof of Corollary 5.17.* The statement is invariant under $t \mapsto \gamma t$ ($\gamma \in \overline{\mathbb{Q}}^\times$) and under $t \mapsto 1/t$ (replace $f_j$ by $tf_j$). By $\Omega\eta \in \overline{\mathbb{Q}}^\times\pi$ and $\Omega'\eta' \in \overline{\mathbb{Q}}^\times\pi$, the classes $\overline{\mathbb{Q}}^\times p_2/\eta$ of Theorem 5.16 are $\overline{\mathbb{Q}}^\times\Omega$, $\overline{\mathbb{Q}}^\times\Omega\Omega'/\pi$, $\overline{\mathbb{Q}}^\times\Omega/\Omega'$ and $\overline{\mathbb{Q}}^\times\Omega P/\pi$. If $\eta \in \overline{\mathbb{Q}}f_1 + \overline{\mathbb{Q}}f_2$, then $tf_1, tf_2 \in \overline{\mathbb{Q}}\mathrm{Fib}_E$ gives

$cp_2 = t\eta \in \overline{\mathbb{Q}}\mathrm{Fib}_E$, which Wüstholz's theorem [52] excludes; otherwise Theorem 5.16 applies. For the example: $t = \varpi/\Omega_\rho \in \overline{\mathbb{Q}}^\times \eta_\rho/\eta_i$ is the class $\overline{\mathbb{Q}}^\times p_2/\eta$ with $p_2 = \eta' = \eta(\Omega_\rho)$ of $E' = E_\rho$, whose CM field $\mathbb{Q}(\rho)$ differs from $\mathbb{Q}(i)$; and $f_j = \zeta_i(u_j) \in \mathrm{Fib}_{E_i}$ ($\kappa_1 = 0$ on $E_i$). By Wüstholz's theorem on $(E_i^\natural)^3$, whose smallest algebraic subgroup through $((\varpi,\eta_i),(u_1,\zeta_i(u_1)),(u_2,\zeta_i(u_2)))$ is $(E_i^\natural)^3$ when $\varpi, u_1, u_2$ are $\mathbb{Q}(i)$-independent, the numbers $\eta_i, \zeta_i(u_1), \zeta_i(u_2)$ are $\overline{\mathbb{Q}}$-independent, so in particular $f_1, f_2$ are. Finally $\overline{\mathbb{Q}}\mathrm{Fib}_{E_i}$ is the $\overline{\mathbb{Q}}$-span of 1 and the $\zeta_i(u)$, $u \in \mathcal{L}_{E_i}$ off the lattice, since $\eta_i = 2\zeta_i(\varpi/2)$ and $\eta(\Lambda) \subset \mathbb{Q}(i)\eta_i$. □

**Remark 5.18.** Without Lemma 5.15 a mixed line $\mathbb{C}(\mu, 1)$ with $\mu \notin K$ would only give $\mathrm{rk}\, Y_{G'} \leq 3n$, which is an obstruction; this was the gap left in Theorem 5.13. The same lemma, with $\psi_{\tilde\sigma}$ in place of $\phi$ and $K \supseteq \tilde\sigma(F)$, gives, by (F1)–(F4), the analogue for second-kind eigen-coordinates of CM abelian varieties (not written out).

## 6. Strong four exponentials at CM periods

This section proves Theorem 1.6 (§6.1), its extension to eigen-periods of CM abelian varieties, Theorem 6.8 (§6.2), and strong elliptic and eigen-slot analogues (§6.3).

6.1. **CM elliptic periods: Theorems SMG, SMG$^\natural$ and S.** Throughout, $E/\overline{\mathbb{Q}}$ has CM by the imaginary quadratic field $k$, $\Omega \neq 0$ is a period, and $\eta = \eta^*(\Omega)$ its conjugate-eigen quasi-period (§4.2); Legendre's relation gives $\eta \in \overline{\mathbb{Q}}^\times \pi/\Omega$ (for $y^2 = 4x^3 - 4x$: $\Omega = \varpi$, $\eta = \pi/\varpi$). $\mathcal{L}_E$ and Fib are as in §4.2; for a number field $K$, $K\mathcal{L}$, $K\mathcal{L}_E$, $K\mathrm{Fib}$ are $K$-spans.

**Lemma 6.1.** *$\Omega \notin \widetilde{\mathcal{L}}$ and $\eta \notin \widetilde{\mathcal{L}}$.*

*Proof.* [Lit: Wüstholz [52]; Baker–Wüstholz [4, Ch. 6]] If $\Omega = \beta_0 + \sum \beta_k \log\alpha_k$, the point $(1, \log\alpha, \Omega)$ of $\mathrm{Lie}(\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times E)$ lies in a $\overline{\mathbb{Q}}$-hyperplane; by Wüstholz the smallest algebraic subgroup through it has tangent space in the hyperplane, but it is a product (P) whose $E$-factor is $E$, and $0 \times 0 \times T(E)$ is not in the hyperplane. For $\eta$ use $(1, \log\alpha, (\Omega, \eta))$ on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times E^\natural$. The torus splits off; the smallest subgroup $R \subset \mathbb{G}_\mathrm{a} \times E^\natural$ maps onto $E$, so its image in $E^\natural$ is $E^\natural$ (as $\mathrm{Hom}(E^\natural, \mathbb{G}_\mathrm{a}) = 0$, $E^\natural$ has no proper subgroup onto $E$); if $R \cap (\mathbb{G}_\mathrm{a} \times 0) = 0$, $R$ would be the graph of a homomorphism $E^\natural \to \mathbb{G}_\mathrm{a}$, which is 0, contradicting the $\mathbb{G}_\mathrm{a}$-coordinate 1. So $R = \mathbb{G}_\mathrm{a} \times E^\natural$, which contains the fibre direction, not in the hyperplane. □

**Theorem 6.2.** *Let $c \in \overline{\mathbb{Q}}^\times$, and let $y_1, y_2 \in \widetilde{\mathcal{L}}$ be such that $2\pi i, y_1, y_2$ are linearly independent over $\overline{\mathbb{Q}}$.*

(a) (Theorem SMG) *If $s = c\Omega/\pi$, then $sy_1$ and $sy_2$ are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$.*

(b) (Theorem SMG$^\natural$) *If $s = c\eta/\pi$ (so $s \in \overline{\mathbb{Q}}^\times/\Omega$), then $sy_1$ and $sy_2$ are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$.*

*Status:* [Proved; formally verified, standard axioms] for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ with $sy_1, sy_2 \in \widetilde{\mathcal{L}}$ (the case used for Theorem 1.6); [Proved modulo the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times \Omega_\rho$ with $sy_1, sy_2 \in \widetilde{\mathcal{L}}$; [Proved modulo published theorems and Lemma A.1] for the mixed second rows ($\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$) and for every other CM period. The paper proof uses Lemma A.1 with its exponent $b > 1$, since $d_0 \geq 1$ (§2.2). Two internal referees (correct with fixes, applied; independent exact enumerations) graded the first write-up as a proof sketch modulo trust point T; a third found every $r' \leq 1$ subgroup listed and nothing missing. Falsity hunt negative (§12.1).

*Provenance.* Both parts are consequences of Theorem 1.1 (§4.4: $\alpha_{K,\Psi} - \kappa = 2$, resp. $2 + 2/n$, against three data). For (a) the printed weight suffices ($\alpha_K(\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times E) = 3$, $\kappa = 1$), and for (b) it does on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^2 \times E^\natural$ read through base and fibre (Remark 11.1). We found no explicit statement of either in print; the proof below applies Theorem 2.3 directly.

*Proof. Field.* Choose a number field $K \supseteq k(i,c)$ containing all coefficients needed to write $y_j \in K + K\mathcal{L}$ and $sy_j$ in $K + K\mathcal{L} + K\mathcal{L}_E$, resp. $K + K\mathcal{L} + K\mathrm{Fib}$, with $[K:\mathbb{Q}] \geq 4$ in (b) (at $n = 2$ the count of (b) fails, with equality, for types Z and F of the table below). The $\overline{\mathbb{Q}}$-independence of $2\pi i, y_1, y_2$ gives their $K$-independence; no base change is needed (and base change would not obviously preserve $K$-independence of elements of $\overline{\mathbb{Q}} + K\mathcal{L}$). Let $n = [K:\mathbb{Q}]$ and $\theta$ a basis of $K/\mathbb{Q}$.

*(a) The group.* $G = \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times E$ with coordinates $(a_1, x, a_2, y, z)$, $\Psi(a_1, x, a_2, y, z) = (a_1 + x,\, a_2 + y + z)$, $V = \mathbb{C}(1, s)$, $\Psi_\theta = \Psi \circ p_\theta$. By (P) and the structure of $E^n$, connected subgroups of $G^n$ are $H \times T_L \times B_S$, and $T \subset \ker \Psi_\theta$ iff $H \subset \{\theta \cdot a_1 = \theta \cdot a_2 = 0\}$, $L = 0$, $\psi(S) = 0$ ($\psi(m) = \sum \theta_j m_j$; $\dim_k \ker \psi = n/2$ by Lemma 4.2). So $G'' = \mathbb{G}_\mathrm{a}^2 \times \mathbb{G}_\mathrm{m}^{2n} \times A''$ with $\dim A'' = n/2$: $d_0 = 2$, $d_1 = 2n$, $d_2 = n/2$, $d = 5n/2 + 2$, $d_1 + 2d_2 = 3n$. $V''$ is a hyperplane and $W'' = \ker \Psi''$ is $\overline{\mathbb{Q}}$-rational of dimension $d - 2$.

*Data.* $w = (2\pi i, 2ic\Omega) \in \Psi(\Omega_K(G))$, and $v_j = (y_j, sy_j) \in V \cap \Psi(\Lambda_K(G))$, writing $y_j = b_j + x_j$ with $b_j \in \overline{\mathbb{Q}} = \Lambda(\mathbb{G}_\mathrm{a})$. Let $Y''$ be generated by lifts of $\theta_h w, \theta_h v_1, \theta_h v_2$; $\Psi''$ is injective on it since $2\pi i, y_1, y_2$ are $K$-independent, so $m = \mathrm{rk}\, Y'' = 3n$, and $\kappa \geq n$ (in fact $\kappa = n$). With $Y_{G'} = \{y \in Y'' : \exp y \in G'\}$, $\lambda = 3n - \mathrm{rk}\, Y_{G'}$, and an obstruction needs $r' \leq 1$ and

$$(\star) \qquad \lambda + \delta_1 + 2\delta_2 \leq (2 - r')(3n - \kappa) \leq (2 - r') \cdot 2n.$$

*Case check.* $r' = 0$ forces $G' = 0$: the preimage of $G'$ in $G^n$ is connected (as $U$ is) with tangent space in $\ker \Psi_\theta$, so it lies in $U$. For $G' = 0$, $(\star)$ reads $6n - \kappa \leq 2(3n - \kappa)$, i.e. $\kappa \leq 0$: excluded, margin $\kappa \geq n$. Let $r' = 1$, $\ell = \Psi''(T_{G'})$, $G' = H \times T_L \times B$ with $h, l, b$ the dimensions. Then $\Psi''(T_H) = T_H$ (so $h \leq 1$, and $h = 1$ forces $\ell = T_H$); $L \neq 0$ forces $\ell$ to be $K$-rational and $l \leq n$; $B \neq 0$ forces $\ell = \mathbb{C}(0,1)$. With $\delta_1 + 2\delta_2 = 3n - l - 2b$, $(\star)$ becomes $\mathrm{rk}\, Y_{G'} + l + 2b \geq 4n$. Since $y \in Y_{G'}$ implies $\Psi''(y) \in \ell + \Omega_\Psi$, $\Omega_\Psi := \Psi(\Omega_K(G)) = K2\pi i \times (K2\pi i + K\Omega)$ (a $K$-space), we have $\mathrm{rk}\, Y_{G'} \leq n \dim_K \mathcal{N}$, $\mathcal{N} = \{\beta \in K^3 : \beta_0 w + \beta_1 v_1 + \beta_2 v_2 \in \ell + \Omega_\Psi\}$.

- $\ell = \mathbb{C}(0,1)$: first coordinates give $\beta_1 y_1 + \beta_2 y_2 \in K2\pi i$, so $\dim \mathcal{N} \leq 1$ and $\mathrm{rk} + l + 2b \leq n + n + n = 3n < 4n$.
- $\ell$ $K$-rational, $\neq \mathbb{C}(0,1)$, $L \neq 0$: $\dim \mathcal{N} \leq 2$ (Lemma 6.4), so $\mathrm{rk} + l \leq 3n < 4n$.
- $L = 0$, $B = 0$, $h = 1$ ($\ell$ any $\overline{\mathbb{Q}}$-line): $\mathrm{rk} \leq 3n$, $l = b = 0$: $3n < 4n$.

No other $G'$ has $r' = 1$. Hence no obstruction, for every $n \geq 2$; the minimum margin is $n$. Condition (2): $5n + 2 \geq 3n + 4$.

*(b) The group.* $G = \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m} \times E^\natural$, $\Psi(a, x, y, (z, w)) = (a + x,\, y + w)$ ($w$ the conjugate-eigen fibre coordinate), $V = \mathbb{C}(1, s)$, $s = c\eta/\pi$. As in the proof of Theorem 5.4, $G'' = \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^{2n} \times Q''$, where the $\mathbb{G}_\mathrm{a}$-part of $U$ is the hyperplane $\{\theta \cdot a = 0\}$ and $Q''$ is the anti-affine extension of $A''$ ($\dim n/2$) by its fibre $\mathbb{G}_\mathrm{a}$, with no splitting over any abelian subvariety $B \neq 0$: $d_0 = 1$, $d_1 = 2n$, $d_2 = n/2 + 1$, $d = 5n/2 + 2$, $d_1 + 2d_2 = 3n + 2$. Although $\mathbb{G}_\mathrm{a}^n \times (E^\natural)^n$ has non-product subgroups, $U$ is as stated: with $U_0 = \{\theta \cdot a = 0\} \times U_E$, where $U_E = B^\natural_{\ker \phi} + \{\theta^\mathsf{T} w = 0\} \subset (E^\natural)^n$ is the group $U$ of the proof of Theorem 5.4, $U/U_0$ is a connected subgroup of $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^{2n} \times Q''$ with tangent space in $\ker \Psi''$, and by Lemma 6.3 it is 0. The data are $w = (2\pi i, 2ic\eta)$ and $v_j = (y_j, sy_j)$; $m = 3n$, $\kappa \geq n$.

**Lemma 6.3.** *A connected subgroup of $G''$ is $T_L \times R$ with $R \subset \mathbb{G}_\mathrm{a} \times Q''$ connected and one of: 0; a $\overline{\mathbb{Q}}$-line $\{(\alpha u, \beta u)\}$ in the vector group $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{a}^\mathrm{fib}$; $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{a}^\mathrm{fib}$; $0 \times Q''_B$; $\mathbb{G}_\mathrm{a} \times Q''_B$ ($B \neq 0$, $Q''_B$ the preimage of $B$).*

*Proof.* There are no nonzero homomorphisms between tori and $\mathbb{G}_\mathrm{a} \times Q''$ or its subgroups, so the torus splits off. A subgroup of $Q''$ mapping onto $B \neq 0$ contains the fibre (otherwise it would split $Q''$ over $B$), so subgroups of $Q''$ are 0, $\mathbb{G}_\mathrm{a}^\mathrm{fib}$, $Q''_B$. If $R$ maps onto $B \neq 0$, its image in $Q''$ is $Q''_B$ and $R \cap (\mathbb{G}_\mathrm{a} \times 0)$ is $\mathbb{G}_\mathrm{a}$ or 0; in the latter case $R$ is the graph of a homomorphism

$Q''_B \to \mathbb{G}_{\mathrm{a}}$, which is 0 since a nonzero one would split $Q''_B$ over $B$. If $B = 0$, $R$ is a $\overline{\mathbb{Q}}$-subspace of $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{a}}^{\mathrm{fib}}$. □

The diagonal lines $\{(\alpha u, \beta u)\}$ are the new feature compared with (a). With $\delta_0 = 1 - \dim \pi_0(R)$ one has $\delta_1 + 2\delta_2 = 3n + 4 - l - 2\dim R - 2\delta_0$, and for $r' = 1$ an obstruction needs $\mathrm{rk}\, Y_{G'} + l + 2\dim R + 2\delta_0 \geq 4n + 2$ (using $\kappa \geq n$). The cases:

| type | $R$ | image | $\mathrm{rk}\, Y_{G'}$ | $l$ | $\dim R$ | $\delta_0$ | total |
|---|---|---|---|---|---|---|---|
| Z | 0 ($G' = 0$) | 0 | | | | | $(\star) \iff \kappa \leq 2$ |
| F | $0 \times Q''_B$ | $\mathbb{C}(0,1)$ | $\leq n$ | $\leq n$ | $\leq n/2 + 1$ | 1 | $\leq 3n + 4$ |
| VF | fibre line | $\mathbb{C}(0,1)$ | $\leq n$ | $\leq n$ | 1 | 1 | $\leq 2n + 4$ |
| T0 | 0, $L \neq 0$ | $\mathbb{C}(0,1)$ | $\leq n$ | $\leq n$ | 0 | 1 | $\leq 2n + 2$ |
| VK | line, $\alpha \neq 0$, $\ell$ $K$-rat. | $\ell$ | $\leq 2n$ | $\leq n$ | 1 | 0 | $\leq 3n + 2$ |
| VQ | line, $\alpha \neq 0$, $\ell$ not $K$-rat. | $\ell$ | $\leq 3n$ | 0 | 1 | 0 | $\leq 3n + 2$ |
| TK | 0, $L \neq 0$, $\ell$ $K$-rat., $\neq \mathbb{C}(0,1)$ | $\ell$ | $\leq 2n$ | $\leq n$ | 0 | 1 | $\leq 3n + 2$ |

Types with $R \supseteq \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{a}}^{\mathrm{fib}}$ or $R = \mathbb{G}_{\mathrm{a}} \times Q''_B$ have $r' = 2$. For VK and TK, Lemma 6.4 applies with $\Omega_\Psi = K2\pi i \times (K2\pi i + K\eta)$. So there is no obstruction for $n \geq 3$, hence for every even $n \geq 4$ (type Z needs $\kappa \geq 3$, type F needs $n \geq 3$); the minimum margin is $n - 2$, attained by types Z and F; for $n = 2$ both obstruct, with equality, which is why $[K : \mathbb{Q}] \geq 4$. Condition (2): $5n + 2 \geq 3n + 2$. □

**Lemma 6.4.** *In the proofs above, if $\ell$ is a $K$-rational line different from $\mathbb{C}(0,1)$, then $\dim_K \mathcal{N} \leq 2$.*

*Proof.* Otherwise $v_1, v_2 \in \ell + \Omega_\Psi$. If $\ell = \mathbb{C}(1,0)$: $sy_j \in K2\pi i + K\omega_0$ ($\omega_0 = \Omega$, resp. $\eta$), so $y_j \in K \cdot 2\pi i + K \cdot \pi^2/\omega_0$ (as $ic \in K$). If $\ell = \mathbb{C}(p,1)$, $p \in K$: $y_j = \lambda_j p + 2\pi i a_j$ and $sy_j = \lambda_j + 2\pi i b_j + \omega_0 c_j$, so $y_j(1 - ps) = 2\pi i(a_j - pb_j) - p\omega_0 c_j$; since $\omega_0 = 2\pi i\, s/(2ic) \in K \cdot 2\pi i s$ (the period datum), $y_j \in 2\pi i(K + Ks)/(1 - ps)$, where $1 - ps \neq 0$ because $s \notin K$. In both cases $y_1, y_2$ lie in a 2-dimensional $K$-space containing $2\pi i$, against the $K$-independence of $2\pi i, y_1, y_2$. Here $s \notin K$: for (a), $s \in \overline{\mathbb{Q}}$ would give $\Omega \in \overline{\mathbb{Q}} \cdot 2\pi i \subset \widetilde{\mathcal{L}}$, against Lemma 6.1; for (b), $s \in \overline{\mathbb{Q}}^\times/\Omega$ is transcendental (Schneider). □

*Proof of Theorem 1.6.* The statement is invariant under $t \mapsto \gamma t$ ($\gamma \in \overline{\mathbb{Q}}^\times$) and under $t \mapsto 1/t$ (replace $y_j$ by $ty_j$), and $1/(c\Omega) \in \overline{\mathbb{Q}}^\times \eta/\pi$; so $t = c\Omega/\pi$ and $t = c\eta/\pi$ suffice. Let $y_1, y_2 \in \widetilde{\mathcal{L}}$ be $\overline{\mathbb{Q}}$-independent with $ty_1, ty_2 \in \widetilde{\mathcal{L}}$. If $2\pi i \in \overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$, then $t \cdot 2\pi i \in \widetilde{\mathcal{L}}$, i.e. $\Omega \in \widetilde{\mathcal{L}}$, resp. $\eta \in \widetilde{\mathcal{L}}$, contradicting Lemma 6.1. Otherwise $2\pi i, y_1, y_2$ are $\overline{\mathbb{Q}}$-independent and Theorem 6.2 applies, since $\widetilde{\mathcal{L}} \subset \widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$. □

**Corollary 6.5.** (i) *$\varpi \log 2$ and $\varpi \log 3$ are not both in $\widetilde{\mathcal{L}}$; $(2e)^\varpi$ and $3^\varpi$ are not both algebraic; $\varpi/\pi$ and $(\varpi/\pi)\log 2$ are not both in $\widetilde{\mathcal{L}}$ (so $e^{\varpi/\pi}$ and $2^{\varpi/\pi}$ are not both algebraic); $e^{1/\varpi}$ and $2^{1/\varpi}$ are not both algebraic. The same holds with $\Omega_\rho = 2^{-4/3}\Gamma(1/3)^3/\pi$ or any CM period in place of $\varpi$.*

(ii) *For every CM period $\Omega$, one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\Omega/\pi}$ is transcendental, and one of $e^{\pi^2}$, $e^{\pi^2/\Omega}$, $e^{1/\Omega}$ is transcendental. More generally, $\pi^2, \pi\Omega, \Omega/\pi$ are not all in $\widetilde{\mathcal{L}}$, and neither are $\pi^2, \pi^2/\Omega, 1/\Omega$.*

(iii) (CM analogue of [42, Cor. 1.3]) *Let $\Lambda_1 \in \overline{\mathbb{Q}}^\times\{\Omega^{\pm 1}, (\Omega/\pi)^{\pm 1}\}$ and let $\Lambda_2, \Lambda_3 \in \widetilde{\mathcal{L}}$ be linearly independent over $\overline{\mathbb{Q}}$. Then $\Lambda_1\Lambda_2$ or $\Lambda_1\Lambda_3$ is not in $\widetilde{\mathcal{L}}$. ($\Lambda_1$ need not lie in $\widetilde{\mathcal{L}}$: $\Lambda_1 \notin \widetilde{\mathcal{L}}$ is proved for $\overline{\mathbb{Q}}^\times\Omega$ and $\overline{\mathbb{Q}}^\times\pi/\Omega$, and open for the other two classes.)*

(iv) (Mixed forms, only at the stated slopes.) *For $s \in \overline{\mathbb{Q}}^\times\Omega/\pi$ and $y_1, y_2 \in \widetilde{\mathcal{L}}$ with $2\pi i, y_1, y_2$ $\overline{\mathbb{Q}}$-independent, $sy_1, sy_2$ are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$; for $s \in \overline{\mathbb{Q}}^\times/\Omega$, not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$.*

*E.g.* $\varpi\log 2/\pi$ *and* $\varpi\log 3/\pi$ *are not both in* $\overline{\mathbb{Q}}+\overline{\mathbb{Q}}\mathcal{L}+\overline{\mathbb{Q}}\mathcal{L}_{E_i}$. *These do not pass to* $1/s$, *since inversion swaps the rows.*

*Status:* [Proved modulo published theorems and Lemma A.1]. (For $\Omega\in\overline{\mathbb{Q}}^{\times}\varpi$, items (i)–(iii) follow in one line from Theorem 1.6 and Theorem 6.2, whose status is stated there, and Baker's theorem; these one-line deductions are not formalised.)

*Proof.* (i) Theorem 1.6 with $(y_1,y_2)=(\log 2,\log 3)$ and $(1+\log 2,\log 3)$ at $t=\varpi$ ($\overline{\mathbb{Q}}$-independent by Baker), and $(1,\log 2)$ at $t=\varpi/\pi$ and at $t=1/\varpi=\eta(\varpi)/\pi$. (ii) Theorem 6.2 with $(y_1,y_2)=(1,\pi^2)$ and $s=\Omega/\pi$, resp. $s=1/\Omega\in\overline{\mathbb{Q}}^{\times}\eta/\pi$; $2\pi i,1,\pi^2$ are $\overline{\mathbb{Q}}$-independent since $\pi$ is transcendental. If $\pi^2\notin\widetilde{\mathcal{L}}$ the statements are trivial. (iii) is Theorem 1.6 with $t=\Lambda_1$, $y=(\Lambda_2,\Lambda_3)$. (iv) is Theorem 6.2; for the example, $2\pi i,\log 2,\log 3$ are $\overline{\mathbb{Q}}$-independent by Baker. □

**Remark 6.6.** (1) The first triple in (ii) and the statement for $e^{\varpi/\pi}$, $2^{\varpi/\pi}$ in (i) are also Corollary 9.7(i), (ii), which §9.4 derives from the count only (proof sketch); the second triple in (ii) is not covered by Corollary 9.7 or by the elliptic results of §5. (2) Rescaling by $x_1$: if $x_1,x_2$ and $y_1,y_2$ are $\overline{\mathbb{Q}}$-independent pairs with $x_2/x_1\in\overline{\mathbb{Q}}^{\times}\{\Omega^{\pm1},(\Omega/\pi)^{\pm1}\}$, then one of $x_iy_j$ is not in $\widetilde{\mathcal{L}}$; this is a strong five exponentials statement with the fifth number $x_1/x_2$ replaced by the CM-ratio hypothesis. (3) The sharp form (no algebraic $\beta_{ij}$ with $x_iy_j-\beta_{ij}\in\mathcal{L}$ for all four pairs) follows and is strictly weaker. (4) The general $2\times 3$ minors statement of [42, Thm. 1.2] at CM data is out of reach of this count: the weight per coordinate is too large (internal note, proof sketch); (iii) is the only specialisation we obtain. (5) The same method gives strong *elliptic* four exponentials on a CM curve $E$, with $\widetilde{\mathcal{L}}_E=\overline{\mathbb{Q}}+\overline{\mathbb{Q}}\mathcal{L}_E$ in place of $\widetilde{\mathcal{L}}$, at $t\in\overline{\mathbb{Q}}^{\times}(\Omega'/\Omega)^{\pm1}$ and $\overline{\mathbb{Q}}^{\times}(\pi/(\Omega\Omega'))^{\pm1}$ ($\Omega'$ a period of a CM curve not isogenous to $E$) and at $t\in\overline{\mathbb{Q}}^{\times}(\pi/\Omega^2)^{\pm1}$; this is now written out and proved in §6.3 (Theorem 6.12, Corollary 6.13), with the eigen-slot generalisation Theorem 6.16.

**Remark 6.7** (Strong six does not give Theorem 1.6)**.** Each inversion pair of slope classes contains one class with $t\notin\widetilde{\mathcal{L}}$ proved ($\overline{\mathbb{Q}}^{\times}\Omega$, $\overline{\mathbb{Q}}^{\times}\pi/\Omega$) and one where membership is open ($\overline{\mathbb{Q}}^{\times}\Omega/\pi$, $\overline{\mathbb{Q}}^{\times}/\Omega$). Theorem 1.6 is not known to follow from the strong six exponentials theorem; it would follow partially (via [42, Cor. 1.3], for pairs with $1,y_1,y_2$ $\overline{\mathbb{Q}}$-independent) only if $\Omega/\pi$ or $1/\Omega$ lay in $\widetilde{\mathcal{L}}$, which is open and expected to be false. Apart from the conjugation overlap described in §11.2, we found no classical few-line proof; it follows in a few lines from Theorem 1.1, through Theorem 6.2 (§4.4, Remark 11.1).

6.2. **CM abelian eigen-periods: Theorem S-A.** Notation is as in §4.1: $A$ is simple with CM by $F$ of primitive type $\Phi$, $z_\sigma$ ($\sigma\in\Phi$) and $w_\sigma$ ($\sigma\notin\Phi$) are the eigen-coordinates on $\mathrm{Lie}(A)$, resp. $\mathrm{Lie}(A^\natural)$, and $P_\sigma\neq 0$ is a period of the slot $(A,\sigma)$. By the reduction to the simple case in §4.1 the following theorem holds for every CM abelian variety over $\overline{\mathbb{Q}}$.

**Theorem 6.8** (Theorem S-A)**.** *Let* $P\neq 0$ *be an eigen-period, of the first or of the second kind, of a CM abelian variety over* $\overline{\mathbb{Q}}$, *and let* $t\in\overline{\mathbb{Q}}^{\times}P\cup\overline{\mathbb{Q}}^{\times}P^{-1}\cup\overline{\mathbb{Q}}^{\times}P/\pi\cup\overline{\mathbb{Q}}^{\times}\pi/P$. *If* $y_1,y_2\in\mathbb{C}$ *are linearly independent over* $\overline{\mathbb{Q}}$, *then* $y_1,y_2,ty_1,ty_2$ *are not all in* $\widetilde{\mathcal{L}}$. *In particular this holds for* $t\in\overline{\mathbb{Q}}^{\times}B(a/N,b/N)^{\pm1}$ *and* $t\in\overline{\mathbb{Q}}^{\times}(B(a/N,b/N)/\pi)^{\pm1}$ ($0<a,b<N$, $a+b\neq N$).

*Status:* [Proved modulo published theorems and Lemma A.1] for $g\geq 2$ (the second-kind slots, and with them (F1)–(F4), enter for every $P$, through $t\leftrightarrow 1/t$); for $g=1$ it is Theorem 1.6, with the status stated there. Two internal referees (one an independent reconstruction with explicit $U$ and $Q''$ for $g=1,2,3$): correct, the corollaries with fixes (applied).
Theorem 1.6 is the case $g=1$. Theorem 6.8 follows from the following affine versions of Theorems 5.7 and 5.9.

**Theorem 6.9** (Theorems SMG-A and SMG$^\natural$-A)**.** *Let $c \in \overline{\mathbb{Q}}^\times$ and $s = cP_\sigma/\pi$. Let $y_1, y_2 \in \widetilde{\mathcal{L}}$ with $2\pi i, y_1, y_2$ linearly independent over $\overline{\mathbb{Q}}$. Then $sy_1$ and $sy_2$ are not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}} z_\sigma(\Lambda(A))$ if $\sigma \in \Phi$ (SMG-A), and not both in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}} w_\sigma(\Lambda(A^\natural))$ if $\sigma \notin \Phi$ (SMG$^\natural$-A).*

*Status:* [Proved modulo published theorems and Lemma A.1].

**Lemma 6.10.** *$P_\sigma \notin \widetilde{\mathcal{L}}$ for every eigen-period $P_\sigma \neq 0$.*

*Proof.* For $\sigma \in \Phi$, a relation $P_\sigma = \beta_0 + \sum \beta_k \log \alpha_k$ puts the vector $(1, \log\alpha, \gamma)$ of $\mathrm{Lie}(\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times A)$ ($z_\sigma(\gamma) = P_\sigma$, $\gamma \in \Omega(A)$) in a $\overline{\mathbb{Q}}$-hyperplane. By Wüstholz's analytic subgroup theorem [52] the smallest algebraic subgroup through it, which is $H \times T \times A$ since $A$ is simple and $\gamma \neq 0$, has its tangent space in that hyperplane; but $z_\sigma$ does not vanish on $\mathrm{Lie}(A)$. For $\sigma \notin \Phi$ argue on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m}^r \times A^\natural$: the torus splits off, and the image in $A^\natural$ of the smallest subgroup maps onto $A$, hence is $A^\natural$ by (F3) (here $W_A$ is the whole fibre); the smallest subgroup is not the graph of a homomorphism $A^\natural \to \mathbb{G}_\mathrm{a}$, since $A^\natural$ is anti-affine and the $\mathbb{G}_\mathrm{a}$-coordinate is 1; so it contains $0 \times A^\natural$, on whose fibre $w_\sigma \neq 0$. □

*Proof of Theorem 6.8.* By §4.1 we may assume $A$ simple of primitive type. Strong four exponentials at $t$ is equivalent to the same statement at $1/t$ (replace $y_j$ by $ty_j$) and is invariant under $t \mapsto \gamma t$, $\gamma \in \overline{\mathbb{Q}}^\times$. By the bilinear relation $P_\sigma P_{\bar\sigma} \in \overline{\mathbb{Q}}^\times \pi$, for a first-kind $Q = P_\sigma$ and $S = P_{\bar\sigma}$ the classes $Q^{\pm1}, (Q/\pi)^{\pm1}$ are the classes $S^{\mp1}, (S/\pi)^{\mp1}$; so it suffices to treat $t = cP_\sigma/\pi$ for slots of both kinds. If $y_1, y_2 \in \widetilde{\mathcal{L}}$ are independent with $ty_1, ty_2 \in \widetilde{\mathcal{L}}$ and $2\pi i \in \overline{\mathbb{Q}} y_1 + \overline{\mathbb{Q}} y_2$, then $2icP_\sigma = t \cdot 2\pi i \in \widetilde{\mathcal{L}}$, against Lemma 6.10. Otherwise Theorem 6.9 applies. For the Beta values use §4.1. □

*Proof of Theorem 6.9.* This is the proof of Theorem 6.2(a) (resp. (b)) with $E$ replaced by $A$ read at $z_\sigma$ (resp. $w_\sigma$). Let $K$ be a number field containing the Galois closure of $F$, $i$, $c$ and all coefficients, of degree $n \geq 4$. For $\sigma \in \Phi$ take $G = \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times A$, $\Psi = (a_1 + x,\ a_2 + y + z_\sigma)$, $V = \mathbb{C}(1, s)$. By Lemma 4.1 the largest connected subgroup of $G^n$ in $\ker \Psi \circ p_\theta$ is $\{\theta \cdot a_1 = \theta \cdot a_2 = 0\} \times 1 \times B_{\ker \psi_\sigma}$, and $G'' = \mathbb{G}_\mathrm{a}^2 \times \mathbb{G}_\mathrm{m}^{2n} \times A''$ with $\dim A'' = n/2$ for every $g$: these are exactly the numbers $d_0 = 2$, $d_1 = 2n$, $d_2 = n/2$ of the elliptic case. Every abelian subvariety $B \neq 0$ of $A''$ has $\Psi''(T_B) = \mathbb{C}(0, 1)$ and $\dim B \leq n/2$. The period datum is $w = (2\pi i, 2icP_\sigma)$, and $P_\sigma = 2\pi i\, s/(2ic) \in K \cdot 2\pi i\, s$, which is all that the line-subtorus step (Lemma 6.4) uses; $s \notin K$ by Lemma 6.10. The case analysis of Theorem 6.2(a) therefore applies verbatim: no connected subgroup satisfies (1), with margin $n$ (also checked by exact enumeration). For $\sigma \notin \Phi$ take $G = \mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times \mathbb{G}_\mathrm{m} \times A^\natural$, $\Psi = (a + x,\ y + w_\sigma)$; by (F1)–(F4) the $A^\natural$-part of $G''$ is an anti-affine extension $Q''$ of $A''$ ($\dim n/2$) by its one-dimensional fibre, with no splitting over any $B \neq 0$, and $\mathrm{Hom}(Q''_B, \mathbb{G}_\mathrm{a}) = 0$, as in the proof of Theorem 5.9. These are the only properties used in Lemma 6.3, the subgroup lemma for $\mathbb{G}_\mathrm{a} \times Q''$ in the proof of Theorem 6.2(b), whose case table then applies verbatim (margin $n - 2$, attained at types Z and F, $n \geq 4$). Condition (2) reads $5n + 2 \geq 3n + 4$, resp. $5n + 2 \geq 3n + 2$. □

**Corollary 6.11.**
(i) *$(2e)^t$ and $3^t$ are not both algebraic for $t = \Gamma(1/5)^2/\Gamma(2/5)$; $e^t$ and $2^t$ are not both algebraic for $t = 1/B(\frac{1}{7}, \frac{1}{7})$ and for $t = B(\frac{1}{5}, \frac{1}{5})/\pi$; $t \log 2$ and $t \log 3$ are not both in $\widetilde{\mathcal{L}}$ for every $t \in \overline{\mathbb{Q}}^\times B(a/N, b/N)^{\pm1}$ with $a + b \neq N$.*

(ii) *For every first-kind eigen-period $P$, at least one of $e^{\pi^2}$, $e^{\pi P}$, $e^{P/\pi}$ is transcendental, and at least one of $e^{\pi^2}$, $e^{\pi^2/P}$, $e^{1/P}$ is transcendental. More strongly, $\pi^2, \pi P, P/\pi$ are not all in $\widetilde{\mathcal{L}}$, and $\pi^2, \pi^2/P, 1/P$ are not all in $\widetilde{\mathcal{L}}$.*

*Status:* [Proved modulo published theorems and Lemma A.1] (as Theorem 6.8; the first triple in (ii) uses only SMG-A).

*Proof.* (i) Theorem 6.8 with $y = (1+\log 2, \log 3)$, resp. $y = (1, \log 2)$, resp. $y = (\log 2, \log 3)$ ($\overline{\mathbb{Q}}$-independent by Baker). (ii) Theorem 6.9 with $y = (1, \pi^2)$ ($2\pi i, 1, \pi^2$ are $\overline{\mathbb{Q}}$-independent since $\pi$ is transcendental; if $\pi^2 \notin \widetilde{\mathcal{L}}$ there is nothing to prove), at $s = cP/\pi$ ($\sigma \in \Phi$) and at $s = cP_{\bar\sigma}/\pi \in \overline{\mathbb{Q}}^\times/P$ ($\bar\sigma \notin \Phi$). □

A pair $y = (1, u)$ has content only at $t \in \overline{\mathbb{Q}}^\times P/\pi$ or $\overline{\mathbb{Q}}^\times/P$: at $t \in \overline{\mathbb{Q}}^\times P$ and $\overline{\mathbb{Q}}^\times \pi/P$ one already has $t \notin \widetilde{\mathcal{L}}$ (Lemma 6.10, applied to $P$ and to $P_{\bar\sigma}$). This is why (i) uses $t = 1/B(\frac{1}{7}, \frac{1}{7})$, not $B(\frac{1}{7}, \frac{1}{7})$. Whether a given Beta slope lies outside the elliptic classes of Theorem 1.6 is decided as in Remark 5.10 ([Heuristic] for the quadratic coincidences). The overlap with complex conjugation is discussed in §11.6.

6.3. **Elliptic and eigen-slot analogues.** In this subsection the strong four exponentials statement is read on a CM curve or on one eigen-coordinate of a CM abelian variety instead of on $\mathbb{G}_{\mathrm{m}}$. Let $E, E'$ be elliptic curves over $\overline{\mathbb{Q}}$ with complex multiplication by $k, k'$, $\Omega, \Omega'$ nonzero periods, $\eta = \eta^*(\Omega)$ and $\eta' = \eta^*(\Omega')$ the conjugate-eigen quasi-periods ($\Omega\eta, \Omega'\eta' \in \overline{\mathbb{Q}}^\times \pi$). Write $\widetilde{\mathcal{L}}_E = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$, and $\mathrm{Fib}_E$, $\mathrm{Fib}_{E'}$ for the sets of fibre values (§4.2; $\mathrm{Fib}_E$ contains $\overline{\mathbb{Q}}$, the values $\zeta^*(u)$, $u \in \mathcal{L}_E$, and $\eta$). Two CM curves are isogenous iff $k = k'$.

**Theorem 6.12** (Theorems SE, SE$^\natural$, SE$_0$)**.** *Let $c \in \overline{\mathbb{Q}}^\times$ and $y_1, y_2 \in \widetilde{\mathcal{L}}_E$ with $\Omega, y_1, y_2$ linearly independent over $\overline{\mathbb{Q}}$.*

1. (SE) *If $k \neq k'$ and $t = c\Omega'/\Omega$, then $ty_1, ty_2$ are not both in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\mathcal{L}_{E'}$.*
2. (SE$^\natural$) *If $k \neq k'$ and $t = c\eta'/\Omega$, then $ty_1, ty_2$ are not both in $\overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\mathrm{Fib}_{E'}$.*
3. (SE$_0$) *If $t = c\eta/\Omega \in \overline{\mathbb{Q}}^\times \pi/\Omega^2$, then $ty_1, ty_2$ are not both in $\overline{\mathbb{Q}}\mathcal{L}_E + \overline{\mathbb{Q}}\mathrm{Fib}_E$.*

*Status:* (1) for $(E, E') = (E_i, E_\rho)$ (every $c \in \overline{\mathbb{Q}}^\times$ and every pair of periods): [Proved modulo the algebraicity of $g_3(\Omega_\rho \mathbb{Z}[\rho])$; formally verified otherwise] (§12.3); (3) for $E = E_i$ (every $c \in \overline{\mathbb{Q}}^\times$ and every nonzero period of $E_i$): [Proved; formally verified, standard axioms]; (3) for $E = E_\rho$: [Proved modulo the algebraicity of $g_3(\Omega_\rho \mathbb{Z}[\rho])$; formally verified otherwise]; (1) for other pairs, (2), and (3) for other CM curves: [Proved modulo published theorems and Lemma A.1] (for other pairs, (1) is checked in Lean conditionally on complex multiplication preserving elliptic logarithms, §12.3). Internal referees: two for (1) and (2) (correct with fixes, applied), two for (3) (correct; one an independent reconstruction with a falsity hunt).

**Corollary 6.13** (Strong elliptic four exponentials)**.** *Let $t \in \overline{\mathbb{Q}}^\times (\pi/\Omega^2)^{\pm 1}$, or, if $k' \neq k$, $t \in \overline{\mathbb{Q}}^\times (\Omega'/\Omega)^{\pm 1} \cup \overline{\mathbb{Q}}^\times (\pi/\Omega\Omega')^{\pm 1}$. If $y_1, y_2 \in \widetilde{\mathcal{L}}_E$ are linearly independent over $\overline{\mathbb{Q}}$, then $ty_1$ and $ty_2$ are not both in $\widetilde{\mathcal{L}}_E$. In particular, if $u_1, u_2 \in \mathcal{L}_E$ are linearly independent over $k$, then $\wp_E(tu_1)$ and $\wp_E(tu_2)$ are not both algebraic.*

*Status:* [Proved modulo published theorems and Lemma A.1] (on $E_i$ at $t \in \overline{\mathbb{Q}}^\times \pi/\varpi^2$ the case $\varpi \notin \overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$ is checked in Lean as a consequence of (3), §12.3).

**Lemma 6.14.** *$\Omega' \notin \widetilde{\mathcal{L}}_E$ and $\eta' \notin \widetilde{\mathcal{L}}_E$ if $k' \neq k$; $\eta \notin \widetilde{\mathcal{L}}_E$; in particular $t \notin \overline{\mathbb{Q}}$ in all cases of Theorem 6.12.*

*Proof.* [Lit: Wüstholz [52]] If $\Omega' = \beta_0 + \sum \beta_i \ell_i$ with $\ell_i \in \mathcal{L}_E$, the vector $(1, \ell, \Omega')$ of $\mathrm{Lie}(\mathbb{G}_{\mathrm{a}} \times E^r \times E')$ lies in a $\overline{\mathbb{Q}}$-hyperplane; the smallest algebraic subgroup through it is $H \times B \times E'$ ($\mathrm{Hom}(E, E') = 0$), whose tangent space is not in the hyperplane. For $\eta'$ use $\mathbb{G}_{\mathrm{a}} \times E^r \times E'^\natural$: the smallest subgroup contains $0 \times 0 \times E'^\natural$ (its image in $\mathbb{G}_{\mathrm{a}} \times E'^\natural$ maps onto $E'$ and has $\mathbb{G}_{\mathrm{a}}$-coordinate 1, so it is $\mathbb{G}_{\mathrm{a}} \times E'^\natural$; the kernel part cannot mix since $\mathrm{Hom}(E', E) = 0$ and $\mathrm{Hom}(\mathbb{G}_{\mathrm{a}}, E^r) = 0$), and the fibre direction is not in the hyperplane. For $\eta$ use $\mathbb{G}_{\mathrm{a}} \times E^r \times E^\natural$, a push-out of $\mathbb{G}_{\mathrm{a}} \times (E^\natural)^{r+1}$:

the smallest subgroup contains the fibre vectors of $B_S^\natural$ for its abelian image $B_S$, whose last projection is nonzero. □

*Proof of Corollary 6.13.* The statement is invariant under $t \mapsto \gamma t$ ($\gamma \in \overline{\mathbb{Q}}^\times$) and under $t \mapsto 1/t$ (replace $y_j$ by $ty_j$), and $\overline{\mathbb{Q}}^\times \pi/\Omega\Omega' = \overline{\mathbb{Q}}^\times \eta'/\Omega$, $\overline{\mathbb{Q}}^\times \pi/\Omega^2 = \overline{\mathbb{Q}}^\times \eta/\Omega$. If $\Omega \in \overline{\mathbb{Q}} y_1 + \overline{\mathbb{Q}} y_2$ then $t\Omega \in \widetilde{\mathcal{L}}_E$, i.e. $\Omega'$, $\eta'$ or $\eta$ lies in $\widetilde{\mathcal{L}}_E$; this is excluded by Lemma 6.14. So Theorem 6.12 applies, since $\widetilde{\mathcal{L}}_E$ lies in each of its targets ($\overline{\mathbb{Q}} \subset \mathrm{Fib}_{E'}$, $\mathrm{Fib}_E$). For the last sentence: $\wp(z) \in \overline{\mathbb{Q}} \cup \{\infty\}$ means $z \in \mathcal{L}_E$, and $k$-independent elliptic logarithms of a CM curve are $\overline{\mathbb{Q}}$-independent (Masser [23]). □

*Proof of Theorem 6.12.* Let $K \supseteq kk'(c)$ (for (3): $K \supseteq k(c)$) contain all coefficients, with $n = [K : \mathbb{Q}] \geq 4$ (automatic if $k \neq k'$; for (3) enlarge $K$ if necessary, since at $n = 2$ the count fails, with equality, at $G' = 0$ and at image $\mathbb{C}(0,1)$). Let $\theta$ be a $\mathbb{Q}$-basis of $K$, and $\psi(m) = \sum \theta_j m_j$, $\phi(m) = \sum \theta_j \bar{m}_j$ on $k^n$, with kernels of $k$-dimension $n/2$ (Lemma 4.2). By Lemma 6.14, $t \notin K$.

(1) $G = \mathbb{G}_\mathrm{a} \times E \times \mathbb{G}_\mathrm{a} \times E \times E'$, $\Psi = (a_1 + z_1,\, a_2 + z_2 + z')$, $V = \mathbb{C}(1,t)$. Then $G'' = \mathbb{G}_\mathrm{a}^2 \times A_1 \times A_2 \times A'$ with each abelian factor of dimension $n/2$ ($d_0 = 2$, $d_1 = 0$, $d_2 = 3n/2$); the data are $w = (\Omega, c\Omega')$ and $v_j = (y_j, ty_j)$, so $m = 3n$ and $\kappa = n$ (a period in $Y''$ has first coordinate in $K\Omega$). Connected subgroups are $H \times B_{12} \times B'$ ($\mathrm{Hom}(E, E') = 0$); for $B_{12} \subset A_1 \times A_2$ the image $\Psi''(T_{B_{12}})$ is $K$-rational, and $\dim B_{12} \leq n/2$ when it is a line. For $r' = \dim \Psi''(T_{G'}) = 1$, (1) reads $\mathrm{rk}\, Y_{G'} + 2 \dim B \geq 4n$ ($B$ the abelian part of $G'$), and for $G' = 0$ it reads $\kappa \leq 0$, excluded with margin $n$. Image $\mathbb{C}(0,1)$ gives $\mathrm{rk}\, Y_{G'} \leq n$ and $\dim B \leq n$; a non-$K$-rational line from $H$ gives $\mathrm{rk}\, Y_{G'} \leq 3n$, $B = 0$; a $K$-rational line $\ell \neq \mathbb{C}(0,1)$ gives $\dim B \leq n/2$ and $\mathrm{rk}\, Y_{G'} \leq 2n$, because $v_j \in \ell + \Psi(\Omega_K(G))$ for $j = 1, 2$ would give $y_j \in K\Omega + K\Omega^2/\Omega'$ if $\ell = \mathbb{C}(1,0)$, and $y_j(1 - pt) \in K\Omega + K\Omega' = \Omega(K + Kt)$ if $\ell = \mathbb{C}(p,1)$ (the period datum is $\Omega' = t\Omega/c$, and $1 - pt \neq 0$ as $t \notin K$); either way $\Omega, y_1, y_2$ would be $K$-dependent. No subgroup satisfies (1); the minimum margin is $n$. Condition (2): $3n + 2 \geq 4 + n$.

(2), (3) $G = \mathbb{G}_\mathrm{a} \times E \times E \times X^\natural$ with $X = E'$, resp. $X = E$, $\Psi = (a + z_1,\, z_2 + w)$, $w$ the conjugate-eigen fibre coordinate of $X^\natural$ (its base coordinate is not read), $V = \mathbb{C}(1,t)$. For $X = E$ all factors are isogenous to powers of $E$, so $G^n$ is a push-out of $\mathbb{G}_\mathrm{a}^n \times (E^\natural)^{3n}$, and its connected subgroups are described by an abelian image $B_S$ ($S \subset k^{3n}$) and a vector part containing the fibre vectors of $B_S^\natural$; since base and fibre parameters of $\mathrm{Lie}(B_S^\natural) = \{(Mu, \bar{M}u')\}$ are independent, the largest subgroup in $\ker \Psi \circ p_\theta$ is, in both cases, the product $\{\theta \cdot a = 0\} \times B_{\ker \psi} \times B_{\ker \psi} \times (B_{\ker \phi}^\natural + \{\theta^\mathsf{T} w = 0\})$ (for $X = E'$, with $\phi$ formed with $k'$ on the last factor). So $G'' = \mathbb{G}_\mathrm{a} \times A_1 \times A_2 \times Q$, $\dim A_i = n/2$, with $Q$ an anti-affine extension of an abelian variety $A_3$ of dimension $n/2$ by its fibre, with no splitting over any abelian subvariety $\neq 0$ ($d_0 = 1$, $d_1 = 0$, $d_2 = 3n/2+1$). The data are $w = (\Omega, c\eta')$, resp. $(\Omega, c\eta)$, and $v_j = (y_j, ty_j)$: $m = 3n$, $\kappa = n$ (as in (1)). A connected $G' \subset G''$ is determined by its abelian image and its vector part in $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{a}^\mathrm{fib}$, and contains the fibre of $Q$ as soon as its abelian image projects nontrivially to $A_3$ (the pull-back of a non-split extension along a surjection stays non-split; this also covers the mixed subgroups between the three $E$-factors when $X = E$). For $r' = 1$, (1) reads $\mathrm{rk}\, Y_{G'} + 2 \dim G' + 2\delta_0 \geq 4n + 2$, and for $G' = 0$ it reads $\kappa \leq 2$. With $r' = 1$ the image of $G'$ is: $\mathbb{C}(0,1)$ ($\mathrm{rk}\, Y_{G'} \leq n$, $\dim G' \leq n + 1$, $\delta_0 = 1$: total $\leq 3n + 4$); or a $K$-rational line coming from $A_1 \times A_2$, possibly with a line of $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{a}^\mathrm{fib}$ ($\dim G' \leq n/2 + 1$, $\mathrm{rk}\, Y_{G'} \leq 2n$ by the period datum $\eta' = t\Omega/c$, resp. $\eta = t\Omega/c$, and $t \notin K$, exactly as in (1): total $\leq 3n + 2$); or a line of $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{a}^\mathrm{fib}$ with trivial abelian image ($\mathrm{rk}\, Y_{G'} \leq 3n$, $\dim G' = 1$: total $\leq 3n + 2$). So no subgroup satisfies (1) for $n \geq 4$; the tight cases are $G' = 0$ and the subgroups with image $\mathbb{C}(0,1)$ containing $A_2$ and all of $Q$, both with margin $n - 2$. Condition (2): $3n + 2 \geq 2 + n$. (All case checks were also rerun by exact enumeration.) □

**Example 6.15.** On $E_i\colon y^2 = 4x^3 - 4x$ (period $\varpi$, $\eta = \pi/\varpi$): for every $u \in \mathcal{L}_{E_i} \setminus \{0\}$, e.g. the logarithm of $(2, \sqrt{24})$, the numbers $\wp_i(\pi/\varpi^2)$ and $\wp_i(\pi u/\varpi^2)$ are not both algebraic, nor are $\wp_i(\varpi^2/\pi)$ and $\wp_i(\varpi^2 u/\pi)$ (Corollary 6.13 in the class $(\pi/\Omega^2)^{\pm 1}$, i.e. from Theorem 6.12(3), with $(y_1, y_2) = (1, u)$; $1, u$ are $\overline{\mathbb{Q}}$-independent since $u \neq 0$ is transcendental). Also $\wp_i(\varpi^2 u_1/\pi)$ and $\wp_i(\varpi^2 u_2/\pi)$ are not both algebraic for $\mathbb{Q}(i)$-independent $u_1, u_2 \in \mathcal{L}_{E_i}$. With $E_\rho\colon y^2 = 4x^3 - 4$ (period $\Omega_\rho$), the class $\Omega'/\Omega$ gives that $\wp_i(\Omega_\rho/\varpi)$ and $\wp_i(\Omega_\rho u/\varpi)$ are not both algebraic; this is a consequence of Theorem 6.12(1).

The content of (3) sits in the $\mathcal{L}_E$ part of the second row: with target $\overline{\mathbb{Q}}\mathrm{Fib}_E$ alone the statement would be linearisable by the third-kind identity (6) ([Heuristic]).
*Provenance.* (1)–(3) are consequences of Theorem 1.1 (§4.4: $\alpha_{K,\Psi} - \kappa = 2$ for (1) and $2 + 2/n$ for (2), (3), against three data), and so is Corollary 6.13 in all its classes. For (1) the printed weight already suffices ($\alpha_K(\mathbb{G}_\mathrm{a} \times E \times \mathbb{G}_\mathrm{a} \times E \times E') = 3$, $\kappa = 1$), and, by the splitting (5), also for (2) and (3) on merged groups such as $\mathbb{G}_\mathrm{a} \times E \times E^\natural$ (Remark 11.1). We found no explicit statement of them in print; the proof above applies Theorem 2.3 directly.
*Eigen-slots.* Let $A/\overline{\mathbb{Q}}$ be simple with CM by $F$ and primitive type $\Phi$, $\sigma \in \Phi$, $P_\sigma \neq 0$ a period of $z_\sigma$, and $\widetilde{\mathcal{L}}_\sigma = \overline{\mathbb{Q}} + \overline{\mathbb{Q}} z_\sigma(\Lambda(A))$.

**Theorem 6.16** (Theorems EA, EA$^\natural$)**.** *Let $(A', \sigma')$ be another first-kind slot ($A' = A$, $\sigma' \neq \sigma$ allowed), resp. $(A', \tilde\sigma)$ a second-kind slot ($\tilde\sigma \notin \Phi'$, read on $A'^\natural$ through $w_{\tilde\sigma}$; $A' = A$ allowed), where an $A'$ isogenous to $A$ is replaced by $A$ (this rescales $P'$ and the target by $\overline{\mathbb{Q}}^\times$; then either $A' = A$ or $\mathrm{Hom}(A, A') = 0$, and we require $\sigma' \neq \sigma$ under the induced identification of the $F$-actions), with period $P'$, and $t = cP'/P_\sigma$, $c \in \overline{\mathbb{Q}}^\times$. If $y_1, y_2 \in \widetilde{\mathcal{L}}_\sigma$ and $P_\sigma, y_1, y_2$ are linearly independent over $\overline{\mathbb{Q}}$, then $ty_1, ty_2$ are not both in $\widetilde{\mathcal{L}}_\sigma + \overline{\mathbb{Q}} z_{\sigma'}(\Lambda(A'))$, resp. in $\overline{\mathbb{Q}} z_\sigma(\Lambda(A)) + \overline{\mathbb{Q}} w_{\tilde\sigma}(\Lambda(A'^\natural))$. Consequently, for $t \in \overline{\mathbb{Q}}^\times (P'/P_\sigma)^{\pm 1}$ and $\overline{\mathbb{Q}}$-independent $y_1, y_2 \in \widetilde{\mathcal{L}}_\sigma$, $ty_1, ty_2$ are not both in $\widetilde{\mathcal{L}}_\sigma$.*

*Status:* [Proved modulo published theorems and Lemma A.1]. Two internal referees (one an independent reconstruction with a falsity hunt): correct with fixes, applied; the one statement-level fix is the slot convention (EA is false as first stated when $A'$ is isogenous to $A$ but $A' \neq A$, since then $t$ can be algebraic).

*Proof.* Take $K$ containing the Galois closures of $F$, $F'$, and $c$ and all coefficients, of degree $n$ divisible by $2g$ and $2g'$, and for EA$^\natural$ also $n \geq 4$ (not automatic when $g = 1$, $A' = E$). Argue as in Theorem 6.12(1) (resp. (3)) with $G = \mathbb{G}_\mathrm{a} \times A \times \mathbb{G}_\mathrm{a} \times A \times A'$, $\Psi = (a_1 + z_\sigma,\, a_2 + z_\sigma + z_{\sigma'})$ (resp. $G = \mathbb{G}_\mathrm{a} \times A \times A \times A'^\natural$, $\Psi = (a + z_\sigma,\, z_\sigma + w_{\tilde\sigma})$). On $\mathrm{Lie}(B_S)$ the coordinates attached to distinct embeddings are independent (S1), so the largest subgroup in the kernel is a product, $G''$ has the elliptic dimensions ($\dim A_i'' = n/2$ for every $g$ by Lemma 4.1; the second-kind factor gives an anti-affine $Q$ of dimension $n/2 + 1$ by (F1)–(F4)), and the image of an abelian subvariety is a $K$-rational subspace from the $\sigma$-coordinates plus $\mathbb{C}(0, 1)$ from the third factor. The case analysis is then that of Theorem 6.12(1) and (3), with the period datum $P' = tP_\sigma/c$ and $t \notin \overline{\mathbb{Q}}$ (Wüstholz's theorem (W), distinct slots); $\kappa = n$ as there, and (2) reads $3n + 2 \geq 4 + n$ for EA ($d_0 = 2$) and $3n + 2 \geq 2 + n$ for EA$^\natural$. For the consequence, $P' \notin \widetilde{\mathcal{L}}_\sigma$ by Wüstholz's theorem, and the statement is symmetric under $t \mapsto 1/t$. □

**Example 6.17.** On $A = \mathrm{Jac}(y^7 = x(1-x))$ with $z_j$ dual to $dx/y^{7-j}$ and $P_j \in \overline{\mathbb{Q}}^\times B(\frac{j}{7}, \frac{j}{7})$ (§8.4), put $t = B(\frac{2}{7}, \frac{2}{7})/B(\frac{1}{7}, \frac{1}{7})$. For every non-torsion $\exp(v) \in A(\overline{\mathbb{Q}})$, the numbers $t$ and $tz_1(v)$ are not both in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}} z_1(\Lambda(A)) + \overline{\mathbb{Q}} z_2(\Lambda(A))$.

Theorem 6.12 is the case $g = 1$ of Theorem 6.16: EA with $A, A'$ non-isogenous CM curves is (1); EA$^\natural$ with $A' = E'$ non-isogenous to $E$ is (2); EA$^\natural$ with $A' = E$ and $\tilde\sigma = \bar\sigma$ is (3). We have

written out the elliptic case in full and give EA, EA$^\natural$ by reduction to it. Like Theorem 6.12, Theorem 6.16 is a consequence of Theorem 1.1; for $g \geq 2$ or $g' \geq 2$ it uses the inner weight (the printed weight $\alpha_K(A) = g$ gives no conclusion there).

## 7. Two CM curves: Theorems M, MX and MX′

7.1. **Theorem M: first proof (the count).** Let $G = E \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$, $\Psi(e,t,a) = (e, t+a)$, $V = \mathbb{C}(\Omega, i\pi)$, $W = 0$, and $K \supseteq k$ containing the algebraic coefficients involved. Then $\alpha_K(G) = 1+1+0 = 2$; $\Psi(\Omega_K(G)) = K\Omega \times K \cdot 2\pi i$ since $K\Lambda_E = K\Omega$ for CM; so $\kappa = 1$. Condition (i): $V$ has a nonzero rational point iff $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false (Masser [23]; cf. [45, Thm. 26]; also Schneider). Corollary 2.2 gives $\dim_K(V \cap \Psi(\Lambda_K)) \leq 1$, and the period $(\Omega, i\pi)$ lies in it. If $i\pi u = \Omega\ell$ then $(u,\ell) \in V \cap \Psi(\Lambda_K)$, so $(u,\ell) = s(\Omega, i\pi)$ with $s \in K$. If $u \in \mathcal{L}_E$, Masser's theorem ([45, Thm. 27]: elements of $\mathcal{L}_E$ that are $k$-independent are $\overline{\mathbb{Q}}$-independent) gives $s \in k$. □

*Controls.* If $E$ has no CM, $\alpha_K(E) = 2$ and the count ties (no conclusion). Replacing $E$ by $\mathbb{G}_{\mathrm{m}}$ ($\Omega \to 2\pi i$) makes $V$ rational, so (i) fails — as it must, since $u = 2\ell$ solves $i\pi u = 2\pi i\ell$. For $e^{\pi^2}$ itself ($G = \mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}^2$, $V = \mathbb{C}(1, i\pi)$) one finds $\kappa = 0$ and an exact tie.

7.2. **Theorem M: second proof for $u \in \mathcal{L}_E$ (Laurent's identity and Wüstholz).** We treat $u \in \mathcal{L}_E$, which covers every item of Corollary 1.11; a version for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ would need several points $Q_j$ and is not written out here. Let $\Lambda_E = \Omega\,\mathbb{Z}[\alpha]$ (a sublattice of finite index suffices), and write Masser's relation [23] as $\eta(\alpha\Omega) = \bar\alpha\eta(\Omega) + c\,\Omega$ with $c \in \overline{\mathbb{Q}}$. Legendre's relation gives $(\alpha - \bar\alpha)\eta(\Omega) - c\Omega = 2\pi i/\Omega$, and $f(z) = \zeta(\alpha z) - \bar\alpha\zeta(z) - cz$ is an elliptic function defined over $\overline{\mathbb{Q}}$. With the third-kind period $\lambda(u,\Omega) = \Omega\zeta(u) - \eta(\Omega)u$ one obtains, for $u \in \mathcal{L}_E$ not a pole of $f$,

$$\frac{2\pi i\, u}{\Omega} = \bar\alpha\,\lambda(u,\Omega) - \lambda(\alpha u,\Omega) + \Omega f(u), \qquad f(u) \in \overline{\mathbb{Q}}. \tag{6}$$

On $E_i$ this reads $2\eta(\varpi)u = i\lambda(iu,\varpi) - \lambda(u,\varpi)$ (cf. [22, §9, p. 160], where the general CM form is derived; the display above is its case $\mu = i$). Hence $i\pi u = \Omega\ell$ with $\ell \in \widetilde{\mathcal{L}}$ is a $\overline{\mathbb{Q}}$-linear relation among 1, logarithms of algebraic numbers, $\Omega$, $\lambda(u,\Omega)$ and $\lambda(\alpha u,\Omega)$. These are coordinates of a point of $\Lambda(\mathcal{G})$ for $\mathcal{G} = G_Q \times G_{\alpha Q} \times \mathbb{G}_{\mathrm{m}}^r \times \mathbb{G}_{\mathrm{a}}$, where $G_Q$ is the extension of $E$ by $\mathbb{G}_{\mathrm{m}}$ attached to $Q = \exp_E(u)$ (Serre [31]) — add a factor $E^\natural$ when algebraic constants enter. By the analytic subgroup theorem [52], the smallest algebraic subgroup $B$ whose tangent space contains this point lies in the hyperplane defined by the relation. $B$ surjects onto the diagonal of $E \times E$, and over it the toric part carries the extension classes $[a + b\alpha]Q$ for characters $(a,b)$; these vanish only for $a = b = 0$ when $Q$ is not torsion. So $B$ contains both fibre tori, on which the relation has nonzero coefficients — a contradiction. Hence $Q$ is torsion, which is the trivial case. □

7.3. **The corollaries of Theorem M.** (M1): $u = c\varpi/\pi$, $\ell = ic$ gives $s = c/\pi \notin \overline{\mathbb{Q}}$. (M2): $\ell = i\log\alpha$ gives $\log\alpha \in \overline{\mathbb{Q}}\pi$, so $\alpha = (-1)^{-is}$ with $s$ algebraic, a root of unity by Gel'fond–Schneider. (M3): if $e^{\pi^2} \in \overline{\mathbb{Q}}$ then $\ell = ic\pi^2 \in \widetilde{\mathcal{L}}$ and $u = c\pi\Omega$ give $s = c\pi \notin \overline{\mathbb{Q}}$. (M3′): $u = \varpi\log\alpha$, $\ell = i\pi\log\alpha$ give $\log\alpha \in \overline{\mathbb{Q}}^\times$, contradicting Hermite–Lindemann; (M3) is the case $\alpha = -1$, using $\wp(iz) = -\wp(z)$. (M$\eta$): $\eta(\varpi) = \pi/\varpi$. On a general CM curve $\eta(\Omega) = \kappa_1\Omega + \kappa_2\pi/\Omega$ with $\kappa_2 \neq 0$, and the statement becomes $(\eta - \kappa_1\Omega)u \notin \widetilde{\mathcal{L}}$.

7.4. **Proof of Theorem 1.7.** The coordinates $(z,w)$ on $E^\natural$, $\eta^*$, $\zeta^*$ and (4) are as in §4.2. Let $G = E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{a}}^2$, let $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ be a number field, $n = [K:\mathbb{Q}]$ (so $4 \mid n$), and let

$$\Psi(x_i, x_\rho, a_1, a_2) = \big(z(x_i) + z(x_\rho) + a_1,\ w(x_i) + w(x_\rho) + a_2\big), \qquad V = \mathbb{C}\,w_1,$$

with $w_1 = (\varpi, \beta\eta_\rho)$, $\eta_i = \eta(\varpi) = \pi/\varpi$, $\eta_\rho = \eta(\Omega_\rho) = 2\pi/(\sqrt{3}\,\Omega_\rho)$, and $w_2 = (\Omega_\rho, \beta\frac{2}{\sqrt{3}}\eta_i)$. By (4), $w_1, w_2 \in \Psi(\Omega_K(G))$ and

$$\Psi(\Omega_K(G)) = \mathcal{P} := (K\varpi + K\Omega_\rho) \times (K\eta_i + K\eta_\rho);$$

$w_1 \parallel w_2$ because $\varpi \cdot \frac{2}{\sqrt{3}}\eta_i = \frac{2\pi}{\sqrt{3}} = \Omega_\rho\eta_\rho$, and $w_1, w_2$ are $K$-independent since $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ (periods of non-isogenous curves are $\overline{\mathbb{Q}}$-independent [52]).

**Proposition 7.1** (Proposition D)**.** *$V \cap \Psi(\Lambda_K(G)) = Kw_1 + Kw_2$.*

*Status:* in its single-point form (one $v \in V \cap \Psi(\Lambda(G))$, with conclusion $v \in \overline{\mathbb{Q}}w_1 + \overline{\mathbb{Q}}w_2$; this is the form used for Theorem 1.7 and Corollary 9.5(ii)): [Proved modulo Wüstholz's theorem in the form $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] ($\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ is proved in Lean, §12.3). The $K$-span form as stated: [Proved modulo published theorems and Lemma A.1].

Granting Proposition 7.1: the point $(X, Y, a, t)$ lies in $\Lambda(G)$ and the hypothesis says that its image lies on $V$, so $z(X) + z(Y) + a = x\varpi + x'\Omega_\rho$ with $x, x' \in K$. By the linear independence of 1 and elliptic logarithms on non-isogenous curves (Wüstholz [52]), $a = 0$, $z(X) = x\varpi$, $z(Y) = x'\Omega_\rho$, and by Masser's theorem [23] $x \in \mathbb{Q}(i)$, $x' \in \mathbb{Q}(\rho)$. □

*Proof of Proposition 7.1.* Suppose $v \in V \cap \Psi(\Lambda_K(G))$ with $v \notin Kw_1 + Kw_2$.

*Step 1: an elementary lemma.* For $L \in \{0,\ \mathbb{C}(1,0),\ \mathbb{C}(0,1)\}$ we have $V \cap (L + \mathcal{P}) = Kw_1 + Kw_2$. Indeed, if $sw_1 \in L + \mathcal{P}$ with $L \neq \mathbb{C}(1,0)$ then $s\varpi \in K\varpi + K\Omega_\rho$, so $s \in K + K\Omega_\rho/\varpi$ and $(\Omega_\rho/\varpi)w_1 = w_2$; if $L = \mathbb{C}(1,0)$ then $s\beta\eta_\rho \in K\eta_i + K\eta_\rho$, so $s \in K + K\eta_i/\eta_\rho$ and $(\eta_i/\eta_\rho)w_1 = (\pi/\eta_\rho, \beta\eta_i) = \frac{\sqrt{3}}{2}w_2$.

*Step 2: the group.* Fix a basis $\theta_1, \dots, \theta_n$ of $K/\mathbb{Q}$, let $p_\theta \colon T_{G^n} \to T_G$, $(x_j) \mapsto \sum \theta_j x_j$, and $\Psi_\theta = \Psi \circ p_\theta$. Write the coordinates of $T_{G^n}$ as $z_i, w_i, z_\rho, w_\rho, a_1, a_2 \in \mathbb{C}^n$ and let $U \subset G^n$ be the vector subgroup $\{z_i = z_\rho = 0,\ \theta^{\mathsf{T}}a_1 = 0,\ \theta^{\mathsf{T}}(w_i + w_\rho + a_2) = 0\}$, of dimension $4n - 2$, defined over $\overline{\mathbb{Q}}$, with $T_U \subset \ker\Psi_\theta$. Then $G'' := G^n/U = \mathbb{G}_{\mathrm{a}} \times G_\ell$, where $G_\ell$ is the extension of $A = E_i^n \times E_\rho^n$ by $\mathbb{G}_{\mathrm{a}}$ obtained from the universal vector extension of $A$ by pushing out along $(w_i, w_\rho) \mapsto \theta^{\mathsf{T}}w_i + \theta^{\mathsf{T}}w_\rho$. In the coordinates $s_1 = \theta^{\mathsf{T}}a_1$, $z_i$, $z_\rho$, $s_2 = \theta^{\mathsf{T}}(w_i + w_\rho + a_2)$ the map induced by $\Psi_\theta$ is

$$\Psi''(s_1, z_i, z_\rho, s_2) = \big(\theta^{\mathsf{T}}z_i + \theta^{\mathsf{T}}z_\rho + s_1,\ s_2\big).$$

We apply [37, Thm. 4.1] (Theorem 2.3) to $G''$ with $d = 2n + 2$, $d_0 = 1$ ($\pi_0 = s_1$), $d_1 = 0$, $G_2 = G_\ell$, $d_2 = 2n + 1$; $V'' = \Psi''^{-1}(V)$, a hyperplane, so $N = 2n + 1$; $W'' = \ker\Psi''$, rational of dimension $2n$; and $Y''$ generated by the images of $3n$ elements of $\Lambda(G^n)$ whose $\Psi_\theta$-images are $D\theta_h w_1$, $D\theta_h w_2$ (taken in $\Omega(G^n)$, $D$ a common denominator) and $\theta_h v$, $h = 1, \dots, n$. Since $U$ is a vector group, $\Omega(G'')$ is the image of $\Omega(G^n)$, so $\Psi''(\Omega(G'')) \subset \mathcal{P}$. By Step 1 with $L = 0$, $\kappa = \mathrm{rk}(Y'' \cap \Omega(G'')) = 2n$, and $\Gamma = \exp Y''$ has rank $n$. The inequality of Theorem 2.3 becomes

$$\lambda + 2\delta_2 \le (\delta - \tau)(2n + 2), \tag{7}$$

and, with $r = \dim\Psi''(T_{G'})$, one has $\tau = \delta + r - 2$; so an obstruction must have $r \le 1$.

*Step 3: the subgroups of $G''$.* Abelian subvarieties of $A$ are $B = B_i \times B_\rho$ ($E_i, E_\rho$ are non-isogenous), and $B_i \subset E_i^n$ corresponds to a $\mathbb{Q}(i)$-subspace $S_i \subset \mathbb{Q}(i)^n$ with $T_{B_i} = \mathbb{C}\,\sigma(S_i)$, $\dim B_i = \dim S_i$; likewise for $E_\rho$. With $\psi, \phi$ as in Lemma 4.2 (for $k = \mathbb{Q}(i)$ and for $k = \mathbb{Q}(\rho)$), $\ker\psi \cap \ker\phi = 0$ and each kernel has $k$-dimension $n/2$. Because $[a]$ acts on $T(E^\natural)$ by $\mathrm{diag}(a, \bar{a})$ and $\mathrm{Hom}(E^\natural, \mathbb{G}_{\mathrm{a}}) = 0$ (the universal vector extension is anti-affine; [25]), the image in $G_\ell$ of the universal extension of $B$ is a splitting $s(B) \cong B$ (with tangent space $T_B \times 0$) if $\phi(S_i) = \phi(S_\rho) = 0$, and is $\pi^{-1}(B)$, a non-split extension, otherwise (a splitting would give a nonzero homomorphism from a quotient of $(E_i^\natural)^{b_i} \times (E_\rho^\natural)^{b_\rho}$ to $\mathbb{G}_{\mathrm{a}}$). A connected subgroup over $B$ either contains the fibre

or is a splitting, and a splitting is unique since $\mathrm{Hom}(B,\mathbb{G}_{\mathrm{a}})=0$. It follows that the connected algebraic subgroups of $G''=\mathbb{G}_{\mathrm{a}}\times G_\ell$ are exactly

| type | $G'$ | condition | $\Psi''(T_{G'})$ |
|---|---|---|---|
| A | $\mathbb{G}_{\mathrm{a}}\times\pi^{-1}(B)$ | — | $\mathbb{C}^2$ |
| B | $\mathbb{G}_{\mathrm{a}}\times s(B)$ | $\phi(S)=0$ | $\mathbb{C}(1,0)$ |
| C | $0\times\pi^{-1}(B)$ | — | $\mathbb{C}(0,1)$ if $\psi(S)=0$, else $\mathbb{C}^2$ |
| D | $0\times s(B)$ | $\phi(S)=0$ | 0 if $B=0$, else $\mathbb{C}(1,0)$ |
| E | $\{(\mu\,\mathrm{pr}_F(g),g):g\in s(B)\times F\}$, $\mu\neq0$ | $\phi(S)=0$ | $\mathbb{C}(\mu,1)$ if $B=0$, else $\mathbb{C}^2$ |

(with $\phi(S)=0$ meaning $\phi(S_i)=\phi(S_\rho)=0$, and similarly for $\psi$). In type E, $F\cong\mathbb{G}_{\mathrm{a}}$ is the fibre of $G_\ell\to A$ (coordinate $s_2$), $\pi^{-1}(B)=s(B)\times F$, and $\mathrm{pr}_F$ is the projection to $F$; so $G'$ is the graph of $\mu\,\mathrm{pr}_F$ into the first factor $\mathbb{G}_{\mathrm{a}}$ (coordinate $s_1$), and for $B=0$ it is $\{s_1=\mu s_2\}$. In every case allowed by the conditions, $b=\dim B\le n$.

*Step 4: no obstruction.* If $\Psi''(T_{G'})\in\{0,\mathbb{C}(1,0),\mathbb{C}(0,1)\}$ then $\lambda=n$: if $\exp(\sum c_h\tilde y_h)\in G'$ for the lifts $\tilde y_h$ of $\theta_h v$, then $xv\in L+\mathcal{P}$ with $x=\sum c_h\theta_h\in K$, so $x=0$ by Step 1. With $\delta_0=0$ for types A, B, E and $\delta_0=1$ for C, D, inequality (7) reads: $5n+2\le4n+4$ for $G'=0$; $5n+2-2b\le2n+2$ for types B and D ($B\neq0$); $5n-2b\le2n+2$ for type C with $r=1$; and $\lambda+4n+2\le2n+2$ for type E with $B=0$. Since $b\le n$, none of these holds for $n\ge3$, and types with $r=2$ fail $\delta>\tau$. So no connected $G'\neq G''$ satisfies the conclusion of Theorem 2.3, a contradiction. □

**Remark 7.2** (The count)**.** Corollary 2.2 predicts Proposition 7.1: with $\kappa=2$ and $\alpha_K(G)=2(2+2/n)$ (Lemma 4.3) it gives $\dim_K(V\cap\Psi(\Lambda_K(G)))\le2+4/n$, which is $<3$ only for $n>4$; with the inner weight $\alpha_{K,\Psi}(G)=\alpha(G'')/n=(4n+2)/n$ (Remark 5.8) one gets $2+2/n<3$ for $n\ge3$, which is the threshold of Step 4. So Proposition 7.1 is a consequence of Theorem 1.1 for $n\ge3$, and of its printed form, Corollary 2.2, for $[K:\mathbb{Q}]\ge8$ (as is Proposition 7.5, with the same weights); its single-point form, which is all that Theorem 1.7 and Corollary 9.5(ii) use, follows for every $K$ (Remark 11.1). For $[K:\mathbb{Q}]=4$ the printed bound is $2+4/4=3$; the $K$-span form there needs, in addition, Masser's theorem for the descent from a field of degree $\ge8$.

A second derivation of Proposition 7.1 applies Theorem 2.5 to $(G'',V'',W'',Y'')$: $d=2n+2$, $d_0=1$, $d_1=0$, $d_2=2n+1$, $t_W=\dim W''=2n$, $m=3n$, $\kappa=2n$, and condition (5.4) of [37] reads $3n>(4n+2-2n)\cdot1$, i.e. $3n>2n+2$ (for $K=\mathbb{Q}(i,\sqrt3)$, $n=4$: $12>18-\kappa=10$). It remains to see that no connected $G'\neq0$ has $T_{G'}\subset V''$. First, $T_{G'}\not\subset\ker\Psi''$. If $G'$ meets the kernel $\mathbb{G}_{\mathrm{a}}\times F$ of $G''\to A$ in a subgroup of positive dimension, $T_{G'}$ contains a vector $(c,0,0,d)\neq0$, whose image is $(c,d)$. Otherwise $G'$ is finite over its image $B$ in $A$, and $B\neq0$. Then $G'$ is an abelian variety, so $G'\subset0\times G_\ell$ ($\mathrm{Hom}(B,\mathbb{G}_{\mathrm{a}})=0$) is a splitting $s(B)$; by Step 3 this forces $\phi(S)=0$, and then $\psi\neq0$ on $S$ (Lemma 4.2), so $\Psi''(T_{G'})=\mathbb{C}(1,0)$. Since $T_{G'}$ is $\overline{\mathbb{Q}}$-rational, it contains a $\overline{\mathbb{Q}}$-rational $x$ with $\Psi''(x)\neq0$. If $T_{G'}\subset V''$, then $\Psi''(x)\in V$ is a nonzero $\overline{\mathbb{Q}}$-rational point of $V=\mathbb{C}w_1$, so $\varpi/(\beta\eta_\rho)\in\overline{\mathbb{Q}}$. This contradicts $\varpi/\eta_\rho\notin\overline{\mathbb{Q}}$, which is Wüstholz's theorem [52] (a period of $E_i$ and a quasi-period of $E_\rho$ are $\overline{\mathbb{Q}}$-linearly independent). The same exclusion can be read off the table in Step 3. Unlike Step 4, this derivation uses $\varpi/\eta_\rho\notin\overline{\mathbb{Q}}$, and it rests on Theorem 2.5 (§3.6).

Corollary 1.8 is the case $X=(u,\zeta_i(u))$, $Y=0$ (the logarithm of the origin), resp. $X=0$; the value 0 of the ratio is excluded separately, since $1,u,\zeta(u)$ are $\overline{\mathbb{Q}}$-linearly independent for non-torsion $u$ on a CM curve [21]; it also follows from [52] applied on $E^\natural\times\mathbb{G}_{\mathrm{a}}$. Note that the base and fibre coordinates of a single logarithm cannot be chosen independently: $w(X)-\zeta^*(z(X))\in\overline{\mathbb{Q}}$.

**Remark 7.3** (Other pairs of curves)**.** Let $E,E'$ be non-isogenous CM curves with lattices $\omega\mathbb{Z}[\alpha]$, $\omega'\mathbb{Z}[\alpha']$ (up to finite index), so that $(\alpha-\bar\alpha)\,\omega\,\eta^*(\omega)=2\pi i=(\alpha'-\bar\alpha')\,\omega'\eta^*(\omega')$. With $V=$

$\mathbb{C}(\omega, \beta\eta^*(\omega'))$ the vectors $w_1 = (\omega, \beta\eta^*(\omega'))$ and $w_2 = \big(\omega', \beta\frac{\alpha-\bar\alpha}{\alpha'-\bar\alpha'}\eta^*(\omega)\big)$ are parallel, and the same proof gives: if $X \in \Lambda(E^\natural)$, $Y \in \Lambda(E'^\natural)$ and $(z(X)+z(Y)+a)\beta\eta^*(\omega') = (w(X)+w(Y)+t)\omega$, then $a = 0$ and $z(X) \in k\omega$, $z(Y) \in k'\omega'$.

**Remark 7.4.** By the explicit formulas for $\varpi$ and $\Omega_\rho$, $\varpi\Omega_\rho/\pi \in \overline{\mathbb{Q}}^\times \cdot \Gamma(1/4)^2\Gamma(1/3)^3/\pi^{5/2}$, and $\varpi/\eta_\rho = \frac{\sqrt{3}}{2\pi}\varpi\Omega_\rho$. If $\varpi/\eta_\rho = c \in \overline{\mathbb{Q}}$, then $X = Y = 0$, $a = 1$, $t = \beta/c$ satisfy the hypothesis of Theorem 1.7 with $a \neq 0$; so Corollary 1.9 is the case $X = Y = 0$. It also follows directly from Wüstholz's theorem [52] (a period of $E_i$ and a quasi-period of $E_\rho$ are $\overline{\mathbb{Q}}$-linearly independent).

7.5. **Controls for Theorem MX.** (1) If either curve has no CM there is no relation making two periods parallel, so $\kappa$ drops to $n$ in Step 2 and $G' = 0$ satisfies the inequality of Theorem 2.3 ($5n+2 \le 6n+4$): no conclusion. The same happens if only one Legendre relation is used. If the fibre action were $\mathrm{diag}(a, a)$ instead of $\mathrm{diag}(a, \bar a)$, subgroups of type D with image 0 would satisfy it; and for a rational line $V = \mathbb{C}(\mu, 1)$ with true data on it ($\kappa = 0$) the type E line does. So the conjugate character, the second Legendre relation and the irrationality of $V$ are all used. (2) With the same curve twice, the antidiagonal of $E^\natural \times E^\natural$ lies in $\ker\Psi$ and the count collapses to the one-curve cell, which only gives $(\zeta(u)+t)/(u+a) \neq \pi/\varpi^2$ — a weak form of Chudnovsky's theorem [11]. (3) Data on the period plane satisfy the relation only trivially.

7.6. **A twisted variant.** Exchanging the roles of base and fibre on $E_\rho$ gives a second statement of the same kind. Let $G = E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{a}}^2$ and

$$\Psi'(X_i, X_\rho, a_1, a_2) = \big(z(X_i)+w(X_\rho)+a_1,\ w(X_i)+z(X_\rho)+a_2\big), \qquad V' = \mathbb{C}\,w_1', \quad w_1' = (\varpi, \beta\Omega_\rho).$$

With $w_2' = (\eta_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ one has $w_2' = (\eta_\rho/\varpi)\,w_1'$ by the two Legendre relations, and for $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ both lie in $\Psi'(\Omega_K(G)) = (K\varpi + K\eta_\rho) \times (K\eta_i + K\Omega_\rho)$. The vectors $w_1'$, $w_2'$ are $K$-independent because $\eta_\rho/\varpi \notin \overline{\mathbb{Q}}$ (Corollary 1.9).

**Proposition 7.5** (Proposition D′)**.** *Let $K \supseteq \mathbb{Q}(i, \sqrt{-3}, \beta)$ be a number field of degree $n \ge 8$. Then $V' \cap \Psi'(\Lambda_K(G)) = Kw_1' + Kw_2'$.*

*Status:* in its single-point form (one $v \in V' \cap \Psi'(\Lambda(G))$, with conclusion $v \in \overline{\mathbb{Q}}w_1' + \overline{\mathbb{Q}}w_2'$, every $\beta \in \overline{\mathbb{Q}}^\times$; this is the form used for Corollary 9.5(i)): [Proved modulo Wüstholz's theorem in the form $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise] ($\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ is proved in Lean, §12.3). The $K$-span form as stated: [Proved modulo published theorems and Lemma A.1]. The formal verification rests on `lean_lsg`'s proof of Theorem 2.3 for the group $G''$ below, a route separate from the one checked against Lemma A.1 (§12.3, Appendix A).

*Proof.* With $\theta$, $p_\theta$ and $\Psi'_\theta = \Psi' \circ p_\theta$ as before, a canonical lift $B^\natural$ of an abelian subvariety $B_{S_i} \times B_{S_\rho}$ has tangent space in $\ker\Psi'_\theta$ only if $\psi(S_\bullet) = \phi(S_\bullet) = 0$, i.e. $B = 0$ (Lemma 4.2). So the largest connected subgroup $U'$ with $T_{U'} \subset \ker\Psi'_\theta$ is the vector subgroup $\{\theta^{\mathsf T}(w_\rho + a_1) = 0,\ \theta^{\mathsf T}(w_i + a_2) = 0\}$, and $G'' = G^n/U'$ is the pushout of $(E_i^n \times E_\rho^n)^\natural$ along $(w_i, w_\rho) \mapsto (\theta^{\mathsf T}w_i, \theta^{\mathsf T}w_\rho)$, an extension of $E_i^n \times E_\rho^n$ by $\mathbb{G}_{\mathrm{a}}^2$ of dimension $2n+2$. It is a quotient of an anti-affine group, so $\mathrm{Hom}(G'', \mathbb{G}_{\mathrm{a}}) = \mathrm{Hom}(G'', \mathbb{G}_{\mathrm{m}}) = 0$: $d_0 = d_1 = 0$ (maximal), $d_2 = d = 2n+2$. Let $\bar\Psi' \colon T_{G''} \to \mathbb{C}^2$ be the map induced by $\Psi'_\theta$, $\bar V' = \bar\Psi'^{-1}(V')$ (a hyperplane) and $W'' = \ker\bar\Psi' \subset \bar V'$, $\overline{\mathbb{Q}}$-rational of dimension $t_W = 2n$. Suppose $v \in V' \cap \Psi'(\Lambda_K(G))$ with $v \notin Kw_1' + Kw_2'$, and let $Y'' \subset \bar V'$ be generated by the images of lifts in $\Lambda(G^n)$ of $D\theta_h w_1'$, $D\theta_h w_2'$ (in $\Omega(G^n)$, $D$ a common denominator) and $\theta_h v$, $h = 1, \dots, n$. Since $w_1', w_2', v$ are $K$-independent, their $\bar\Psi'$-images are $\mathbb{Q}$-independent, so $m = \mathrm{rk}\,Y'' = 3n$; the $2n$ period lifts give $\kappa \ge 2n$. The $\kappa$-refined Theorem 1.1 of [37, §5d] (Theorem 2.5) applies to $(G'', \bar V', W'', Y'')$: condition (5.4), $m > (d_1 + 2d_2 - \kappa)(d - 1 - t_W)$, holds since $(4n+4-\kappa)\cdot 1 \le 2n+4 < 3n$ for $n > 4$, hence for $n \ge 8$ ($4 \mid n$). So $\bar V'$ contains $T_{G'}$ for some

connected algebraic subgroup $G' \neq 0$ defined over $\overline{\mathbb{Q}}$. Then $\bar{\Psi}'(T_{G'})$ is a $\overline{\mathbb{Q}}$-rational subspace of $V'$, which has no nonzero $\overline{\mathbb{Q}}$-rational point ($\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$); so $T_{G'} \subset \ker \bar{\Psi}'$, the preimage of $G'$ in $G^n$ is a connected subgroup with tangent space in $\ker \Psi'_\theta$, and $G' = 0$ by the maximality of $U'$, a contradiction. (A full case check against Theorem 2.3, with the analogue of Step 1 of the proof of Theorem 1.7, should give the same threshold; we have not written it out.) □

**Theorem 7.6** (Theorem MX′)**.** *Let $X \in \Lambda(E_i^\natural)$, $Y \in \Lambda(E_\rho^\natural)$, $a, b \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^\times$. If*

$$\big(z(X) + w(Y) + a\big) \cdot \beta\,\Omega_\rho = \big(w(X) + z(Y) + b\big) \cdot \varpi,$$

*then $z(X) = z(Y) = 0$ and both sides vanish.*

*Proof (sketch).* By Proposition 7.5, $z(X) + w(Y) + a = x\varpi + x'\eta_\rho$ and $w(X) + z(Y) + b = x\beta\Omega_\rho + x'\frac{2\beta}{\sqrt{3}}\eta_i$ with $x, x' \in K$. The analytic subgroup theorem gives $z(X) = p\varpi$, $z(Y) = q\Omega_\rho$ with $p \in \mathbb{Q}(i)$, $q \in \mathbb{Q}(\rho)$; then $\zeta_i(p\varpi) \in \bar{p}\,\eta_i + \overline{\mathbb{Q}}$ and $\zeta_\rho(q\Omega_\rho) \in \bar{q}\,\eta_\rho + \overline{\mathbb{Q}}$, and the $\overline{\mathbb{Q}}$-independence of $1, \varpi, \eta_\rho$ and of $1, \eta_i, \Omega_\rho$ gives $x = p$, $x' = \bar{q}$, $q = p\beta$, $\bar{p} = \frac{2\beta}{\sqrt{3}}\bar{q}$, and vanishing constant terms. If $p \neq 0$ this gives $\sqrt{3}|p|^2 = 2|q|^2$ with $|p|^2, |q|^2 \in \mathbb{Q}_{>0}$, which is impossible. □

**Corollary 7.7.** *For every nonzero $u \in \mathcal{L}_{E_i}$ (off the lattice) and $a, b \in \overline{\mathbb{Q}}$: $(\zeta_i(u) + b)/(u + a) \notin \overline{\mathbb{Q}}^\times \cdot \Omega_\rho/\varpi = \overline{\mathbb{Q}}^\times \cdot \Gamma(1/3)^3/(\sqrt{\pi}\,\Gamma(1/4)^2)$.*

Together with Corollary 1.8, the ratio $(\zeta_i(u) + t)/(u + a)$ avoids both $\overline{\mathbb{Q}}\,\pi/(\varpi\Omega_\rho)$ and $\overline{\mathbb{Q}}^\times\Omega_\rho/\varpi$. *Status:* Theorem 7.6 and Corollary 7.7: [Proof sketch]. The step from Proposition 7.5 (whose status is stated there) to Theorem 7.6 rests on Masser's and Wüstholz's theorems and is sketched, not written out in full; it is not formalised. Corollary 7.7 is the case $X = (u, \zeta_i(u))$, $Y = 0$.

7.7. **The abelian-surface analogue.** A simple CM abelian surface $A$ has two eigen-pairs of periods with $P_1P_4,\ P_2P_3 \in \pi\,\overline{\mathbb{Q}}^\times$ (Riemann's bilinear relations). For $A = \mathrm{Jac}(y^2 = x^5 + 1)$, $P_a = (1 - \zeta_5^a)B(\frac{a}{5}, \frac{1}{2} - \frac{a}{5})/5$, and the reflection formula gives these relations exactly. With $G = A^\natural \times \mathbb{G}_{\mathrm{a}}^2$, $\Psi = (z_1 + z_2 + a_1,\ w_3 + w_4 + a_2)$ and $V = \mathbb{C}(P_1, \beta P_3)$, the same argument (with $\alpha_K(A^\natural) = 4 + 4/n$) shows: for a non-torsion point of $A(\overline{\mathbb{Q}})$ with logarithm $v$, $(z_1(v) + z_2(v) + a)\beta P_3 \neq (w_3(v) + w_4(v) + t)P_1$. We state this only as a proof sketch; the steps not written out are: the computation $\alpha_K(A^\natural) = 4 + 4/n$ (no abelian part in $H$, since the four coordinates carry four distinct characters of $\mathbb{Q}(\zeta_5)$); condition (i) via Wüstholz on $A^\natural$, which uses that $A^\natural$ has no proper algebraic subgroup surjecting onto $A$; the description of $\Lambda(A^\natural)$; and the final deduction that $v$ is torsion. This sketch uses Corollary 2.2. The direct argument of §7.4 should carry over with the same numbers ($G'' = \mathbb{G}_{\mathrm{a}} \times G_\ell$ of dimension $2n + 2$, abelian subvarieties given by $\mathbb{Q}(\zeta_5)$-subspaces and four conditions from the four embeddings), but we have not written it out.

7.8. **Single-partner limits for $e^{\pi^2}$.** The three exponentials conjecture asks that $\lambda_1\lambda_2 = \gamma\lambda_3$ with $\lambda_j \in \mathcal{L}$, $\gamma \in \overline{\mathbb{Q}}^\times$ force $\lambda_3 = 0$; $e^{\pi^2} \notin \overline{\mathbb{Q}}$ is the instance $\lambda_1 = \lambda_2 = i\pi$. In the count above, a configuration of rank-one shape gives a theorem only if $\kappa$ exceeds what the true data need. For lines with a single coordinate per factor, $\kappa \leq 1$, and one CM factor supplies exactly one unit of weight. On the period side, Laurent's identity (6) and its abelian analogue show that these configurations are linearisable: multiplying by $\pi$ turns them into linear relations among 1-periods, decided by the analytic subgroup theorem.

To get $\kappa = 2$ on a line one needs two independent periods with parallel images, i.e. a true *homogeneous* degree-two relation $p_ap_d = p_bp_c$ among periods. The CM Legendre relations $(\alpha - \bar{\alpha})\,\omega\,\eta^*(\omega) = 2\pi i$ of two non-isogenous CM curves provide one; so do two eigen-pairs of a CM abelian variety. The multiplicative group cannot take part: its only period is $2\pi i$, and the corresponding relation would pair $2\pi i$ with 1, which is not a period (for $E^\natural \times \mathbb{G}_{\mathrm{m}}$ with $\mathbb{G}_{\mathrm{m}}$

added to either coordinate, parallel periods force a $\overline{\mathbb{Q}}$-linear relation among $1, \omega, \eta^*$, so $\kappa \le 1$). Under the hypothesis $e^{\pi^2} \in \overline{\mathbb{Q}}$ the number $\pi^2$ is a logarithm, not a period, so it adds data, not $\kappa$. This is why Theorem 1.7 exists while the same count cannot reach $e^{\pi^2}$. The $\mathbb{G}_{\mathrm{m}}$-analogue of Theorem 1.7 (replace $\Omega_\rho$ by $i\pi$ and the $E_i$-data by logarithms) is $i\pi \cdot \log\beta = \gamma \log\alpha$, an open case of the conjecture.

With two partners the mechanism does reach $e^{\pi^2}$: Corollary 9.1 (one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$) uses $\pi^2$ and $\pi^2/\Omega$ as data and the period $(2\pi i, 2ic\Omega)$ for $\kappa$. In the configurations we enumerated (lines in $\mathbb{C}^2$, at most three factors per coordinate; not refereed), no single-partner disjunction "one of $e^{\pi^2}$, $X$" appears beyond (M3) and the known strong-six partners $e^{c/\pi}$, $e^{c\pi^3}$ (from the strong six exponentials theorem [27, Cor. 2] with $x = (1, \pi)$ and $y = (1/\pi, 1, \pi)$, resp. $y = (1, \pi, \pi^2)$).

[Heuristic] Within the monomial model (not refereed), a slack count explains this. On a line $V = \mathbb{C}(1, r)$, each factor contributes monomials: its period, plus the constant 1 for $\mathbb{G}_{\mathrm{a}}$ and for the fibre. So $\mathbb{G}_{\mathrm{a}}$ gives $\{1\}$, $\mathbb{G}_{\mathrm{m}}$ gives $\{\pi\}$, the base of $E$ gives $\{\Omega\}$, and the fibre of $E^\natural$ gives $\{\pi/\Omega, 1\}$. Their inner weights $\alpha_{K,\Psi}$ in the count (Remark 5.8; not the printed weights of Lemma 4.3, which give $2 + 2/n$ for a fibre factor) are 0, 1, 1, $1 + 2/n$, and the true relations are homogeneous for the grading $\deg\pi = 2$, $\deg(\text{1-period}) = 1$. A line count is then a matching between the monomials of the two coordinates. The hypothesis $e^{\pi^2} \in \overline{\mathbb{Q}}$ adds the monomial $\pi^2$ (degree 4) to the $\mathbb{G}_{\mathrm{m}}$-factors, and a partner hypothesis adds one monomial $m_X$. A single-partner disjunction needs the partner to sit opposite $\pi^2$, so $m_X = \pi^2 r^{\pm 1}$, or the $e^{\pi^2}$-edges alone to fill the slack. Running through the cases leaves exactly $m_X = \pi \cdot p$ with $p$ a period of the factor carrying the partner, i.e. $\wp(c\pi\Omega)$ (the case (M3)), together with the strong-six partners $e^{c/\pi}$ and $e^{c\pi^3}$. An exhaustive enumeration in a bounded range (at most four factors per coordinate) confirms this. On lines of type MX the slope has degree 0, so a $\mathbb{G}_{\mathrm{m}}$-factor can never be matched; $e^{\pi^2}$ can enter such a count only together with a second hypothesis. On lines in $\mathbb{C}^3$, where Theorem 2.3 has to be applied subgroup by subgroup, there is one more single partner: $\wp_E(c/\pi + b)$ on a CM curve (Theorem 10.4).

## 8. First-kind period relations: Theorem X

8.1. **Set-up.** A *first-kind slot* is an eigen-slot of the first kind (§4.1), written $j = (A_j, \sigma_j)$ with $A_j/\overline{\mathbb{Q}}$ a simple CM abelian variety (CM field $F_j$, primitive type $\Phi_j$) and $\sigma_j \in \Phi_j$; $z_j$ is the corresponding eigen-coordinate on $\mathrm{Lie}(A_j)$ and $P_j = z_j(\gamma_j) \neq 0$ for a fixed $\gamma_j \in \Omega(A_j)$, so $z_j(\Omega(A_j)) \subset \sigma_j(F_j)P_j$. As in §4.1, two slots are equal iff the abelian varieties are isogenous and the embeddings correspond; distinct slots of one $A$ have distinct embeddings. By (W), periods of distinct slots are $\overline{\mathbb{Q}}$-linearly independent. Let $a, b, c, d$ be four pairwise distinct first-kind slots (distinct in the sense just defined, i.e. up to isogeny) with a *first-kind relation*

$$P_a P_b = \mu\, P_c P_d, \qquad \mu \in \overline{\mathbb{Q}}^\times. \tag{8}$$

Let $G = \prod_A A \times \mathbb{G}_{\mathrm{a}}^2$, the product over the isogeny classes among $A_a, A_b, A_c, A_d$, each taken once (so the factors $A$ are pairwise non-isogenous), let $\beta \in \overline{\mathbb{Q}}^\times$ and

$$\Psi = \big(z_a + z_d + q_1,\ z_c + z_b + q_2\big)\colon T_G \to \mathbb{C}^2, \qquad V = \mathbb{C}w, \quad w = (P_a, \beta P_c),$$

where $q_1, q_2$ are the $\mathbb{G}_{\mathrm{a}}$-coordinates. Then $w' := (P_d, \beta P_b/\mu) = (P_d/P_a)\, w$ by (8). Let $K$ be a number field containing the Galois closures of all $F_j$, $\mu$ and $\beta$, and $n = [K : \mathbb{Q}]$. By (Ch) and $z_j(\Omega(A_j)) \subset \sigma_j(F_j)P_j$,

$$\Psi(\Omega_K(G)) = \mathcal{P} := (KP_a + KP_d) \times (KP_c + KP_b),$$

so $w, w' \in \Psi(\Omega_K(G))$, and they are $K$-independent since $P_d/P_a \notin \overline{\mathbb{Q}}$.

**Proposition 8.1.** $V \cap \Psi(\Lambda_K(G)) = Kw + Kw'$.

**Theorem 8.2** (Theorem X)**.** *In the set-up above, let $v_j \in \overline{\mathbb{Q}}\Lambda(A_j)$ for $j = a, b, c, d$ and $q_1, q_2 \in \overline{\mathbb{Q}}$ satisfy*

$$\big(z_a(v_a) + z_d(v_d) + q_1\big) \cdot \beta P_c = \big(z_c(v_c) + z_b(v_b) + q_2\big) \cdot P_a. \tag{9}$$

*Then $q_1 = q_2 = 0$ and $z_j(v_j) \in \overline{\mathbb{Q}}P_j$ for each $j$. If $v_j \in \Lambda(A_j)$, then $\exp_{A_j}(v_j)$ is a torsion point.*

*Status:* Proposition 8.1 and Theorem 8.2: [Proved modulo published theorems and Lemma A.1] (three internal referees, one with a blind classification of the subgroups; independent exact enumerations; minimum excess $n$). The deduction of Theorem 8.2 from the proposition is the standard analytic-subgroup step, as in §7.4 for Theorem 1.7, written out below.
When several of the slots lie on the same $A$ (as in §8.4), the $v_j$ may be logarithms of different points: by (Ch), a single element of $\Lambda_K(A)$ has slot coordinates $z_j(v_j)$ for any prescribed $v_j$.

8.2. **Proof of Proposition 8.1.** Suppose $v \in V \cap \Psi(\Lambda_K(G))$, $v \notin Kw + Kw'$. Let $\theta$, $p_\theta$ and $\Psi_\theta = \Psi \circ p_\theta$ be as before.

*The group.* For each isogeny class $A$ let $\Sigma_A$ be the set of slots on $A$ used by $\Psi$. By (P) and (S1), the largest connected subgroup with tangent space in $\ker \Psi_\theta$ is $U = \prod_A B_{\ker(\psi_\sigma)_{\sigma\in\Sigma_A}} \times \{\theta^{\mathsf{T}} q_1 = \theta^{\mathsf{T}} q_2 = 0\}$, and $G'' = G^n/U = \mathbb{G}_{\mathrm{a}}^2 \times A''$ with $\dim A'' = \sum_A |\Sigma_A|\, n/2 = 2n$ by Lemma 4.1. So $d = 2n + 2$, $d_0 = 2$, $d_1 = 0$, $d_2 = 2n$. The induced $\Psi''\colon T_{G''} \to \mathbb{C}^2$ is onto, $V'' = \Psi''^{-1}(V)$ is a hyperplane ($N = 2n + 1$), and $W'' = \ker \Psi'' \subset V''$ is $\overline{\mathbb{Q}}$-rational of dimension $2n$.

*The data.* Let $Y''$ be generated by the images of lifts of $D\theta_h w$, $D\theta_h w'$ (in $\Omega(G^n)$) and $\theta_h v$, $h = 1, \dots, n$. Since $U$ is connected, $\Omega(G'')$ is the image of $\Omega(G^n)$, so $\Psi''(\Omega(G'')) \subset \mathcal{P}$, and $V \cap \mathcal{P} = Kw + Kw'$ (Lemma 8.4 with $L = 0$). Hence $\operatorname{rk} Y'' = 3n$, $\kappa = 2n$ exactly, and $\Gamma = \exp Y''$ has rank $n$. Since $d_1 + 2d_2 - \kappa = 2n$ and $d - N = 1$, (1) reads

$$\lambda + 2\delta_2 \le (2 - r') \cdot 2n. \tag{10}$$

**Lemma 8.3** (Lemma S′)**.** *Every connected algebraic subgroup of $G''$ is $L \times B$ with $L$ a $\overline{\mathbb{Q}}$-subspace of $\mathbb{G}_{\mathrm{a}}^2$ and $B$ an abelian subvariety of $A''$. $\Psi''(T_B)$ is spanned by the target directions of the slots on which $B$ is not killed: it is $0$ iff $B = 0$; it is $\subset \mathbb{C}(1,0)$ iff the slots $c, b$ are killed, and then $\dim B \le n$; similarly for $\mathbb{C}(0,1)$ (slots $a, d$ killed); otherwise it is $\mathbb{C}^2$.*

*Proof.* The product decomposition is (P). By (S1) the slot forms $\theta^{\mathsf{T}} z_j$ on $T(B_S)$ involve independent vectors $u_\sigma$, and each slot feeds exactly one target coordinate, so $\Psi''(T_B)$ is a coordinate subspace; it is 0 only for $B = 0$ by maximality of $U$. If only the slots of one target survive, $S/\ker$ embeds into $K^2$ via the two surviving $\psi_\sigma$, so $\dim B \le 2 \cdot n/2 = n$ (Lemma 4.1). □

**Lemma 8.4** (Lemma L′)**.** *For $L \in \{0,\ \mathbb{C}(1,0),\ \mathbb{C}(0,1)\}$ one has $V \cap (L + \mathcal{P}) = Kw + Kw'$.*

*Proof.* If $L \ne \mathbb{C}(1,0)$ and $\xi w \in L + \mathcal{P}$, then $\xi P_a \in KP_a + KP_d$, so $\xi = x + yP_d/P_a$ $(x, y \in K)$ and $\xi w = xw + yw'$. If $L = \mathbb{C}(1,0)$, then $\xi\beta P_c \in KP_c + KP_b$, so $\xi = x + yP_b/P_c = x + y\mu P_d/P_a$ and $\xi w = xw + y\mu w'$. □

*The case check.* Let $G' = L \times B$ with $r' = \dim \Psi''(T_{G'}) \le 1$, $e = \dim B$. Then $\delta_2 = 2n - e$. If $\Psi''(T_{G'}) \in \{0, \mathbb{C}(1,0), \mathbb{C}(0,1)\}$, then $\lambda = n$: if $\exp(\sum c_h \tilde{y}_h) \in G'$ for the lifts $\tilde{y}_h$ of $\theta_h v$, then $xv \in \Psi''(T_{G'}) + \mathcal{P}$ with $x = \sum c_h \theta_h \in K$, so $x = 0$ by Lemma 8.4.

| $G'$ | $r'$ | $\lambda$ | (10) reads | holds? |
|---|---|---|---|---|
| $0$ | 0 | $n$ | $5n \le 4n$ | never |
| $B \ne 0$, image an axis ($e \le n$), $L \in \{0, \text{same axis}\}$ | 1 | $n$ | $5n - 2e \le 2n$ | never ($e \le n$) |
| $L$ an axis, $B = 0$ | 1 | $n$ | $5n \le 2n$ | never |
| $L$ a non-axis line, $B = 0$ | 1 | $\ge 0$ | $\lambda + 4n \le 2n$ | never |
| $r' = 2$ | | | $\delta > \tau$ fails | |

So no connected $G' \neq G''$ satisfies the conclusion of Theorem 2.3, for any $n$; the minimum excess (left side minus right side) is $n$, attained at $G' = 0$ and at axis subgroups with $e = n$. Condition (2) reads $4n+2 \geq 2n+4$, and $d_0 = 2 \geq 1$ is covered by the final paragraph of Lemma A.1. This contradiction proves Proposition 8.1. □

**Remark 8.5** (Second derivation)**.** Waldschmidt's [37, §5d] ($\kappa$-refined Theorem 1.1, hyperplane case) applies to $(G'', V'', W'', Y'')$ directly: condition (5.4) there reads $m = 3n > (d_1 + 2d_2 - \kappa)(d - 1 - t) = 2n$, and $V''$ contains no nonzero Lie algebra of an algebraic subgroup, by Lemma 8.3 and $P_c/P_a \notin \overline{\mathbb{Q}}$. Here $m = \operatorname{rk} Y''$; p. 394 of [37] prints "$\kappa = \ell - m$", which must be read $m - \ell$ ($\ell = \operatorname{rk}\Gamma$, cf. pp. 392–393), as noted after Theorem 2.5.

*Relation to* [29, Théorème (iii)]. With the weights of §4.3, $\alpha_K(G) = \sum_A \dim A \geq 4$, with equality exactly when the four slots are all the first-kind slots of the varieties involved. In that case (levels 7 and 9 below) Corollary 2.2 with $\kappa = 2$ gives $\dim_K(V \cap \Psi(\Lambda_K(G))) \leq (4-2)\cdot 1 = 2$, so Proposition 8.1 already follows from it (Remark 11.1); otherwise (level 24 below) the printed weight on $G$ itself is larger, and Theorem 1.1 with the inner weight $\alpha_{K,\Psi} = 4$ is used (§4.4). But the level-24 quadruple consists of two of the four pairs of a full family of Theorem 8.15, so Theorem 8.2 at level 24 follows from that family by the slot-splitting step of §8.3 (the paragraph after the proof of Theorem 8.13).

8.3. **Proof of Theorem 8.2 from Proposition 8.1.** Enlarge $K$ to contain $q_1, q_2$ and the coefficients of the $v_j$, so that $v_j \in K\Lambda(A_j)$; the proposition holds for every such $K$. When several slots lie on one $A$, (Ch) gives a single element of $\Lambda_K(A)$ with the prescribed slot coordinates $z_j(v_j)$. So hypothesis (9) says that the point $\Psi(v_a, \dots, v_d, q_1, q_2)$ lies in $V \cap \Psi(\Lambda_K(G))$, and Proposition 8.1 gives $x, y \in K$ with

$$z_a(v_a - x\gamma_a) + z_d(v_d - y\gamma_d) + q_1 = 0,$$

and similarly in the second coordinate. *Slot splitting.* Write $v_a - x\gamma_a$ and $v_d - y\gamma_d$ as $K$-combinations of finitely many $u_1, \dots, u_m \in \Lambda(A)$ (for each isogeny class $A$ involved). The point $\xi = (1, (u_i)_i) \in \operatorname{Lie}(\mathbb{G}_{\mathrm{a}} \times \prod_A A^m)$ has $\exp\xi$ algebraic, and the displayed relation is a $\overline{\mathbb{Q}}$-linear form $\ell$ vanishing at $\xi$. By Wüstholz's theorem [52], the smallest algebraic subgroup $H$ whose Lie algebra contains $\xi$ has $T_H \subset \ker\ell$. By (P) and (S1), $H = L \times \prod_A B_{S_A}$, and $L = \mathbb{G}_{\mathrm{a}}$ because the $\mathbb{G}_{\mathrm{a}}$-coordinate of $\xi$ is 1; so $(1, 0) \in T_H$ and $q_1 = 0$. On $T(B_S)$ the slot forms involve independent vectors $u_\sigma$ (S1), and the slots $a$, $d$ are distinct; so each slot form vanishes on $T_H$ separately, and $z_a(v_a) = xP_a$, $z_d(v_d) = yP_d$. Likewise $q_2 = 0$, $z_c(v_c) \in KP_c$ and $z_b(v_b) \in KP_b$. *Torsion.* Let $v_j \in \Lambda(A_j)$ with $z_j(v_j) = x_jP_j$. If $x_j = 0$, Wüstholz on the simple $A_j$ gives $v_j = 0$. Otherwise the smallest algebraic subgroup of $A_j^2$ whose Lie algebra contains $(v_j, \gamma_j)$ is a proper abelian subvariety $B_S$ (its tangent space lies in $\ker(z_j \oplus -x_jz_j)$) that projects onto the second factor. So $\dim_F S = 1$ and $B_S$ is the graph of some $\alpha \in F_j = \operatorname{End}^0(A_j)$; hence $v_j \in F_j\gamma_j \subset \mathbb{Q}\,\Omega(A_j)$ and $\exp(v_j)$ is torsion. □

8.4. **The level-seven instance.** Let $C\colon y^7 = x(1-x)$. It has genus 3 (branch points $0, 1, \infty$), and $dx/y^k$ is holomorphic exactly for $k = 4, 5, 6$ (orders $6-k$ at 0 and 1, $2k-8$ at $\infty$). The automorphism $y \mapsto \zeta y$ ($\zeta = e^{2\pi i/7}$) multiplies $dx/y^6, dx/y^5, dx/y^4$ by $\zeta, \zeta^2, \zeta^3$, so $A = \operatorname{Jac}(C)$ has CM by $\mathbb{Q}(\zeta_7)$ with type $\{1, 2, 3\}$, which has trivial stabiliser in $(\mathbb{Z}/7)^\times$; so $A$ is simple. Let $z_1, z_2, z_3$ be the eigen-coordinates dual to $dx/y^6, dx/y^5, dx/y^4$. On the closed cycle $\gamma =$ (lift of $[0, 1]$ to the sheet $y > 0$) minus (its image under $y \mapsto \zeta y$),

$$\int_\gamma \frac{dx}{y^{7-j}} = (1 - \zeta^j)\, B(\tfrac{j}{7}, \tfrac{j}{7}) =: P_j \qquad (j = 1, 2, 3).$$

Let $E = E_{-7}$ be an elliptic curve over $\overline{\mathbb{Q}}$ with CM by $\mathbb{Q}(\sqrt{-7})$; by Chowla–Selberg [Lit] [18] it has a period $\Omega \in \overline{\mathbb{Q}}^{\times} B(\frac{1}{7},\frac{2}{7})$. The Beta cocycle $B(x,y)B(x+y,z) = B(y,z)B(x,y+z)$ at $(\frac{1}{7},\frac{1}{7},\frac{2}{7})$ and one reflection give, exactly,

$$B(\tfrac{1}{7},\tfrac{1}{7})B(\tfrac{2}{7},\tfrac{2}{7}) = B(\tfrac{1}{7},\tfrac{2}{7})B(\tfrac{1}{7},\tfrac{3}{7}), \qquad B(\tfrac{1}{7},\tfrac{3}{7}) = \mu\, B(\tfrac{3}{7},\tfrac{3}{7}),$$

$$\mu = \frac{\Gamma(\frac{1}{7})\Gamma(\frac{6}{7})}{\Gamma(\frac{3}{7})\Gamma(\frac{4}{7})} = \frac{\sin(3\pi/7)}{\sin(\pi/7)} = 1 + 2\cos\tfrac{2\pi}{7},$$

so $P_1P_2 \in \overline{\mathbb{Q}}^{\times} P_3\Omega$, a relation (8) with $a = z_1$, $b = z_2$, $c = z_3$, $d = E$. (In Koblitz–Ogus labels: $\{1,2,3\}+\{1,4,5\} = \{1,3,5\}+\{1,2,4\}$; the label $\{1,2,4\}$ is induced from $\mathbb{Q}(\sqrt{-7})$.) No use of Deligne's theorem is needed at this level.

**Theorem 8.6** (Theorem X7)**.** *Let $v_1, v_2, v_3 \in \Lambda(A)$, $u \in \mathcal{L}_E$, $q_1, q_2 \in \overline{\mathbb{Q}}$ and $\beta \in \overline{\mathbb{Q}}^{\times}$. If*

$$\big(z_1(v_1) + u + q_1\big) \cdot \beta P_3 = \big(z_3(v_3) + z_2(v_2) + q_2\big) \cdot P_1,$$

*then $q_1 = q_2 = 0$ and $\exp(v_1), \exp(v_2), \exp(v_3)$, $\exp_E(u)$ are torsion points.*

This is Theorem 8.2 (with $\Omega$ normalised so that $P_1P_2 = \mu' P_3\Omega$ and $\mu' \in K$); here $n$ is a multiple of 6 and the margin is $n$. Since $z_1, z_2, z_3$ are all the first-kind slots of $A$, Proposition 8.1 for these slots already follows from Corollary 2.2 ($\alpha_K(A \times E \times \mathbb{G}_{\mathrm{a}}^2) = 3+1$, $\kappa = 2$; Remark 8.5, Remark 11.1).
*Status:* [Proved modulo published theorems and Lemma A.1].

**Corollary 8.7.** (a) *For every non-torsion point $\exp(v) \in A(\overline{\mathbb{Q}})$: $z_2(v)/z_1(v) \notin \overline{\mathbb{Q}} \cdot P_3/P_1 = \overline{\mathbb{Q}} \cdot B(\frac{3}{7},\frac{3}{7})/B(\frac{1}{7},\frac{1}{7})$. In particular Corollary 1.12(a) holds.*

(b) *For every non-torsion $u \in \mathcal{L}_E$ and every $v \in \Lambda(A)$: $z_3(v)/u \notin \overline{\mathbb{Q}}^{\times} \cdot P_3/P_1$.*

(c) *$P_1P_2$, $P_1P_3$, $P_1$, $P_3$ are $\overline{\mathbb{Q}}$-linearly independent; equivalently, Corollary 1.12(b).*

*Status:* [Proved modulo published theorems and Lemma A.1] ((a) and (c) contain Corollary 1.12). That (c) is not a consequence of the analytic subgroup theorem is [Heuristic] (Remark 8.8).

*Proof.* (a) Data $v_1 = v_2 = v$, $v_3 = 0$, $u = 0$, $q_1 = q_2 = 0$, $\beta = \gamma$: if $z_2(v) = \gamma\, z_1(v) P_3/P_1$ then the hypothesis of Theorem 8.6 holds, so $\exp(v)$ is torsion. The values 0 and $\infty$ of the ratio do not occur for $v \neq 0$: $z_j(v) = 0$ forces $v = 0$ by Wüstholz on the simple $A$. For $O, Q \in C(\overline{\mathbb{Q}})$, $z_j$ of the Abel–Jacobi logarithm of $Q - O$ is $\int_O^Q$ of the dual differential. (b) Data $v_3 = v$, $u$, $\beta = \gamma \in \overline{\mathbb{Q}}^{\times}$, all others 0. (c) Substituting (8) into $\alpha_1P_1P_2 + \alpha_2P_1P_3 + \alpha_3P_1 + \alpha_4P_3 = 0$ gives $(\alpha_2P_1 + \alpha_1\mu'\Omega + \alpha_4)\cdot P_3 = (-\alpha_3)\cdot P_1$. So, with $\beta = 1$ and $K \ni \alpha_i$, $v = (\alpha_2P_1 + \alpha_1\mu'\Omega + \alpha_4,\ -\alpha_3) \in V \cap \Psi(\Lambda_K(G))$, and Proposition 8.1 gives $v = (xP_1 + y\Omega,\ xP_3 + yP_2/\mu')$ with $x, y \in K$. The $\overline{\mathbb{Q}}$-independence of $1, P_2, P_3$ (W) gives $x = y = \alpha_3 = 0$, and then that of $1, P_1, \Omega$ gives $\alpha_1 = \alpha_2 = \alpha_4 = 0$. Dividing the four numbers by $P_1$ gives $P_2, P_3, 1, P_3/P_1$, which are $B(\frac{2}{7},\frac{2}{7}), B(\frac{3}{7},\frac{3}{7}), 1, B(\frac{3}{7},\frac{3}{7})/B(\frac{1}{7},\frac{1}{7})$ up to factors in $\mathbb{Q}(\zeta_7)^{\times}$. The Gamma form follows from $B(\frac{1}{7},\frac{1}{7}) = \Gamma(\frac{1}{7})^2/\Gamma(\frac{2}{7})$ (the definition) and, by the reflection formula, $B(\frac{2}{7},\frac{2}{7}) = \Gamma(\frac{2}{7})^2\Gamma(\frac{3}{7})\sin(\frac{3\pi}{7})/\pi$ and $B(\frac{3}{7},\frac{3}{7}) = \Gamma(\frac{3}{7})^2\Gamma(\frac{1}{7})\sin(\frac{\pi}{7})/\pi$, after multiplying by $\pi$. □

*No hidden symmetry.* $C$ is hyperelliptic: $(2x-1)^2 = 1 - 4y^7$, and $x \mapsto 1 - x$ is the hyperelliptic involution, acting as $[-1]$ on $A$ and multiplying $z_1, z_2, z_3$ by $-1$ together. $\mathrm{Aut}(C) \cong \mathbb{Z}/14$ and $\mathrm{End}^0(A) = \mathbb{Q}(\zeta_7)$ acts diagonally, $z_j([x]v) = \sigma_j(x) z_j(v)$; so no symmetry links $z_1, z_2, z_3$ of one point in a way that could produce $P_3/P_1$ in (a).

8.5. **Remarks: linearisation, other relations, and $e^{\pi^2}$.**

**Remark 8.8** (Not linearisations)**.** [Heuristic] By the abelian analogue of Laurent's identity (internal note, refereed internally, not reproduced), each $\ell_j := \pi z_j(v)/P_j$ is a $\overline{\mathbb{Q}}$-combination of 1-periods; it linearises only products $P_{\bar\sigma} z_\sigma$ of a period with the logarithm of the conjugate slot on the same variety. Dividing (9) by $P_a P_c$ and using (8) gives

$$\ell_c - \beta\ell_a + \frac{P_d}{P_a}\,(\mu\ell_b - \beta\ell_d) + \pi\Big(\frac{q_2}{P_c} - \frac{\beta q_1}{P_a}\Big) = 0,$$

where the ratio $P_d/P_a$ of periods of two different slots multiplies a combination of 1-periods. Unless $\mu\ell_b = \beta\ell_d$ this is not a $\overline{\mathbb{Q}}$-linear relation among 1-periods, so the analytic subgroup theorem does not apply as in §7.2. For (c): $P_3/P_1 \in \overline{\mathbb{Q}}^{\times} P_3 P_{\bar 1}/\pi$ has Koblitz–Ogus function $\mathbf{1}_{\{1,3,5\}} - \mathbf{1}_{\{1,2,3\}}$, which takes three values, while every slot period and $\pi$ have functions with at most two values; under the Rohrlich–Lang (or period) conjecture, $P_3/P_1$ is not in the $\overline{\mathbb{Q}}$-span of 1-periods, and Wüstholz's theorem does not decide (c). Chudnovsky's theorem gives the algebraic independence of $\pi$ and $\Omega_{-7} \sim \Gamma(\frac17)\Gamma(\frac27)/\Gamma(\frac37)$, which does not suffice, since (a)–(c) involve $P_1, P_2, P_3$. By the period conjecture (the Mumford–Tate group of $A$ is a 4-dimensional torus and $\Omega$ adds no transcendence degree), Theorem 8.6 is a consequence of the period conjecture.

**Remark 8.9** (Where first-kind relations occur)**.** *Elliptic slots.* Among CM elliptic slots (any number of non-isogenous curves) the relations of Deligne's theorem are only the Legendre relations [Proved]: the indicators are $(1\pm\chi_k)/2$ with $\chi_k$ odd quadratic characters, which are linearly independent. (That there are no other relations is the Rohrlich–Lang conjecture, [Heuristic].) So for products of CM elliptic curves, Theorems 1.7 and 7.6 (two Legendre relations on a line) are the only constructions of this kind.

*Labels.* [Proved] (elementary combinatorics). For CM types of $\mathbb{Q}(\zeta_N)$ containing 1, a first-kind partner of $(\Phi_a, \Phi_b)$ exists iff $\Phi_a, \Phi_b$ differ on at least two $\pm$-pairs: with $\Delta = \Phi_a \triangle \Phi_b$, the partners are $(\Phi_a \cap \Phi_b) \cup X$ and $(\Phi_a \cap \Phi_b) \cup (\Delta \setminus X)$, $X$ one element from each $\pm$-pair in $\Delta$, $X \notin \{\Phi_a \setminus \Phi_b, \Phi_b \setminus \Phi_a\}$. Hence first-kind relations among labels exist at every level $N$ with $\varphi(N)/2 \geq 3$. The passage from labels to periods is Deligne's theorem [Lit] (§4.1 for Beta values; in general [12]).

*Examples.* The smallest levels are $N = 7$ and $N = 9$, with one relation each: at $N = 7$ the instance above; at $N = 9$ one with slots of a simple CM 3-fold with CM by $\mathbb{Q}(\zeta_9)$ and of $E_\rho$, $B(\frac19,\frac19)B(\frac19,\frac49) = 3^{1/3}\frac{\sin(2\pi/9)}{\sin(\pi/9)}B(\frac19,\frac29)B(\frac13,\frac13)$ ([Proved], Lemma 8.12, from Gauss's multiplication formula and the reflection formula). An exact enumeration of label relations (four pairwise distinct first-kind labels) gives 100, 720, 12, 12, 12, 720, 12 relations at $N = 11, 13, 15, 16, 20, 21, 24$. At $N = 24$ there is one whose slots are two CM elliptic curves (CM by $\mathbb{Q}(\sqrt{-6})$ and $\mathbb{Q}(i)$) and two slots of a simple CM 4-fold: $B(\frac{1}{24},\frac{1}{24})B(\frac{1}{24},\frac14) \in \overline{\mathbb{Q}}^{\times} B(\frac{1}{24},\frac{1}{12})B(\frac{1}{24},\frac18)$ ([Proved], Lemma 8.12). Both give instances of Theorem 8.2 (§8.6).

**Remark 8.10** (Several pairs)**.** With all CM types of $\mathbb{Q}(\zeta_N)$, slope classes containing $\kappa = 2^{g-2}$ parallel first-kind pairs occur ($g \geq 3$). The case check of Proposition 8.1 with $\kappa$ slots in each target has margin $n$ for every $\kappa$; this is now Theorem 8.15 (§8.6), which gives a degree-two independence statement for 1, the common slope and $\kappa$ periods.

**Remark 8.11** ($e^{\pi^2}$)**.** [Heuristic] The multiplicative group cannot enter a parallel pair: its only period is $2\pi i$, and a relation $p_a p_d = p_b p_c$ involving it would pair $2\pi i$ with 1, which is not a period (this is argued in §7.8). So we do not expect Theorem 8.2 to reach $e^{\pi^2}$.

8.6. **Levels 9 and 24, and several pairs.**

**Lemma 8.12.** (1) $B(\frac{1}{9},\frac{1}{9})\,B(\frac{1}{9},\frac{4}{9}) = 3^{1/3}\,\frac{\sin(2\pi/9)}{\sin(\pi/9)}\,B(\frac{1}{9},\frac{2}{9})\,B(\frac{1}{3},\frac{1}{3})$, *and* $B(\frac{1}{9},\frac{1}{9})\,B(\frac{4}{9},\frac{4}{9}) = 16\sin(\frac{\pi}{9})\sin^3(\frac{2\pi}{9})\,B(\frac{2}{9},\frac{2}{9})\,B(\frac{1}{3},\frac{1}{3})$.

(2) $B(\frac{1}{24},\frac{1}{24})\,B(\frac{1}{24},\frac{1}{4}) = \mu_{24}\,B(\frac{1}{24},\frac{1}{12})\,B(\frac{1}{24},\frac{1}{8})$, *where*

$$\mu_{24} = 2^{-1/12}\,3^{1/8}\,\frac{\sin(7\pi/24)\sin(9\pi/24)}{\sin(10\pi/24)\sin(6\pi/24)}.$$

*Status:* [Proved]. (Also checked numerically to 60 digits; $\mu_{24} = 1.16200378885386328607\ldots$)

*Proof.* We use Euler's reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$ and Gauss's multiplication formulas $\Gamma(z)\Gamma(z+\frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z)$ and $\Gamma(z)\Gamma(z+\frac{1}{3})\Gamma(z+\frac{2}{3}) = 2\pi\,3^{1/2-3z}\Gamma(3z)$.

(1) With $g_k = \Gamma(k/9)$ and $s_k = \sin(k\pi/9)$, the case $m=3$ at $z = 1/9, 2/9$ gives $g_1g_4g_7 = 2\pi\,3^{1/6}g_3$ and $g_2g_5g_8 = 2\pi\,3^{-1/6}g_6$, and reflection gives $g_1g_8 = \pi/s_1$, $g_2g_7 = \pi/s_2$. Hence $g_4 = 2\cdot 3^{1/6}s_2\,g_2g_3/g_1$ and $g_5 = 2\cdot 3^{-1/6}s_1\,g_1g_6/g_2$, and the first ratio equals $g_1^2g_4g_6/(g_2^2g_3g_5) = 3^{1/3}s_2/s_1$. The second identity follows from the first combined with $B(\frac{1}{9},\frac{2}{9})/B(\frac{2}{9},\frac{2}{9}) = 2\cdot 3^{1/6}s_2$ and $B(\frac{1}{9},\frac{4}{9})/B(\frac{4}{9},\frac{4}{9}) = s_4/s_1$ (consequences of the same formulas) and $8s_1s_2s_4 = \sqrt{3}$.

(2) With $g_k = \Gamma(k/24)$ the left side divided by $B(\frac{1}{24},\frac{1}{12})B(\frac{1}{24},\frac{1}{8})$ is $g_1g_4g_6/(g_2^2g_7)$. Write $R_k$ for reflection at $z = k/24$, $D_z$ for duplication at $z$ and $T_z$ for triplication at $z$. The combination $R_6 - R_7 - R_9 + R_{10} - D_{2/24} + D_{3/24} + T_{1/24} - T_{2/24}$ (exponents of the $g_k$ added) has exponent vector $e_1 = e_4 = e_6 = 1$, $e_2 = -2$, $e_7 = -1$, all others 0; its constants multiply to $\mu_{24}$ (the powers of $\pi$ cancel; the powers of 2 give $2^{-5/6+3/4} = 2^{-1/12}$, those of 3 give $3^{3/8-1/4} = 3^{1/8}$). □

*The slots* [Lit] (Koblitz–Ogus labels [19]; §4.1; Chowla–Selberg [18]). At level 9, $B(\frac{1}{9},\frac{1}{9})$, $B(\frac{2}{9},\frac{2}{9})$, $B(\frac{4}{9},\frac{4}{9})$ are, up to factors in $\mathbb{Q}(\zeta_9)^\times$, periods of three distinct first-kind eigen-differentials $dx/y^8$, $dx/y^7$, $dx/y^5$ of the curve $C_9\colon y^9 = x(1-x)$ (genus 4), whose Jacobian is isogenous to $A_9 \times E_\rho$ with $A_9$ a simple CM threefold (CM by $\mathbb{Q}(\zeta_9)$, type $\{1,2,4\}$), and $B(\frac{1}{3},\frac{1}{3}) \in \overline{\mathbb{Q}}^\times\Omega_\rho$ is the period of $dx/y^6$, which comes from $E_\rho$. At level 24, $B(\frac{1}{24},\frac{1}{24}) \in \overline{\mathbb{Q}}^\times\Omega_{-6}$ and $B(\frac{1}{24},\frac{1}{4}) \in \overline{\mathbb{Q}}^\times\varpi$ are periods of CM curves $E_{-6}$, $E_i$ with CM by $\mathbb{Q}(\sqrt{-6})$ and $\mathbb{Q}(i)$, and $B(\frac{1}{24},\frac{1}{12})$, $B(\frac{1}{24},\frac{1}{8})$ are, up to $\overline{\mathbb{Q}}^\times$, periods $P_c$, $P_d$ of two distinct first-kind eigen-coordinates $z_c, z_d$ of a simple CM fourfold $A_{24}$ (CM by $\mathbb{Q}(\zeta_{24})$). By Lemma 8.12, $P_1P_4 \in \overline{\mathbb{Q}}^\times P_2\Omega_\rho$ and $\Omega_{-6}\varpi \in \overline{\mathbb{Q}}^\times P_cP_d$, so Theorem 8.2 applies to both quadruples.

**Theorem 8.13** (Theorems X9 and X24)**.** (1) (Level 9.) *On $A_9$ let $z_1$, $z_2$, $z_4$ be the coordinates dual to the eigen-differentials $dx/y^8$, $dx/y^7$, $dx/y^5$, with periods $P_j \in \overline{\mathbb{Q}}^\times B(\frac{j}{9},\frac{j}{9})$. If $v_1, v_2, v_4 \in \Lambda(A_9)$, $u \in \mathcal{L}_{E_\rho}$, $q_1, q_2 \in \overline{\mathbb{Q}}$, $\beta \in \overline{\mathbb{Q}}^\times$ and $(z_1(v_1) + u + q_1)\beta P_2 = (z_2(v_2) + z_4(v_4) + q_2)P_1$, then $q_1 = q_2 = 0$ and all of $\exp(v_j)$, $\exp_{E_\rho}(u)$ are torsion.*

(2) (Level 24.) *If $u \in \mathcal{L}_{E_{-6}}$, $u' \in \mathcal{L}_{E_i}$, $v_c, v_d \in \Lambda(A_{24})$, $q_1, q_2 \in \overline{\mathbb{Q}}$, $\beta \in \overline{\mathbb{Q}}^\times$ and $(u + z_d(v_d) + q_1)\beta P_c = (z_c(v_c) + u' + q_2)\Omega_{-6}$, then $q_1 = q_2 = 0$ and all exponentials are torsion.*

*Status:* [Proved modulo published theorems and Lemma A.1]. Two internal referees (correct, with fixes applied).

**Corollary 8.14.** (1) *For every non-torsion point $\exp(v) \in A_9(\overline{\mathbb{Q}})$, the ratio $z_4(v)/z_1(v)$ is not in $\overline{\mathbb{Q}}\,B(\frac{2}{9},\frac{2}{9})/B(\frac{1}{9},\frac{1}{9})$.*

(2) *The numbers 1, $B(\frac{4}{9},\frac{4}{9})$, $B(\frac{2}{9},\frac{2}{9})$, $B(\frac{2}{9},\frac{2}{9})/B(\frac{1}{9},\frac{1}{9})$ are linearly independent over $\overline{\mathbb{Q}}$.*

(3) *For non-torsion $u \in \mathcal{L}_{E_{-6}}$ and every $u' \in \mathcal{L}_{E_i}$: $u'/u \notin \overline{\mathbb{Q}}^\times B(\frac{1}{24},\frac{1}{12})/B(\frac{1}{24},\frac{1}{24})$.*

(4) *The numbers 1, $\varpi$, $B(\frac{1}{24},\frac{1}{12})$, $B(\frac{1}{24},\frac{1}{12})/B(\frac{1}{24},\frac{1}{24})$ are linearly independent over $\overline{\mathbb{Q}}$.*

*Status:* [Proved modulo published theorems and Lemma A.1]. That (2) and (4) are not consequences of the analytic subgroup theorem is [Heuristic], as for Corollary 8.7(c) (Remark 8.8).

*Proof of Theorem 8.13 and Corollary 8.14.* As for Theorem 8.6 and Corollary 8.7, taking for $(a,b,c,d)$ the slots $(z_1,z_4,z_2,E_\rho)$, resp. $(E_{-6},E_i,z_c,z_d)$ in the relation (8) and in (9). The case analysis of Proposition 8.1 uses only that the four slots are distinct and of the first kind, the relation (8), and $\dim A''=2n$, which holds here ($3n/2+n/2$, resp. $n+n/2+n/2$, by Lemma 4.1). In (3), Theorem 8.13(2) makes $u$ torsion, so no condition on $u'$ is needed. □

At level 9 the slots $z_1,z_2,z_4$ are all the first-kind slots of $A_9$, so, as at level 7, Proposition 8.1 for them already follows from Corollary 2.2 ($\alpha_K=3+1$, $\kappa=2$; Remark 8.5). At level 24 only two of the four first-kind slots of $A_{24}$ are used ($\alpha_K=4+1+1$), so the printed weight on its own group gives the bound $(6-2)\cdot 1=4$ and no conclusion; Theorem 1.1 with the inner weight applies (§4.4). Moreover its two pairs are two of the four pairs of the full family $T(11)$ at $N=24$ of Theorem 8.15 below: in Koblitz–Ogus labels [19], $B(\frac{1}{24},\frac{1}{24})\leftrightarrow\{1,5,7,11\}\mapsto B(\frac{1}{24},\frac{1}{12})\leftrightarrow\{1,5,7,13\}$ ($E_{-6}\to A_{24}$) and $B(\frac{1}{24},\frac{1}{8})\leftrightarrow\{1,5,11,17\}\mapsto B(\frac{1}{24},\frac{1}{4})\leftrightarrow\{1,5,13,17\}$ ($A_{24}\to E_i$); the other two pairs are $\{1,7,11,19\}\mapsto\{1,7,13,19\}$ ($A_{24}$ to a curve with CM by $\mathbb{Q}(\sqrt{-3})$) and $\{1,11,17,19\}\mapsto\{1,13,17,19\}$ (a curve with CM by $\mathbb{Q}(\sqrt{-2})$ to $A_{24}$). So it follows from item (iv) of Remark 11.1 by the slot-splitting step of §8.3: (Ch) moves a point of $\Lambda_K$ of the level-24 group into $\Lambda_K$ of the group of $T(11)$ with the same $\Psi$-image (after rescaling the second target by a constant in $\overline{\mathbb{Q}}^\times$, which keeps it an instance), Theorem 8.15 applies, and Wüstholz's theorem splits the slots, which kills the coefficients of the two unused pairs. This is the same kind of input as the Wüstholz and Masser steps of item (iii) of Remark 11.1. Hence Theorem 8.13(2) and Corollary 8.14(3) also follow from the printed form [29, Théorème (iii)] (with Wüstholz's theorem, and Deligne's theorem for the common slope of $T(11)$ as in the proof of Theorem 8.15), and Corollary 8.14(4) is part of the independence statement of Theorem 8.15 for $T(11)$ (there $r\in\overline{\mathbb{Q}}^\times B(\frac{1}{24},\frac{1}{12})/B(\frac{1}{24},\frac{1}{24})$, and the $P_{\Phi'}$ include $B(\frac{1}{24},\frac{1}{12})$ and $B(\frac{1}{24},\frac{1}{4})\in\overline{\mathbb{Q}}^\times\varpi$). The same applies to every proper subfamily of Theorem 8.15. What is ours at level 24 is the elementary proof of the relation (Lemma 8.12(2)), which avoids Deligne's theorem.

**Theorem 8.15** (Theorem X$^\kappa$: several pairs)**.** *Let $N\geq 3$, $G_N=(\mathbb{Z}/N)^\times$ of order $2g$, $\tau\in G_N\setminus\{\pm1\}$, and let $T(\tau)$ be the set of CM types $\Phi\subset G_N$ with $1,\tau\in\Phi$ (so $\kappa:=|T(\tau)|=2^{g-2}$). For $\Phi\in T(\tau)$ put $\Phi'=(\Phi\setminus\{\tau\})\cup\{-\tau\}$, and let $P_\Phi$, $P_{\Phi'}$ be periods of the corresponding first-kind slots (the eigen-slot at the embedding 1 of a CM abelian variety with CM by $\mathbb{Q}(\zeta_N)$ of type $\Phi$, resp. $\Phi'$). Then the $2\kappa$ slots are distinct, there are $\beta_\Phi\in\overline{\mathbb{Q}}^\times$ and $r$ with $\beta_\Phi P_{\Phi'}=rP_\Phi$ for all $\Phi$, and with $G=\mathbb{G}_{\mathrm{a}}^2\times\prod A$ (each isogeny class once), $\Psi=(\sum_\Phi z_\Phi+q_1,\ \sum_\Phi\beta_\Phi z_{\Phi'}+q_2)$, $V=\mathbb{C}(1,r)$ and every number field $K\supseteq\mathbb{Q}(\zeta_N)$ containing the $\beta_\Phi$:*

$$V\cap\Psi(\Lambda_K(G))=\bigoplus_{\Phi\in T(\tau)}K\,P_\Phi(1,r).$$

*Consequently 1, $r$ and the $\kappa$ numbers $P_{\Phi'}$ are linearly independent over $\overline{\mathbb{Q}}$. The same holds for every subfamily of $T(\tau)$.*

*Status:* [Proved modulo published theorems and Lemma A.1]. Two internal referees (correct with fixes, applied; no error in the case check).

*Proof. Distinctness.* $\Phi\ni\tau$, $\Phi'\not\ni\tau$, and $\Phi\mapsto\Phi'$ is injective, so the $2\kappa$ labels are distinct; distinct labels give distinct slots, since $A_{\Phi_2}\sim A_\Phi$ forces $\Phi_2=\Phi u$ and then the 1-eigenline of $A_{\Phi_2}$ is the $u^{\mp1}$-eigenline of $A_\Phi$, another embedding unless $u\in\operatorname{Stab}\Phi$. *Common slope.* $\mathbf{1}_{\Phi'}-\mathbf{1}_\Phi=\delta_{-\tau}-\delta_\tau$ for all $\Phi$, so $P_{\Phi_1'}P_{\Phi_2}\in\overline{\mathbb{Q}}^\times P_{\Phi_1}P_{\Phi_2'}$: for Fermat labels by Deligne's theorem [12] in the form of [19] (§4.1), and in general by the linearity in $\Phi$ of Shimura's period symbol, proved by Blasius [9] (via Deligne's theorem on absolute Hodge cycles) and, for abelian CM fields, by Anderson [2]. *The count.* The proof of Proposition 8.1 goes through with $\kappa$ pairs: $G''=\mathbb{G}_{\mathrm{a}}^2\times A''$ with $\dim A''=\kappa n$

(Lemma 4.1); the periods contribute $\kappa n$ to $\mathrm{rk}(Y'' \cap \ker \exp)$; one extra datum gives $m = (\kappa+1)n$; every slot occurs in one target only, and distinct slots (several on one $A$ are distinct embeddings) carry independent coordinates on $T_B$ (S1), so $\Psi''(T_B) \in \{0, \mathbb{C}(1,0), \mathbb{C}(0,1), \mathbb{C}^2\}$; and for an axis image $\ell$, $V \cap (\ell + \Psi(\Omega_K(G))) = \bigoplus_\Phi KP_\Phi(1,r)$ (as in Lemma 8.4), whence $\lambda = n$. Condition (1) then reads $n + 2\kappa n \le 2\kappa n$ for $G' = 0$, $n + 2(\kappa n - e) \le \kappa n$ with $e \le \kappa n/2$ for axis images, and $\lambda + 2\kappa n \le \kappa n$ for a non-axis line in $\mathbb{G}_{\mathrm{a}}^2$: never (margin $n$). Condition (2): $2\kappa n + 2 \ge \kappa n + 4$. *Independence.* A relation $a_0 + a_1 r + \sum \gamma_\Phi P_{\Phi'} = 0$ ($K$ containing the coefficients) gives the point $(a_1, -a_0 - \sum \gamma_\Phi P_{\Phi'}) = a_1(1,r)$ of $V \cap \Psi(\Lambda_K(G))$, whose first coordinate must lie in $\sum KP_\Phi$; so $a_1 = 0$ by (W), and then $\gamma = 0$, $a_0 = 0$ by (W). □

**Remark 8.16.** For one pair ($\kappa = 1$, i.e. $g = 2$, or a one-pair subfamily) the independence statement follows from Wüstholz's theorem alone (multiply by $\pi/P_{\Phi'}$, which makes every member a 1-period); for $\kappa \ge 2$ no common Gamma-monomial multiplier does this, and we do not know a linear proof. (That there is none is [Heuristic], as in Remark 8.8.) For the whole family $T(\tau)$ the theorem already follows from the printed form [29, Théorème (iii)] [Proved]: every CM type contains exactly one of $\tau, -\tau$, so the $2\kappa$ types $\Phi, \Phi'$ are all the CM types of $\mathbb{Q}(\zeta_N)$ containing 1, and the $2\kappa$ slots are all the first-kind slots of the simple varieties involved; hence $\alpha_K(G) = \sum_A \dim A = 2\kappa$, and Corollary 2.2 gives $\dim_K \le (2\kappa - \kappa) \cdot 1 = \kappa$ (§11.3). This covers the example below. For a proper subfamily the printed weight on its own group can be larger, but the subfamily follows from its full family $T(\tau)$ by the slot-splitting step of §8.3, and its independence statement is part of that of the full family (as at level 24, after the proof of Theorem 8.13).

**Example 8.17.** $N = 15$, $\tau = 2$: the four pairs $B(\frac{1}{15}, \frac{2}{15}) \mapsto B(\frac{4}{15}, \frac{4}{15})$, $B(\frac{1}{15}, \frac{3}{15}) \mapsto B(\frac{1}{15}, \frac{7}{15})$, $B(\frac{2}{15}, \frac{2}{15}) \mapsto B(\frac{2}{15}, \frac{6}{15})$, $B(\frac{1}{15}, \frac{1}{15}) \mapsto B(\frac{1}{3}, \frac{1}{3})$ form the whole family $T(2)$ ($\kappa = 4$; their common slope class was also checked exactly), so $1$, $B(\frac{4}{15}, \frac{4}{15})/B(\frac{1}{15}, \frac{2}{15})$, $B(\frac{4}{15}, \frac{4}{15})$, $B(\frac{1}{15}, \frac{7}{15})$, $B(\frac{2}{15}, \frac{6}{15})$, $B(\frac{1}{3}, \frac{1}{3})$ are linearly independent over $\overline{\mathbb{Q}}$.

## 9. Corollaries

### 9.1. **Corollaries MG1 and MG3.**

**Corollary 9.1** (MG1)**.** *For every CM curve $E$, every period $\Omega$ and all $a, b, c \in \overline{\mathbb{Q}}^\times$, at least one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\pi^2/\Omega}$ is transcendental. On $E_i$: one of $e^{\pi^2}$, $e^{\pi\varpi}$, $e^{\pi\eta(\varpi)}$ is transcendental.*

*Status:* [Proved modulo Wüstholz's theorem in the form $\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}}$; formally verified otherwise] for $\Omega \in \overline{\mathbb{Q}}^\times \varpi$ (every $a, b, c \in \overline{\mathbb{Q}}^\times$); [Proved modulo published theorems and Lemma A.1] for every other CM period.

*Proof.* Take $s = \Omega/\pi$ (up to $\overline{\mathbb{Q}}^\times$), $x_1 = \pi^2$, $x_2 = \pi^2/\Omega$: if all three exponentials are algebraic then $x_1, x_2 \in \overline{\mathbb{Q}}\mathcal{L}$ and $sx_1 = \pi\Omega$, $sx_2 = \pi$ lie in $\overline{\mathbb{Q}}\mathcal{L}$. The numbers $2\pi i, \pi^2, \pi^2/\Omega$ are $\overline{\mathbb{Q}}$-independent. A relation without the $\pi^2/\Omega$ term contradicts the transcendence of $\pi$. Otherwise $\pi/\Omega = a_1 + b_1\pi$ with $a_1, b_1 \in \overline{\mathbb{Q}}$. If $b_1 = 0$ then $\Omega/\pi \in \overline{\mathbb{Q}}$, which is false. If $b_1 \ne 0$ then $\pi\Omega = (\pi - a_1\Omega)/b_1$; since $\pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$ (as $e^{b\pi\Omega} \in \overline{\mathbb{Q}}$) and $\pi \in \overline{\mathbb{Q}}\mathcal{L}$, this gives $a_1\Omega \in \overline{\mathbb{Q}}\mathcal{L}$, so $a_1 = 0$ because $\Omega \notin \overline{\mathbb{Q}}\mathcal{L}$ (proof of Corollary 5.2); then $\Omega = 1/b_1 \in \overline{\mathbb{Q}}$, which is false. (Chudnovsky's algebraic independence of $\pi$ and $\Omega$ [11] would give this at once, but is not needed.) On $E_i$, $\pi/\varpi = \eta(\varpi)$. □

For a general CM curve, $\eta(\Omega) = \kappa_1\Omega + \kappa_2\pi/\Omega$ with $\kappa_1, \kappa_2 \in \overline{\mathbb{Q}}$ and $\kappa_2 \ne 0$ (Legendre); $\kappa_1 = 0$ when $j(E) \in \{0, 1728\}$. Since $\pi\Omega, \pi\eta \in \overline{\mathbb{Q}}\mathcal{L}$ imply $\pi^2/\Omega = (\pi\eta - \kappa_1\pi\Omega)/\kappa_2 \in \overline{\mathbb{Q}}\mathcal{L}$, the statement "one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\pi\eta}$ is transcendental" holds for every CM period as well.

**Corollary 9.2** (MG3)**.** *If $e^{\pi^2}$ and $e^{c\pi\Omega}$ ($c \in \overline{\mathbb{Q}}^\times$) are algebraic, then $\alpha^{\Omega/\pi} \notin \overline{\mathbb{Q}}$, for every determination, for every $\alpha \in \overline{\mathbb{Q}}^\times$ such that $2\pi i, \pi^2, \log\alpha$ are linearly independent over $\mathbb{Q}$.*

*Status:* [Proved modulo Baker's theorem; formally verified otherwise] for $\Omega = \varpi$ (the exponent $\varpi/\pi$); [Proved modulo published theorems and Lemma A.1] for every other CM period, including the other elements of $\overline{\mathbb{Q}}^\times\varpi$ (the constant cannot be absorbed into $\log\alpha$).

*Proof.* With $s = \Omega/\pi$, take $x_1 = \pi^2$ (then $sx_1 = \pi\Omega \in \overline{\mathbb{Q}}\mathcal{L}$) and $x_2 = \log\alpha$ (then $sx_2 \in \mathcal{L}$ if $\alpha^{\Omega/\pi} \in \overline{\mathbb{Q}}$). These are $\mathbb{Q}$-independent logarithms together with $2\pi i$, hence $K$-independent, and Theorem 5.1 gives a contradiction. The hypothesis is needed: for $c = 1$ and $\alpha = e^{\pi^2}$ ($\log\alpha = \pi^2$) one has $\alpha^{\Omega/\pi} = e^{\pi\Omega} \in \overline{\mathbb{Q}}$. (Corollary 9.2 is also the case $\beta = e^{\pi^2}$ of the second sentence of Corollary 5.2.) □

9.2. **Corollary A2.**

**Corollary 9.3.** *Let $P = P_\sigma$, $\sigma \in \Phi$, be a first-kind eigen-period and $a, b, c \in \overline{\mathbb{Q}}^\times$. At least one of $e^{a\pi^2}$, $e^{b\pi P}$, $e^{c\pi^2/P}$ is transcendental. Since $\pi^2/P \in \overline{\mathbb{Q}}^\times\pi P_{\bar\sigma}$, this is "one of $e^{\pi^2}$, $e^{\pi P_\sigma}$, $e^{\pi P_{\bar\sigma}}$", up to algebraic factors in the exponents.*

*Status:* [Proved modulo published theorems and Lemma A.1]; for $P = (1-\zeta_5)B(\frac{1}{5},\frac{1}{5})$: [Proved modulo Wüstholz's theorem in the form $P \notin \widetilde{\mathcal{L}}$; formally verified otherwise].

*Proof.* Suppose all three are algebraic. Take $s = P/\pi$ (Theorem 5.7 with its constant equal to 1), $x_1 = \pi^2$, $x_2 = \pi^2/P$: then $x_1, x_2 \in \overline{\mathbb{Q}}\mathcal{L}$, $sx_1 = \pi P \in \overline{\mathbb{Q}}\mathcal{L}$, $sx_2 = \pi \in \overline{\mathbb{Q}}\mathcal{L}$. It remains to show that $2\pi i, \pi^2, \pi^2/P$ are $\overline{\mathbb{Q}}$-independent. A relation without the $\pi^2/P$ term contradicts the transcendence of $\pi$. Otherwise $\pi/P = a_1 + b_1\pi$ with $a_1, b_1 \in \overline{\mathbb{Q}}$. If $b_1 = 0$ then $P/\pi \in \overline{\mathbb{Q}}$, contradicting (W). If $b_1 \neq 0$ then $P(a' + \pi) = c''\pi$ with $a', c'' \in \overline{\mathbb{Q}}$, so $\pi P = c''\pi - a'P$. Since $\pi P \in \overline{\mathbb{Q}}\mathcal{L}$ and $\pi \in \overline{\mathbb{Q}}\mathcal{L}$, this gives $a'P \in \overline{\mathbb{Q}}\mathcal{L}$, so $a' = 0$ by (W), and then $P = c'' \in \overline{\mathbb{Q}}$, again contradicting (W). □

**Remark 9.4.** In Corollary 9.1 (one of $e^{\pi^2}$, $e^{\pi\Omega}$, $e^{\pi^2/\Omega}$ for a CM elliptic period $\Omega$), the $K$-independence of $2\pi i, \pi^2, \pi^2/\Omega$ follows at once from Chudnovsky's algebraic independence of $\pi$ and $\Omega$ [11], which is not available for abelian periods. The argument of Corollary 9.3, with $g = 1$, replaces it by the analytic subgroup theorem under the hypothesis $e^{\pi\Omega} \in \overline{\mathbb{Q}}$; this is the proof given for Corollary 9.1 above.

9.3. **Two degree-two period statements.** Propositions 7.1 and 7.5 also apply to data that are periods plus algebraic constants; their single-point forms suffice. This gives statements about products of periods of the two curves.

**Corollary 9.5.** (i) *The numbers $\pi$, $\varpi$, $\Omega_\rho$, $\varpi\Omega_\rho$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are $\pi, \eta_i, \eta_\rho, \eta_i\eta_\rho$.*

(ii) *The numbers $1$, $\varpi$, $\Omega_\rho$, $\varpi\Omega_\rho/\pi$ are linearly independent over $\overline{\mathbb{Q}}$; equivalently, so are $\pi, \varpi, \eta_\rho, \varpi\eta_\rho$.*

*Status:* (i) and (ii): [Proved modulo Wüstholz's theorem in the form that $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent, and the algebraicity of $g_3(\Omega_\rho\mathbb{Z}[\rho])$; formally verified otherwise]. Part (i) is proved from Proposition 7.5, part (ii) from Proposition 7.1, in the paper and in Lean alike; neither uses Theorem 7.6.

*Proof.* The equivalences follow by multiplying a relation by $\eta_i\eta_\rho/\pi$, resp. by $\eta_\rho$, and using $\varpi\eta_i = \pi$, $\Omega_\rho\eta_\rho = 2\pi/\sqrt{3}$. We use that $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are linearly independent over $\overline{\mathbb{Q}}$ (analytic subgroup theorem [52] on $E_i^\natural \times E_\rho^\natural \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$).

(i) A relation with zero coefficient on $\varpi\Omega_\rho$ is excluded by that independence. A relation $c_1\varpi + c_2\Omega_\rho + c_3\varpi\Omega_\rho = 0$ with zero coefficient on $\pi$ is excluded too: multiplying it by $\pi/(\varpi\Omega_\rho)$ and using $\pi/\varpi = \eta_i$, $\pi/\Omega_\rho = \frac{\sqrt{3}}{2}\eta_\rho$ gives $c_3\pi + c_2\eta_i + \frac{\sqrt{3}}{2}c_1\eta_\rho = 0$, so $c_1 = c_2 = c_3 = 0$. So suppose $\pi = A\varpi\Omega_\rho + B\varpi + C\Omega_\rho$ with $A \neq 0$. In the set-up of §7.6 take $\beta = A$, $X = (\varpi, \eta_i) \in \Lambda(E_i^\natural)$, $Y = 0$, $a = C/A$, $b = -B$. The point $v = \Psi'(X, Y, a, b) = (\varpi + C/A,\ \eta_i - B)$ of $\Psi'(\Lambda(G))$ lies on $V' = \mathbb{C}(\varpi, A\Omega_\rho)$: $(\varpi + C/A)\, A\Omega_\rho = \pi - B\varpi = (\eta_i - B)\varpi$. By Proposition 7.5, $v = x(\varpi, A\Omega_\rho) + x'(\eta_\rho, \frac{2A}{\sqrt{3}}\eta_i)$ with $x, x' \in \overline{\mathbb{Q}}$. The first coordinate, $\varpi + C/A = x\varpi + x'\eta_\rho$, and the independence of $1, \varpi, \eta_\rho$ give $C = 0$, $x = 1$, $x' = 0$. Then $\eta_i - B = A\Omega_\rho$, which contradicts the independence of $1, \eta_i, \Omega_\rho$.

(ii) Suppose $\varpi\Omega_\rho/\pi = a + b\varpi + c\Omega_\rho$ (a relation with zero left-hand coefficient is excluded by the independence above). If $c = 0$, dividing by $\varpi$ and using $\pi/\varpi = \eta_i$, $\Omega_\rho = 2\pi/(\sqrt{3}\eta_\rho)$ gives $\Omega_\rho = a\eta_i + b\pi$. If $b = 0$, multiplying by $\eta_\rho$ gives $\frac{2}{\sqrt{3}}\varpi = a\eta_\rho + \frac{2c}{\sqrt{3}}\pi$. Both contradict the independence above, so $b, c \neq 0$. Put $t = -1/c$, $\beta = -\sqrt{3}\, b/(2c)$, $a_1 = a/b$, and take $X = (\varpi, \eta_i + t) \in \Lambda(E_i^\natural)$, $Y = 0$. Then $v = (\varpi + a_1, \eta_i + t) \in \Psi(\Lambda(G))$ lies on $V = \mathbb{C}(\varpi, \beta\eta_\rho)$: expanding $(\varpi + a_1)\beta\eta_\rho = (\eta_i + t)\varpi$ with $\varpi\eta_i = \pi$ and $\eta_\rho = 2\pi/(\sqrt{3}\Omega_\rho)$ gives exactly the assumed relation. Proposition 7.1 gives $v = x(\varpi, \beta\eta_\rho) + x'(\Omega_\rho, \frac{2\beta}{\sqrt{3}}\eta_i)$ with $x, x' \in \overline{\mathbb{Q}}$. The first coordinate and the independence of $1, \varpi, \Omega_\rho$ give $a_1 = 0$, $x = 1$, $x' = 0$; then $\eta_i + t = \beta\eta_\rho$, contradicting the independence of $1, \eta_i, \eta_\rho$. □

[Heuristic] Both statements are genuinely of degree two: rewriting with the Legendre relations always leaves a product of periods of the two different curves, so under the period conjecture they are not consequences of the analytic subgroup theorem. They are predicted by the period conjecture (by Chowla–Selberg [18] they concern $\pi$, $\Gamma(1/4)$, $\Gamma(1/3)$, whose algebraic independence is open), and they do not follow from known algebraic independence results. They have the shape of a nonvanishing $2 \times 2$ determinant of periods, like the theorem of Barré-Sirieix, Diaz, Gramain and Philibert [5] (see also [45]). We found them nowhere in print, but they are consequences of Theorem 1.1, already of its printed form [29, Théorème (iii)] (through Propositions 7.1 and 7.5, Remark 11.1), and Wüstholz's theorem; they are among the claims flagged for an expert check in the introduction and in §12.2. Neither statement contains the constant 1 together with $\varpi\Omega_\rho$, so they do not give $\varpi\Omega_\rho \notin \overline{\mathbb{Q}}$, a case of a conjecture of Waldschmidt that we do not claim (§11.3).

9.4. **The affine version and strong four exponentials.** Adding one factor $\mathbb{G}_{\mathrm{a}}$ to each coordinate of $\Psi$ (so $G = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2 \times E$ with coordinates $(a, a', x, y, z)$ and $\Psi(a, a', x, y, z) = (x + a,\ y + z + a')$) gives the following statement. In the quotient $G'' = \mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ one has $d_0 = 2$, which does not enter the left-hand side of Theorem 2.3. The only new subgroups are $\mathbb{G}_{\mathrm{a}}$-lines $\{(\mu a, a)\}$ with $\mu \notin K$, which are excluded with margin $n$. So the count is the same as for Theorem 5.1; Lemma A.1 applies since $5n + 2 \geq 3n + 4$. This was checked at proof-sketch level by both internal referees of Theorem 5.1 and by a third; the subgroup list is not written out here. The base change in $K$ now uses Baker's inhomogeneous theorem.

**Theorem 9.6** (Affine MG; proof sketch)**.** *With $E$, $\Omega$, $c$, $s = c\Omega/\pi$ and $K$ as in Theorem 5.1,*

$$\dim_K\{x \in K + K\mathcal{L} :\ s\,x \in K + K\mathcal{L} + K\mathcal{L}_E\} \leq 2.$$

**Corollary 9.7** (proof sketch)**.** (i) *For $a, b, c \in \overline{\mathbb{Q}}^\times$, one of $e^{a\pi^2}$, $e^{b\pi\Omega}$, $e^{c\Omega/\pi}$ is transcendental.*

(ii) *For $\alpha \in \overline{\mathbb{Q}}^\times$ not a root of unity, $e^{\Omega/\pi}$ and $\alpha^{\Omega/\pi}$ (any determination) are not both algebraic.*

(iii) *(Strong four exponentials at ratio $t \in \overline{\mathbb{Q}}^\times\Omega/\pi$.) If $y_1, y_2$ are linearly independent over $\overline{\mathbb{Q}}$, then $y_1$, $y_2$, $ty_1$, $ty_2$ are not all in $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. By $y \mapsto ty$ the same holds at $t \in \overline{\mathbb{Q}}^\times\pi/\Omega$. (The strong statement is invariant under $t \mapsto ct$, $c \in \overline{\mathbb{Q}}^\times$, and $\overline{\mathbb{Q}}^\times\Omega/\pi$ contains a real number. So this is a statement at real $t$, where the $\widetilde{\mathcal{L}}$-version of the conjugation argument* [46, Thm. 4.1] *applies when $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-independent. The content not reached by conjugation is the remaining $y$; the count itself already follows from* [29, Théorème (iii)]*, Remark 11.1.)*

*Sketch.* (i) Take $x_1 = 1$ ($s \cdot 1 \in \overline{\mathbb{Q}}\mathcal{L}$ if $e^{c\Omega/\pi}$ is algebraic, with $s$ adjusted by a constant) and $x_2 = \pi^2$. With $2\pi i$ these are $K$-independent, because $1, \pi, \pi^2$ are $\overline{\mathbb{Q}}$-independent (Lindemann). (ii) Take $x_1 = 1$ and $x_2 = \log\alpha$; then $1, 2\pi i, \log\alpha$ are $K$-independent by Baker's theorem. (iii) Take $K$ containing all the coefficients. Then $y_1, y_2 \in T_K := \{x \in K+K\mathcal{L} :\ sx \in K+K\mathcal{L}+K\mathcal{L}_E\}$, which has dimension $\le 2$ and contains $2\pi i$. Since $y_1, y_2$ are $K$-independent, $2\pi i \in Ky_1 + Ky_2$. Then $2ic\Omega = s \cdot 2\pi i \in \widetilde{\mathcal{L}}$, which contradicts the analytic subgroup theorem ($\Omega \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$). □

The subgroup list for the count of Theorem 9.6 is written out in §6 in the closely related form of Theorem 6.2(a) (with $\overline{\mathbb{Q}}$-independent data and $K$ chosen to contain all coefficients, so that no base change in $K$ is needed). Accordingly Corollary 9.7(i) is contained in the first triple of Corollary 6.5(ii) ($\pi^2, \pi\Omega, \Omega/\pi$ are not all in $\widetilde{\mathcal{L}}$), (ii) follows from Theorem 1.6 with $(y_1, y_2) = (1, \log\alpha)$ at $t = \Omega/\pi$, and (iii) is Theorem 1.6 at $t \in \overline{\mathbb{Q}}^\times(\Omega/\pi)^{\pm 1}$; these are proved in §6, with the labels stated there. The $K$-dimension form of Theorem 9.6 itself remains a proof sketch.

## 10. Four exponentials on $E \times \mathbb{G}_{\mathrm{m}}$, and applications in codimension $\ge 2$

This section adds four results. Theorem 10.1 is a consequence of the hyperplane case of Theorem 1.1, like the results of §§5–9. Theorems 10.4 and 10.5, and Theorem 10.8 except its part (U2) for $r = 2$ (that is, (U1), (U3), (U4), and (U2) for $r \ge 3$), live on subspaces $V''$ of codimension $\ge 2$, where Theorem 2.3 applied subgroup by subgroup is stronger than the uniform bound of Theorem 1.1: at $G' = 0$, (1) divides the threshold by $d - N$ ([37, (4.2)]; the end of the proof of Proposition 3.11). Theorem 1.1, applied in the configurations we use, gives no conclusion or a weaker one (bound 3 against 2 data for Theorems 10.4 and 10.5, $k \le g - 2$ for (U1), and condition (i) fails when $K$ contains the imaginary quadratic field of Theorem 10.8), and we know no derivation of these results from Theorem 1.1; we prove them as applications of Theorem 2.3 completed by Lemma A.1. The subgroup-by-subgroup count is classical (the six exponentials theorem is a line in $\mathbb{G}_{\mathrm{m}}^3$, where Theorem 1.1 gives only rank $\le 3$). Part (U2) of Theorem 10.8 for $r = 2$ is the hyperplane case of Theorem 1.1 at $K = \mathbb{Q}$. Each result was checked by two internal referees (correct with fixes, applied) and by exact enumeration of its case table; none is formalised. Notation: $E/\overline{\mathbb{Q}}$ has CM by $k$, $\Omega \ne 0$ is a period, $\eta^*$ the eigen quasi-period (§4.2), so $\eta^*(\Omega) \in \overline{\mathbb{Q}}^\times\pi/\Omega$; $\mathrm{Fib}_E$ is the set of fibre values; $\widetilde{\mathcal{L}} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}$. A CM curve $E'$ "not isogenous to $E$" is defined over $\overline{\mathbb{Q}}$ with CM field $k' \ne k$, and $\Omega'$ is a nonzero period of $E'$.

10.1. **Conjecture 42 at CM ratios.** Waldschmidt's Conjecture 42 [45, p. 173] asks that a rank-one matrix $\left(\begin{smallmatrix} u_1 & \log\alpha_1 \\ u_2 & \log\alpha_2 \end{smallmatrix}\right)$ with $u_j \in \mathcal{L}_E$, $\log\alpha_j \in \mathcal{L}$ have $\mathbb{Q}$-proportional rows; equivalently, the middle inequality $\delta(e^z, \wp(tz)) \le 1$ of Ramachandra's $\delta$-conjecture in the form of [47, §2(a)]. Theorem 1.10 is its CM case with a period row.

**Theorem 10.1** (Theorem C42)**.** *Let $\rho \in R_E := \overline{\mathbb{Q}}^{\times} \cdot \{\Omega,\ \Omega',\ \pi/\Omega',\ \Omega'/\Omega,\ \pi/\Omega^2,\ \pi/(\Omega\Omega')\}$. If $u_1, u_2 \in \mathcal{L}_E \setminus \{0\}$ satisfy $\rho u_1, \rho u_2 \in \widetilde{\mathcal{L}}$, then $u_2/u_1 \in k$. If moreover $\rho u_j \in \mathcal{L}$ $(j = 1, 2)$, then $u_2/u_1 \in \mathbb{Q}$.*

*Status:* [Proved modulo published theorems and Lemma A.1].

**Corollary 10.2.** (a) *Conjecture 42 of* [45] *holds for every rank-one matrix whose column ratio $\log \alpha_j/u_j$ lies in $R_E$.*

(b) *If $\log\alpha, \log\beta$ are linearly independent over $\mathbb{Q}$ and*

$$t \in \overline{\mathbb{Q}}^{\times} \cdot \{1/\Omega,\ 1/\Omega',\ \Omega'/\pi,\ \Omega/\Omega',\ \Omega^2/\pi,\ \Omega\Omega'/\pi\},$$

*then $\wp_E(t\log\alpha)$ and $\wp_E(t\log\beta)$ are not both algebraic (a pole counts as algebraic). For instance $\wp_i(\log 2/\varpi)$ and $\wp_i(\log 3/\varpi)$ are not both algebraic, nor are $\wp_i(\Omega_\rho \log 2/\pi)$ and $\wp_i(\Omega_\rho \log 3/\pi)$.*

(c) *For $\rho \in R_E$ and $k$-independent $u_1, u_2 \in \mathcal{L}_E$, $\rho u_1$ and $\rho u_2$ are not both in $\widetilde{\mathcal{L}}$; e.g. $e^{\varpi u_1}$ and $e^{\varpi u_2}$ are not both algebraic for $\mathbb{Q}(i)$-independent $u_1, u_2 \in \mathcal{L}_{E_i}$, nor are $e^{\pi u_j/\varpi^2}$, nor $e^{\Omega_\rho u_j/\varpi}$.*

*Status:* [Proved modulo published theorems and Lemma A.1].

*Proof of Corollary 10.2.* (a) and (c) are restatements. For (b) put $\rho = 1/t$ and $u_j = t\log\alpha_j$; if both values are algebraic then $u_j \in \mathcal{L}_E$ and $\rho u_j \in \mathcal{L}$, so $\log\beta/\log\alpha = u_2/u_1 \in \mathbb{Q}$. □

*The line form.* Each class of $\rho$ is a cell $\sigma_1 \,|\, \sigma_2 + \sigma_3$: a factor $\sigma_1$ read into the first coordinate of $\mathbb{C}^2$, two factors $\sigma_2, \sigma_3$ read into the second, a slope $r = c\,p_2/p_1$ with $p_i$ the period of $\sigma_i$, and the set

$$\mathcal{S}_K = \{\lambda \in \Lambda_K(\sigma_1) :\ r\lambda \in \Lambda_K(\sigma_2) + \Lambda_K(\sigma_3)\},$$

where $\Lambda_K(\mathbb{G}_{\mathrm{m}}) = K\mathcal{L}$, $\Lambda_K(E) = K\mathcal{L}_E$ and $\Lambda_K(\text{fibre}) = K \cdot \text{Fib}$.

| row | $\sigma_1$ | $\sigma_2$ | $\sigma_3$ | slope $r$ | $p_1 \mapsto rp_1$ |
|---|---|---|---|---|---|
| (i) | $\mathbb{G}_{\mathrm{m}}$ | fibre of $E$ | $E$ | $c/\Omega$ | $2\pi i \mapsto \overline{\mathbb{Q}}^{\times}\eta^*(\Omega)$ |
| (ii) | $\mathbb{G}_{\mathrm{m}}$ | fibre of $E'$ | $E$ | $c/\Omega'$ | $2\pi i \mapsto \overline{\mathbb{Q}}^{\times}\eta^{*\prime}(\Omega')$ |
| (iii) | $\mathbb{G}_{\mathrm{m}}$ | $E'$ | $E$ | $c\Omega'/\pi$ | $2\pi i \mapsto 2ic\Omega'$ |
| (iv) | $E$ | $E'$ | $\mathbb{G}_{\mathrm{m}}$ | $c\Omega'/\Omega$ | $\Omega \mapsto c\Omega'$ |
| (v) | $E$ | fibre of $E$ | $\mathbb{G}_{\mathrm{m}}$ | $c\pi/\Omega^2$ | $\Omega \mapsto \overline{\mathbb{Q}}^{\times}\eta^*(\Omega)$ |
| (vi) | $E$ | fibre of $E'$ | $\mathbb{G}_{\mathrm{m}}$ | $c\pi/(\Omega\Omega')$ | $\Omega \mapsto \overline{\mathbb{Q}}^{\times}\eta^{*\prime}(\Omega')$ |

with $\rho = 1/r$ in rows (i)–(iii) and $\rho = r$ in rows (iv)–(vi).

**Proposition 10.3** (Line form)**.** *In each row, for every number field $K \supseteq kk'(i, c)$ with $[K : \mathbb{Q}] \geq 4$, $\dim_K \mathcal{S}_K \leq 2$, and $p_1 \in \mathcal{S}_K$. The same holds with a factor $\mathbb{G}_{\mathrm{a}}$ added to the coordinate that carries the ordinary logarithms (target 1 in rows (i)–(iii), target 2 in rows (iv)–(vi)), i.e. with $K\mathcal{L}$ replaced by $K + K\mathcal{L}$.*

*Proof. Template: row (iv).* Let $G = E \times E' \times \mathbb{G}_{\mathrm{m}}$ with coordinates $(z, z', x)$, $\Psi(z, z', x) = (z, z' + x)$, $V = \mathbb{C}(1, r)$, $n = [K : \mathbb{Q}]$, $\theta$, $p_\theta$, $\Psi_\theta$ as in §5.1. Connected algebraic subgroups of $E^n \times E'^n \times \mathbb{G}_{\mathrm{m}}^n$ are products $B_S \times B'_{S'} \times T_L$ ($\mathrm{Hom}(E, E') = 0$), and such a subgroup has tangent space in $\ker\Psi_\theta$ iff $\psi(S) = 0$, $\psi'(S') = 0$ and $L = 0$. So $U = B_{\ker\psi} \times B'_{\ker\psi'}$, and $G'' = G^n/U = A'' \times A''' \times \mathbb{G}_{\mathrm{m}}^n$ with $\dim A'' = \dim A''' = n/2$: $d_0 = 0$, $d_1 = n$, $d_2 = n$, $d = 2n$, weight $3n$. $\Psi''$ is onto $\mathbb{C}^2$, $V''$ is a hyperplane and $W'' = \ker\Psi''$. The period datum is $w = (\Omega, c\Omega') \in \Psi(\Omega_K(G))$. Suppose $\Omega, u_1, u_2 \in \mathcal{S}_K$ are $K$-independent, put $v_j = (u_j, ru_j)$, and let $Y''$ be generated by lifts of $D\theta_h w$, $\theta_h v_1$, $\theta_h v_2$: then $m = 3n$, and $\kappa = n$ (a period of $G''$ has first coordinate in $K\Omega$, which kills the $v$-parts). With $d - N = 1$ and $\delta - \tau = 2 - r'$, $r' = \dim\Psi''(T_{G'})$,

an obstruction needs $(m-\operatorname{rk} Y_{G'})+(3n-\operatorname{wt} G')\le(2-r')\cdot 2n$, where $Y_{G'}=Y''\cap(T_{G'}+\Omega(G''))$. Since $U$ is maximal, a nonzero $B''$ has image $\mathbb{C}(1,0)$ and a nonzero $B'''\times T_L$ has image containing $\mathbb{C}(0,1)$; so $r'\le 1$ leaves: $G'=0$ ($5n\le 4n$ fails, margin $n$); $G'=B'''\times T_L\neq 0$ with image $\mathbb{C}(0,1)$ ($\operatorname{rk} Y_{G'}\le n$, so the left side is $\ge 3n>2n$); $G'=B''\neq 0$ with image $\mathbb{C}(1,0)$: an element of $Y_{G'}$ has $\Psi''$-image $(X,rX)$ with $rX\in K\Omega'+K\cdot 2\pi i$, i.e. $X\in K\Omega+K\cdot 2\pi i\Omega/(c\Omega')$, a 2-dimensional $K$-space, so $\operatorname{rk} Y_{G'}\le 2n$ and the left side is $\ge n+2n>2n$. No $G'$ satisfies (1). (2) reads $2(2n-1)\ge 2n$.

*The other rows.* Weights per $n$: $\mathbb{G}_{\mathrm{m}}$ and a CM base $n$ each, a CM eigen-fibre $n+2$ (the quotient $Q''$ of $(E^{\natural})^n$, anti-affine of dimension $n/2+1$, whose subgroups are 0, the fibre and the $\pi^{-1}(B)$, as in the proof of Theorem 5.4). Rows (i), (ii), (v), (vi) have one fibre slot: weight $3n+2$, $\kappa=n$, and the margin at $G'=0$ is $n-2$, whence $[K:\mathbb{Q}]\ge 4$. There are no mixed subgroups between $\sigma_1$ and $\sigma_3$ (different kinds, Hom = 0). The same curve appears twice (base and fibre) only in rows (i) $(\sigma_2,\sigma_3)$ and (v) $(\sigma_1,\sigma_2)$; there every connected subgroup of $(E^{\natural})^n\times E^m$ over $B_S$ contains $B_S^{\natural}$, so the conditions for $T_H\subset\ker\Psi_\theta$ separate, $U=U_{\mathrm{base}}\times U_{\mathrm{fibre}}$, and every subgroup with $r'\le 1$ has a coordinate line as image. In row (i) the mixed subgroups lie in target 2 (rank $\le n$); in row (v) a mixed subgroup has $r'=2$ unless it is a product. In each row the analogue of the step for $B''$ puts $X$ in a 2-dimensional $K$-space, spanned by $p_1$ and $p_3'/r$ with $p_3'$ the target-2 period. The case tables were run by exact enumeration for even $n$ up to 40: no obstruction for even $n\ge 4$ (rows (iii), (iv): $n\ge 2$); for the fibre rows $n=2$ is obstructed at $G'=0$, with equality, as for Theorem 5.4.

*Affine variant.* The extra $\mathbb{G}_{\mathrm{a}}$ gives $d_0=1$ in rows (i)–(iv) and merges into the fibre of $Q''$ in rows (v), (vi); $d_1+2d_2$ is unchanged and (2) holds with a wide margin. The new subgroups are the $\mathbb{G}_{\mathrm{a}}$ factor (type $B''$ in rows (i)–(iii), with $X\in K+K\mathcal{L}$ in the same 2-dimensional count; in target 2 with rank $\le n$ in rows (iv)–(vi)), and a slanted vector line in $\mathbb{G}_{\mathrm{a}}\times$ (fibre) in rows (i), (ii) ($r'=1$, weight 0), whose left side is $\ge(m-2n)+\operatorname{wt} G''$, more than the right side. The case $\delta=\tau$ uses the exponent $b>1$ of Lemma A.1. The margins are unchanged. □

*Proof of Theorem 10.1.* Take $K$ large (containing the coefficients needed below). *Rows (i)–(iii).* $x_j:=\rho u_j\in\widetilde{\mathcal{L}}\subset K+K\mathcal{L}$ and $rx_j=u_j\in K\mathcal{L}_E$, so $2\pi i,x_1,x_2\in\mathcal{S}_K$ (affine variant) and $a\cdot 2\pi i+bx_1+c'x_2=0$ with $(a,b,c')\neq 0$. If $a\neq 0$, then $bu_1+c'u_2=-2\pi ia/\rho$ is a nonzero element of $\overline{\mathbb{Q}}^{\times}\eta^*(\Omega)$, $\overline{\mathbb{Q}}^{\times}\eta^{*\prime}(\Omega')$ or $\overline{\mathbb{Q}}^{\times}\Omega'$ lying in $\overline{\mathbb{Q}}\mathcal{L}_E$. This is impossible by Wüstholz's theorem [52]: $\overline{\mathbb{Q}}\mathcal{L}_E\cap\overline{\mathbb{Q}}\eta^*(\Omega)=0$ (on $E^{\natural}$: a logarithm $(u,\zeta^*(u)+t)$ with $u\in\overline{\mathbb{Q}}\mathcal{L}_E$ equal to a multiple of the fibre period would lie in a $\overline{\mathbb{Q}}$-rational line that is not the Lie algebra of a subgroup), $\overline{\mathbb{Q}}\mathcal{L}_E\cap\overline{\mathbb{Q}}\eta^{*\prime}=0$ (on $E\times E'^{\natural}$) and $\overline{\mathbb{Q}}\mathcal{L}_E\cap\overline{\mathbb{Q}}\Omega'=0$ (on $E\times E'$). So $a=0$. *Rows (iv)–(vi).* $\Omega,u_1,u_2\in\mathcal{S}_K$ (affine variant in target 2), so $a\Omega+bu_1+c'u_2=0$; if $a\neq 0$ then $\rho(bu_1+c'u_2)=-a\rho\Omega\in\widetilde{\mathcal{L}}$, but $\rho\Omega\in\overline{\mathbb{Q}}^{\times}\{\Omega',\eta^*(\Omega),\eta^{*\prime}(\Omega')\}$, none of which lies in $\widetilde{\mathcal{L}}$ (Wüstholz on $\mathbb{G}_{\mathrm{a}}\times\mathbb{G}_{\mathrm{m}}^N\times E'$, $\times E^{\natural}$, $\times E'^{\natural}$). So $a=0$. In all rows $u_1,u_2$ are $\overline{\mathbb{Q}}$-dependent, so $u_2/u_1\in k$ by Masser's theorem [23]. If $\rho u_j=\log\alpha_j$ and $u_2=\gamma u_1$ with $\gamma\in k$, then $\log\alpha_2=\gamma\log\alpha_1\neq 0$, and the Gel'fond–Schneider theorem forces $\gamma\in\mathbb{Q}$. □

*Controls.* A non-CM curve, the period term deleted, one datum only, or $E'$ isogenous to $E$ (where the line form is false and graph subgroups have image $V$) each give an obstruction. *Provenance.* By the table of §4.4, Proposition 10.3 is a consequence of Theorem 1.1 ($\alpha_{K,\Psi}-\kappa=2$ or $2+2/n$ against 3 data). Rows (iii), (iv) already follow from Corollary 2.2 on $\mathbb{G}_{\mathrm{a}}\times\mathbb{G}_{\mathrm{m}}\times E\times E'$ ($\alpha_K=3$, $\kappa=1$), and rows (i), (v) from Corollary 2.2 on $\mathbb{G}_{\mathrm{a}}\times\mathbb{G}_{\mathrm{m}}\times E^{\natural}$ read through base and fibre ($\alpha_K=3+2/n$, $\kappa=1$, by (5)). The linearisation filter: on a CM curve only $\eta^*\cdot u$ and $\Omega\cdot\zeta^*(u)$ are periods of 1-motives (Laurent), so $\rho\in\overline{\mathbb{Q}}^{\times}\pi/\Omega$ (Theorem 1.10) is the unique linearisable CM ratio, and we do not expect Theorem 10.1 to follow from the analytic subgroup theorem [Heuristic].

10.2. **A single elliptic partner for $e^{\pi^2}$.**

**Theorem 10.4** (Theorem P)**.** *Let $E/\overline{\mathbb{Q}}$ have complex multiplication. If $\pi^2 \in \overline{\mathbb{Q}}\mathcal{L}$ (for instance if $e^{a\pi^2}$ is algebraic for some $a \in \overline{\mathbb{Q}}^\times$), then $1/\pi \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$. In particular, for $a, c \in \overline{\mathbb{Q}}^\times$ and $b \in \overline{\mathbb{Q}}$, $e^{a\pi^2}$ and $\wp_E(c/\pi + b)$ are not both algebraic; e.g. one of $e^{\pi^2}$, $\wp_i(1/\pi)$ is transcendental on $y^2 = 4x^3 - 4x$, and one of $e^{\pi^2}$, $\wp_\rho(1/\pi)$ on $y^2 = 4x^3 - 4$.*

*Status:* [Proved modulo published theorems and Lemma A.1] (also using Schneider's theorem [30] and Chudnovsky's theorem (1976; see [11]) that $\pi$ and $\Omega$ are algebraically independent).

*Proof.* Let $K \supseteq k(i, a)$ contain the coefficients expressing $\pi^2 \in \overline{\mathbb{Q}}\mathcal{L}$ and (for a contradiction) $1/\pi \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$, $n = [K : \mathbb{Q}]$. Let $G = \mathbb{G}_{\mathrm{m}} \times (\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}) \times (\mathbb{G}_{\mathrm{a}} \times E)$ and let $\Psi$ read the slots into $\mathbb{C}^3$: coordinate 1 from $\mathbb{G}_{\mathrm{m}}$, coordinate 2 from $\mathbb{G}_{\mathrm{a}} + \mathbb{G}_{\mathrm{m}}$, coordinate 3 from $\mathbb{G}_{\mathrm{a}} + E$; $V = \mathbb{C}(1, 1/\pi, 1/\pi^2)$, $W = 0$. After the quotient by the largest subgroup in $\ker \Psi_\theta$, $G'' = \mathbb{G}_{\mathrm{m}}^n \times (\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}}^n) \times (\mathbb{G}_{\mathrm{a}} \times A'')$ with $\dim A'' = n/2$: $d_0 = 2$, $d_1 = 2n$, $d_2 = n/2$, weight $3n$; $V''$ has codimension 2 and $W'' = \ker \Psi''$. The period term is $\kappa = 0$: a period vector $(\pi a', \pi b', \Omega c')$ $(a', b', c' \in K$, up to the factor $2i)$ lies on $V$ only if $\pi b' = \pi a'/\pi$, which forces $a' = b' = c' = 0$. The data are $H\colon \lambda = \pi^2 \mapsto (\pi^2, \pi, 1)$ and $X\colon \lambda = \pi \mapsto (\pi, 1, 1/\pi)$, which are $K$-independent, so $m = 2n$. Modulo periods (multiples of $\pi$ in coordinates 1, 2), $H \equiv (\pi^2, 0, 1)$ and $X \equiv (0, 1, 1/\pi)$; since $\Omega \notin \overline{\mathbb{Q}}$ (Schneider) and $\pi\Omega \notin \overline{\mathbb{Q}}$ (Chudnovsky), $1 \notin K\Omega$ and $1/\pi \notin K\Omega$, so the third coordinates are not periods.

Every connected subgroup is (vector part)×(subtorus)×(abelian part), and its image $U'$ is the sum of a $\overline{\mathbb{Q}}$-rational subspace of coordinates $2, 3$, a $K$-rational subspace of coordinates $1, 2$ and possibly axis 3; with $\Psi''$ onto $\mathbb{C}^3$ and $W'' = \ker \Psi''$, $\delta - \tau = 3 - \dim U'$, $d - N = 2$, and (1) reads $(\lambda_{U'} + 3n - \operatorname{wt} G') \cdot 2 \le (3 - \dim U') \cdot 3n$, where $\lambda_{U'} = n \cdot \#\{\text{data not in } U'\text{+periods}\}$ (in this cell the $K$-rank of the combinations $\alpha H + \beta X$ in $U'$+periods agrees with the datum-by-datum count). The twelve image types:

| $G'$ (image $U'$) | wt | $\lambda_{U'}$ | left | right |
|---|---|---|---|---|
| 0 | 0 | $2n$ | $10n$ | $9n$ |
| torus of coordinate 1 or 2, or the abelian part (an axis) | $n$ | $2n$ | $8n$ | $6n$ |
| $\mathbb{G}_{\mathrm{a}}$ of coordinate 2 or 3 (an axis) | 0 | $2n$ | $10n$ | $6n$ |
| $K$-line in the $(1,2)$-plane (a subtorus) | $n$ | $2n$ | $8n$ | $6n$ |
| $\overline{\mathbb{Q}}$-line in the $(2,3)$-plane (a vector subgroup) | 0 | $2n$ | $10n$ | $6n$ |
| plane $(1,2)$ | $2n$ | $2n$ | $6n$ | $3n$ |
| plane $(1,3)$ | $2n$ | $n$ ($H$ inside) | $4n$ | $3n$ |
| plane $(2,3)$ | $2n$ | $n$ ($X$ inside) | $4n$ | $3n$ |
| $K$-line in $(1,2)$ + axis 3 | $\le 2n$ | $2n$ | $\ge 6n$ | $3n$ |
| axis 1 + $\overline{\mathbb{Q}}$-line in $(2,3)$; $K$-line in $(1,2)$ + $\overline{\mathbb{Q}}$-line in $(2,3)$ | $\le n$ | $\ge n$ | $\ge 6n$ | $3n$ |
| $\dim U' = 3$ | | | | $\delta \le \tau$ |

(On a $K$-line of the $(1,2)$-plane, $H$ and $X$ modulo periods project to slopes 0 and $\infty$, with nonzero third coordinates; the axis cases use the two transcendence inputs.) No $G'$ satisfies (1). Condition (2) with $N = d - 2$, $d = 5n/2 + 2$: $5n \ge 2n + 4$. The exponent $b > 1$ is not needed: $\delta = \tau$ would force $G' \supseteq \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ with vector part inside $V$, and $\lambda = 0$ would put the $\mathbb{G}_{\mathrm{a}}$-components of all lifts in that vector part, which they span; so [37, (7.2)] with $a = b = 1$ suffices. So the two data cannot coexist. For the second sentence, if $e^{a\pi^2} \in \overline{\mathbb{Q}}$ and $\wp_E(c/\pi + b) \in \overline{\mathbb{Q}} \cup \{\infty\}$, then $a\pi^2 \in \mathcal{L}$ and $c/\pi + b \in \mathcal{L}_E$, so $1/\pi \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\mathcal{L}_E$. □

*Controls.* Each datum alone is consistent ($8n \le 9n$ at $G' = 0$); each projection to $\mathbb{C}^2$ gives $4n \le 4n$; a non-CM curve gives $G' = 0$ at equality; replacing $E$ by $\mathbb{G}_{\mathrm{m}}$ gives the classical partner $e^{1/\pi}$ of the strong six exponentials theorem [14, Cor. 1(3)]. Corollary 2.2 on this line allows

three data and gives nothing. $\wp_E(c/\pi) \notin \overline{\mathbb{Q}}$ alone is not known to us. Elliptic logarithms of algebraic points are periods in the sense of Kontsevich–Zagier, so Theorem 10.4 proves, under $\pi^2 \in \overline{\mathbb{Q}}\mathcal{L}$, a special case of the conjecture that $1/\pi$ is not a period. Illustration [Numerical]: $\wp_i(1/\pi) = 9.889882511\ldots$, $\wp_\rho(1/\pi) = 9.871070987\ldots$.

10.3. **Collinearity of period points.**

**Theorem 10.5** (Theorem CL4)**.** *Let $E_1, E_2, E_3$ be CM elliptic curves over $\overline{\mathbb{Q}}$ with pairwise distinct CM fields $k_1, k_2, k_3$, and $\Omega_j$ a nonzero period of $E_j$. If $s, t \in \mathbb{C}$ satisfy*

$$s + 2\pi i\, t \in \overline{\mathbb{Q}}\mathcal{L} \quad \text{and} \quad s + \Omega_j t \in \overline{\mathbb{Q}}\mathcal{L}_{E_j}\ (j = 1, 2, 3),$$

*then $s = 0$. Equivalently, if the four points $(2\pi i, x)$, $(\Omega_j, y_j)$ of $\mathbb{C}^2$ with $x \in \overline{\mathbb{Q}}\mathcal{L}$, $y_j \in \overline{\mathbb{Q}}\mathcal{L}_{E_j}$ lie on one line, the line passes through the origin. The same holds with $\mathbb{G}_{\mathrm{m}}$ replaced by a fourth CM curve $E_0$ with a CM field distinct from the others.*

*Status:* [Proved modulo published theorems and Lemma A.1].

**Corollary 10.6.** *Put $X = \Omega_1(2\pi i - \Omega_2)/(\Omega_1 - \Omega_2)$ and $Y = \Omega_1(\Omega_3 - \Omega_2)/(\Omega_1 - \Omega_2)$. Then $e^X$ and $\wp_{E_3}(Y)$ are not both algebraic.*

*Proof.* Take $y_1 = \Omega_1$, $y_2 = 0$: then $t = \Omega_1/(\Omega_1 - \Omega_2)$ and $s = -\Omega_2 t \neq 0$, and the theorem forbids $x = X \in \mathcal{L}$ together with $y_3 = Y \in \mathcal{L}_{E_3}$. (Some such proviso is needed: $y_j = \Omega_j/2$ gives $s = 0$, $x = \pi i$.) □

*Proof of Theorem 10.5.* Let $K \supseteq k_1k_2k_3(i)$, $n = [K : \mathbb{Q}] \geq 8$, $G = \mathbb{G}_{\mathrm{m}} \times E_1 \times E_2 \times E_3$, $\Psi = \mathrm{id}$ (one slot per coordinate). Subgroups of $G^n$ are products, $U = 0 \times B_{\ker \psi_1} \times B_{\ker \psi_2} \times B_{\ker \psi_3}$, and $G'' = \mathbb{G}_{\mathrm{m}}^n \times A_1'' \times A_2'' \times A_3''$ with $\dim A_j'' = n/2$: $d = 5n/2$, $d_0 = 0$, $d_1 = n$, $d_2 = 3n/2$, weight $4n$. Put $w = (1, 1, 1, 1)$, $p = (2\pi i, \Omega_1, \Omega_2, \Omega_3)$, $V = \mathbb{C}w \oplus \mathbb{C}p$, $W = \mathbb{C}w$ ($\overline{\mathbb{Q}}$-rational). So $V'' = \Psi''^{-1}(V)$ has codimension 2 and $W'' = \Psi''^{-1}(W)$ is $\overline{\mathbb{Q}}$-rational of dimension $d - 3$. A datum $v = sw + tp$ with $s \neq 0$ lies in $\Psi(\Lambda_K(G))$; $Y''$ is generated by lifts of $D\theta_h p$ and $\theta_h v$, so $m = 2n$.

$\kappa = n$. A $K$-period vector on $V$ is $s'w + t'p$ with $s' + 2\pi i t' \in K \cdot 2\pi i$ and $s' + \Omega_j t' \in K\Omega_j$. Writing $s' = 2\pi i(a - t') = \Omega_j(b_j - t')$ and eliminating $t'$ between $j = 1, 2$ gives $(b_1 - b_2)/(2\pi i) - (b_1 - a)/\Omega_2 + (b_2 - a)/\Omega_1 = 0$; as $1/\Omega_j \in \overline{\mathbb{Q}}^\times \eta_j^*/\pi$, this is a $\overline{\mathbb{Q}}$-linear relation among $1, \eta_1^*, \eta_2^*$, trivial by Wüstholz's theorem on $\mathbb{G}_{\mathrm{a}} \times E_1^\natural \times E_2^\natural$. So $b_1 = b_2 = a$, $s' = 0$, and $V \cap \Psi(\Omega_K(G)) = Kp$.

*Lemma S: if three of the four coordinates of a datum $v = sw + tp$ are $K$-periods, then $s = 0$.* If they are the $E_j$-coordinates, $s = \Omega_j(b_j - t)$ for $j = 1, 2, 3$; eliminating $t$ pairwise gives $(b_2 - b_3)/\Omega_1 + (b_1 - b_2)/\Omega_3 + (b_3 - b_1)/\Omega_2 = 0$, a $\overline{\mathbb{Q}}$-relation among $\eta_1^*, \eta_2^*, \eta_3^*$ (after multiplying by $\pi$), trivial by Wüstholz on $E_1^\natural \times E_2^\natural \times E_3^\natural$; so $b_1 = b_2 = b_3 = t$ and $s = 0$. If they include the $\mathbb{G}_{\mathrm{m}}$-coordinate, the computation of $\kappa$ applies. (The eliminations use $2\pi i \notin \overline{\mathbb{Q}}\Omega_j$ and $\Omega_1/\Omega_2 \notin \overline{\mathbb{Q}}$.)

*Obstructions.* Connected subgroups are $G' = T_L \times B_1 \times B_2 \times B_3$; every nonzero factor maps onto its coordinate line ($U$ is maximal), so $\Psi''(T_{G'}) = \mathbb{C}^S$ for the set $S$ of nonzero factors, and $\delta - \tau = 3 - |S| + [S = \{0, 1, 2, 3\}]$. (1) reads $(\lambda + 4n - \mathrm{wt}\, G') \cdot 2 \leq (\delta - \tau) \cdot 3n$: for $S = \emptyset$, $\lambda = n$ and $10n \leq 9n$ fails; for one slot, $\mathrm{wt} \leq n$ and $\lambda = n$ (by Lemma S, the datum's other three coordinates are not all periods), so $8n \leq 6n$ fails (without Lemma S, $\lambda = 0$ would give $6n \leq 6n$); for two slots the left side is $\geq 4n > 3n$; for three or four slots $\delta > \tau$ fails. Condition (2): $2(5n/2 - 2) \geq 2n$. So no datum with $s \neq 0$ exists. The four-curve version has the same table (weight $4n$, Lemma S with any three of the four $\eta^*$'s, (2) $4n - 4 \geq n$). □

*Controls.* Three points (a line in $\mathbb{C}^3$, codimension 1) give $e \leq 1$, the four exponentials shape, in agreement with Theorem 1.1; with $W = 0$, or without $\kappa$, one datum is allowed; a non-CM $E_3$ gives $G' = 0$ at equality. Corollary 2.2 on this plane ($\alpha_K = 4$, $\kappa = 1$, $\dim V/W = 1$) gives

the bound 3 against 2 vectors, hence nothing. For one isogenous pair with $\Omega_2 = \mu\Omega_1$, $\mu \neq 1$, the statement still holds [Proof sketch]; for $\mu = 1$ it becomes the three-point statement, which is open.

10.4. **A unitary source of the period term.** Let $\mathbf{K} \subset \mathbb{C}$ be imaginary quadratic, $\iota$ the inclusion, and $A/\overline{\mathbb{Q}}$ an abelian variety of dimension $g$, simple over $\overline{\mathbb{Q}}$, with $\mathcal{O}_{\mathbf{K}} \subset \mathrm{End}(A)$. Then $\mathrm{Lie}\, A = L_\iota \oplus L_{\bar\iota}$, where $[a]$ acts by $\iota(a)$ on $L_\iota$ and by $\bar\iota(a)$ on $L_{\bar\iota}$; both are $\overline{\mathbb{Q}}$-rational. The signature is $(r,s) = (\dim L_\iota, \dim L_{\bar\iota})$; for $A$ simple and $r \geq 2$ we have $s \geq 1$. Examples: Picard curves $y^3 = f_4(x)$ (signature $(2,1)$; $dx/y^2$, $x\,dx/y^2$ span one eigenspace) and $y^3 = f_6(x)$ (signature $(3,1)$), whose Jacobians are simple for generic $f$ (simplicity must be checked for each curve). These are generically not of CM type. For $x \in \mathrm{Lie}\, A$ write $x = x_\iota + x_{\bar\iota}$.

**Lemma 10.7.** *For a nonzero period $\omega$ of $A$, $\omega_\iota \neq 0$ and $\omega_{\bar\iota} \neq 0$. If $s = 1$, or $r > s$, and $\omega, \omega'$ are periods with $\omega'_\iota \in \mathbb{C}\omega_\iota$, then $\omega' \in \mathbf{K}\omega$ in $\Omega(A) \otimes \mathbb{Q}$.*

*Proof.* If $\omega \in L_{\bar\iota}$, then $(\mathbf{K}\otimes\mathbb{R})\omega$ is a complex line spanned by the rank-2 lattice $\mathcal{O}_{\mathbf{K}}\omega$, an elliptic curve in $A$, impossible for $A$ simple of dimension $\geq 2$. For the second claim with $s = 1$: if $\omega, \omega'$ are $\mathbf{K}$-independent with $\omega'_\iota = t\omega_\iota$, take $\theta_t \in \mathbf{K}\otimes\mathbb{R}$ with $\iota(\theta_t) = t$; then $0 \neq \omega' - \theta_t\omega \in U_{\mathbb{R}} \cap L_{\bar\iota}$ for $U = \mathbf{K}\omega + \mathbf{K}\omega'$, and since $U_{\mathbb{R}}$ is stable under the complex structure and $\dim L_{\bar\iota} = 1$, $L_{\bar\iota} \subset U_{\mathbb{R}}$. As $g \geq 3$, $U$ lies in the kernel of a nonzero $\mathbf{K}$-linear $\varphi\colon \Omega(A)_{\mathbb{Q}} \to \mathbf{K}$, whose extension kills $L_{\bar\iota}$ and is complex-linear on $L_\iota$, hence is a holomorphic 1-form with periods in $\iota(\mathbf{K})$; the $\mathbf{K}$-span of $\mathrm{Tr}\circ\varphi$ is a 2-dimensional rational sub-Hodge structure of $H^1$, i.e. an elliptic quotient of $A$, a contradiction. For $s \geq 2$, $r > s$: Theorem 2.3 with $Y = \mathcal{O}_{\mathbf{K}}\omega + \mathcal{O}_{\mathbf{K}}\omega'$ as in the proof below ($\kappa = 4$, $\lambda = 0$) gives $2g(r-1) \leq r(2g-4)$, i.e. $r \leq s$, a contradiction; here (2) holds strictly. □

**Theorem 10.8** (Theorem U)**.** *Let $A$ be as above, of signature $(r,s)$ with $r \geq 2$, and $\omega$ a nonzero period.*

- (U1) *If $r \geq s+2$ and $u \in \Lambda(A)$ has $u_\iota \in \mathbb{C}\omega_\iota$, then $\exp_A(u)$ is torsion.*
- (U2) *The $u \in \Lambda(A)$ with $u_\iota \in \mathbb{C}\omega_\iota$ form a group whose image in $A(\overline{\mathbb{Q}})/\mathrm{tors}$ is an $\mathcal{O}_{\mathbf{K}}$-module of rank $\leq \lfloor s/(r-1) \rfloor$ ($\leq 1$ for Picard curves).*
- (U3) *If $r \geq s+1$ and $u \in \Lambda(A)$ has $u_\iota = t\,\omega_\iota$ with $2\pi i t \in \mathcal{L} + \iota(\tau)\mathcal{L}$ ($\mathcal{O}_{\mathbf{K}} = \mathbb{Z}[\tau]$), then $t \in \mathbf{K}$ and $\exp_A(u)$ is torsion; in particular $u_\iota = (\log\alpha/2\pi i)\,\omega_\iota$ with $\alpha \in \overline{\mathbb{Q}}^\times$ forces $\alpha$ to be a root of unity.*
- (U4) *If $r \geq s+1$, $E$ is a CM curve with CM by $\mathcal{O}_{\mathbf{K}}$ (acting on $\mathrm{Lie}\, E$ through $\iota$) with period $\Omega$, and $u \in \Lambda(A)$, $v \in \mathcal{L}_E$ satisfy $u_\iota = (v/\Omega)\,\omega_\iota$, then $\exp_E(v)$ and $\exp_A(u)$ are torsion.*

*Status:* [Proved modulo published theorems and Lemma A.1] (Lemma 10.7 for $s = 1$ uses only Hodge theory). Lemma A.1 is used here for an abelian variety $A$ of dimension $r + s \geq 3$, in general not isogenous to a product of elliptic curves, where it is at the level of the proof sketch (Remark A.3(a)). Concretely, on a simple Picard Jacobian,

$$\Big(\int^P \frac{dx}{y^2}, \int^P \frac{x\,dx}{y^2}\Big) = \frac{\log\alpha}{2\pi i}\Big(\int_\gamma \frac{dx}{y^2}, \int_\gamma \frac{x\,dx}{y^2}\Big)$$

only for $P$ torsion and $\alpha$ a root of unity: the unitary analogue of $\wp(\varpi\log\alpha/\pi) \notin \overline{\mathbb{Q}}$ (Corollary 1.11(M2)).

*Proof.* Let $G = A \times E^a \times T^b$ ($a, b \in \{0,1\}$; $T = \mathbb{G}_{\mathrm{m}} \otimes \mathcal{O}_{\mathbf{K}}$, a split torus of dimension 2), so $d = g+a+2b$, $d_0 = 0$, $d_1 = 2b$, $d_2 = g+a$, weight $2g+2a+2b$. $L_\iota(G) = L_\iota(A) \oplus \mathrm{Lie}\, E^a \oplus \mathbb{C}_\iota^b$ has dimension $R = r+a+b$, and $L_{\bar\iota}(G) = L_{\bar\iota}(A) \oplus \mathbb{C}_{\bar\iota}^b$ has dimension $S = s+b$; both are $\overline{\mathbb{Q}}$-rational. Let $w = (\omega, \Omega^a, (2\pi i \otimes 1)^b)$ be the period vector, and

$$V = \mathbb{C}w_\iota \oplus L_{\bar\iota}(G), \quad W = L_{\bar\iota}(G), \quad N = 1+S, \quad t = S, \quad d - N = R - 1.$$

Since $[a]$ acts on $L_\iota(G)$ by the scalar $\iota(a)$, $\mathcal{O}_{\mathbf{K}}w \subset V$, and every datum below lies in $V$.

*Data.* A datum is $y \in \Lambda(G)$ with $y_\iota = t'w_\iota$; put $Y = \mathcal{O}_{\mathbf{K}}w + \sum_j \mathcal{O}_{\mathbf{K}}y_j$. If a component of $y$ is torsion, the datum is trivial: e.g. if its torus component is torsion then $t' \in \iota(\mathbf{K})$ and $y - [c]w \in W$ for some $c \in \mathbf{K}$, a $\overline{\mathbb{Q}}$-rational subspace containing a logarithm of an algebraic point, so by Wüstholz's theorem it lies in the Lie algebra of an algebraic subgroup inside $L_{\bar\iota}(G)$, which is 0; if the $A$-component is torsion and $r > s$, Lemma 10.7 gives $t' \in \mathbf{K}$. So for nontrivial data each orbit $\mathcal{O}_{\mathbf{K}}y_j$ has rank 2 modulo $\mathcal{O}_{\mathbf{K}}w$, $\kappa = 2$, and for $G' \subset G$, $\lambda(G') \geq 2$ per independent datum unless $G'$ contains all factors of the datum (a rational line $l \subset T$ meets an orbit $\mathcal{O}_{\mathbf{K}}z$ of a non-torsion $z$ in rank $\leq 1$).

*Subgroups.* $A$ and $E$ are simple, $\mathrm{Hom}(A, E) = 0$, and there are no homomorphisms between abelian varieties and tori; so $G' = A' \times E' \times T_L$ with $A' \in \{0, A\}$, $E' \in \{0, E\}$, $T_L \in \{0, T, \text{a rational line } l\}$, and $\tau = S - \dim(W \cap T_{G'})$ with $W \cap T_{G'} = L_{\bar\iota}(A') + (\mathbb{C}_{\bar\iota}$ if $T_L = T)$.

*The inequality.* $G'$ obstructs iff $\delta > \tau$ and $(\lambda + \delta_1 + 2\delta_2)(R-1) \leq (\delta - \tau)(2b + 2g + 2a - 2)$. For $G = A$ ($a = b = 0$), $G' = 0$ and $k$ data: $\lambda = 2k$, $\delta = g$, $\tau = s$, and the inequality is $(2k + 2g)(r-1) \leq r(2g-2)$, i.e. $k(r-1) \leq s$. So $k$ data are excluded iff $k > s/(r-1)$; with $G' = A$ one has $\delta = 0$. This gives (U1) ($k = 1$, $r \geq s + 2$) and (U2). For (U3), (U4) ($G = A \times T$, resp. $A \times E$, one datum) all proper subgroup types were tabulated for every $(r, s)$: for $r \geq s + 1$ none obstructs. For Picard $(2, 1)$ and $A \times T$: $G' = 0$: $20 > 18$; $G' = A$: $8 > 6$; $G' = T$: $16 > 12$; $G' = A \times l$: $\delta = \tau$; $G' = l$: $18 > 12$. For $A \times E$: $G' = 0$: $20 > 18$; $G' = A$: $8 > 6$; $G' = E$: $16 > 12$. In general $G' = 0$ and $G' = A$ obstruct exactly when $r \leq s$, and $T$, $l$, $E$ never do. Condition (2): $2 + 2S \geq 2b + 2$, strict as $s \geq 1$; $d_2 \geq 1$. For (U3), $t \in \mathbf{K}$ and $\log \alpha = 2\pi i(c_1 + c_2\sqrt{-D})$ give $|\alpha| = e^{-2\pi c_2\sqrt{D}}$, and Gel'fond–Schneider forces $c_2 = 0$. □

*Controls.* With the period term deleted, or with $W = 0$, the count obstructs; for Picard $(2, 1)$ one datum is allowed ($8 \leq 8$ at $G' = 0$); for a non-simple $A = B \times E_\rho$ of signature $(3, 1)$, (U1) is false and the count finds the obstruction $G' = B$. *Provenance.* (U2) for $r = 2$ (Picard curves included) is Theorem 1.1 at $K = \mathbb{Q}$, $G = A$, $V$ a hyperplane ($\alpha = 2g$, $\kappa = 2$, so $2 + 2k \leq 2g - 2$, i.e. $k \leq s$). For $r \geq 3$, and for (U1), (U3), (U4), $V$ has codimension $r - 1 \geq 2$ in $T_G$ (resp. $R - 1$), and the subgroup-by-subgroup count gives more than Theorem 1.1; for $K \supseteq \mathbf{K}$ condition (i) fails, since $L_{\bar\iota} \in S_K(A)$. The relation $u_\iota \parallel \omega_\iota$ is bilinear in a logarithm and a period of the same abelian variety; it is $\overline{\mathbb{Q}}$-linear only for algebraic $t$ [Heuristic: not linearisable].

## 11. Credit, literature, and overlap

This section gives the position of Theorem 1.1 (§11.1) and compares the applications with the literature.

11.1. **Theorem R.** *Credit.* The statement of Theorem 1.1 with the printed weight is Roy and Waldschmidt's [29, Théorème, p. 363], and the inner weight is the inequality inside their step (b) [29, p. 364]; the transcendence content is Waldschmidt's Theorem 4.1 [37]. The parts in print are listed in §2.4: hyperplanes [37, §5d]; linear groups in every codimension [40, §11.7.3] and [27, Thms. 1, 2]; $\kappa = 0$ in every codimension [28, §6, remarque (i)] (through [29, Remarque 2]). Waldschmidt describes the period refinement of his effective theorem as "accessible" [39, §7(f)].

*Our claim.* To our knowledge, no complete written proof of the $\kappa$-refined statement in codimension $\geq 2$ exists for groups with an abelian part; we could not consult [28]. Nor do we give one in full: our proof in that range is complete relative to [37, Thm. 4.1] as printed (level (R-W)); at level (R-P) groups with a torus in codimension $\geq 2$ outside (NR$^*$) remain open (Remark 3.12), and Lemma A.1 is a proof sketch outside the groups isogenous to $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b \times \prod_j E_j$ (Remark A.3).

We also write out the multiplicity step of [37, Thm. 4.1] for $W \neq 0$, $b > 1$, which [37, p. 396] gives as a pointer (Lemma A.1). The arguments of §3 are short, and none of them is a new method: they are the induction over subgroups that Waldschmidt uses in [37, §5], arranged so that every node is a hyperplane node (§3.6) or a reduced node (§3.7).

*Evidence for the claim.* We read: RW93 (the text and the scan, pp. 358–367); W88 (pp. 374–399); W83; Roy's [27]; Waldschmidt's book [40] (Ch. 11; its preface restricts it to linear groups, and its remark that the method of §11.7 "works more generally for commutative algebraic groups" comes without a proof for non-linear groups); Waldschmidt [39] (§2(d), which restates the subgroup theorem with $\kappa$ citing [37, Thm. 4.1], and §7(f)); Waldschmidt's Kingston survey (OUP 1993), whose Theorem 10 is the bound of [29] without $\kappa$ and without proof; the zbMATH reviews of [29] (Zbl 0813.11041) and of [28] (Zbl 0763.11031, which states its main inequality without a period term); and the works citing [29] or [28] that we could find (Philippon, JTNB 7 (1995); Waldschmidt's Afrika Mat. paper (2011); Kahn (2015), who uses only the appendix of [29]; Huber–Wüstholz, arXiv:1805.10104, App. A). None proves the $\kappa$-refined statement in codimension $\geq 2$ for non-linear groups or says where such a proof is. Of Baker–Wüstholz [4] we saw only the table of contents (the analytic subgroup theorem and its quantitative forms, without counting $\dim_{\mathbb{Q}}(V \cap \Lambda)$ or a period term). Not accessed: [28] itself (the residual risk: Remarque 2 suggests that the function $a'$ is already defined in [28, §6], and [28, §6] may verify Roy's Proposition 2.1 for it; an inter-library request is pending), the MathSciNet citation lists of [29] and [28], Waldschmidt's Bolyai paper [38], and the Roy–Waldschmidt papers of 1997 on linear groups.

*What the applications owe to Theorem R.* Every application in §§5–9 and Theorem 10.1 is a consequence of the hyperplane case of Theorem 1.1, which, for $K = \mathbb{Q}$, is in print [37, §5d]; so for these results Theorem 1.1 adds no new transcendence input, and their content is the choice of group, map and line, the period datum, the computation of $H_\Psi$, and the explicit statements (§11.3). The results of §10 in codimension $\geq 2$ (Theorems 10.4, 10.5, and 10.8 except (U2) for $r = 2$) are applications of Theorem 4.1 itself.

11.2. **Complex conjugation.** Waldschmidt's [46, Cor. 2.7] (after Diaz [13]) says: if $x_1, x_2 \in \mathbb{R} \cup i\mathbb{R}$ are linearly independent over $\mathbb{Q}$ and $y_1, y_2, \bar{y}_2$ are linearly independent over $\mathbb{Q}$, then one of the four numbers $e^{x_i y_j}$ is transcendental. Its $\widetilde{\mathcal{L}}$-version [46, Thm. 4.1], a consequence of Roy's strong six exponentials theorem and complex conjugation, says: if $x_1, x_2 \in \mathbb{R} \cup i\mathbb{R}$ are $\overline{\mathbb{Q}}$-linearly independent and $y_1, y_2, \bar{y}_2$ are $\overline{\mathbb{Q}}$-linearly independent, then one of the numbers $x_i y_j$ is not in $\widetilde{\mathcal{L}}$.

*Every slope class contains a real number.* For a CM elliptic period, $\bar{E}$ has CM by the same field, so it is isogenous to $E$ over $\overline{\mathbb{Q}}$ and $\bar{\Omega} \in \overline{\mathbb{Q}}^{\times}\Omega$; and $\pi$ is real. For an eigen-period of a CM abelian variety $A$, let $A^c$ be the complex conjugate of $A$, with the conjugate $F$-action; it has CM type $\bar{\Phi}$. Composing that action with complex conjugation of $F$ gives type $\Phi$, so $A^c$ with the twisted action is $F$-isogenous to $A$ (two CM abelian varieties of the same type $(F, \Phi)$ are isogenous; a standard fact, see [32]). The conjugate $\bar{\omega}_\sigma$ of $\omega_\sigma$ is a $\sigma$-eigen class for the twisted action. Its pull-back along such an isogeny is a nonzero $\sigma$-eigen class of $A$, so it lies in $\overline{\mathbb{Q}}^{\times}\omega_\sigma$, and comparing periods gives $\bar{P}_\sigma \in \overline{\mathbb{Q}}^{\times} P_\sigma$. Hence $e^{2i \arg P_\sigma} \in \overline{\mathbb{Q}}$, and the same holds for non-simple $A$ by the reduction in §4.1. So every $t$ in Theorems 1.2, 1.3 and 1.6 is $\gamma r$ with $\gamma \in \overline{\mathbb{Q}}^{\times}$ and $r \in \mathbb{R}$.

*Four exponentials.* If $\alpha^t, \beta^t \in \overline{\mathbb{Q}}$, then $y = (\log\alpha, \log\beta)$ satisfies $y_j$, $ry_j = \gamma^{-1} t y_j \in \widetilde{\mathcal{L}}$. The $\widetilde{\mathcal{L}}$-version [46, Thm. 4.1] at the real point $r$ then excludes every pair with $\log\alpha, \log\beta, \overline{\log\beta}$ linearly independent (over $\overline{\mathbb{Q}}$, which for logarithms is the same as over $\mathbb{Q}$ by Baker's theorem), and, with the roles of $\alpha$ and $\beta$ exchanged, every pair with $\log\beta, \log\alpha, \overline{\log\alpha}$ linearly independent, whatever the argument of $t$. For the remaining pairs both $\overline{\log\alpha}$ and $\overline{\log\beta}$ lie in $\mathbb{Q}\log\alpha + \mathbb{Q}\log\beta$, so they have a conjugation-stable $\mathbb{Q}$-span of logarithms, which has a basis in $\mathbb{R} \cup i\mathbb{R}$ (for $(\alpha, \bar{\alpha})$,

for instance, $\log|\alpha|^2$ and $\log(\alpha/\bar\alpha)$). For them, [46, Cor. 2.7] with the roles exchanged ($x$ that basis, $y = (1, t, \bar t)$) applies as soon as $t \notin \mathbb{R} \cup i\mathbb{R}$: then $1, t, \bar t$ are $\mathbb{Q}$-independent, because $t$ is transcendental and $\bar\gamma/\gamma \notin \mathbb{Q}$ (e.g. $t = e^{i\pi/5}\varpi/\pi$). So Theorems 1.2 and 1.3 (and the four exponentials statements of Corollaries 5.2 and 5.5) are already known at every $t \notin \mathbb{R} \cup i\mathbb{R}$, and at every $t$ for pairs whose logarithms span a space that is not conjugation-stable (e.g. $(2, 1+2i)$). What conjugation does not reach is real or purely imaginary $t$ — as for $\varpi$, $\varpi/\pi$, $\Omega_\rho$, and the Beta values $B(a/N, b/N)^{\pm1}$ and $(B(a/N, b/N)/\pi)^{\pm1}$ themselves — with a conjugation-stable span of logarithms, such as $(2, 3)$ or $(5, (1 + 2i)/(1 - 2i))$; Corollaries 9.1 and 9.3; and the dimension statements themselves. (Independently of conjugation, all of these are consequences of Theorem 1.1; Remark 11.1.)

*Strong four exponentials.* The strong statement at $t$ depends only on the class $\overline{\mathbb{Q}}^\times t$. Hence Theorem 1.6 is *already known whenever* $\overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$ *is not stable under complex conjugation*, whatever the argument of $t$. What conjugation leaves open is the case of a conjugation-stable span (equivalently, a span with a real basis), e.g. $(\log 2, \log 3)$, $(1+\log 2, \log 3)$, $(1, \log 2)$, $(1, \pi^2)$: this contains all our examples. (Unlike for four exponentials, the second orientation does not help here: since $\bar t \in \overline{\mathbb{Q}}^\times t$, the numbers $1, t, \bar t$ are never $\overline{\mathbb{Q}}$-independent.) The same applies to Corollary 9.7(iii): $\overline{\mathbb{Q}}^\times\Omega/\pi$ contains a real number, so it is a statement at real $t$, where [46, Thm. 4.1] applies when $y_1, y_2, \bar y_2$ are $\overline{\mathbb{Q}}$-independent. The mixed forms SMG, SMG$^\natural$ (second row in $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$) are not reached by conjugation, since no *mixed* analogue of the strong six exponentials theorem (with rows in $\widetilde{\mathcal{L}}$ and $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathcal{L}_E$, resp. $\widetilde{\mathcal{L}} + \overline{\mathbb{Q}}\mathrm{Fib}$) is in print; the purely elliptic strong six exponentials theorem on $\mathbb{G}_\mathrm{a} \times E$ is [29, Cor. 1]. Independently of conjugation, Theorems 6.2 and 1.6 are consequences of Theorem 1.1 (Remark 11.1).

11.3. **Waldschmidt's "Variations" papers and slides, Roy, and explicit consequences of Theorem R.** The closest published theorem to Theorem 1.2 is Waldschmidt's [44, Thm. 4.2] ("Variations on the six exponentials theorem", Hyderabad 2003), a mixed statement on $\mathbb{G}_\mathrm{a} \times \mathbb{G}_\mathrm{m} \times E$ without CM whose second row consists of elliptic logarithms; it needs seven columns (one of six numbers $\ell_1 u_j - \ell_j u_1$ is transcendental) in general, and five columns (one of four) in its period-column case. CM and the mixing map bring the count down to two data with ordinary logarithms. The nearest published statement to Theorem 1.7 we know is [44, Thm. 4.3]: a bilinear statement for two elliptic curves (non-isogenous, or equal with $k$-independent first column), but for $2 \times 9$ matrices of elliptic logarithms, without quasi-periods and without a $2 \times 2$ case. The case $W = 0$ of Theorem 2.5 is [44, Thm. 5.1], and the strong four exponentials conjecture is [44, Conj. 1.5]. Waldschmidt's "Further variations" [42] concerns $2 \times 2$ minors with entries in $\widetilde{\mathcal{L}}$ only (no elliptic functions); Corollary 6.5(iii) is a CM analogue of its Cor. 1.3, and its general $2 \times 3$ minors statement [42, Thm. 1.2] at CM data is out of reach of our count (Remark 6.6(4)). In lecture slides [41] (we cite them from our notes and have not rechecked the slide), Waldschmidt conjectures that $\omega_1\omega_2^* - \omega_1^*\omega_2$ is transcendental for non-isogenous curves. For $E_i, E_\rho$ this would give $\varpi\Omega_\rho \notin \overline{\mathbb{Q}}$, since $\varpi \cdot \rho\Omega_\rho - \Omega_\rho \cdot i\varpi = (\rho - i)\varpi\Omega_\rho$; neither statement of Corollary 9.5 gives it (§9.3), and we do not claim it.

The six exponentials theorem (through [46, Cor. 2.7]) and Roy's strong six exponentials theorem [27, Cor. 2] (through [46, Thm. 4.1]) are the sources of the conjugation overlap above; we do not know Theorem 1.6 to follow from it (Remark 6.7). For Theorem 1.3 with $g \geq 2$ the printed weight $\alpha_K(A) = g$ gives no conclusion, and the inner weight of Theorem 1.1 gives Theorem 5.7 at once (Remark 5.8). We present Theorems 5.1, 5.4, 1.7, 5.7, 5.9, 6.2 and 8.2 (and the results of §§6.2, 6.3, 8.6, 10 and Theorem 5.16) as applications of Theorem 1.1, or of Theorem 4.1, to explicit groups, not as a new method.

**Remark 11.1** (Explicit consequences of Theorem R)**.** Every main result of §§5–9 and Theorem 10.1 is a consequence of the hyperplane case of Theorem 1.1 (§4.4), and we credit them as explicit consequences of Theorem 1.1, that is, of [29] and [37] with §3, with proofs written in full given those and Lemma A.1. For many of them the printed weight $\alpha_K$ of Corollary 2.2 already suffices, on a product of standard groups ($\mathbb{G}_{\mathrm{a}}$, $\mathbb{G}_{\mathrm{m}}$, CM curves, their universal vector extensions, CM abelian varieties) with the weights of §4.3:

(i) Theorem 5.1 ($\mathbb{G}_{\mathrm{m}}^2 \times E$: $\alpha_K = 3$, $\kappa = 1$), hence Theorem 1.2(b), Corollaries 5.2, 1.4, 9.2, and Theorem 6.2(a) ($\mathbb{G}_{\mathrm{a}}^2 \times \mathbb{G}_{\mathrm{m}}^2 \times E$), hence Theorem 1.6 at $t \in \overline{\mathbb{Q}}^{\times}(\Omega/\pi)^{\pm 1}$ and Theorem 9.6;

(ii) Theorem 6.12(1) ($\mathbb{G}_{\mathrm{a}}^2 \times E^2 \times E'$), hence Corollary 6.13 in the classes $(\Omega'/\Omega)^{\pm 1}$ and Theorem 6.16 for $g = g' = 1$;

(iii) Propositions 7.1 (for $[K : \mathbb{Q}] \geq 8$) and 7.5 ($E_i^{\natural} \times E_\rho^{\natural} \times \mathbb{G}_{\mathrm{a}}^2$: $\alpha_K = 4 + 4/n$, $\kappa = 2$; Remark 7.2), hence Theorem 1.7, Corollaries 1.8 and 9.5, and Theorem 7.6 modulo its sketched step;

(iv) Proposition 8.1 when the four slots are all the first-kind slots of the varieties involved (levels 7 and 9: $\alpha_K = 4$, $\kappa = 2$), and Theorem 8.15 for a whole family $T(\tau)$ ($\alpha_K = 2\kappa$), hence, through the family $T(11)$ at $N = 24$ and the slot splitting of §8.3, Theorem 8.13(2), Corollary 8.14(3), (4) and every proper subfamily of Theorem 8.15 (the paragraph after the proof of Theorem 8.13);

(v) Theorem 10.1, rows (iii), (iv) ($\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E \times E'$).

By the splitting (5), one factor $E^{\natural}$ can serve a base slot in one coordinate and a fibre slot in the other; with this, internal checks find that Theorem 5.4 ($\mathbb{G}_{\mathrm{m}}^2 \times E^{\natural}$ read through base and fibre: $\alpha_K = 4 + 2/n$, $\kappa = 1$, against 4 data including the vector $(\Omega, s\Omega)$), Theorem 6.2(b), Theorem 6.12(2), (3) (so also the example $\wp_i(\pi/\varpi^2)$, $\wp_i(\pi u/\varpi^2)$), EA$^{\natural}$ for $g = g' = 1$ and Theorem 10.1, rows (i), (v), also follow from the printed weight (rows (ii), (vi) at proof-sketch level; the cases other than Theorem 5.4 were checked in one internal report each). The results on CM abelian varieties of dimension $g \geq 2$ read at fewer than $g$ slots (Theorem 1.3 for $g \geq 2$, Theorems 5.7, 5.9, 6.8, 6.9, Corollary 9.3, Theorem 6.16 for $\max(g, g') \geq 2$) and Theorem 5.16 use the inner weight. Theorem 1.10, whose first proof is Corollary 2.2 on $E \times \mathbb{G}_{\mathrm{m}} \times \mathbb{G}_{\mathrm{a}}$, is classical in substance (§11.4).

We found none of these consequences stated in print. What is ours there is the choice of group, map and line; the period datum (for (iii) and (iv), a second period relation placed on the line); the computation of the quotient $G''$ (Lemma K′, (F1)–(F4), Lemma 5.15); the explicit statements and examples; the direct proofs, which apply Theorem 2.3 to $G''$ and exclude every subgroup; and, at level 24, the elementary proof of the relation (Lemma 8.12(2)), which avoids Deligne's theorem.

11.4. **Theorem M and the two-curve results.** With $\ell \in \mathcal{L}$, Theorem 1.10 is the CM case, with one row of Waldschmidt's matrix a period of $E \times \mathbb{G}_{\mathrm{m}}$, of his Conjecture 42 in [45] (four exponentials for $E \times \mathbb{G}_{\mathrm{m}}$). Its special case $\ell \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}} \cdot \pi i$, $u \in \mathcal{L}_E$, follows in one line from Laurent's [22, Cor. 3, p. 146]: for any CM lattice, a period $\omega \neq 0$ and $u \notin \mathbb{Q}\Lambda_E$ with $\wp(u) \in \overline{\mathbb{Q}}$, the numbers $1, \omega, \eta(\omega), 2\pi i, \omega H(u), 2\pi i u/\omega$ are $\overline{\mathbb{Q}}$-linearly independent, where $H = \zeta - \gamma z$ and $\gamma = 0$ when $g_2 g_3 = 0$. (M1) is one line from it ($2\pi i u/\Omega = 2ic$), and so is (M$\eta$) modulo $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\varpi + \overline{\mathbb{Q}}\eta(\varpi) + \overline{\mathbb{Q}}\pi i$. The remaining items of Corollary 1.11 (which only use $u \in \mathcal{L}_E$) follow quickly from Laurent's identity and the analytic subgroup theorem (§7.2); we found none of them stated in print (Abe [1] claimed (M1); the preprint was withdrawn). The theorem of Masser that $1, \omega, \eta, 2\pi i, \log \sigma$ are $\overline{\mathbb{Q}}$-linearly independent on a CM curve, for $\sigma \in \overline{\mathbb{Q}}^{\times}$ not a root of unity ([24]), is linear and gives none of these; nor do the algebraic independence results of Brownawell–Kubota [10],

which have six-exponentials strength, or the third-kind results surveyed in [48] (Wüstholz's linear independence of $1, \omega, \eta, \lambda(u_i, \omega)$).

Theorem 1.2, Corollary 1.4, Theorem 1.7 and Corollary 1.8 we found stated nowhere in the literature (apart from what follows from [46] as shown in §11.2); the four exponentials problem is listed as open in [8, p. 27] and [49], and [49], which we read in full, mentions no case at $t$ of these kinds. All of them are consequences of Theorem 1.1 (with Wüstholz's theorem), indeed of the printed form [29, Théorème (iii)] (Remark 11.1); for them what is new is the explicit statement. Laurent [22] never mixes two curves; Corollary 1.8 is the cross-curve analogue of his Cor. 3, with $\eta(\Omega_\rho)$ in place of $\eta(\varpi)$. For Corollary 1.9 see the remark after it (§1); for Corollary 9.5, the remark after its proof (§9.3). Algebraic independence of $\pi, \Gamma(1/4), \Gamma(1/3)$ is open, so neither Corollary 1.8 nor Corollary 9.5 follows from known algebraic independence results.

11.5. **CM abelian varieties and first-kind relations.** Apart from what follows from [46] as shown in §11.2 (all pairs at every $t \notin \mathbb{R} \cup i\mathbb{R}$), we found no $t$ of the kind of Theorem 1.3 (an abelian eigen-period, a Beta value, or such a value divided by $\pi$, up to $\overline{\mathbb{Q}}^\times$) at which the four exponentials conjecture was stated in the literature for all pairs, and no real $t$ of this kind at which it was known for a single pair with conjugation-stable span of logarithms, such as $(2, 3)$. We looked in Waldschmidt's survey [49] (read in full), in [44, 42, 46, 35, 43] and elsewhere. Nor did we find any use of a first-kind (Deligne–Koblitz–Ogus) relation as the period term of a subgroup theorem. Once such a relation is placed on the line, Proposition 8.1 and Theorem 8.15 are consequences of Theorem 1.1, and already of its printed form at levels 7, 9, 24 and for the families $T(\tau)$ (Remark 11.1(iv)); Theorem 1.3 for $g \geq 2$ uses the inner weight. What is ours at level 24 is the elementary proof of the relation (Lemma 8.12(2)), which avoids Deligne's theorem.

Corollary 1.12(b) lies in territory that Gao–Ullmo–Yafaev [17] and Gao–Ullmo [16] describe as open: [17] write that almost nothing is known about algebraic relations among holomorphic CM periods beyond the linear ones (apart from cases reducible to the linear case, such as $\theta\theta'/\pi \notin \overline{\mathbb{Q}}$ [17, Cor. 3.1]), and ask whether all quadratic relations are elementary [17, Question 1.3]; the relation used in Theorem 8.2 is an elementary quadratic relation in their sense. Corollary 1.12(b) is therefore, with Corollary 9.5, among the claims flagged for an expert check in the introduction and in §12.2. Wolfart–Wüstholz [51] determine the $\overline{\mathbb{Q}}$-linear relations among Beta values; Corollary 1.12(b) involves the quotient $B(\frac{3}{7}, \frac{3}{7})/B(\frac{1}{7}, \frac{1}{7})$, which is not known to be a Beta value up to $\overline{\mathbb{Q}}^\times$, so it is not covered (its Koblitz–Ogus function takes three values, so under the Rohrlich–Lang conjecture it is not one; §8.5). Tosi [33] proves algebraic independence results of Gel'fond–Chudnovsky type for abelian periods and their exponentials; they do not give Theorem 1.3 or Corollary 1.5. Grinspan and Vasil'ev (two of $\pi, \Gamma(1/5), \Gamma(2/5)$ are algebraically independent, see [43, §3]) do not imply Theorem 1.3: they concern $\pi$ and Gamma values only and contain no statement about $\alpha^t$ for algebraic $\alpha$.

11.6. **Strong abelian and elliptic forms, quasi-logarithms, levels 9 and 24.** This subsection covers the results of §§6.2, 6.3, Theorem 5.16 and §8.6. Our search for them consisted of the literature already cited in this section, our own notes, and two web searches (2026-10-31: "elliptic four exponentials" with complex multiplication and period ratios; "four exponentials" with Beta values, Gamma values or abelian periods). The searches found Waldschmidt's elliptic surveys, where elliptic four exponentials is a conjecture and six exponentials the closest theorem, and papers on the elliptic period conjecture. Nothing found states a four exponentials statement, elliptic or exponential, at a period-ratio slope or at a Beta value ([Lit], search summaries only).

*Complex conjugation.* By §11.2, $\bar{P}_\sigma \in \overline{\mathbb{Q}}^\times P_\sigma$ for every eigen-period, and $\bar{\Omega} \in \overline{\mathbb{Q}}^\times \Omega$, $\bar{\eta} \in \overline{\mathbb{Q}}^\times \eta$ for CM elliptic periods. So every slope class in Theorem 6.8, Corollary 6.13, Theorem 6.16 and Corollary 5.17 is $\overline{\mathbb{Q}}^\times$ times a real number. The strong statements depend only on the class $\overline{\mathbb{Q}}^\times t$.

Diaz's argument gives the strong four exponentials statement of Theorem 6.8 whenever $\overline{\mathbb{Q}}y_1+\overline{\mathbb{Q}}y_2$ is not stable under complex conjugation: this is [46, Thm. 4.1], after [13], a consequence of Roy's strong six exponentials theorem. The same argument gives Corollary 6.13 for non-conjugation-stable spans: $\widetilde{\mathcal{L}}_E$ is conjugation-stable for CM $E$, and the elliptic strong six exponentials theorem [29, Cor. 1] on $\mathbb{G}_{\mathrm{a}}\times E$ is in print. It also gives the consequence in Theorem 6.16 for such spans, by [29, Cor. 1] on $\mathbb{G}_{\mathrm{a}}\times A$ ([Proof sketch]). What conjugation leaves open is the case of conjugation-stable spans, which contains all our examples, such as $(1+\log 2, \log 3)$, $(1, \log 2)$, $(1, \pi^2)$, and $(1, u)$ for the logarithm $u$ of a real point such as $(2, \sqrt{24})$ on $E_i$.

*Relation to Theorem 1.1.* Independently of conjugation, all results of this subsection are consequences of Theorem 1.1, several already of its printed form (Remark 11.1); we found no explicit statement of them in print. Our examples lie where conjugation does not reach (conjugation-stable spans), such as $\wp_i(\pi/\varpi^2)$, $\wp_i(\pi u/\varpi^2)$ (Theorem 6.12(3)) and $\wp_i(\Omega_\rho/\varpi)$, $\wp_i(\Omega_\rho u/\varpi)$ (Theorem 6.12(1)). As far as we know, conjugation does not reach the mixed forms, since no mixed strong six exponentials theorem is in print.

*Elliptic and eigen-slot four exponentials.* The elliptic four exponentials conjecture (a $2\times 2$ matrix, independence over $\mathrm{End}(E)$) is open with or without CM. The CM statement in Waldschmidt's survey [45] is of six exponentials type $(2\times 3)$, and the mixed $E\times\mathbb{G}_{\mathrm{m}}$ version is [45, Conj. 42] (§11.4). The closest published results we know are [44, Thms. 4.2, 4.3] and the elliptic analogue in [41] (two non-isogenous curves, nine data, no CM). They need many more data and do not involve a period slope. We found no statement of Theorems 6.12 (2), (3) or 6.16 (for $g\geq 2$ or $g'\geq 2$) in print. All of these, like Theorems 5.1 and 1.6, are consequences of Theorem 1.1.

*Theorem 6.8.* Before Theorem 1.3 we found no four exponentials statement at a Beta value or an abelian period. Waldschmidt's Fermat-curve paper [35] and Wolfart–Wüstholz [51] give linear statements only, and we found no strong ($\widetilde{\mathcal{L}}$-level) four exponentials statement at a transcendental slope other than Theorem 1.6. Theorem 6.8 contains Theorem 1.6 ($g=1$). Whether a given Beta slope lies outside the elliptic classes is decided by Wüstholz's theorem for the linear coincidences, and only under the Rohrlich–Lang conjecture for the quadratic ones (Remark 5.10, [Heuristic]).

*Quasi-logarithms.* Laurent's results [22] on $1, \omega, \eta, u, \zeta(u), \dots$ are of degree one. So is Theorem 5.16 in the class $\overline{\mathbb{Q}}^{\times}\Omega^{\pm 1}$ ($p_2 = 2\pi i$): there, by Laurent's identity (6), its hypothesis is a $\overline{\mathbb{Q}}$-linear relation among periods of 1-motives, within reach of the analytic subgroup theorem (the remark after Corollary 5.17), and we keep that case only as a calibration. In the other three classes, $\Omega\Omega'/\pi$, $\Omega/\Omega'$ and $\Omega P/\pi$, a period of another slot multiplies a fibre value (e.g. $(\varpi/\Omega_\rho)\cdot\zeta_i(u)$, the example of Corollary 5.17), and Theorem 5.16 is bilinear. It is a consequence of Theorem 1.1 with the inner weight (with two factors $E^\natural$ the printed weight gives no conclusion). No strong six exponentials theorem for quasi-logarithms is in our reading list, so conjugation gives nothing here ([Heuristic]: we have not checked every corollary of [29]). We found no trace of Theorem 5.16 in print.

*Levels* 9 *and* 24*, several pairs.* The identities of Lemma 8.12 are instances of Deligne's theorem on Koblitz–Ogus relations [12, 19], and, as proved here, of the classical multiplication and reflection formulas. They are not new as identities. What is new is their use as the period term (§11.5). Theorems 8.13 and 8.15 are new instances of Theorem 8.2, and consequences of Theorem 1.1 (already of its printed form, through the full families $T(\tau)$; Remark 11.1(iv)). What is ours at level 24 is the elementary proof of the relation (Lemma 8.12(2)), which avoids Deligne's theorem. Wolfart–Wüstholz [51] controls only $\overline{\mathbb{Q}}$-linear relations among Beta values. Corollary 8.14(2), (4) and Theorem 8.15 involve ratios such as $B(\frac{2}{9},\frac{2}{9})/B(\frac{1}{9},\frac{1}{9})$ or the slope $r$, which are not Beta values under the Rohrlich–Lang conjecture: their Koblitz–Ogus functions take three values. So [51] does not decide them ([Heuristic], as for Corollary 8.7(c)).

*Caveat.* None of these searches accessed Waldschmidt's Bolyai paper [38] (Colloq. Math. Soc. János Bolyai 51, 1990), the most likely place for worked examples with a period term, or the MathSciNet citation lists of [29] and [37]. Every application is a specialisation of the same period-refined subgroup theorem, so the residual novelty risk is shared by all of them. Before submission we recommend one targeted search for "elliptic four exponentials" with "complex multiplication" and "period", and a check of those citation lists.

11.7. **The results of §10.** *Theorem 10.1.* All six classes satisfy $\bar\rho/\rho \in \overline{\mathbb{Q}}^\times$ and $\bar E$ is isogenous to $E$, so a datum $u$ gives a second datum $\bar u$. Write $V(\alpha,\beta)$ for the $\mathbb{Q}$-span of $\log\alpha, \log\beta, \overline{\log\alpha}, \overline{\log\beta}$. If $\dim V(\alpha,\beta) = 4$, Corollary 10.2(b) already follows from the Lang–Ramachandra mixed six exponentials theorem [45, Thm. 24] plus conjugation (Ramachandra; Diaz [13]); if $\dim V(\alpha,\beta) = 3$, it is a one-line consequence of Corollary 2.2 on $E \times \mathbb{G}_{\mathrm{m}}$ ($\alpha_K = 2$, $\kappa = 0$) plus conjugation. We found no trace of the conjugation-stable case $\dim V(\alpha,\beta) = 2$ (for instance $\alpha,\beta$ real), which contains all our examples. Corollary 10.2(b) is the inequality $\delta(e^z, \wp(tz)) \le 1$ of Ramachandra's $\delta$-conjecture as repaired in [47, §2(a)], at six CM classes of $t$; the best bounds listed there are $\delta \le 3$ in general and $\delta \le 2$ at the slope of Theorem 1.10, with no case of $\delta \le 1$. Not accessed: Waldschmidt, Acta Arith. 23 (1973), §3.5, the likeliest place for an explicit mixed CM corollary.

*Theorem 10.4.* The strong six exponentials theorem gives that one of $e^{\pi^2}$, $e^{1/\pi}$ is transcendental [14, Cor. 1(3)]; on lines in $\mathbb{C}^2$ the only other single partners of $e^{\pi^2}$ reachable by these counts are the values $\wp(c\pi\Omega)$ of Corollary 1.11(M3). Theorem 10.4 replaces the $\mathbb{G}_{\mathrm{m}}$ slot carrying $e^{1/\pi}$ by a CM elliptic slot, on a line in $\mathbb{C}^3$. We found no trace of it in print, and it is not implied by a published corollary we know; Waldschmidt's "Variations" [44] list elliptic analogues as future work (its Thm. 4.2 needs seven points).

*Theorem 10.5.* The count is classical ([37, (4.2)], Cor. 4.3); its use with a CM $K$-structure and a period on $V$, and the statement, we found nowhere (searches in [29, 27, 44] and the 2023–26 arXiv; MathSciNet not accessed). The numbers in Corollary 10.6 are artificial; the content is the mechanism.

*Theorem 10.8.* We found no statement of (U1)–(U4), or of the device of putting the conjugate eigenspace into $W$, in [37, 36, 29] or in the work on periods of Picard curves we know (Shiga, Wolfart, Huber–Wüstholz). Bertrand's [6], the one plausible earlier source, was not read; it must be checked before any external claim.

## 12. Consistency checks, status, and formal verification

12.1. **Consistency checks.** The main results are consequences of a published theorem (completed by Lemma A.1 in the case used) applied to explicit groups, so an error would most likely sit in the bookkeeping. We checked that the same bookkeeping does not prove false statements, and that it gives no conclusion in the control configurations below, where it should not.

- *Rational lines.* If $V$ contains a $\overline{\mathbb{Q}}$-rational point, the statement is false (e.g. $\mathbb{G}_{\mathrm{m}}$ in place of $E$, $s = 2$, $x = \log 2$). In each configuration we checked (MG, MG$^\natural$, MX) the count finds an obstruction, typically a subtorus or a $\mathbb{G}_{\mathrm{a}}$-line; for MG and MG$^\natural$ it occurs at equality in the inequality of Theorem 2.3.
- *No CM, wrong field, one Legendre relation, wrong coordinates.* An obstruction appears (the trivial subgroup, or a type-D subgroup) in each of these cases: a non-CM curve; $k \not\subset K$; only one Legendre relation on an MX-type line; for MX, $\mathrm{diag}(a,a)$ in place of $\mathrm{diag}(a,\bar a)$ on $T(E^\natural)$; for MG$^\natural$, a non-eigen fibre coordinate. So no conclusion is claimed in these cases. (For MG$^\natural$ the conjugation itself is not load-bearing; what matters is that $\Psi$ reads an eigen-coordinate.)

- *Pure periods.* Data consisting of periods alone are true, so they test the period term by itself. In all the configurations we built (semi-abelian $G$ with $V$ a hyperplane containing a corank-one sublattice of periods, $\mathbb{G}_{\mathrm{m}}^{d_1} \times E^g$, the groups of §§5.1, 7.4), an obstruction exists.
- *Sharpness of the period term.* For the graph of complex multiplication in $E \times E$ ($\kappa = 2$), the diagonal in $\mathbb{G}_{\mathrm{m}}^2$ ($\kappa = 1$), and the Weil-restricted graph of complex multiplication with maximal $\kappa = n$, the obstruction occurs exactly at equality. A period term larger by one would prove false statements there. The applications in §§5.1, 7.4 use $\kappa = \mathrm{rk}(V'' \cap \ker \exp)$ exactly. This is calibration evidence, not a proof of [37, Prop. 6.1].
- $\mathbb{G}_{\mathrm{m}}$ *alone* [Proved]. For $G = \mathbb{G}_{\mathrm{m}}^r$ the period space $\Psi(\Omega_K)$ is $2\pi i$ times a $K$-rational subspace. So a line $V = \mathbb{C}(1, s)$ meets it only if $s \in K$, where the statement is false and the method is silent. With a logarithm in place of the elliptic period ($\kappa = 0$), the same count reproduces the six exponentials theorem (a calibration).
- *Special exponents.* Values of $t$ that make one power algebraic ($2^t = 3$, $2^t = 1$) are not known to lie outside $\overline{\mathbb{Q}}^{\times}\Omega/\pi$. They lie inside only under degree-two relations such as $\pi \log 3 \in \overline{\mathbb{Q}}\,\Omega \log 2$ or $\Omega \log 2 \in \overline{\mathbb{Q}}\pi^2$, which are conjecturally false. If they did lie inside, the resulting consequences of Corollary 5.2 would be conjecturally true and would contradict no known result.
- *The results of §§5.3, 6, 8.* All case checks were rerun by exact integer enumeration, each by at least two scripts that share no code; all controls below obstruct, as they must.
- *Theorems MG-A, MG$^\natural$-A.* Exact tables for $g = 1, \dots, 6$ and all $n$ divisible by $2g$ up to 60: margins $n$ and $n - 2$, the latter obstructing only at $n = 2$. One referee built $U$, $A''$, $Q''$ concretely as subspaces ($g = 2$: $K = \mathbb{Q}(\zeta_{20})$, $\mathbb{Q}(\zeta_{40})$; $g = 3$: $K = \mathbb{Q}(\zeta_{28})$). Controls: $\kappa$ deleted; one datum; a non-eigen first-kind coordinate $z_{\phi_1} + \mu z_{\phi_2}$ (abelian part of dimension $n$); a non-CM $A$; a fictitious rational $V$ ($P/\pi \in \overline{\mathbb{Q}}$); $\mathbb{G}_{\mathrm{m}}$ in place of $A$; for MG$^\natural$-A, $\kappa$ deleted, a non-eigen fibre coordinate, and a rational $V$.
- *Theorem X.* Minimum excess $n$ for $n \le 120$. Controls: no relation or wrong pairing ($\kappa = n$); dropping slot $d$ (equality obstruction: the fourth slot is load-bearing); a non-CM slot, with or without $\kappa = 2$ imposed; $\kappa$ deleted with three data; $\lambda$ forced to 0 (Lemma 8.4 deleted: equality obstructions, so the lemma is load-bearing); a slot-weight sweep (equality obstruction at weight $5n/8$ per slot, so the weight $n/2$ of Lemma 4.1, which needs $K \supseteq$ the Galois closures, is load-bearing). The same code reproduces a known linearisable slot-reading cell with no obstruction, and obstructs its non-CM twin.
- *Theorem SMG, SMG$^\natural$.* No obstruction for $n = 2, \dots, 40$ (resp. $4, \dots, 40$), margins $n$ and $n - 2$. Controls: non-CM curve; $\kappa$ deleted (so the CM period is load-bearing; for SMG the margin is exactly $\kappa$); one datum only; $s \in K$ (equality obstruction, as it must, since the strong statement is false at $t = 2$); SMG$^\natural$ at $n = 2$.
- *The results of §§6.2, 6.3, 8.6 and Theorem 5.16.* The case checks were rerun by exact integer enumeration (project scripts), and independently by the two internal referees of each result (one of each pair a reconstruction with a falsity hunt; for Lemma 5.15, an exact search over about 500 subspaces). The controls obstruct as they must, for instance $t \in K$ (the obstruction is at the $K$-line $V$), $\kappa$ deleted, and one datum. Four statement-level errors were caught and corrected, three by the internal referees: Theorem 6.16 as first stated was false for $A'$ isogenous to but different from $A$; in Theorem 5.16 with $X = E'^{\natural}$ the two fibres read in the second target must be merged; and an example at $t = B(\frac{1}{7}, \frac{1}{7})$ with $y = (1, \log 2)$ was vacuous, since there $t \notin \widetilde{\mathcal{L}}$ already, and was replaced by $t = 1/B(\frac{1}{7}, \frac{1}{7})$. The fourth was caught when the results were integrated into this paper: Corollary 5.17 is false in the class $\Omega/\Omega'$ without $k' \neq k$ (for $E' = E$ the

slope is algebraic), and that hypothesis was added; a misprinted bound in the proof of Theorem 5.16 was corrected at the same time. A later internal referee found that the first example given for Corollary 5.17 (in the class $\Omega^{\pm 1}$) is linearisable; it was replaced (the remark after Corollary 5.17). The relations of Lemma 8.12 were also checked numerically (60 digits and more), as was the common slope class of the example after Theorem 8.15.

- *Falsity hunts.* [Numerical] For Theorem 1.6: 64 targets $t \cdot y$ ($\Omega \in \{\varpi, \Omega_\rho\}$, four slope classes, eight $y$'s) against real $\widetilde{\mathcal{L}}$-bases at 210 digits, PSLQ and LLL: no relation. No corollary is implied by Nesterenko's theorem (which controls only $e^{\pi\sqrt{d}r}$) or by Chudnovsky's. For Corollary 8.7, all computations use the real numbers $B_j := B(\frac{j}{7}, \frac{j}{7})$ in place of $P_j$ (they differ by the algebraic factors $1 - \zeta^j$; a $\mathbb{Q}(\zeta_7)$-linear relation among real numbers gives a $\mathbb{Q}(\mu)$-linear one). With $Q = (x_0, (x_0(1-x_0))^{1/7})$, $O = (0,0)$ and $x_0 = 1/3, 1/5, 2/7$ (220 digits), $(z_2/z_1)/(B_3/B_1)$ satisfies no integer polynomial of degree $\leq 12$ with coefficients $\leq 10^{10}$. At $x_0 = 1/2$ (a torsion point) $z_j = B_j/2$ by the symmetry $x \mapsto 1 - x$, and the pipeline returns $(z_2/z_1)/(B_2/B_1) = 1$ to working precision (with the $P_j$ of §8.4 this ratio is $1/(1+\zeta)$). PSLQ on $\{1, B_2, B_3, B_3/B_1\} \otimes \{1, \mu, \mu^2\}$ finds no relation with coefficients $\leq 10^{14}$, while the positive control $\{B_1B_2, B_3B(\frac{1}{7}, \frac{2}{7}), B_1, B_3\}$ finds (8) at once.
- *Period-conjecture plausibility.* [Heuristic] Under the period conjecture, $y, ty \in \widetilde{\mathcal{L}}$ already forces $y = 0$ for $t$ in a CM slope class (every monomial on one side contains $\Omega$), so Theorem 1.6 is strictly weaker than predicted; and Theorem 8.6 is a consequence of the period conjecture. A counterexample would have to refute the period conjecture.
- *Theorem 1.1.* The arguments of §3 were checked by one internal referee, who also tried to break them on small groups: hyperplanes in $\mathbb{G}_{\mathrm{m}}^2 \times E$ and in $\mathbb{G}_{\mathrm{a}} \times \mathbb{G}_{\mathrm{m}} \times E$ (every node listed; $\kappa \geq 3$, resp. $\kappa \geq 2$, forces $T_E \subset V$), a line $\mathbb{C}(1, \beta)$ in $\mathbb{G}_{\mathrm{m}}^2$ with $\beta \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$ (Gel'fond–Schneider for $W = V$, six exponentials for $W = 0$), a line in $\mathbb{G}_{\mathrm{m}}^3$ (the bound 3, weaker than six exponentials, as it must be), and abelian $G$ in every codimension. No counterexample was found. The referee also found three errors in our first draft, all corrected here: the linear non-corner nodes (§3.5) were missing; the first form of (NR$^*$) failed for $H = \mathbb{G}_{\mathrm{a}}$; and the derivation of the analytic subgroup theorem was placed outside the range of Lemma A.1, although the trivial decomposition puts it inside.
- *The results of §10.* Each case table was rerun by exact enumeration, by the project scripts and by both internal referees (for Theorem 10.4, three provably false configurations were obstructed when the full subgroup list and the $K$-rank $\lambda$ were used; for Theorem 10.5, removing $W$, $\kappa$ or Lemma S creates obstructions; for Theorem 10.8, a non-simple $A$ of signature $(3,1)$ is a genuine counterexample to (U1), and the count finds the obstruction). Statement-level fixes made by the referees: in Theorem 10.1, $E'$ must be defined over $\overline{\mathbb{Q}}$ and the eigen-coordinate lemma is needed for rows (i), (v); in Theorem 10.4, the slanted planes were missing from the first table, and both transcendence inputs are load-bearing; in Corollary 10.6, the proviso $s \neq 0$.

12.2. **Dependencies and status.** The table lists the main results with their labels ((a), (b), (c) as in the conventions of §1; "(b): X" names the classical input) and what they depend on.

| result | label | depends on |
|---|---|---|
| Thm. 1.1, level (R-W) | Proved, given [37, Thm. 4.1] as printed | Thm. 2.3; Props. 3.4, 3.11, 3.5; Lemmas 3.1, 3.2 |

| result | label | depends on |
|---|---|---|
| Thm. 1.1, level (R-P) $(\alpha)$–$(\gamma)$ | (c) (Lemma A.1 formally verified for $G$ isogenous to $\mathbb{G}_{\mathrm{a}}^{a} \times \mathbb{G}_{\mathrm{m}}^{b} \times \prod_j E_j$, a proof sketch otherwise); open for a torus in codimension $\geq 2$ outside (NR$^*$) | Lemma A.1 ([36], [31], Bertrand, Masser's interpolation theorem, [26]); Lemma 3.8 ([40, Thm. 11.21]); Baker; Props. 3.9, 3.11; Lemmas 3.6, 3.10 |
| Thm. 2.5; Prop. 3.9 | (c) | as level (R-P)$(\alpha)$ |
| Prop. 3.5 | Proved, given Thm. 2.3 (trivial decomposition) | Thm. 2.3; also [52] |
| Thm. 1.2; Cor. 5.5; Cor. 5.2, first sentence | (a) at $\varpi$; (b): $g_3$ at $\Omega_\rho$; (c) otherwise | Thms. 5.1, 5.4; Wüstholz or Remarks 5.3, 5.6 |
| Cor. 5.2, second sentence | (c) | Thm. 5.1, Baker |
| Thms. 5.1, 5.4 ($K$-forms) | (a) at $\varpi$ for the $K$ stated there; (c) otherwise | Thm. 2.3, Lemma A.1, Lemma 4.2, (F1)–(F4) for MG$^\natural$, Baker, transcendence of $\Omega$ and $\Omega/\pi$ |
| Thm. 5.7 (MG-A) | (c) | Thm. 2.3, Lemma A.1, (S1), (P), (W), Baker |
| Thm. 5.9 (MG$^\natural$-A) | (c) | as above, plus (F1)–(F4) |
| Thm. 1.3 ($g \geq 2$) | (c) | Thms. 5.7, 5.9; (W); (F1)–(F4) (through Thm. 5.9); Beta dictionary (Lit) |
| Cor. 9.3 | (c) | Thm. 5.7 (first kind only; no (F1)–(F4)), (W) |
| Thm. 1.3 ($g = 1$) | as Thm. 1.2; (c) for $t \in \overline{\mathbb{Q}}^{\times}/\Omega \cup \overline{\mathbb{Q}}^{\times}\pi/\Omega$ | Thm. 1.2, $t \leftrightarrow 1/t$ |
| Thm. 6.2 (SMG, SMG$^\natural$) | (a) at $\varpi$ with rows in $\widetilde{\mathcal{L}}$; (b): $g_3$ at $\Omega_\rho$; (c) otherwise | Thm. 2.3, Lemma A.1 (exponent $b > 1$, $d_0 \leq 2$), (P), Lemma 4.2 |
| Thm. 1.6 | (b): Wüstholz $(\Omega, \eta \notin \widetilde{\mathcal{L}})$ at $\varpi$, also $g_3$ at $\Omega_\rho$; (c) otherwise | Thm. 6.2, Lemma 6.1 |
| Cor. 6.5 | (c) | Thms. 1.6, 6.2, Baker |
| Thm. 1.10 | (a) at $\varpi$; (b): Masser for $s \in k$ at $\varpi$; (b): $g_3$ at $\Omega_\rho$, (c) for $s \in k$ there; (c) otherwise | Cor. 2.2 (hyperplane case of Thm. 1.1); for $u \in \mathcal{L}_E$ also Laurent's identity and Wüstholz |
| Cor. 1.11 | by item (status line) | Thm. 1.10; Gel'fond–Schneider (M2), Hermite–Lindemann (M3′), Masser (M$\eta$) |
| Prop. 7.1 (single point), Thm. 1.7 | (b): Wüstholz, Masser (for Thm. 1.7), $g_3$; Prop. 7.1 as stated ($K$-span): (c) | Thm. 2.3, Lemma A.1, Lemma 4.2, (F1)–(F4) for $g = 1$, Wüstholz, Masser |
| Cor. 1.8 | (b): as Thm. 1.7, and Laurent (on $E_i$); (c) on $E_\rho$ | Thm. 1.7, [21] |
| Cor. 1.9 | (c) | Wüstholz |
| Prop. 7.5 (single point) | (b): Wüstholz ($\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$), $g_3$; as stated ($K$-span): (c) | Thm. 2.5 (Prop. 3.9; in print as [37, §5d]), Wüstholz, Lemma 4.2, (F1)–(F4) for $g = 1$; in Lean: `lean_lsg`'s Thm. 2.3 for this group |

| result | label | depends on |
|---|---|---|
| Thm. 7.6, Cor. 7.7 | Proof sketch (from Prop. 7.5) | Prop. 7.5, Masser, Wüstholz |
| Prop. 8.1 | (c) | Thm. 2.3, Lemma A.1, (S1), (P), (Ch) |
| Thm. 8.2, Thm. 8.6 | (c) | Prop. 8.1; Wüstholz, (P), (S1), (Ch) |
| Cor. 8.7 | (c) | Thm. 8.6, Prop. 8.1, (W); Beta cocycle, reflection, Chowla–Selberg |
| Cor. 9.1 | (b): Wüstholz $(\varpi, \eta(\varpi) \notin \widetilde{\mathcal{L}})$ at $\varpi$; (c) otherwise | Thm. 5.1, Wüstholz ($\Omega \notin \overline{\mathbb{Q}}\mathcal{L}$), Masser ($\Omega/\pi \notin \overline{\mathbb{Q}}$) |
| Cor. 9.2 | (b): Baker for $\Omega = \varpi$; (c) otherwise | Thm. 5.1, Baker |
| Cor. 9.5(i) | (b): Wüstholz (six-independence), $g_3$ | Prop. 7.5, Wüstholz |
| Cor. 9.5(ii) | (b): Wüstholz (six-independence), $g_3$ | Prop. 7.1, Wüstholz |
| Thm. 5.16, Cor. 5.17 | (c) | Thm. 2.3, Lemma A.1, Lemma 5.15, Lemma 4.2, (F1)–(F4) for $g = 1$ (and (S1), Lemma 4.1 for $X = A$), Wüstholz |
| Lemma 5.15 | Proved (given (F3) for $g = 1$) | universality of $E^\natural$, Lemma 4.2 |
| Thm. 6.9 (SMG-A, SMG$^\natural$-A) | (c) | Thm. 2.3, Lemma A.1 (exponent $b > 1$), Lemma 4.1, (P), (S1), Lemmas 6.4, 6.3, 6.10; (F1)–(F4) for SMG$^\natural$-A |
| Thm. 6.8 (S-A), Cor. 6.11 | (c) ($g \geq 2$); $g = 1$: Thm. 1.6 | Thm. 6.9, Lemma 6.10 (Wüstholz), bilinear relations, Beta dictionary, Baker; (F1)–(F4) ($g \geq 2$) |
| Thm. 6.12 (SE, SE$^\natural$, SE$_0$), Cor. 6.13 | (b): $g_3$ for (1) at $(E_i, E_\rho)$; (a) for (3) on $E_i$; (b): $g_3$ for (3) on $E_\rho$; (c) otherwise | Thm. 2.3, Lemma A.1, Lemma 4.2, (F1)–(F4) for $g = 1$, Lemma 6.14 (Wüstholz), Masser |
| Thm. 6.16 (EA, EA$^\natural$) | (c) | as Thm. 6.12, with Lemma 4.1, (S1), (W); (F1)–(F4) for EA$^\natural$ |
| Lemma 8.12 | Proved | Gauss multiplication, reflection |
| Thm. 8.13, Cor. 8.14 | (c) | Prop. 8.1 (its proof), Lemma 8.12, Koblitz–Ogus labels, Chowla–Selberg, (W) |
| Thm. 8.15 | (c) | proof of Prop. 8.1, Lemma 4.1, (S1), (W); Deligne / Blasius / Anderson for the common slope |
| Thms. 9.6, 5.13, Cors. 5.14, 9.7, §7.7 | Proof sketch (Cor. 5.14 is now contained in Thm. 6.12) | as indicated there |
| non-linearisability (Rems. 5.11, 8.8, §9.3, Cor. 8.14(2), (4)); new instances (Rem. 5.10); §7.8 | Heuristic | period conjecture, Rohrlich–Lang |
| (F1)–(F4) | Proved (given [25]) | Hodge sequence, Poincaré reducibility, universality |
| Thm. 10.1, Cor. 10.2 | (c) | Thm. 2.3, Lemma A.1, Lemma 4.2, (F1)–(F4) for $g = 1$, Wüstholz, Masser, Gel'fond–Schneider |
| Thm. 10.4 | (c) | Thm. 2.3 (codimension 2), Lemma A.1 ($b = 1$), Schneider, Chudnovsky |

| result | label | depends on |
|---|---|---|
| Thm. 10.5, Cor. 10.6 | (c) | Thm. 2.3 (codimension 2), Lemma A.1, Wüstholz (Lemma S, $\kappa$) |
| Thm. 10.8, Lemma 10.7 | (c); Lemma 10.7 for $s = 1$: Proved (Hodge theory) | Thm. 2.3, Lemma A.1, Wüstholz, Gel'fond–Schneider (U3), Masser (U4) |

In more detail, the main results rest on the following.

- Theorem 2.3 [37, Thm. 4.1]. Its proof [37, §§6–7] uses:
  - the auxiliary function with periods and the $\kappa$-saving [36, Prop. 2.1, 2.4];
  - multiplicities along $W$ (Lemma A.1; a proof sketch, checked by two independent internal referees, not peer-reviewed; formalised for the groups listed at the end of Appendix A). Its published inputs are [36, Prop. 2.1, Lemma 2.2], Serre's embedding and height bound [31], Bertrand's [34, App. I, Lemme 7 and its Corollaire, with Lemme 1] (cited in [37, p. 396] as "Lemma 7"), and the lower bound of [36, p. 640];
  - Philippon's zero estimate [26];
  - a complete system of addition laws of one fixed bidegree for the projective embedding of $G_2$. This exists for any quasi-projective commutative group variety by an elementary argument (Remark A.2); for abelian varieties see also Lange–Ruppert [20]. (Serre's appendix [31] gives the compactification with its $G$-action, but not addition laws; cf. also [15], which we have not checked.)
- For Proposition 7.5 (and so for the sketch of Theorem 7.6) and the second derivations (Remarks 5.3, 5.6, 7.2, 8.5): Theorem 2.5, proved in §3.6 (in print as [37, §5d]).
- For Theorem 1.1, in addition to Theorem 2.3 and Lemma A.1: Baker's theorem [3] at corners, Waldschmidt's linear subgroup theorem [40, Thm. 11.21] at linear nodes, and the analytic subgroup theorem (Proposition 3.5, or [52]).
- Legendre's relation; Wüstholz's analytic subgroup theorem [52]; Masser [23]; Baker; the transcendence of $\Omega$ and of $\Omega/\pi$ (Schneider, Masser [23]).
- Standard facts on CM abelian varieties and universal vector extensions: (S1), (F1)–(F4) (§4.1); the Beta dictionary, and the Beta cocycle and Chowla–Selberg [18] at level 7.

For $u \in \mathcal{L}_E$ (all of Corollary 1.11), Theorem 1.10 follows from Laurent's identity and the analytic subgroup theorem; for $u \in \overline{\mathbb{Q}}\mathcal{L}_E$ it rests on Corollary 2.2 in the hyperplane case (Proposition 3.9). Proposition 8.1 and Theorem 6.2(a) use $d_0 = 2$ (as do Theorems 6.9 (SMG-A), 6.12(1), 6.16 (EA), 8.13 and 8.15); Theorem 2.3 allows any $d_0$; [37] itself uses $d_0 = 2$ (Cor. 2.1) and $d_0 = n$ (Cor. 3.3, second proof); and Lemma A.1 is uniform in $d_0$ (by a two-sentence step that is part of trust point T), its hypotheses (2) and $d_2 \geq 1$ being checked in each application (§2.2).

What most needs an expert read, in order: the proof of Theorem 1.1 (§3; checked by two internal referees); Corollaries 8.7(c) and 9.5, which say more about $\Gamma(\frac{1}{7}), \Gamma(\frac{2}{7}), \Gamma(\frac{3}{7}), \pi$, resp. $\pi, \Gamma(1/4), \Gamma(1/3)$, than any known algebraic-independence result (the first lies where [17, 16] say little is known; Corollary 9.5(i) and (ii) rest on Propositions 7.5 and 7.1, which are formally verified in their single-point forms modulo named inputs, §12.3, so what needs the expert read is their mathematical content and the group models); the use of a first-kind relation as the period term (Proposition 8.1); Lemma A.1 (the one step of [37, Prop. 6.1] that is only indicated in print); and the subgroup classifications in §§5, 6, 7 and 8. Among the results added last: the subgroup classifications for the mixed vector-extension groups of Theorems 6.9 (SMG$^\natural$-A, the quotient $Q''$ of $A^\natural$), 6.12(2), (3) and 6.16 (EA$^\natural$), and Lemma 5.15, on which Theorem 5.16 rests; the slot convention of Theorem 6.16; the common-slope input and the distinctness of slots in Theorem 8.15; the slot identifications at levels 9 and 24 (§8.6); and, for §10, the subgroup tables

of Theorems 10.4, 10.5 and 10.8, Lemma S, and Lemma 10.7. The claims that MX, the degree-two statements and Corollaries 1.12, 8.14(2), (4) are not linearisations, and the discussion in §7.8, are heuristic (they use the period conjecture).

Sources we could not consult, any of which might contain these statements: Bertrand [6] (the source of the $K$-structure used in [29]); Waldschmidt [38] (the general form of Theorem 2.3); Baker–Wüstholz [4, §6.2]; Bertrand–Laurent [7]; Faltings–Wüstholz [15]; Roy [28] (decisive for the novelty claim on Theorem 1.1; an inter-library request is pending); Chapters 12–13 of [40] (Chapter 11 was read); Waldschmidt, Acta Arith. 23 (1973) (for Theorem 10.1); any published version of [33]; Bertrand's papers on 1-motives; §§2–7 of [16]; MathSciNet/zbMATH citation lists of [29], [37], [51]. Not checked against the printed source: [32] as the precise source of (S1) and of the isogeny criterion used in §11.2. Numerical values were computed with mpmath (50–60 digits unless stated otherwise); they illustrate and prove nothing. The results were found with substantial use of AI tools; all checks reported here are internal to the project and none is peer review.

12.3. **Formal verification.** A Lean 4 certificate, `lean_mstar` (Lean and Mathlib v4.35.0), checks deductions of this paper from explicitly stated hypotheses. Every declaration depends only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`); there is no `sorry`, no `axiom` declaration and no `native_decide`. Each use of the subgroup theorem is a hypothesis about an explicit *group model* (the tangent space of $G''$ with its $\overline{\mathbb{Q}}$-structure, the logarithms and periods used, and the family of connected algebraic subgroups); named classical inputs enter as further explicit hypotheses, never as axioms.

*Theorem 1.1.* Neither Theorem 1.1 nor its proof in §3 is formalised. What is formalised are instances of Theorem 2.3 and of its hyperplane consequence Theorem 2.5 for the groups listed below, and the deductions of the applications from them.

*The subgroup theorem.* The companion project `lean_lsg` proves Theorem 2.3 under (2), including the multiplicity step of Lemma A.1 and the $\kappa$-saving of [36, Prop. 2.1], for $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ ($E_j$ elliptic curves over $\overline{\mathbb{Q}}$), and for vector extensions of such groups in the form its §5d consequence needs: vector part of dimension $r \le 1$ with $V \cap T(\text{fibre}) = 0$, which the §5d deduction guarantees in this range, under a hypothesis `LiftAlg` (algebraic points of abelian subvarieties lift to algebraic points of their universal vector extensions); and, for $d_0 = 0$, vector part of dimension $r \le 2$ with $\dim(V \cap T(\text{fibre})) \le 1$ (`Theorem41Vec2`, which covers the group of Proposition 7.5), under `LiftAlg` and a hypothesis `PerTanRat` ($T(B^\natural)$ is $\overline{\mathbb{Q}}$-rational for every abelian subvariety $B$). These two hypotheses are true for all elliptic curves over $\overline{\mathbb{Q}}$ but are proved in Lean only for the CM blocks used here. It also uses [37, Thm. 4.1] for linear groups in Roy's form, which is proved in Lean (in the formalisation of Roy's strong six exponentials theorem, a dependency of `lean_lsg` that uses only the standard axioms), not assumed. The internal comparison of Lemma A.1 with `lean_lsg`'s proofs covered the groups $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ and the vector extensions with $r \le 1$; it did *not* cover the $r = 2$ route used for Proposition 7.5 (a separate Lean proof of Theorem 2.3 for that group, Appendix A), whose formal verification therefore rests on the Lean kernel check and on the internal review of its group model only. From these it proves the hyperplane statement Theorem 2.5 for $\mathbb{G}_{\mathrm{m}}^{d_1} \times E$, for the standard model of $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E^m$ (one curve repeated, every period pair with algebraic invariants), for $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ with pairwise equal or non-isogenous curves (used for Theorem 6.12(1)), and for the vector-extension groups below. The Lean proofs therefore follow the route through [37, §5d] (Theorem 2.5; Remarks 5.3, 5.6, 7.2; for Theorem 1.10, the reduction of §3.2 written out and §5d applied to a hyperplane), not the direct application of Theorem 2.3 to the quotient groups in our main proofs; the labels refer to the statements, not to the text of those proofs. Since `lean_lsg` derives the §5d statements from its own Theorem 4.1, handling

the corners inside the proof (as in §3.6), the second application of Theorem 4.1 inside [37, §5d] (§2.3) raises no issue for these groups.

*The bridges.* The groups $\mathbb{G}_{\mathrm{m}}^{2n} \times A''$, $\mathbb{G}_{\mathrm{a}}^{2} \times \mathbb{G}_{\mathrm{m}}^{2n} \times A''$ and $A'' \times \mathbb{G}_{\mathrm{m}}^{n} \times \mathbb{G}_{\mathrm{a}}$ of Theorems 5.1, 6.2(a) and 1.10 are reduced to the standard model by the isomorphism $A'' = E^n/B_{\ker\psi} \cong E^{n/2}$, and the models with elliptic logarithms among the data by the arithmetic of elliptic logarithms: the addition and duplication formulas make $\mathcal{L}_E$ a group, and $\wp(iz) = -\wp(z)$, $\wp(\rho z) = \rho\wp(z)$ make it stable under complex multiplication for $\varpi\mathbb{Z}[i]$ and $\Omega_\rho\mathbb{Z}[\rho]$. All of this is proved in Lean. The vector-extension groups of Theorems 5.4, 6.2(b) and 1.7 (vector part of dimension 1) and of Proposition 7.5 (vector part of dimension 2) are covered by `lean_lsg`'s theorems for vector extensions. So the hyperplane statements (instances of Theorem 2.5) that the Lean proofs use are proved in Lean for these groups, for the group of Proposition 7.5 by the separate route of Appendix A. (Trust point T, the direct use of Theorem 2.3 in our main proofs, is not what Lean checks.) Also proved in Lean: $g_2(\varpi\mathbb{Z}[i]) = 4$, $g_3(\varpi\mathbb{Z}[i]) = 0$, $g_2(\Omega_\rho\mathbb{Z}[\rho]) = 0$, Schneider's theorem ($\varpi$ and $\varpi/\pi$ are transcendental), the transcendence of $\pi$, and, assuming that $g_3(\Omega_\rho\mathbb{Z}[\rho])$ is algebraic, $\Omega_\rho/\varpi \notin \overline{\mathbb{Q}}$ (by the hyperplane statement on $E_i \times E_\rho$); so this form of Wüstholz's theorem is no longer a named input of the results below. Unconditionally, $\pi/\varpi^2$ is transcendental (`Bridge.transcendental_pi_div_varpi_sq`, from the hyperplane statement for the vector extension of $E_i$).

*Formally verified results* (labels (a), (b)). With no hypothesis beyond Lean's axioms (label (a)): Theorem 1.2, both parts, for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ (every algebraic $c \neq 0$; in particular $2^{\varpi}, 3^{\varpi}$ and $2^{\varpi/\pi}, 3^{\varpi/\pi}$ are not both algebraic); Theorem 1.3 at the classes $t \in \overline{\mathbb{Q}}^{\times}B(\frac{1}{5},\frac{1}{5})^{\pm1}$ and $\overline{\mathbb{Q}}^{\times}(B(\frac{1}{5},\frac{1}{5})/\pi)^{\pm1}$ (in particular $2^t, 3^t$ are not both algebraic at $t = \Gamma(\frac{1}{5})^2/\Gamma(\frac{2}{5})$ and at $t = \Gamma(\frac{1}{5})^2/(\pi\Gamma(\frac{2}{5}))$, `Bridge.two_three_beta5'_uncond`, `Bridge.two_three_beta5_uncond`), and the transcendence of $B(\frac{1}{5},\frac{1}{5})/\pi$ and $B(\frac{4}{5},\frac{4}{5})/\pi$, from `lean_lsg`'s hyperplane statement for $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times A^r$, $A$ the Jacobian of $y^5 = x(1-x)$, for toric-injective hyperplanes $V$ and data whose $A$-components are periods, which is all that these proofs use (`CM4E/Bridge/AbVar.lean`; Lean presents $A$ by its period lattice in the coordinates $\int dX/Y$, $\int X\,dX/Y$ on the model $Y^2 = 4X^5+1$ and computes these periods as Beta values; the identification of this torus with the Jacobian is classical and not formalised, and the final statements, about explicit numbers, do not depend on it); by the same route, the $K$-form of Theorem 5.7 at $P = (1-\zeta_5)B(\frac{1}{5},\frac{1}{5})$ with $z_\sigma(\Lambda(A))$ replaced by $\mathbb{Z}P$ (`Bridge.mgA5_K_uncond`). For the classes $\overline{\mathbb{Q}}^{\times}B(\frac{1}{5},\frac{1}{5})^{\pm1}$ the hyperplane statement is `lean_lsg`'s for $\mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times Q''$, $Q''$ the push-out of $r$ copies of a one-dimensional vector extension of $A$ (the second-kind eigenline), again for toric-injective $V$ (no nonzero period of $Q''$ alone in $V$; for the hyperplanes used, this follows from the $F$-independence of $b$ and the fact that the fibre character $\zeta \mapsto \zeta^{N-1}$ is a field embedding) and for data whose $A$-components are periods, the fibre components being arbitrary algebraic; by the same route, the $K$-form of Theorem 5.9 at $P = (1-\zeta_5^4)B(\frac{4}{5},\frac{4}{5})$ with $w_\sigma(\Lambda(A^\natural))$ replaced by $\mathbb{Z}P$ (`Bridge.mgAn5_K_uncond`). Also with label (a): the $K$-forms of Theorems 5.1 and 5.4 at $\varpi$, for the $K$ stated in their status lines; Theorems SMG and SMG$^\natural$ at $\varpi$ with second rows in $\widetilde{\mathcal{L}}$; Theorem 1.10 for $\Omega \in \overline{\mathbb{Q}}^{\times}\varpi$ (first conclusion) with (M1) and (M3); Theorem 6.12(3) ($\mathrm{SE}_0$) for $E = E_i$, every $c \in \overline{\mathbb{Q}}^{\times}$ and every nonzero period of $E_i$ (`Bridge.thmSE0_varpi`, `Bridge.thmSE0_paper`; no form of Theorem 2.5 assumed beyond what `lean_lsg` proves for the vector extension of $E_i$; its group model was refereed, with no fatal or major finding). Formally verified modulo named classical inputs (label (b)): Theorem 1.6 at $\varpi$ (Wüstholz's theorem in the form $\Omega, \eta \notin \widetilde{\mathcal{L}}$, Lemma 6.1); Corollary 9.1 on $E_i$ (the same form of Wüstholz's theorem); Corollary 9.2 for the exponent $\varpi/\pi$, i.e. $\Omega = \varpi$ (Baker's theorem); (M3′) (the Hermite–Lindemann theorem); $s \in \mathbb{Q}(i)$ in Theorem 1.10 at $\varpi$ and (M$\eta$) (Masser's theorem); Proposition 7.1 in its single-point form (one $v \in V \cap \Psi(\Lambda(G))$,

$v \in \overline{\mathbb{Q}}w_1 + \overline{\mathbb{Q}}w_2$) and Theorem 1.7 (Wüstholz's theorem in the form $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$ and, for Theorem 1.7, the linear independence of 1 and elliptic logarithms of $E_i$, $E_\rho$, and Masser's theorem); Corollary 1.8 on $E_i$ (also Laurent's theorem [21]); Corollary 9.5(ii) (Wüstholz's theorem: $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent); Theorem 6.12(1) for $(E, E') = (E_i, E_\rho)$ (every $c \in \overline{\mathbb{Q}}^\times$ and every pair of periods; no further named input, and no form of Theorem 2.5 assumed beyond what `lean_lsg` proves; its group model was refereed); Theorem 6.12(3) for $E = E_\rho$; and the example $\wp_i(\pi/\varpi^2)$, $\wp_i(\pi u/\varpi^2)$ of Example 6.15 (the case $t = \pi/\varpi^2$ of Corollary 6.13 with $\varpi \notin \overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2$, `Bridge.thmSE0_varpi_Ltilde`; its only input is the $\overline{\mathbb{Q}}$-linear independence of $\varpi, 1, u$, a case of Wüstholz's theorem); Corollary 9.3 for $B(\frac{1}{5}, \frac{1}{5})$ (Wüstholz's theorem in the form $P \notin \widetilde{\mathcal{L}}$, $P = (1-\zeta_5)B(\frac{1}{5}, \frac{1}{5})$; `Bridge.corA2_beta5_uncond`). Every statement at $\Omega_\rho$, and Proposition 7.1, Theorem 1.7, Corollaries 1.8 and 9.5(ii) and Theorem 6.12(1), (3) on $E_\rho$, additionally assume that $g_3(\Omega_\rho\mathbb{Z}[\rho])$ is algebraic ($g_3 = 4$ is classical but not formalised).

Also formally verified modulo named classical inputs, by the $r = 2$ route of `lean_lsg` described above: Proposition 7.5 in its single-point form (one $v \in V' \cap \Psi'(\Lambda(G))$, $v \in \overline{\mathbb{Q}}w_1' + \overline{\mathbb{Q}}w_2'$, every $\beta \in \overline{\mathbb{Q}}^\times$; Wüstholz's theorem in the form $\varpi\Omega_\rho/\pi \notin \overline{\mathbb{Q}}$) and Corollary 9.5(i) (Wüstholz's theorem: $1, \pi, \varpi, \eta_i, \Omega_\rho, \eta_\rho$ are $\overline{\mathbb{Q}}$-linearly independent; including the case of a zero coefficient on $\pi$), both also assuming that $g_3(\Omega_\rho\mathbb{Z}[\rho])$ is algebraic. The Lean proof of Corollary 9.5(i) goes through Proposition 7.5 directly, as does ours. The step from Proposition 7.5 to Theorem 7.6 (Masser's and Wüstholz's theorems) is not written out in Lean.

*Results with label (c) checked in Lean conditionally* on Theorem 2.5 for CM abelian varieties of dimension 2 and 3 and their vector extensions (these instances are not formalised): the $K$-forms of Theorems 5.7 and 5.9 for the stated instances (the first-kind eigen-periods $(1-\zeta_N)B(\frac{1}{N}, \frac{1}{N})$ of $y^N = x(1-x)$, $N = 5, 7$, resp. the second-kind eigen-period of $y^5 = x(1-x)$), for every number field $K \supseteq \mathbb{Q}(\zeta_N, i, c)$ (for $N = 5$ with logarithms of $A$, resp. $A^\natural$; the part with $\mathbb{Z}P$ in place of $z_\sigma(\Lambda(A))$, $w_\sigma(\Lambda(A^\natural))$ has label (a), above); Theorem 1.3 at the class $t \in \overline{\mathbb{Q}}^\times B(\frac{1}{7}, \frac{1}{7})/\pi$ (the classes at $B(\frac{1}{5}, \frac{1}{5})$ have label (a), above). This Beta-value instance assumes only the subgroup-theorem hypothesis `Thm11KappaA` at $N = 7$: the input $P/\pi \notin \overline{\mathbb{Q}}$ (Wüstholz's theorem) is derived in Lean from it, and `Thm11KappaA`, `Thm11KappaAn` are stated only for toric-injective hyperplanes (no nonzero period of $A^r$, resp. $Q''$, alone in $V$), which is weaker than the full hyperplane statement; Proposition 8.1 and Theorem 8.2 in general (for any four pairwise distinct first-kind slots of simple CM abelian varieties with a relation (8) taken as hypothesis, as in the paper: Proposition 8.1 for $[K : \mathbb{Q}] \geq 5$, and Theorem 8.2 without its torsion clause; (Ch) is proved in Lean; conditional on the abelian-variety instance of Theorem 2.5 for the block groups); Theorem 8.6 at level seven, including the torsion clause (the torsion step a named input); and Corollary 8.7(a)–(c) ((a), (b) for Abel–Jacobi integrals along paths avoiding the branch points). The level-seven relation $P_1P_2 = \mu' P_3\Omega$ and $B(\frac{1}{5}, \frac{1}{5})B(\frac{4}{5}, \frac{4}{5}) \in \pi\overline{\mathbb{Q}}^\times$ are proved in Lean from the Beta cocycle and Euler's reflection formula. Also checked in Lean, conditionally on its hypotheses but with no form of Theorem 2.5 assumed: Theorem 6.12(1) in the form stated (any two CM lattices with algebraic invariants and CM fields $k \neq k'$, arbitrary nonzero periods $\Omega, \Omega'$), assuming that complex multiplication preserves the elliptic logarithms of algebraic points of both curves (proved in Lean for $\varpi\mathbb{Z}[i]$ and $\Omega_\rho\mathbb{Z}[\rho]$ only, which gives the case $(E_i, E_\rho)$ above).

*Not formalised at all:* Theorem 1.3 for other eigen-periods (each needs its own period data), and the torsion clause of Theorem 8.2 in general; Theorem 1.2 for CM periods outside $\overline{\mathbb{Q}}^\times\varpi \cup \overline{\mathbb{Q}}^\times\Omega_\rho$, and its $t \leftrightarrow 1/t$ extension to $\overline{\mathbb{Q}}^\times/\Omega$ and $\overline{\mathbb{Q}}^\times\pi/\Omega$; the passage to arbitrary $K \supseteq k$ by Baker's theorem; the second sentence of Corollary 5.2; (M2); $s \in k$ in Theorem 1.10 at $\Omega_\rho$; the mixed second rows of SMG and SMG$^\natural$; Corollary 9.2 for $\Omega \neq \varpi$; Propositions 7.1 and 7.5 in their $K$-span forms; the step from Proposition 7.5 to Theorem 7.6; Corollaries 6.5 and 1.9 (the

latter is a named hypothesis); the proof sketches; Lemmas 4.2 and 4.1; Lemma A.1 for general $G_2$; the results of §6.2 for $g \geq 2$ (for $g = 1$ they are Theorems 1.6 and 6.2(a), with the labels stated there); the results of §6.3 except Theorem 6.12(1), (3) (above); the slot identifications and Lemma 8.12 of §8.6 (Theorem 8.13 without its torsion clause is an instance of Theorem 8.2, checked in Lean conditionally in general as stated above, but its level-9 and level-24 data are not formalised), Theorem 8.15; and Lemma 5.15, Theorem 5.16 and Corollary 5.17; Theorem 1.1 and §3; and the results of §10.

*What is not machine-checked.* The group models are hand-written. They were refereed (by internal AI referees), not machine-derived from algebraic geometry: that a model's logarithms, periods and subgroup family are true of the group is checked by these reviews, not by Lean. The same holds for the facts that the Beta expressions are periods of the stated cycles (Weil, Rohrlich) and for Chowla–Selberg, which are checked numerically. No human expert has reviewed the definitions or the proofs. Details, with the Lean name of each statement: the certificate's `TRUST.md` (§§1a, 2, 5, 8).

## Appendix A. Multiplicities along $W$ in the auxiliary function

We use the notation of [37, §6]. $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times G_2$ over a number field $K_0$ with $d_2 \geq 1$ and $G_2$ embedded in $\mathbb{P}_M$ by Serre's embedding (coordinates $\psi_0, \dots, \psi_M$, entire of order $\leq 2$; $\xi = \psi/\psi_0$). $V \subset T_G$ has dimension $N < d$, $W \subset V$ is $K_0$-rational of dimension $t$, $y_1, \dots, y_m \in V$ with $\exp_G(y_j) \in G(K_0)$, $Y(S) = \{\sum h_j y_j : 0 \leq h_j \leq S\}$, $0 \leq \kappa \leq \mathrm{rk}_{\mathbb{Z}}(V \cap \ker \exp_G)$, and $\Delta^{d-N} = C\, S^{d_1+2d_2-\kappa}(\log S)^{bd_0}$, $T(\log S)^a = D_0(\log S)^b = D_1 S = D_2 S^2 = \Delta$.

**Lemma A.1.** *Assume $2N \geq d_1 + 2d_0 + \kappa$, $d_2 \geq 1$, $a, b \geq 1$. For $\nu$ large and $S \geq S_0(\nu)$ there is a polynomial $P_S \in K_0[U, V, W]$ of degree $\leq D_0$ in $U$, $\leq D_1$ in $V$, homogeneous of degree $\leq D_2$ in $W$, not vanishing identically on $G$, which vanishes with multiplicity $\geq T$ along $W$ at every point of $\exp_G(Y(S))$. Hence* [37, Prop. 6.1] *holds under these assumptions.*

*Sketch. Step 1* is [36, Prop. 2.1 and p. 640] unchanged. With $r = \nu^2 S$, $R = \nu^3 S$, it gives a nonzero $Q_S \in \mathbb{Z}[U, V, W]$ of degrees $\leq D_i/\nu$ and coefficients $\leq e^{\Delta/\nu}$ with $|Q_S(\lambda, \mu, \psi)(z)| \leq e^{-\nu^6\Delta}$ for $z \in V$, $|z| \leq \nu^2 S$. The hypothesis $2N \geq d_1 + 2d_0 + \kappa$ is the condition $U \geq r^2 \log(R/r)$ of Prop. 2.1 (printed with $-\kappa$ on p. 641, a misprint). No condition along $W$ enters the linear system, so the saving $r^\kappa$ in the hypothesis of [36, Prop. 2.1, p. 633] is untouched. We also use the lower bound $\max_i |\psi_i(t+z)| \geq e^{-c(\nu S)^2}$ for $t \in Y(\nu S) \cap \mathcal{T}$ and $|z| \leq \rho$, where $Y = \mathcal{T} + F$ with $F$ finite ([36, p. 640], [34, Lemme 1.2.2]; the $\psi_i$ have no common zero [31, Prop. 8]). In Step 2 it bounds the scalar $\mu_j(z)$ in $E^{(j)}(\xi(\gamma), \xi(z)) = \mu_j(z)\psi(t+z)$. (The polynomial variables $U, V, W$ below are not the subspaces $V, W$.)

*Step 2* (derivatives). Fix $t \in Y(\nu S) \cap \mathcal{T}$, $\gamma = \exp_G(t)$, and an addition law $E^{(j)}$ of the embedding with $E^{(j)}(\xi(\gamma), \xi(0)) \neq 0$. For $z = \sum s_i w_i \in W$ small put $F_t(s) = Q_S(\lambda(\gamma) + \lambda(z), \mu(\gamma)\mu(z), E^{(j)}(\xi(\gamma), \xi(z)))$. By homogeneity, $F_t$ is a unit times $Q_S(\lambda, \mu, \psi)(t+z)/\psi_0(t+z)^{D^\circ}$, so it has the same order along $W$. Its Taylor coefficients $\theta'$ of order $< T$ satisfy $|\theta'| \leq e^{-\nu^6\Delta + c\nu\Delta}$ (Cauchy). They are polynomials, with no denominator depending on $\gamma$, in the coefficients of $Q_S$, the coordinates of $\gamma$, and the Taylor coefficients of $\lambda, \mu, \xi$ along $W$ at the origin. The latter are $K_0$-rational with house and denominator $\leq c^{|\tau|+1}(|\tau|!)^c$: by Bertrand's Corollaire to his Lemme 7 [34, App. I, p. 177] the coordinates of the exponential map have finite arithmetic functional order, every point being regular, and his Lemme 1 [34, App. I, pp. 166–167] turns this into such bounds for derivatives ([37, p. 396] cites this as "Lemma 7"). Since $|\tau| < T$ and $T \log T \leq c\Delta$ (as $a \geq 1$ and $\log \Delta \leq c \log S$), Serre's height bounds for $\gamma$ then give $h(\theta') \leq c\nu\Delta$ with $c$ independent of $\nu$. Liouville's inequality forces $\theta' = 0$ once $\nu^5$ exceeds a

constant. (Dividing by a power of $E_0^{(j)}(\xi(\gamma),\xi(0))$ instead would cost a height $T(\nu S)^2$, which is too large.)

*Step 3* (translation). Write $Y(S) \subset (Y(\nu S)\cap\mathcal{T}) + F'$ with $F'$ finite. For $f \in F'$ take the product over the addition laws $E^{(j)}$ with $E^{(j)}(\cdot,\xi(-\exp f)) \not\equiv 0$ of $Q_S$ translated by $-\exp f$; then take the product over $f \in F'$. Translations commute with invariant derivations, and each factor is a unit times the translate on a dense open set, so the product vanishes with multiplicity $\geq T$ along $W$ on $\exp_G(Y(S))$. It is not identically zero on $G$, and its degrees are $\leq D_i$ once $\nu$ exceeds (number of laws)·(bidegree)·$|F'|$. No norm to $\mathbb{Q}$ is taken, since [37, Prop. 6.1] allows $\overline{\mathbb{Q}}$-coefficients.

The exponent $b \geq 1$ changes only the number of unknowns, which is compensated exactly by the factor $(\log S)^{bd_0}$ of $\Delta^{d-N}$, and lowers the height term $(D_0/\nu)\log S$. In the proof of [37, Thm. 4.1], $b > 1$ is used only in the case $\delta = \tau$, which requires $d_0 \geq 1$ (Theorems 1.7 and 6.2, Proposition 8.1, the results of §§6.2, 6.3, 8.6, Theorem 10.1, and the non-linear nodes of §3 with $d_0 \geq 1$). □

**Remark A.2** (Addition laws)**.** Let $G_2 \subset \mathbb{P}_M$ be locally closed over $K_0$, and $X = G_2 \times G_2 \subset \mathbb{P}_M \times \mathbb{P}_M$. The local ring of $X$ at a point $p$ is a quotient of that of $\mathbb{P}_M \times \mathbb{P}_M$, whose elements are ratios of bihomogeneous forms of equal bidegree. Clearing denominators, near $p$ the group law $X \to G_2 \subset \mathbb{P}_M$ is given by bihomogeneous forms $(A_0,\dots,A_M)$ of one bidegree, not all zero at $p$. By quasi-compactness finitely many such systems suffice, and multiplying each by suitable monomials $x_i^a y_l^b$ gives a complete system of addition laws of one bidegree $(\beta,\beta)$, with coefficients in a finite extension of $K_0$. (This replaces $K_0$ by a finite extension in Step 2, which only changes $[K_0:\mathbb{Q}]$ in the Liouville inequality.) Step 2 uses these laws only on $G_2(K_0) \times \exp(\text{a neighbourhood of 0 in } W)$, and Step 3 only with a fixed second argument.

**Remark A.3** (Scope of Lemma A.1; what Theorem R needs)**.** [Proof sketch] Steps 1–3 use only the following properties of $G_2$: entire coordinates of order $\leq 2$ with a theta denominator, and their height bounds (Serre's embedding [31, Prop. 5, Prop. 8], valid for every connected commutative $G_2$); the lower bound for $\max_i|\psi_i|$ near $Y(\nu S)\cap\mathcal{T}$ ([36, Lemme 2.2, p. 640]); the derivative bounds of Bertrand [34, App. I, Lemme 7 and Corollaire, Lemme 1], stated there for an arbitrary connected commutative group over a number field; and addition laws of one bidegree (Remark A.2). Step 1, [36, Prop. 2.1], needs only that $\Omega$ be discrete (its periods are $\mathbb{R}$-independent); for abelian blocks it contains Masser's interpolation theorem [36, Lemme 2.3]. So the sketch applies, as written, to:

(a) *abelian* $G_2$ of any dimension (the CM abelian varieties of §§5.3, 6.2, 8), with Masser's theorem as a black box;

(b) *vector extensions* (push-outs of $(E_1\times\cdots\times E_m)^\natural$ or $A^\natural$ along algebraic linear maps), with the fibre coordinates $w-\zeta(z)$ made entire by multiplying with $\sigma$ (resp. theta functions); $V$ may meet the fibre (Proposition 7.5, Theorem 6.12(2), (3), Theorem 5.16);

(c) *every* $d_0$, by the last paragraph of the proof (the exponent $b$);

(d) $G_2$ *containing factors* $\mathbb{G}_{\mathrm{a}}$ *or* $\mathbb{G}_{\mathrm{m}}$, as in the trivial decomposition $d_0 = d_1 = 0$, $G_2 = G$ of Proposition 3.5, where (2) reads $2N \geq \kappa$ and always holds; Serre's embedding of such a $G_2$ is again [31];

(e) *arbitrary connected commutative* $G_2$, in particular non-split semi-abelian groups (such as a nontrivial extension of an elliptic curve by $\mathbb{G}_{\mathrm{m}}$) and mixed extensions of an abelian variety by $\mathbb{G}_{\mathrm{a}}^a \times \mathbb{G}_{\mathrm{m}}^b$, which Theorem 1.1($\alpha$), ($\gamma$) cover. Each property used holds for them: Serre's embedding and its height bounds [31, Prop. 5, Prop. 8] are stated for every connected commutative group variety over a number field; [36, Prop. 2.1, Lemme 2.2] are analytic statements that use nothing about $G_2$ beyond such an embedding, and W83 applies them in its Prop. 2.4 to an arbitrary commutative $G_2$; Bertrand's [34, App. I,

Lemme 7] is stated for an arbitrary connected commutative group; and addition laws of one bidegree exist by Remark A.2.

It does not apply to linear $G$ ($d_2 = 0$): there Theorem 1.1 uses Waldschmidt's complete proof for linear groups (Lemma 3.8), and the corners use Baker's theorem (Lemma 3.7). Outside (2) it does not apply (Remark 3.12). Lemma A.1 is formally verified only for $G$ isogenous to $\mathbb{G}_{\mathrm{a}}^{a} \times \mathbb{G}_{\mathrm{m}}^{b} \times E_1 \times \cdots \times E_m$ (and for the vector extensions listed below); this class is closed under the connected subgroups and quotients that occur in the proof of Theorem 1.1, by (P) of §4.1. Writing Steps 1–3 in full for general $G_2$ (cases (a), (b), (d), (e)) is the main remaining writing task at level (R-P) of Theorem 1.1; we expect no new mathematics there [Heuristic].

*What is formalised.* The companion Lean project `lean_lsg` (§12.3) proves Lemma A.1, in a form at least as strong for these groups (with $a = 1$, integer $b \geq 1$, some constant $C$, and a Weierstrass–Segre model in place of Serre's embedding), for $G = \mathbb{G}_{\mathrm{a}}^{d_0} \times \mathbb{G}_{\mathrm{m}}^{d_1} \times E_1 \times \cdots \times E_m$ (`LSG.W88.prop61_b`), including the $d_0$-part of the last paragraph of the proof; and for push-outs of $(E_1 \times \cdots \times E_m)^{\natural}$ along an algebraic linear map, under the additional hypothesis $V \cap T(\mathrm{fibre}) = 0$ (`LSG.AuxV.prop61V_b`). For the group of Proposition 7.5 ($d_0 = 0$, vector part of dimension 2 meeting $V$) `lean_lsg` proves Theorem 2.3 by a separate route (its `Theorem41Vec2`); that route has not been compared with Lemma A.1 line by line, and the internal comparison of Lemma A.1 with `lean_lsg` did not cover it. Lemma A.1 as stated, for general $G_2$, is not formalised.

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